



XR872 User Manual

*High Performance Wireless MCU for Artificial Intelligence
and Internet of Things (AIoT) Applications*

Revision 1.0

Dec 26, 2019

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Revision History

Revision	Data	Summary of Changes
1.0	2019-12-26	Initial Version

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Chapter 1 About This Documentation

1.1 Documentation Overview

This documentation provides an overall description of the Xradio XR872 wireless application processor, which will provide instructions to programmers from several sections, including application system, memory, network and interfaces.

1.2 Acronyms and Abbreviations.

The table below contains acronyms and abbreviations used in this document.

A		
AES	Advanced Encryption Standard	A specification for the encryption of electronic data established by the U.S. National Institute of Standards and Technology (NIST) in 2001
AGC	Automatic Gain Control	An adaptive system found in electronic devices that automatically controls the gain of a signal: the average output signal level is fed back to adjust the gain to an appropriate level for a range of input signal levels
AHB	AMBA High-speed Bus	A bus protocol introduced in Advanced Microcontroller Bus Architecture version 2 published by ARM Ltd company
APB	Advanced Peripheral Bus	APB is designed for low bandwidth control accesses, which has an address and data phase similar to AHB, but a much reduced, low complexity signal list (for example no bursts)
C		
CIR	Consumer IR	The CIR (Consumer IR) interface is used for remote control through infra-red light
CRC	Cyclic Redundancy Check	A type of hash function used to produce a checksum in order to detect errors in data storage or transmission
CSI	CMOS Sensor Interface	The hardware block that interfaces with different image sensor interfaces and provides a standard output that can be used for

		subsequent image processing
D		
DES	Data Encryption Standard	A previously predominant algorithm for the encryption of electronic data
DLL	Delay-Locked Loop	A digital circuit similar to a phase-locked loop (PLL), with the main difference being the absence of an internal voltage-controlled oscillator, replaced by a delay line
DVFS	Dynamic Voltage and Frequency Scaling	Dynamic voltage scaling is a power management technique where the voltage used in a component is increased or decreased, depending on circumstances. Dynamic frequency scaling is a technique whereby the frequency of a microprocessor can be automatically adjusted on the fly so that the power consumption or heat generated by the chip can be reduced. These two are often used together to save power in mobile devices.
E		
eMMC	Embedded Multi-Media Card	An architecture consisting of an embedded storage solution with MMC interface, flash memory and controller, all in a small BGA package.
I		
I2S	IIS	An electrical serial bus interface standard used for connecting digital audio devices together
L		
LSB	Least Significant Bit	The bit position in a binary integer giving the units value, that is, determining whether the number is even or odd. It is sometimes referred to as the right-most bit, due to the convention in positional notation of writing less significant digits further to the right.
M		
MAC	Media Access Control	A sublayer of the data link layer, which provides addressing and channel access control mechanisms that make it possible for several terminals or network nodes to communicate within a multiple access network that incorporates a shared medium, e.g. Ethernet.

MSB	Most Significant Bit	The bit position in a binary number having the greatest value, which is sometimes referred to as the left-most bit due to the convention in positional notation of writing more significant digits further to the left
P		
PCM	Pulse Code Modulation	A method used to digitally represent sampled analog signals
S		
SPI	Synchronous Peripheral Interface	A synchronous serial data link standard that operates in full duplex mode. Devices communicate in master/slave mode where the master device initiates the data frame.

Chapter 2 Overview

2.1 General Description

XR872 delivers a highly integrated high performance Wi-Fi wireless System-on-Chip (SoC) solution designed for Artificial Intelligence and Internet of Things (AIOT) applications, such as smart audio, face detection and recognition, Machine-to-Machine (M2M), Smart Home, Cloud Connectivity and Smart Energy.

The XR872 application subsystem is powered by an ARM Cortex-M4F CPU that operates up to 384MHz. It supports an integrated 416KB SRAM and 160KB ROM, a QSPI interface to SIP up to 16MB Flash, a OPI interface to SIP 4MB PSRAM. Integrated I-cache enables Execute In Place (XIP) from flash and PSRAM, and integrated D-cache enables read and write from PSRAM. It also includes many peripherals, including UART, TWI, SPI, DMIC, AUDIO CODEC, PWM, CIR (T/R), CSI, SDIO and auxiliary ADC.

The Wi-Fi subsystem contains the 802.11b/g/n baseband, MAC and radio with integrated PA, LNA, Switch and harmonic filter, which is design to meet both the low power, high integration and high performance network application. A novel digital RF transmitter is design using XRADIOTECH's MPD™ technology to deliver higher output power and maintain higher efficiency, and also to keep the chip not sensitive to antenna mismatch but always have good EVM at different VSWR.

The SoC is optimized for low-power operation by using several low-power state and fast wake-up times from hardware to software. Multiple power domains and clocks can be shut down individually. The application subsystem and Wi-Fi subsystem can be put into low-power states independently, supporting a variety of application use cases. Also an optional external DC-DC regulator can provide voltage from 1.5V to 2.5V for whole VDD_ANA power domain with on chip DC-DC control signal, without sacrificing standby power consumption even using high quiescent current DC-DC.

2.2 Features

- General System Features

Table 2-1 XR872 Features

chip List	Description	XR872AT	XR872ET
Package	Trays and tape-in-reel	6x6mm ²	5x5mm ²
		QFN52	QFN40
Supply voltage	Power supply from system	1.8~5.5V	
PMU	LDO for external device (EXT LDO)	Yes	
External Clock	Reference High frequency clock	24/26/40	

MCU Core	Core Type		Cortex-M4F	
	Core clock maximum frequency		384MHz	240MHz
Memory	Internal ROM		160KB	
	Internal RAM		416KB	416KB
	Internal Flash		/	/
	External Flash with XIP		128Mb	
	Cache (can be configured to system ram)		40KB	40KB
	PSRAM		4MB	/
Backup register	Backup register for power save		16B	
Secure boot			Yes	
Crypto Engine	AES/DES/3DES/SHA/MD5/CRC		Yes	
TRNG	Provide random number seed		Yes	
Watchdog reset protection	Protect specified peripherals been reset by watchdog reset		Yes	
BOR	BOR Detection		Yes	
Wi-Fi	802.11 b/g/n		Yes	
Peripheral	GPIO	General Purpose	35	27
	UART		3	3
	SPI	Master and slave	2	2
	TWI	Max.400Kbps	2	2
	SDIO	Master	1	1
	GPADC	VBAT	1	1
		Normal	7	2
	DMA	8Channel	1	1
	I2S		1	0
	DMIC		1	0
	CODEC	x1 ADC	1	1

			x1 DAC	1	1
			x1 Line-in	1	0
		CSI+JPEG		1	1
		RTC		1	1
		Timer	Watchdog	1	1
			Normal	2	2
			Wakeup	1	1
		PWM	Output	8	8
			Input capture	8	8
		CIR	Receiver	1	1
			Transmitter	1	1
		32K	Internal RCOSC	1	1
			External XTAL	1	0
		WAKEUP IO	From RTC wakeup	9	9

● Audio Subsystem

- 1 Digital Microphone Controller with left and right channel voice input
- 1 24bit audio digital-to-analog(DAC) channel, support sample rates from 8KHz to 192KHz
- 1 24bit audio analog-to-digital(ADC) channel for microphone input, support sample rates from 8KHz to 48KHz
- 1 24bit audio-to-digital(ADC) channel for line-in, support sample rates from 8KHz to 48KHz
- DMA and interrupt support

● Video Subsystem

- a CMOS Sensor Interface Controller(CSIC) which is an image or video input control module to receive image or video data via digital camera(DC)interface, CCIR656 interface
- Supports JPEG base line
- Supports online encoding mode with CSI
- Supports nv12 input format in offline encoding mode
- Supports configurable picture resolutions

.1. Minimum picture resolution: 32x32

- .2. Offline maximum picture resolution: 1920x1088
- .3. Online maximum picture resolution: 1920x1088
- Supports online 640x480@60fps or online 1280x720@40fps
- Supports offline 640x480@30fps or offline 1280x720@20fps
- Supports bitrate control through quantization table
- **Wi-Fi Subsystem**
- IEEE 802.11b/g/n, 1x1 SISO 2.4GHz
- Integrated MAC, BB, RF and Embedded TCP/IP Stack
- Integrated T/R switch, harmonic filter, PA and LNA
- Antenna diversity
- Station, AP Modes
- Smart-Config Technology for Autonomous and Fast Wi-Fi Connections
- Security support for WEP, WPA/WPA2 personal, WPS2.0
- Industry-Standard BSD Socket Application Programming Interfaces (APIs)
- **Power Management**
- Integrate highly flexibility power management unit by several LDOs and external DC-DC controller
- Wide range power supply: 1.8-5.5V
- **Miscellaneous**
- Integrates 1Kbit eFuse to store device specific information and RF calibration data

2.3 Application

- | | |
|--------------------|---------------------------|
| – Smart Audio | – Smart Home |
| – Smart Video | – Access Control |
| – Security Systems | – Cloud Connectivity |
| – Smart Energy | – Industrial Control |
| – Internet Gateway | – IP Network Sensor Nodes |

2.4 Block Diagram

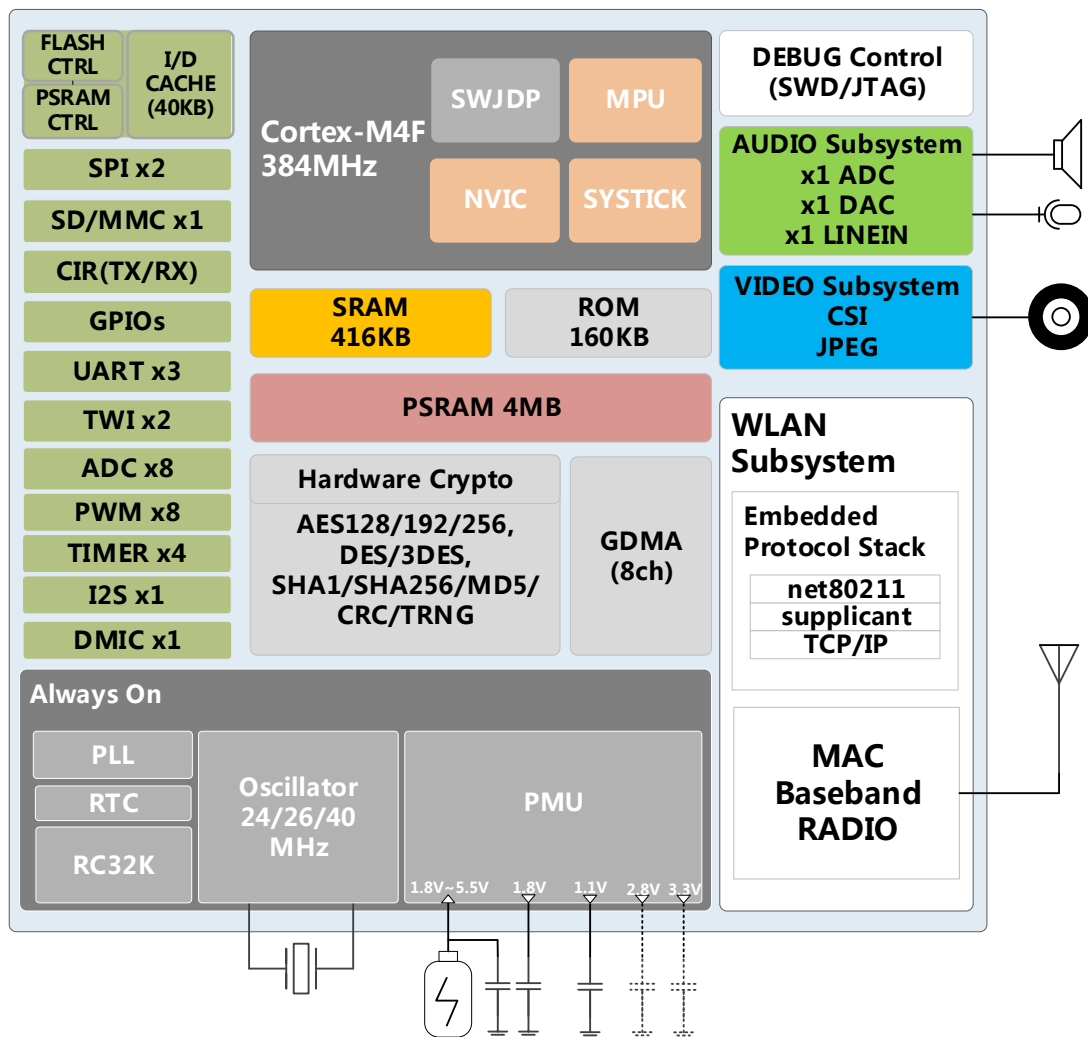


Figure 2-1 XR872 Functional Block Diagram

Chapter 3 System

3.1 Power Management

A single 1.8~5.5V power supply is required for the XR872. It could be from an AC-DC converter, USB to supply to 5V or a DC-DC converter to convert higher voltage supply to 3.3V or even lower. It could be from a battery directly too, no matter it is lithium, single 3V button or 2 serial NI-MH battery.

The Power Management Unit (PMU) contains a DC-DC control interface, several Low Drop-out Regulators (LDOs), and a reference band-gap circuit. The circuits are optimized for low quiescent current, low drop-out voltage, load regulation, high ripple rejection, and low output noise. The PMU integrates several LDOs for different circuits: TOP LDO, RTC LDO, SoC LDO, EXT LDO, as shown in Figure 2-1. They have different operating conditions and features:

- **TOP LDO** provide programmable voltage from 1.4V to 3.6V with maximum 350mA load current, for Analog and PSRAM, also SoC LDO input. Normally, make sure VBAT voltage is higher than this programmable output voltage setting.
- **RTC LDO** is the main supply only for RTC domain to optimize power consumption at HIBERNATION state.
- **SoC LDO** is the main supply for whole chip digital circuit with programmable voltage from 0.6V to 1.35V to let DVFS operate effectively.
- **EXT LDO** is main power supply for external device in application, and also can be provide to VDDIO, GPADC and internal CODEC. It has maximum 200mA load current. The output voltage is limited to 3.3/3.1V (by register configure setting), when VBAT is lower than the value, it will automatically switch to bypass mode to let output voltage follow VBAT.

When using external DC-DC to further reduce power consumption, the connection can be set as figure 2-1. PA23 GPIO pin is specified for external DC-DC pup, the detail software setting and flow is integrated in our SDK. When external DC-DC is used, the output voltage on VDD18 should be at least 0.1V step higher than TOP LDO setting voltage to have DC-DC be operating normally.

There are four power domains in the system: RTC domain, OA domain, Digital Core domain and Wi-Fi domain. They mainly used for different scenario to maintain ultra-low power application. We define XR872 into **ACTIVE**, **STANDBY**, **HIBERNATION** and **SHUTDOWN** power management states, is shown in table 2-1.

Table 3-1 Power Management States

POWER MODE	CM4F	Wi-Fi	EXT LDO	RTC LDO	TOP LDO	SoC LDO	DCXO	Description
							/DPLL	
ACTIVE	ACTIVE	ACTIVE	ON	ON	ON	ON	ON	All CPU active
	ACTIVE	OFF	ON	ON	ON	ON	ON/OFF	APP CPU active

STANDBY	SLEEP	ACTIVE	ON	ON	ON	ON	ON	APP CPU goes to sleep, Wi-Fi DTIMx state
	SLEEP	SLEEP	ON	ON	LP	LP	OFF	
	SLEEP	OFF	ON	ON	LP	LP	OFF	APP CPU goes to sleep, Wi-Fi power off
HIBERNATE	OFF	OFF	ON/OFF	ON	OFF/LP	OFF	OFF	Only RTC on, waiting for timer or wakeup IO to interrupt
SHUTDOWN	OFF	OFF	OFF	OFF	OFF	OFF	OFF	CHIP_PWD pin keep low level

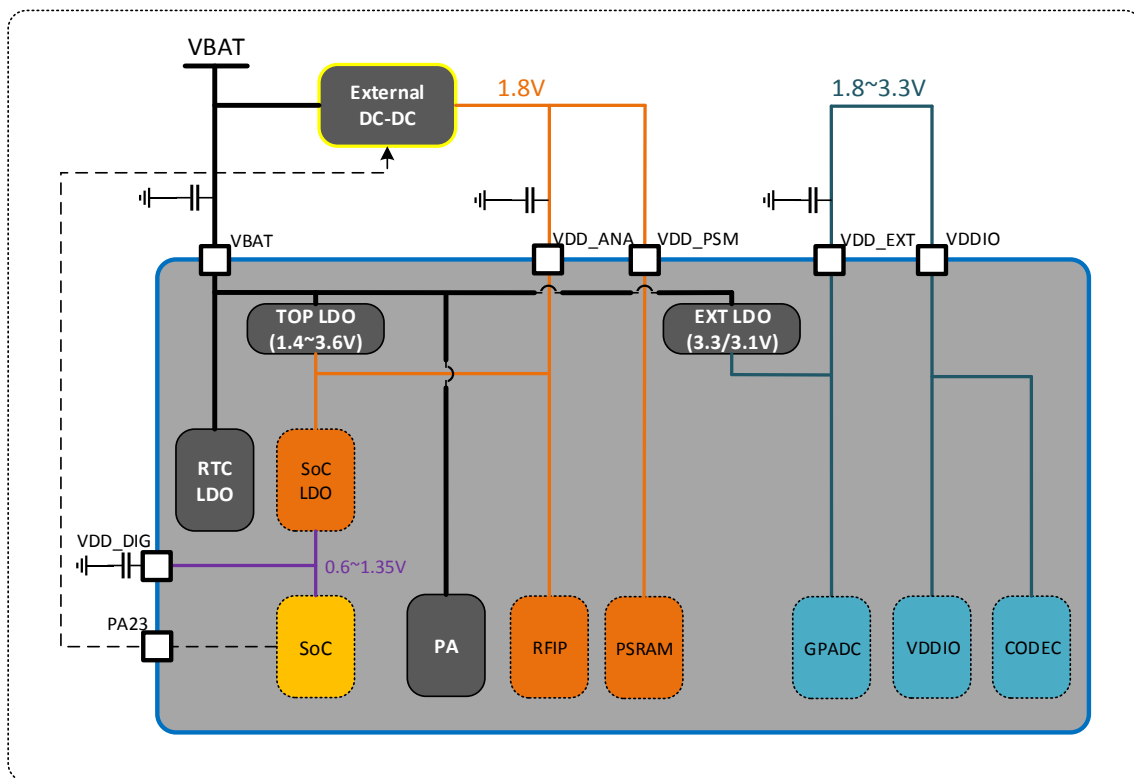


Figure 3-1 Power Architecture

3.2 Clock

The clock management system can source the system clocks from a range of internal or external high and low frequency oscillators and distribute them to modules based upon module's individual requirements. The system generates two different clocks: a high frequency clock HFCLK and a low frequency clock LFCLK.

The system supports two LFCLK clock sources, the 32.768 KHz crystal oscillator and the 32.768 KHz RC oscillator. The 32.768 KHz crystal oscillator requires an external AT-cut quartz crystal to be connected to the LXTAL1 and LXTAL2 pins. The LFCLK clock and all of the available LFCLK sources are switched off by default when the system is powered up. The LFCLK clock can be started by selecting the preferred clock source in PRCM register. It is used for each subsystem to achieve lower current consumption for different running mode. In addition, the LFCLK is also

used in RTC circuit to achieve accuracy timing.

There is only one clock source for HFCLK, generated by the 24MHz, 26MHz, or 40MHz crystal oscillator. The HFCLK is enabled automatically when the system is powered up and can be switched off when all subsystems won't use it anymore in some low power modes.

The HFCLK is used to generate the clock source for Digital PLL, which is used to generate the clock sources for Cortex-M4F core, Wi-Fi and peripherals. There is also an Audio PLL used to generate the clock source for Audio Subsystem.

The following figure shows the clock control block diagram.

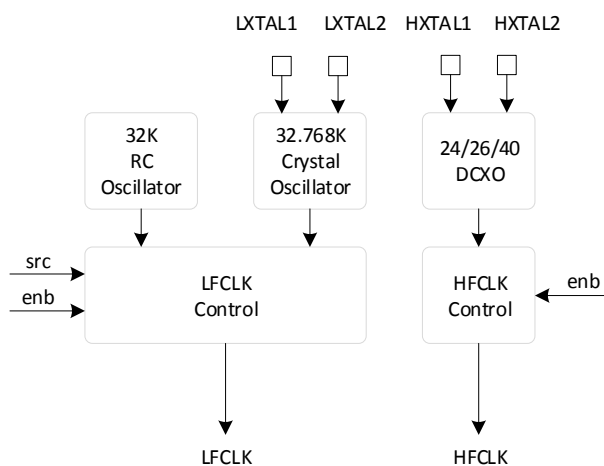


Figure 3-2 XR872 Clock Control

3.3 Power State and Power Sequence

3.3.1 Power-on Sequences

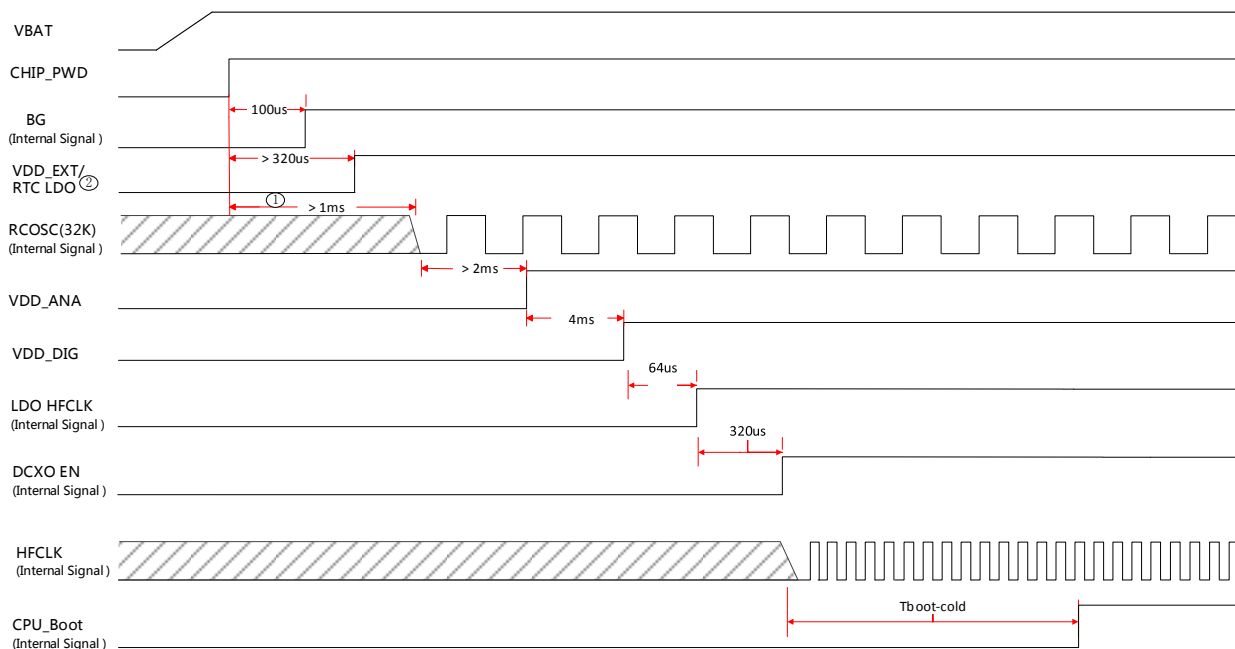


Figure 3-3 Power-on Sequence

3.3.2 Wakeup from Standby

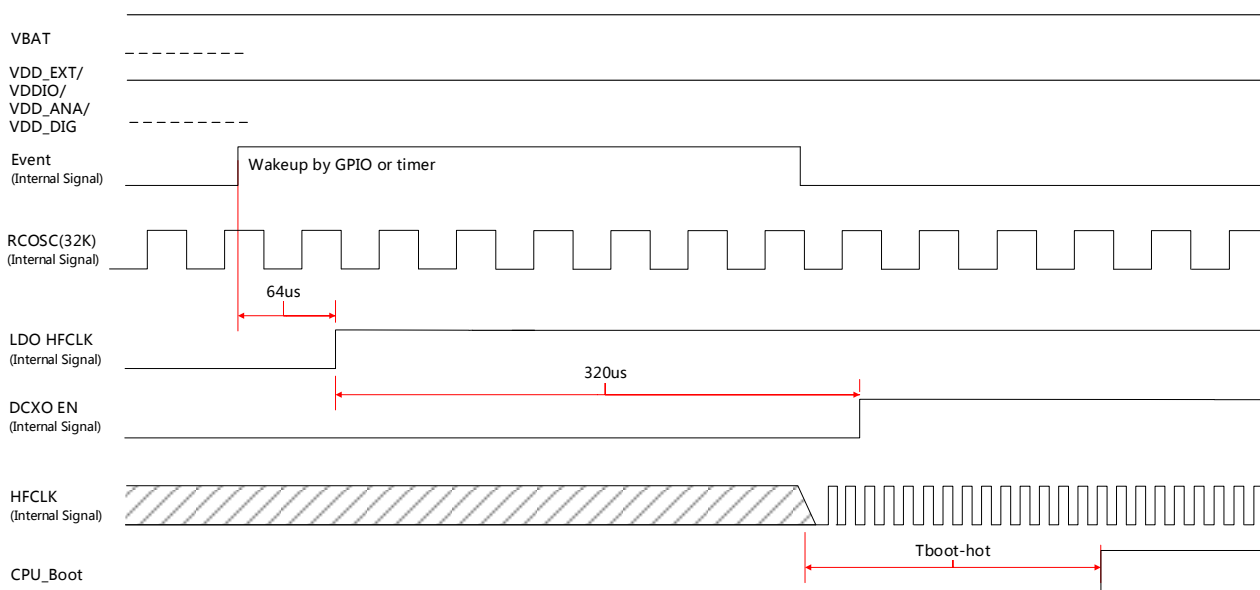


Figure 3-4 Wakeup from standby Sequence

3.3.3 Wakeup from Hibernation

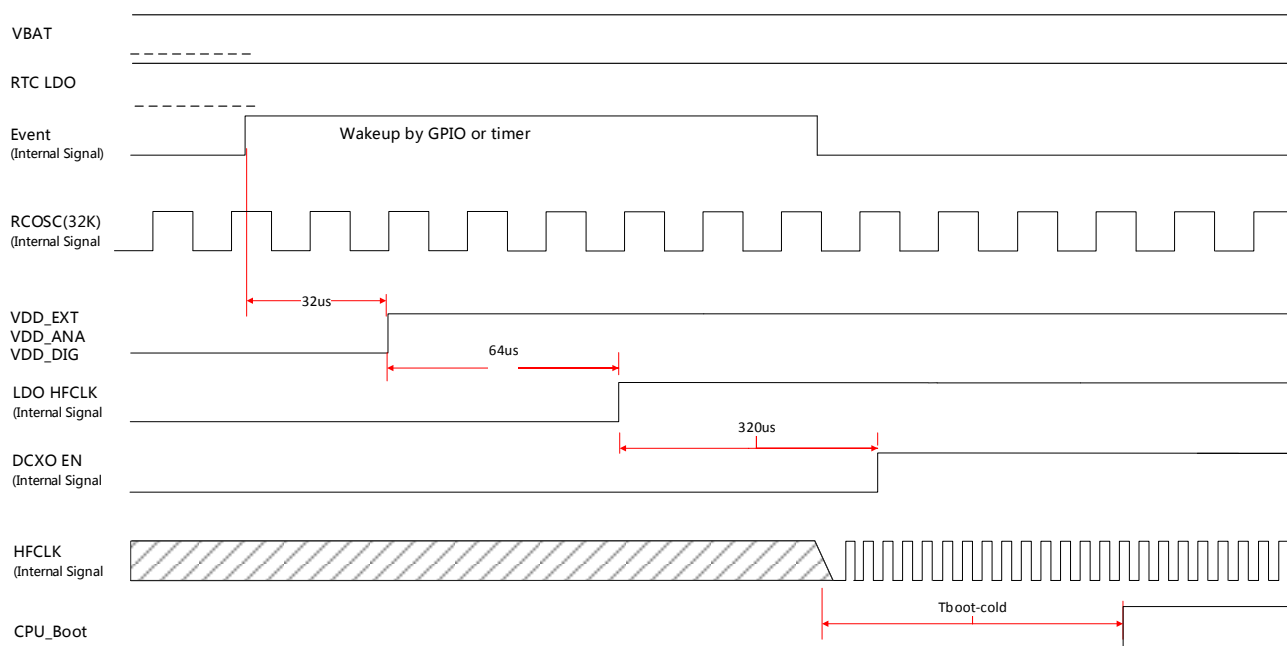


Figure 3-5 Wakeup from hibernation Sequence

3.3.4 Shutdown Sequence

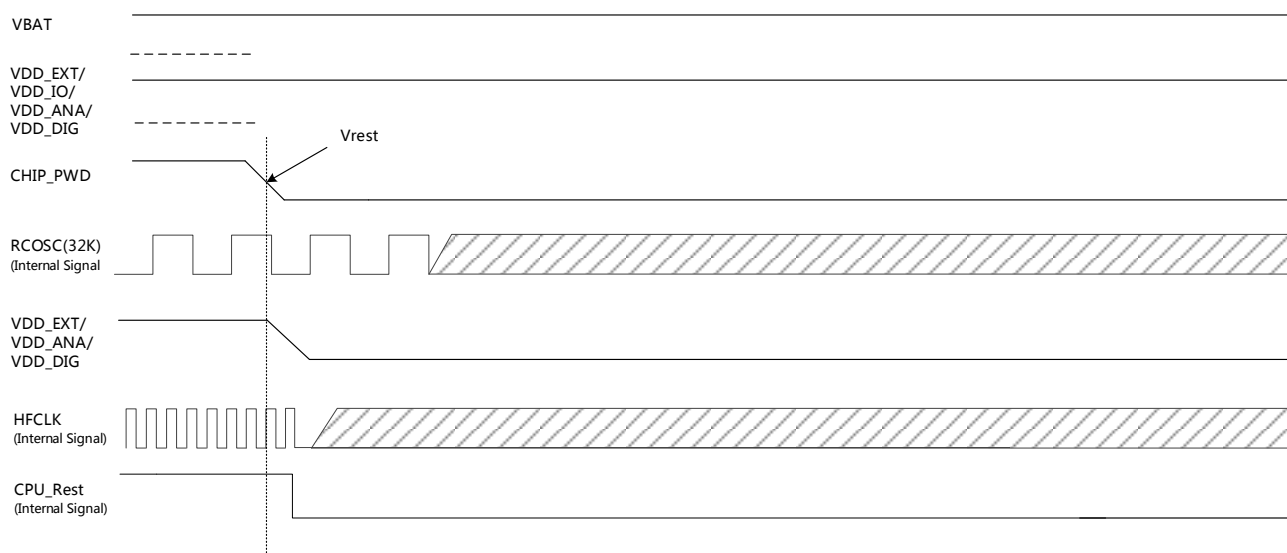


Figure 3-6 Shutdown Sequence

Notes:

- ① RCOSC is inaccurate, and allowing 32k start-up point with deviation.
- ② RTC LDO shared capacitance with SOC LDO(VDD_DIG).

3.4 Memory Mapping

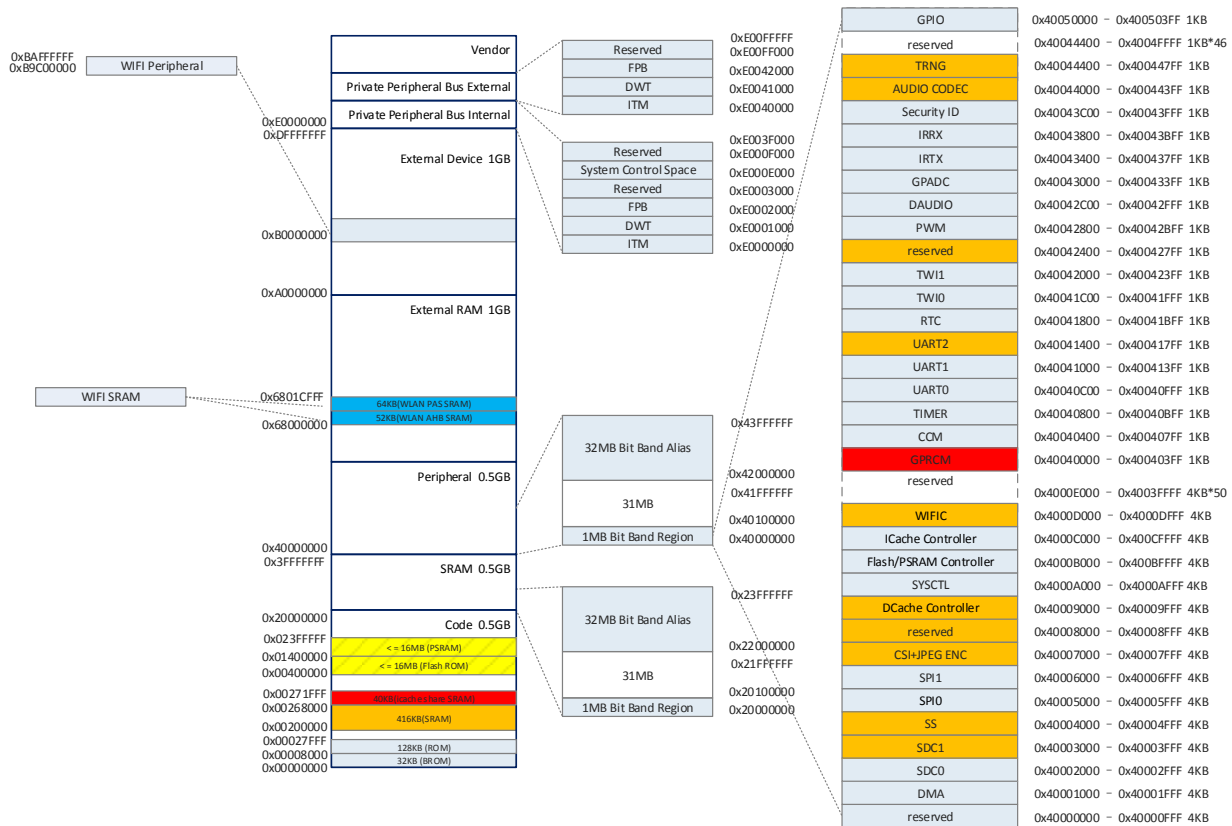


Figure 3-7 XR872 Memory Mapping

3.5 CPU System

XR872 features an ARM Cortex-M4F processor, which is the most energy efficient ARM processor available. It supports the clock rates from 32KHz up to 384MHz. The processor provides a low-cost platform that meets the needs of minimal memory implementation, reduced pin count, and low power consumption.

The ARM Cortex-M4 core has low-latency interrupt processing with the following features:

- Thumb-2 instruction set for optimal performance and code size
- Handler and thread modes
- Memory Protection Unit (MPU) for memory protection features
- Floating Point Unit (FPU) to support DSP related function
- Nested Vectored Interrupt Controller (NVIC) to achieve low latency interrupt processing
- Three Advanced High-Performance bus AHB-Lite interfaces: ICode, DCode and system bus
- Bit-band support for memory and select peripheral that include atomic bit-band write and read

operations

- Wake-up Interrupt Controller (WIC) providing ultra-low power sleep mode support

Note: Detail information of Cortex-M4, NVIC and MPU please refer to 'ARMv7-M Architecture Reference Manual' and 'ARM Cortex-M4 Processor Technology Reference Manual'.

3.5.1 Register List

Module Name	Base Address	
Nested Vectored Interrupt Controller	0xE000E000	
Register Name	Offset Address	Description
	0x004	
SYSTICK_CTRL	0x010	SysTick Control and Status Register
SYSTICK_RELOAD	0x014	SysTick Reload Value Register
SYSTICK_CURVAL	0x018	SysTick Current Value Register
NVIC_ISER0 – NVIC_ISER7	0x100 – 0x11C	Interrupt Set-Enable Registers
NVIC_ICER0 – NVIC_ICER7	0x180 – 0x19C	Interrupt Clear-Enable Registers
NVIC_ISPR0 – NVIC_ISPR7	0x200 – 0x21C	Interrupt Set-Pending Registers
NVIC_ICPR0 – NVIC_ICPR7	0x280 – 0x29C	Interrupt Clear-Pending Registers
NVIC_IABR0 – NVIC_IABR7	0x300 – 0x31C	Interrupt Active Bit Registers
NVIC_IPR0 – NVIC_IPR59	0x400 – 0x4CF	Interrupt Priority Registers
SCB	0xD08 – 0xD3f	System Control Block

3.5.2 Interrupt Source Description

IRQ Number	Vector	Interrupt Source	Priority	Description
0	0x00			Stack top is loaded from first entry of vector table on reset.
1	0x04		-3	Reset
2	0x08		-2	NMI
3	0x0C		-1	Hardware Fault
4	0x10		configurable	Memory Management
5	0x14		configurable	Bus Fault
6	0x18		configurable	Usage Fault
7	0x1C			-
8	0x20			-
9	0x24			-
10	0x28			-
11	0x2C		configurable	SVCall

12	0x30		configurable	Debug Monitor
13	0x34			-
14	0x38		configurable	PendSV
15	0x3C		configurable	System Tick
16	0x40	DMA	configurable	Direct Memory Access Controller
17	0x44	GPIOA	configurable	GPIO Controller Port A
18	0x48	SDC	configurable	SDMMC/SDIO Controller
19	0x4C			
20	0x50	UART0	configurable	UART Controller 0
21	0x54	UART1	configurable	UART Controller 1
22	0x58	SPI0	configurable	SPI Controller 0
23	0x5C	SPI1	configurable	SPI Controller 1
24	0x60	TWI0	configurable	TWI Controller 0
25	0x64	TWI1	configurable	TWI Controller 1
26	0x68	WDT	configurable	Watchdog
27	0x6C	TIMER0	configurable	Timer 0
28	0x70	TIMER1	configurable	Timer 1
29	0x74	ALARM0	configurable	Alarm 0
30	0x78	ALARM1	configurable	Alarm 1
31	0x7C	CSI_JPE	configurable	CSI_JPE Controller
32	0x80	DAUDIO	configurable	Digital Audio Controller
33	0x84	PWM/ECT	configurable	PWM/ECT Controller
34	0x88	SS	configurable	Secure System Controller
35	0x8C	GPADC	configurable	GPADC Controller
36	0x90	GPIOB	configurable	GPIO Controller Port B
37	0x94			
38	0x98	IRRX	configurable	IR Receiver
39	0x9C	IRTX	configurable	IR Transmitter
40	0xA0			
41	0xA4	A-WUP	configurable	APP Wake Up(Timer & IO)
42	0xA8	FLASHC	configurable	FLASH Controller
43	0xAC	UART2	configurable	UART2 Controller2
44	0xB0	SDC1	configurable	SDMMC1/SDIO1 Controller
45	0xB4	WLAN	configurable	WLAN
46	0xB8	CODEC DAC	configurable	AUDIO CODEC DAC
47	0xBC	CODEC ADC	configurable	AUDIO CODEC ADC
48	0xB0	AVS	configurable	AVS psensor
49	0xB4	GPIOC	configurable	GPIO Controller Port C
50	0xB8	PSRAMCTRL	configurable	PSRAM controller

1.1 System Control (SYSCTRL)

1.1.1 Overview

1.1.2 Register List

Module Name	Base Address	
System Control	0x4000A000	
Register Name	Offset Address	Description
SIP_FLASH_TEST_MAP	0x44	SIP FLASH Test Mapping Register
WLAN_SRAM_SHARE	0x48	WLAN SRAM share control Register
PS_CTL_REG	0x50	Psensor control Register
PS_CNT_REG	0x54	Psensor counter Register
GENERAL_DBG0_REG	0xB0	General Debug 0 Register
GENERAL_DBG1_REG	0xB4	General Debug 1 Register

1.1.3 Register Description

1.1.3.1 SIP FLASH Test Mapping Register

Address : 0x0044			Name: SIP_FLASH_TEST_MAP Default: 0x0000_0000
Field	Type	Default	Description
31:16	RW	0x0	0x429d: SIP_FLASH_TEST/ SIP_PSRAM_TEST filed is writable other: SIP_FLASH_TEST/ SIP_PSRAM_TEST filed is read only Note: The field of SIP_FLASH_TEST/SIP_PSRAM_TEST only can be changed when these bits are set to 0x429D. In one write operation, the GRPCM will check this field whether equals to 0x429D. If not, the value of SIP_FLASH_TEST/SIP_PSRAM_TEST will be ignored. This field is always read with value 0x0000. Examples: - After write 0x429D0001, then read value = 0x00000001 - After write 0x11110000, then read value = 0x00000001 - After write 0x429D0000, then read value = 0x00000000
15:2	/	/	/
1	RW	0x0	SIP_PSRAM_TEST 0: SIP PSRAM IOs mapping is disabled 1: SIP PSRAM IOs are mapped to the external PINs

0	RW	0x0	SIP_FLASH_TEST 0: SIP FLASH IOs mapping is disabled 1: SIP FLASH IOs are mapped to the external flash PINs
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1.1.3.2 WLAN SRAM share control Register

Address : 0x0048			Name: WLAN_SRAM_SHARE Default: 0x0000_0000
Field	Type	Default	Description
31:17	/	/	/
16	RW	0x0	PINMUX_FUNC3_MCU_MOD PINMUX mapping to MCU mode when it is func3. 0: not mcu mode 1: mcu mode
15:10	/	/	/
9:8	RW	0x0	CSI_JPE_SHARE_SRAM if image width <=720, use share 32k SRAM case, else use share 64k SRAM case. 00: not share 01: share 32k bytes SRAM 10: share 64k bytes SRAM 11: reserved
7:3	/	/	/
2:0	RW	0x0	WLAN_SRAM_SHARE_CTRL 000: APP CPU access the 128K SRAM 001: WLAN AHB access the 32K SRAM 010: WLAN AHB access the 64K SRAM 011: WLAN AHB access the 96K SRAM 100: WLAN AHB access the 128K SRAM

1.1.3.3 Psensor control Register

Address : 0x0050			Name: PS_CTL_REG Default: 0x0000_0000
Field	Type	Default	Description
31:17	/	/	/
16	RW1C	0x0	CLK250M_CNT_RDY clk250M counter ready 0: not finish 1: finish
15:11	/	/	/
10:8	R/W	0x0	PS_DEV_SEL psensor device sel

			000: select psensor_0 001: select psensor_1 Others: reserved
7:6	/	/	/
5:4	R/W	0x0	PS_EN_SEL OSC select 00:disable(no dynamic power) 01 : rvt_cascode ring osc 10: lvt_cascode ring osc 11:rvt_normal ring osc
3:1	R/W	0x0	PS_N_PRD clk num period 0: 4 --- 2/13 us (26M HOSC), 0.1us (40M HOSC) 1: 8 --- 4/13 us (26M HOSC) , 0.2us (40M HOSC) 2: 16---8/13 us (26M HOSC) , 0.4us (40M HOSC) 3: 32---16/13us (26M HOSC) , 0.8us (40M HOSC) 4: 64---32/13 us (26M HOSC) , 1.6us (40M HOSC) 5: 128---64/13 us (26M HOSC) , 3.2us (40M HOSC) 6: 256---128/13us (26M HOSC) , 6.4us (40M HOSC) 7: 512---256/13us (26M HOSC) , 12.8us (40M HOSC)
0	R/W	0x0	PS_EN psensor Enable 0: disable 1: psensor enable

1.1.3.4Psensor counter Register

Address : 0x0054			Name: PS_CNT_REG Default: 0x0000_0000
Field	Type	Default	Description
31:16	/	/	/
15:0	R	0x0	CLK250M_CNT clk 250M counter

1.1.3.5General Debug 0 Register

Address : 0x00B0			Name: GENERAL_DBG0 Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0x0	GENERAL_DBG0 General debug register 0 is used for storing flag or some data.

1.1.3.6 General Debug 1 Register

Address : 0x00B4			Name: GENERAL_DBG1 Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0x0	GENERAL_DBG1 General debug register 1 is used for storing flag or some data.

3.6 Global Power, Reset and Clock Manager (GPRCM)

3.6.1 Overview

The GPRCM module manages the power, reset and input clocks for this system. The GPRCM is placed in always on power domain.

The GPRCM module includes the following features:

- Manage the power of this system
- Manage the reset of each system
- Manage the OSC clock
- Manage the wakeup source

Note: All register should be placed in RTC power domain and will keep the value unchanged when the system is powered off by disabling the internal LDO. The system can automatically start up when detecting external events like GPIO or power events. The value of this register should be read correctly after the system starts up.

3.6.2 Register List

Module Name	Base Address	
Power, Reset and Clock Manager	0x40040000/0xa0040000	
Register Name	Offset Address	Description
SYS_TOP_LDO_CTRL	0x0000	System TOP_LDO Control
SYS_LDO_SW_CTRL	0x0004	System LDO & Switch Control
SYS_LFCLK_CTRL	0x0008	System Low Frequency Clock Control
SYS_HOSC_TYPE	0x000C	System HOSC Type
SYS_RCOSC_CAL_CTRL	0x0010	System RC-OSC Calibration Control
SYS_PLL_CTRL	0x0020	System PLL Control
SYS_CLK1_CTRL	0x0024	System Clock 1 Control
SYS_CLK2_CTRL	0x0028	System Clock 2 Control
SYS_CLK3_CTRL	0x002C	System Clock 3 Control

AUDIO_PLL_CTRL	0x0030	Audio PLL Control
DEV_CLK_CTRL	0x0034	Device Clock Control
AUDIO_PLL_BIAS	0x0044	
AUDIO_PATTERN	0x0054	Audio Pattern Control
DCXO_CTRL	0x0058	DCXO control
HXTALOUT_CTRL	0x005c	HXTALOUT enable control
SYS1_CTRL	0x080	System 1 Control
SYS1_STATUS	0x084	System 1 Status
SYS2_CTRL	0x088	System 2 Control
SYS2_STATUS	0x08C	System 2 Status
SYS3_CTRL	0x090	System 3 Control
SYS3_STATUS	0x094	System 3 Status
SYS1_WAKEUP_CTRL	0x098	System 1 wakeup control
SYS2_WAKEUP_CTRL	0x09C	System 2 wakeup control
SYS1_SLEEP_CTRL	0x0A0	System 1 Sleep control
SYS2_SLEEP_CTRL	0x0A4	System 2 Sleep control
DCXO_STABLE_REF_TIME	0x0A8	DCXO Stable Reference Time
DPLL_STABLE_REF_TIME	0x0AC	DPLL Stable Reference Time
LDO_STABLE_REF_TIME	0x0B0	LDO Stable Reference Time
DIGITAL_SWITCH_REF_TIME	0x0B4	Digital Switch Reference Time
SRAM_VOL_CTRL	0x0B8	SRAM Voltage Control
BANDGAP_STABLE_REF_TIME	0x0BC	Band Gap Stable Reference Time
TOP_LDO_STABLE_REF_TIME	0x0C0	TOP_LDO Stable Reference Time
AR400A_BOOT_FLAG	0x0100	AR400A boot flag
AR400A_BOOT_ADDR	0x0104	AR400A boot address
AR400A_BOOT_ARG	0x0108	AR400A boot argument
AR400N_PRIV_REG0	0x010C	AR400A Private Register
AR400N_PRIV_REG1	0x0110	AR400A Private Register
AR400N_PRIV_REG2	0x0114	AR400A Private Register
AR400A_PRIV_REG3	0x0118	AR400A Private Register
AR400A_PRIV_REG4	0x011c	AR400A Private Register
AR400A_WAKE_TIMER_CNT	0x0120	AR400A wakeup timer counter
AR400A_WAKE_TIMER_CTRL	0x0124	AR400A wakeup timer control
AR400A_PRIV_REG5	0x0128	AR400A Private Register
AR400A_PRIV_REG6	0x012c	AR400A Private Register
AR400A_IO_WAKE_EN	0x0130	AR400A IO Wakeup Enable
AR400A_IO_WAKE_MOD	0x0134	AR400A IO Wakeup Mode

AR400A_IO_WAKE_ST	0x0138	AR400A IO Wakeup Status
AR400A_IO_HOLD_CTRL	0x013c	AR400A IO Hold Control
AR400A_IO_WUP_GEN	0x0140	AR400A IO Wakeup Global Enable
LDO_MODE_SW_SEL	0x0144	LDO Mode Software Select
TOP_LDO_PARAM	0x0200	TOP_LDO Parameter Register
ANA_BANDGAP	0x0204	ANA Bandgap Control Register
CLK_LDO_PARAM	0x0208	Clock LDO Parameter Register
DIG_LDO_PARAM	0x020C	Digital LDO Parameter Register
DPLL_ST	0x0210	DPLL Status Register
CFG_IO_STATUS	0x214	Config IO Status
CPU_RST_SRC	0x218	CPU reset source register
WLAN_HIF_OV_CTRL	0x21c	WLAN HIF override control register
SRAM_BIST_CFG	0x220	SRAM configure register

3.6.3 Register Description

3.6.3.1 System TOP_LDO Control Register

Address : 0x0000			Name: SYS_TOP_LDO_CTRL0 Default: 0x0008_0001
Field	Type	Default	Description
31:20	/	/	/
24	R/W	0x0	TOP_LDO_FORCE_ACTIVE 0: don't force TOP_LDO mode 1: TOP LDO always in active mode when it is enable (most high priority)
19:16	R/W	0x00	TOP_LDO_VOLTAGE_SEL (1.4V – 3.6V) N: 0~15 0: 1.8V 1: 1.4V 2: 1.7V 3: 1.8V 4: 1.9V 5: 2.0V 6: 2.1V 7: 2.4V 8: 2.5V 9: 2.6V 10: 2.9V 11: 3.0V

			12: 3.1V 13: 3.3V 14: 3.4V 15: 3.6V the default is 1800mV Note: This bit will only be reset to the default value in power-on-reset operation. When the system is waked up from the deep sleep mode by some external events, this bit will keep the value which is set by the last write operation.
15:1	/	/	/
0	R	1	SYS_TOP_LDO_EN 1: Write 1 to enable the system TOP_LDO. 0: write 0 to disable the system TOP_LDO. Note: This bit will only be reset to the default value in power-on-reset operation. When the system is waked up from the deep sleep mode by some external events, this bit should be set to 1 automatically to make sure that the CPU can start up correctly.

3.6.3.2 System LDO & Switch Control Register

Address : 0x0004			Name: SYS_LDO_SW_CTRL Default: 0x0000_0707
Field	Type	Default	Description
31:27	/	/	/
27:24	R/W	0	LDO1_RET_VOL_SEL LDO1 retention voltage select 0 ~ 12: N 0: 1125mV 1: 1025mV 2: 925mV 3: 825mV 4: 725mV 5: 625mV 8: 1175mV 9: 1225mV 10: 1275mV 11: 1325mV 12: 1375mV

			others: reserved <i>the default is 1125mV</i>
23:19			
18	R/W	0	EXT_LDO_VOL_SEL EXT_LDO voltage select 0: 3.3V 1: 3.1V <i>the default is 3.3V</i>
17:16	R/W	0x2	EXT_LDO_MODE EXT_LDO mode 00: always off 01: on/off same as TOP_LDO 1x: always on
15	R	0	SR_SW3_EN WLAN System SRAM Power Switch Enable 1: enable 0: disable <i>Note: this bit will only be set by WLAN system.</i>
14	R	0	SW5_EN WLAN System Power Switch 2 Enable 1: enable 0: disable <i>Note: this bit will only be set by WLAN system.</i>
13	R	0	SW4_EN WLAN System Power Switch 1 Enable 1: enable 0: disable <i>Note: this bit will only be set by WLAN system.</i>
12:9	/	/	/
8	R	1	SW1_STATUS AR400A and Application System Power Switch status 1: SW1 is on 0: SW1 is off <i>Note: this bit will only be set by POWERCTRL module.</i>
7:4	R/W	0	LDO1_VOL_SEL 0 ~ 12: N

			0: 1125mV 1: 1025mV 2: 925mV 3: 825mV 4: 725mV 5: 625mV 8: 1175mV 9: 1225mV 10: 1275mV 11: 1325mV 12: 1375mV others: reserved <i>the default is 1125mV</i>
3	/	/	/
2	R	1	LDO1_STATUS 1: LDO1 is on. 0: LDO1 is off. <i>Note: this bit will only be set by POWERCTRL module.</i>
1	R/W	1	PLL_LDO_EN 1: Write 1 to enable the PLL_LDO_EN. 0: write 0 to disable the PLL_LDO_EN.
0	R	1	LDO1_EN 1: Write 1 to enable the LDO1. 0: write 0 to disable the LDO1. <i>Note: this bit will only be set by POWERCTRL module.</i>

3.6.3.3 System LFCLK Control Register

Address : 0x0008			Name: SYS_LFCLK_CTRL Default: 0xc000_000f
Field	Type	Default	Description
31	R/W	1	OSC32K_EN 1: enable the 32768 Oscillator 0: disable the 32768 Oscillator
30	R/W	1	RC32K_EN 1: enable the 32768 RC Oscillator

			0: disable the 32768 RC Oscillator
29:25	/	/	/
24	R/W	0	LFCLK_SRC_SEL Low frequency clock source select 0: source the LFCLK from the internal 32768 RC oscillator. 1: source the LFCLK from the external 32768 crystal oscillator
23:8	R/W	0	PAD_CLOCK_OUT_FACTOR_M factor M, M= Factor + 1 Output clock frequency Fout = Fsrc/M
2:1	R/W	0	PAD_CLOCK_OUT_SEL GPIO PAD 32KOSCO clock source select 0: source the LFCLK from the internal 32768 RC oscillator. 1: source the LFCLK from the external 32768 crystal oscillator 2,3: source the HFCLK from the external crystal oscillator
0	R/W	0	PAD_CLOCK_OUT_EN GPIO PAD 32KOSCO clock output enable. 0: disable 1: enable

3.6.3.4 System HOSC Type Register

Address : 0x000C			Name: SYS_HOSC_TYPE Default: 0x0000_0000
Field	Type	Default	Description
31:2	/	/	/
1:0	R/W	0	HOSC_TYPE 0: System uses 26MHz HOSC 1: System uses 40MHz HOSC 2: System uses 24MHz HOSC 3: System uses 52MHz HOSC <i>Note: This bit is set by application software and must be consistent with the real situation. A wrong configuration will cause the RCOSC calibration fail.</i>

3.6.3.5 System RCOSC Calibration Control Register

Address : 0x0010	Name: SYS_RCOSC_CAL_CTL L Default: 0x0000_0001
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Field	Type	Default	Description
31:28	/	/	/
27:8	R	0xXXX	RC-OSC Frequency V=FREQ/10
7:1	/	/	/
0	R/W	1	RC-OSC Calibration Enable

3.6.3.6 System PLL Control Register

Address : 0x0020			Name: SYS_PLL_CTRL Default: 0x6000_0501 @24MHz OSC, 0x1B13_B491 @26MHz OSC, 0x6000_0301 @40MHz OSC, 0x1D89_D241 @52MHz OSC
Field	Type	Default	Description
31	R/W	0	PLL_ENABLE Enable or disable the system PLL. 0: Disable the system PLL 1: Enable the system PLL Note: The PLL frequency $F_{output} = (F_{hosc} * N.f) / M$, the output clock must be in range of 960~1920MHz. The default value is set for 1920MHz with 24MHz OSC.
30:29	RW	0	DPLL_DITHER_DISABLE Enable or disable the PLL dither. 0: Disable the system PLL dither 1: Enable the system PLL dither
28:13	RW	0x7627	DPLL_FRAC Note : $V_{FRAC} = f * 2^{16}$
12	RW	1	DPLL_FRAC_CTRL 0: Disable 1: Enable
11:4	RW	18	DPLL_NDIV PLL factor N
3:0	RW	1	FACTOR_M PLL factor M (1 ~ 8). M = Factor Note: when the factor is set to 0, M will be set to 1

3.6.3.7 System Clock 1 Control Register

Address : 0x0024			Name: SYS_CLK1_CTRL Default: 0x0000_000b
Field	Type	Default	Description
31:18	/	/	/
24	R/W	0x0	ROM_READ_TWO_CYCLES_MODE 0: one cycle ROM read. 1: two cycles ROM read
23:18	/	/	/
17:16	R/W	0x0	CPU_CLK_SRC_SEL CPU clock source select 00: HFCLK(default) 01: LFCLK 1x: SYSCLK1
15:11	/	/	/
10:8	RW	0x1	FACTOR_M PLL factor M (5~9) 0~4: M= Factor + 5 5~6: reserved 7: M=5.5
7:4	/	/	/
3:0	RW	0x3	FACTOR_N PLL factor N (1 ~ 16). Factor = 0~15 N = Factor + 1 Note: $F_{sysclk1} = 1920\text{MHz} / M/N$, the default value is 80MHz.

3.6.3.8 System Clock 3 Control Register

Address : 0x002c			Name: SYS_CLK3_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31	RW	0	SYS_CLK3_EN 1: enable the clock of system 3 0: disable the clock of system 3 Note: $F_{sysclk} = F_{src} / N$, the clock is fixed to 160MHz.
30:0	/	/	/

3.6.3.9 Audio PLL Control Register

Address : 0x0030			Name: AUDIO_PLL_CTRL Default: 0x0103_0601@26MHz, 0x0103_0100@40MHz
Field	Type	Default	Description
31	RW	0	PLL_EN 1: enable the PLL 0: disable the PLL Note: <i>This PLL is used for Audio application.</i> $AUDIO_PLL = (HFCLK * N) / (M * P)$ $AUDIO_PLL_8X = (HFCLK * N * 2) / (M * P)$ $3 \leq N / M \leq 21$ $HFCLK * N / P$ must be in the range of 74MHz ~ 504MHz AUDIO_PLL default is 24.576MHz
30:29	/	/	/
28	R	x	LOCK 0: Unlocked 1: Locked (means the pll is stable)
27:25	/	/	/
24	RW	1	PLL_SDM_EN 0: Disable 1: Enable In this case, the PLL_FACTOR_N only low 4 bits are valid (1~16)
23:20	/	/	/
19:16	RW	0x3	PLL_POSTDIV_P Post-div factor (P = Factor + 1) the range if from 1 to 16
15	/	/	/
14:8	RW	0x06	PLL_FACTOR_N N = Factor + 1 the range if from 1 to 128
7:5	/	/	/
4:0	RW	0x01	PLL_PRETDIV_M Pre-div factor (M = Factor + 1) the range if from 1 to 32

3.6.3.10 Device Clock Control Register

Address : 0x0034			Name: DEV_CLK_CTRL Default: 0x0000_0004
Field	Type	Default	Description
31	RW	0x0	DEV2_CLK_EN DEVCLK2 enable 0: disable 1: enable
30:27	/	/	/
26:24	RW	0x0	DEV2_FACTOR_M PLL factor M (5~9) 0~4: M= Factor + 5 5~7: reserved
23:20	/	/	/
19:16	RW	0x1	DEV2_FACTOR_N PLL factor N (1 ~ 16). Factor = 0~15 N = Factor + 1 <i>Note: $F_{devclk2} = 1920\text{MHz} / M/N$, the default value is 192MHz. this clock is used as a source clock only for PSRAM OPI controller.</i>
15:11			
10:8	RW	0x0	DEV1_FACTOR_M PLL factor M (5~9) 0~4: M= Factor + 5 5~7: reserved
7:4	/	/	/
3:0	RW	0x1	DEV1_FACTOR_N PLL factor N (1 ~ 16). Factor = 0~15 N = Factor + 1 <i>Note: $F_{devclk1} = 1920\text{MHz} / M/N$, the default value is 192MHz. this clock is used as a source clock for some devices like SDC, SPI, CSI, Specail APB etc.</i>

3.6.3.11 Audio PLL Bias Register

Address : 0x044			Name: AUDIO_BIAS Default: 0x1010_0000
Field	Type	Default	Description
31:29	/	/	/

28:24	RW	0x10	PLL_VCO_BIAS_CTRL PLL VCO Bias Control[4:0]
23:21	/	/	/
20:16	RW	0x10	PLL_CUR_BIAS_CTRL PLL Current Bias Control[4:0]
15:0	/	/	/

3.6.3.12 Audio PLL Pattern Control Register

Address : 0x0054			Name: AUDIO_PLL_PATTERN_CTRL Default: 0xC001_1FAA@26MHz, 0xC000_EA4A@40MHz
Field	Type	Default	Description
31	RW	1	DIG_DELT_PAT_EN Sigma-Delta Pattern Enable
30:29	RW	0x2	SPR_FREQ_MODE Spread Frequency Mode 00: DC = 0 01: DC = 1 1X: Triangular
28:20	RW	0	WAVE_STEP Wave Step
19	/	/	/
18:17	RW	0	FREQ 00: 31.5KHz 01: 32KHz 10: 32.5 KHz 11: 33 KHz
16:0	RW	0x11faa	WAVE_BOT Wave Bottom

3.6.3.13 DCXO Control Register

Address : 0x058			Name: DCXO_CTRL0 Default: 0x9409_0b00
Field	Type	Default	Description
31	RW	1	DCXO_EN DCXO Enable 0: disable 1: enable
30	/	/	/

29:28	RW	1	CLK_MODE_SEL Clock Mode Select (When AUTO_DET_EN is disable) 00: Internal Clock 01: Crystal 10: Analog 11: Digital
27:20	RW	0x40	FRE_TRI Frequency Trimming
19	RW	1	ALC_EN ALC Enable 0: disable 1: enable
18:17	/	/	/
16:12	RW	0x10	ICTRL BIAS Current Control
11	RW	1	RF_DIG_REF_CLK_EN 0: disable 1: enable
10	/	/	/
9	RW	1	BUFFER_DPLL_EN 0: disable 1: enable
8:6	/	/	/
5:4	R	xx	DCXO_DET 00: Internal clock 01: Crystal 10: Analog 11: Digital
3:1	/	/	/
0	RW	0	AUTO_DET_EN Auto Detect Enable 0: disable 1: enable

3.6.3.14 HXTALOUT Enable Control Register

Address : 0x05c			Name: HXTALOUT_EN_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/

0	RW	0	HXTALOUT_EN enable HXTALOUT buffer 0: disable 1: enable
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3.6.3.15 System 1 Control Register

Address : 0x080			Name: SYS1_CTRL Default: 0x0000_0003
Field	Type	Default	Description
31:2	/	/	/
1	R	1	AR400A_RST CPU AR400A Reset 1: release the reset signal of AR400A 0: hold the global reset signal of AR400A
0	R	1	SYS1_RST system 1 Reset 1: release the global reset signal of the system 1 0: hold the global reset signal of the system 1

3.6.3.16 System 1 Status Register

Address : 0x0084			Name: SYS1_STA Default: 0x0000_0000
Field	Type	Default	Description
31:4	/	/	/
3	W/R	0	AR400N_SLEEP_LOCK 1: means the AR400N has not go to the DEEPSLEEP mode 0: ignore <i>Note: This bit is used as a software flag for AR400N's system. If this bit is set to 1, the software must keep the power of the AR400N and system2 on so that the AR400A can access the resources of the system2. At this time the AR400N can only go into the SLEEP mode with clock gated.</i>
2	R	0	CPU_SLEEP 1: the CPU of system 1 is in SLEEP mode 0: the CPU of system 1 is not in SLEEP mode
1	R	0	CPU_SLEEPDEEP 1: the CPU of system 1 is in SLEEPDEEP mode

			0: the CPU of system 1 is not in SLEEPDEEP mode
0	R	0	SUBSYS1_ALIVE 1: system 1 is alive 0: system 1 is down

3.6.3.17 System 2 Control Register

Address : 0x0088			Name: SYS2_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:3	/	/	/
2	RW	0	SYS2_ISOLATION_CTRL Isolation control for AR400N and system2 1: disable the isolation for system 2 0: enable the isolation for system2 <i>Note: set this bit to 0 to cut off the connection with the whole circuits of system2. So you must set this bit before disabling the SW3 and SR_SW2</i>
1	RW	0	AR400N_RST CPU AR400N Reset 1: release the reset signal of AR400N 0: hold the global reset signal of AR400N
0	RW	0	SYS2_RST System 2 Reset 1: release the global reset of the system 2 0: hold the global reset of the system 2

3.6.3.18 System 3 Control Register

Address : 0x0090			Name: SYS3_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	RW	0	SYS3_RST System 3 Reset 1: release the global reset of the system 3 0: hold the global reset of the system 3

3.6.3.19 System 3 Status Register

Address : 0x0094			Name: SYS3_STA Default: 0x0000_0000
Field	Type	Default	Description
31:3	/	/	/
2	R	0	CPU_SLEEP 1: the CPU of system 3 is in SLEEP mode 0: the CPU of system 3 is not in SLEEP mode
1	R	0	CPU_SLEEPDEEP 1: the CPU of system 3 is in SLEEPDEEP mode 0: the CPU of system 3 is not in SLEEPDEEP mode
0	R	0	SYS3_ALIVE 1: system 3 is alive 0: system 3 is down

3.6.3.20 System 1 Wakeup Control Register

Address : 0x0098			Name: SYS1_WAKEUP_CTRL Default: 0x0000_000f
Field	Type	Default	Description
31:6	/	/	/
5	RW	0	SYS1_WAKEUP_SW3_ON 1: SW3 will be enabled when waking up the system 1 0: keep the power switch with its current state
4	RW	0	SYS1_WAKEUP_SR_SW2_ON 1: SR_SW2 will be enabled when waking up the system 1 0: keep the power switch with its current state
3	RW	1	SYS1_WAKEUP_SR_SW1_ON 1: SR_SW1 will be enabled when waking up the system 1 0: keep the power switch with its current state
2	RW	1	SYS1_WAKEUP_SW2_ON 1: SW2 will be enabled when waking up the system 1 0: keep the power switch with its current state
1	RW	1	SYS1_WAKEUP_SW1_ON

			1: SW1 will be enabled when waking up the system 1 0: keep the power switch with its current state
0	RW	1	SYS1_WAKEUP_LDO1_ON 1: LDO1 will be enabled when waking up the system 1 0: keep the LDO with its current state

3.6.3.21 System 1 Sleep Control Register

Address : 0x00A0			Name: SYS1_SLEEP_CTRL Default: 0x0000_000e
Field	Type	Default	Description
31:29	/	/	/
28	RW	0	WLAN_SRAM_116K_SWM5_MODE WLAN SRAM(116k bytes) retention power switch (SWM5) mode 0: always on 1: SRAM switch maybe off
27	RW	0	SRAM_32K_SWM4_MODE SRAM(32k bytes) switch (SWM4) mode 0: always on 1: SRAM switch maybe off
26	RW	0	SRAM_32K_SWM3_MODE SRAM(32k bytes) switch (SWM3) mode 0: always on 1: SRAM switch maybe off
25	RW	0	SRAM_352K_SWM2_MODE SRAM(352k bytes) switch (SWM2) mode 0: always on 1: SRAM switch maybe off
24	RW	0	CACHE_SRAM_SWM1_MODE Cache SRAM switch (SWM1) mode 0: always on 1: SRAM switch maybe off
23:22	/	/	/
21	RW	0	CACHE_SRAM_PWR_CTRL_MODE

			Cache SRAM power control mode 0: deep-sleep mode 1: shutdown mode
20:8	RW	0	SRAM_PWR_CTRL_MODE 32k bytes SRAM0~SRAM12 power control mode, one bit control one 32k SRAM. 0: deep-sleep mode 1: shutdown mode
7:6	/	/	/
5	RW	0	SYS1_SLEEP_SW3_OFF 1: SW3 will be disabled when the system 1 goes into the deep-sleep mode 0: keep the power switch with its current state
4	RW	0	SYS1_SLEEP_SR_SW2_OFF 1: SR_SW2 will be disabled when the system 1 goes into the deep-sleep mode 0: keep the power switch with its current state
3	RW	1	SYS1_SLEEP_SR_SW1_OFF 1: SR_SW1 will be disabled when the system 1 goes into the deep-sleep mode 0: keep the power switch with its current state
2	RW	1	SYS1_SLEEP_SW2_OFF 1: SW2 will be disabled when the system 1 goes into the deep-sleep mode 0: keep the power switch with its current state
1	RW	1	SYS1_SLEEP_SW1_OFF 1: SW1 will be disabled when the system 1 goes into the deep-sleep mode 0: keep the power switch with its current state
0	RW	0	SYS1_SLEEP_LDO1_OFF 1: LDO1 will be disabled when the system 1 goes into the deep-sleep mode 0: keep the power switch with its current state

3.6.3.22 DCXO Stable Reference Time Register

Address : 0x00A8			Name: DCXO_STABLE_REF_TIME Default: 0x0000_0100
Field	Type	Default	Description
31:10	/	/	/
9:0	RW	0x100	DCXO Stable Reference Time $REF_TIME = T_{LFCLK} * n$ Typical Value $\leq 1.5ms$ Set Value = $32us * 256 = 8ms$

3.6.3.23 DPLL Stable Reference Time Register

Address : 0x00AC			Name: DPLL_STABLE_REF_TIME Default: 0x0000_0010
Field	Type	Default	Description
31:8	/	/	/
7:0	RW	0x10	DPLL Stable Reference Time $REF_TIME = T_{LFCLK} * n$ Typical Value $\leq 150us$ Set Value = $32us * 16 = 512us$

3.6.3.24 LDO Stable Reference Time Register

Address : 0x00B0			Name: LDO_STABLE_REF_TIME Default: 0x0000_000a
Field	Type	Default	Description
31:8	/	/	/
7:0	RW	0xa	LDO Stable Reference Time $REF_TIME = T_{LFCLK} * n$ Typical Value $\leq 100us$ Set Value = $32us * 10 = 320us$

3.6.3.25 Digital Switch Reference Time Register

Address : 0x00B4			Name: DIGITAL_SWITCH_REF_TIME Default: 0x1000_0002
Field	Type	Default	Description
31:4	/	/	/
29:16	RW	0x1000	Reset Up Reference Time $REF_TIME = T_{HFCLK} * n$ Typical Value ≥ 4096 cycles

			<i>Set Value = 25ns * 4096 = 100us (@40MHz OSC)</i>
15:14	/	/	/
13:0	RW	0x2	Digital Switch Reference Time $REF_TIME = T_{HFCLK} * n$ <i>Typical Value >= 2 cycles</i> <i>Set Value = 25ns * 2 = 50ns (@40MHz OSC)</i>

3.6.3.26 RTC LDO Voltage Control Register

Address : 0x00B8			Name: RTC_LDO_VOL_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:3	/	/	/
18:16	RW	0	RTC LDO work Voltage Control 0: 1.075V 1: 0.975V 2: 0.875V 3: 0.775V 4: 0.675V 5: 0.60V 6: 1.175V 7: 1.275V
15:3	/	/	/
2:0	RW	0	RTC LDO Retention Voltage Control 0: 1.075V 1: 0.975V 2: 0.875V 3: 0.775V 4: 0.675V 5: 0.60V 6: 1.175V 7: 1.275V

3.6.3.27 BANDGAP Stable Reference Time Register

Address : 0x00BC			Name: BG_STABLE_REF_TIME Default: 0x0000_000a
Field	Type	Default	Description
31:8	/	/	/
7:0	RW	0xa	LDO Stable Reference Time $REF_TIME = T_{LFCLK} * n$ <i>Typical Value <= 100us</i> <i>Set Value = 32us * 10 = 320us</i>

3.6.3.28 TOP_LDO Stable Reference Time Register

Address : 0x00C0			Name: TOP_LDO_STABLE_REF_TIME Default: 0x0080_0010
Field	Type	Default	Description
31:26	/	/	/
25:16	RW	0x80	LDO to PWM Mode Stable Reference Time $REF_TIME = T_{LFCLK} * n$ Typical Value $\leq 1ms$ Set Value $= 32us * 128 = 4ms$
15:10	/	/	/
9:0	RW	0x10	Other Work Mode Switch Stable Reference Time $REF_TIME = T_{LFCLK} * n$ Typical Value $\leq 200us$ Set Value $= 32us * 16 = 512us$

3.6.3.29 AR400A Boot Flag Register

Address : 0x0100			Name: BOOT_FLAG Default: 0x0000_0000
Field	Type	Default	Description
31:16	RW	0	BOOT_FLAG_WRITE_LOCK 0x429B: AR400A_BOOT_FLAG filed is writable other: AR400A_BOOT_FLAG filed is read only Note: The field of AR400A_BOOT_FLAG only can be changed when this bit is set to 0x429B . In one write operation, the GRPCM will check this field whether equals to 0x429B. If not, the value of AR400A_BOOT_FLAG will be ignored. This field is always read with value 0x0000. Examples: 4. After write 0x429B0001, then read value = 0x00000001 5. After write 0x11110000, then read value = 0x00000001 6. After write 0x429B0000, then read value = 0x00000000
15:4	/	/	/
3:0	RW	0	AR400A_BOOT_FLAG 0: boot from cold reset(default) 1: boot from deep sleep 2: boot for system update 3: boot for recovery

		others: reserved
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3.6.3.30 AR400A Boot Address Register

Address : 0x0104			Name: BOOT_ADDR Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0	AR400A_BOOT_ADDR

3.6.3.31 AR400A Boot Argument Register

Address : 0x0108			Name: BOOT_ARG Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0	AR400A_BOOT_ARG

3.6.3.32 AR400A Private Register0

Address : 0x010c			Name: AR400A_PRIV_REG0 Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0	AR400A_PRIV_REG0

3.6.3.33 AR400A Private Register1

Address : 0x0110			Name: AR400A_PRIV_REG1 Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0	AR400A_PRIV_REG1

3.6.3.34 AR400A Private Register2

Address : 0x0114			Name: AR400A_PRIV_REG2 Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0	AR400A_PRIV_REG2

3.6.3.35 AR400A Private Register3

Address : 0x0118			Name: AR400A_PRIV_REG3 Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0	AR400A_PRIV_REG3

3.6.3.36 AR400A Private Register4

Address : 0x011c			Name: AR400A_PRIV_REG4 Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0	AR400A_PRIV_REG4

3.6.3.37 AR400A Wakeup Timer Counter Register

Address : 0x0120			Name: AR400A_WAKE_TIMER_CNT Default: 0x0000_0000
Field	Type	Default	Description
31	RW	0	AR400A_WAKE_TIMER_EN
30:0	R	0	AR400A_WAKE_TIMER_CNT <i>Note: This is a free running counter and when the value of this counter equals to the value of AR400A_WAKE_TIMER_VAL, a wakeup signal will be issued to the Wakeup Interrupt Controller. If AR400A_WAKE_TIMER_EN is set to '0', this counter will be reset to 0 and stop counting.</i>

3.6.3.38 AR400A Wakeup Timer Value Register

Address : 0x0124			Name: AR400A_WAKE_TIMER_VAL Default: 0x0000_0000
Field	Type	Default	Description
31	RW	0	AR400A_WAKEUP_TIMER_IRQ_STATUS 0: the ar400a wakeup timer irq is not pending 1: the ar400a wakeup timer irq is pending <i>Note: this is a write-1-to-clear register. This is an asynchronous operation and this bit will not become to '0' immediately after writing '1' to clear the irq state. We should read the register until the value of this bit becomes '0', which means the irq pending state is actually cleared.</i>
30:0	RW	0	AR400A_WAKE_TIMER_VAL

3.6.3.39 AR400A Private Register5

Address : 0x0128			Name: AR400A_PRIV_REG5 Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0	AR400A_PRIV_REG5

3.6.3.40 AR400A Private Register6

Address : 0x012C			Name: AR400A_PRIV_REG6 Default: 0x0000_0000
Field	Type	Default	Description
31:0	RW	0	AR400A_PRIV_REG6

3.6.3.41 AR400A IO Wakeup Enable Register

Address : 0x0130			Name: AR400A_WAKE_IO_EN Default: 0x0000_0000
Field	Type	Default	Description
31:16	RW	0x0000	AR400A_WAKEUP_IO0_3_DEB_CYCLES bit19-bit16: IO0 debounce cycles. bit23-bit20: IO1 debounce cycles. bit27-bit24: IO2 debounce cycles. bit31-bit28: IO3 debounce cycles. First must select clock for debounce (configure by register AR400A_WAKE_IO_MODE), finally the IO input debounce clock cycles is (DEBC_CYCLE[3:0]+1)+16.
15:10	/	/	/
9:0	RW	0	AR400A_WAKEUP_IOx_ENB 1: IOx wakeup detection enable 0: IOx wakeup detection disable bit0- bit9: WUPIO0 – WUPIO9

3.6.3.42 AR400A IO Wakeup Mode Register

Address : 0x0134			Name: AR400A_WAKE_IO_MODE Default: 0x0000_0000
Field	Type	Default	Description
31:28	RW	0x0	AR400A_WAKEUP_DEB_CLK_SET1 IO wakeup debounce clk1 select, the DEB_CLK1 frequency $CLK_SEL1 \leq 14$, $Freq = FCLK / (2^{CLK_SEL1})$; $CLK_SEL1 = 15$, reserved.
27:24	RW	0x0	AR400A_WAKEUP_DEB_CLK_SET0 IO wakeup debounce clk0 select, the DEB_CLK0 frequency $CLK_SEL0 \leq 14$, $Freq = FCLK / (2^{CLK_SEL0})$; $CLK_SEL0 = 15$, reserved.
23:22	/	/	/
21:12	RW	0	AR400A_WAKEUP_IOx_CLK_SEL 0: IOx select DEB_CLK0

			1: IOx select DEB_CLK1
11:10	/	/	/
9:0	RW	0	AR400A_WAKEUP_IOx_MODE 1: Positive Edge 0: Negative Edge bit0- bit9: WUPIO0 – WUPIO9

3.6.3.43 AR400A IO Wakeup Status Register

Address : 0x0138			Name: AR400A_WAKE_IO_STA Default: 0x0000_0000
Field	Type	Default	Description
31:10	/	/	/
9:0	RW	0	AR400A_WAKEUP_IOx_ST 1: wakeup event is detected on IOx 0: Nothing bit0- bit9: WUPIO0 – WUPIO9 <i>Note: Write 1 to clear the status</i>

3.6.3.44 AR400A IO Hold Control Register

Address : 0x013C			Name: AR400A_IO_HODL_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:10	/	/	/
9:0	RW	0	AR400A_WAKEUP_IOx_ST 1: IO Configure Hold Enable 0: IO Configure Hold Disable bit0- bit9: WUPIO0 – WUPIO9 <i>Note: when enable the IO hold function, the configuration(IEN, OEN, PULL, ODAT) of the relative IO will be hold to the current state even if the GPIO controller is powered down or the origin configuration is modified.</i>

3.6.3.45 AR400A IO Wakeup Global Enable Register

Address : 0x0140			Name: AR400A_IO_WUP_GEN Default: 0x0000_0000
Field	Type	Default	Description
31:8	RW	0x000000	AR400A_WAKEUP_IO4_9_DEB_CYCLES bit11-bit8: IO4 debounce cycles. bit15-bit12: IO5 debounce cycles. bit19-bit16: IO6 debounce cycles. bit23-bit20: IO7 debounce cycles.

			bit27-bit24: IO8 debounce cycles. bit31-bit28: IO9 debounce cycles. First must select clock for debounce (configure by register AR400A_WAKE_IO_MODE), finally the IO input debounce clock cycles is (DEBC_CYCLE[3:0]+1)+16.
0	RW	0	AR400A_IO_WUP_GEN 1: IO Wakeup Enable 0: IO Wakeup Disable

3.6.3.46 LDO mode software select Register

Address : 0x0144			Name: LDO_MODE_SW_SEL Default: 0x0000_0000
Field	Type	Default	Description
31:3	/	/	/
2	RW	0	TOP_LDO_LQ_MODE 1: TOP_LDO select active mode 0: TOP_LDO select active mode
1	RW	0	SYS_LDO_LQ_MODE 1: SYS_LDO select lq mode 0: SYS_LDO select active mode
0	RW	1	LDO_MODE_SW_SEL_ENABLE 1: ldo active or lq mode select by this register 0: ldo active or lq mode select by power control

3.6.3.47 TOP_LDO Parameter Control Register

Address : 0x0200			Name: TOP_LDO_PARAM_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:29	RW	0	TOP_LDO_OCP TOP_LDO over current protection threshold tuning default: 400mA
28:24	RW	0	TOP_LDO_BGTR [4:1]: for TOP_LDO BANDGAP trimming [0]: for OCP disable
23	RW	0	TOP_LDO_DEEPSLEEP_EN top_ldo enable control in deepsleep mode. 0, disable; 1, enable;
22:16	RW	0	TOP_LDO_FTR [4:0]: for switching frequency tuning [6:5]: for option to decrease frequency

15:8	RW	0	TOP_LDO_TEST_ITEM_SEL
7:4	RW	0	TOP_LDO_TEST_MODE
3:1	/	/	/
0	R	0	TOP_LDO_WORK_MODE 0: PWM Mode 1: PFM Mode <i>Note: the TOP_LDO starts up from PWM mode by default.</i> <i>This field is read-only and can only be changed by POWERCTRL Module.</i>

3.6.3.48 ANA BANDGAP Register

Address : 0x0204			Name: ANA_BG Default: 0x0010_000x
Field	Type	Default	Description
31:21	RW	0	
20:16	RW	0x10	BG Output Voltage Trimming
15:9	/	/	/
8	R	0	TEST_EN <i>Note: this field is read-only and will be set by TAM module, this bit shows the enable/disable state.</i>
7:1	/	/	/
0	R	x	ANA_BG_PND_1V8 BG Power Up Enable <i>Note: this field can be only set by POWERCTRL module.</i>

3.6.3.49 Clock LDO Parameter Register

Address : 0x0208			Name: CLKLDO_PARAM Default: 0x0000_0004
Field	Type	Default	Description
31:3	/	/	/
2	R	1	LDO_WORK_MODE_SEL 0: LDO in light load mode 1: normal mode <i>Note: the work mode is fixed to normal mode.</i>
1	R	0	BYPASS_EN Bypass mode enable. when set to 1, $V_{out} = V_{DDIN}$
0	R	0	TEST_EN <i>Note: this field is read-only and will be set by TAM module, this bit shows the enable/disable state.</i>

3.6.3.50 Digital LDO Parameter Register

Address : 0x020c			Name: DLDO_PARAM Default: 0x0808_0000
Field	Type	Default	Description
31:28	/	/	/
27	RW	1	TESTPEAK_SRAMLDO enable the auxiliary circuit to avoid output large ripple during select LDO work 1: enable 0: disable <i>Note: this field is set by POWERCTRL module</i>
26	R	0	STARTICC_SRAM decrease over current protection threshold at start up 1: enable 0: disable <i>Note: this field is set by POWERCTRL module</i>
25	RW	0	TESTICC_SRAM over current protection enable 1: enable 0: disable <i>Note: this field is set by POWERCTRL module</i>
24	R	0	SRAM_LQ_MODE_EN SRAM LQ mode enable 1: enable 0: disable <i>Note: this field is set by POWERCTRL module</i>
23:20	/	/	/
19	RW	1	TESTPEAK_LDO1 enable the auxiliary circuit to avoid output large ripple during select LDO work 1: enable 0: disable <i>Note: this field is set by POWERCTRL module</i>
18	R	0	STARTICC_LDO1 decrease over current protection threshold at start up 1: enable

			0: disable <i>Note: this field is set by POWERCTRL module</i>
17	RW	0	TESTICC_LDO over current protection enable 1: enable 0: disable <i>Note: this field is set by POWERCTRL module</i>
16	R	0	LDO1_LQ_MODE_EN LDO1 LQ mode enable 1: enable 0: disable <i>Note: this field is set by POWERCTRL module</i>
15:12	/	/	/
11:8	RW	0	BG_OUTPUT_TRIM BG output trimming <i>Note: eFuse override?</i>
7:1	/	/	/
0	R	0	BG_PWR 0: disable 1: enable <i>Note: This field is set by POWERCTRL module and must be enable before enabling LDO and LDO_SRAM.</i>

3.6.3.51 Config IO Status Register

Address : 0x0214			Name: CFG_IO_ST Default: 0x0000_000x
Field	Type	Default	Description
31:3	RW	0	
/	/	/	/
1	R	x	SIP IO 0: no flash sip 1: with flash sip in package
/	/	/	/

3.6.3.52 CPU Reset Source Register

Address : 0x0218			Name: CPU_RST_SRC Default: 0x0000_000x
Field	Type	Default	Description
10:9	RW	0	IS_WATCHDOG_CPU_RST

			The CPU reset source is watchdog(only reset CPU) output. Write 2'b11 clear these bits. 00: no 01: one time watchdog(only reset CPU) 10: two times watchdog(only reset CPU) 11: three times watchdog(only reset CPU)
8	RW	0	IS_WATCHDOG_ALL_RST The CPU reset source is watchdog(reset all system) output. Write 1 clear this bit. 0: no 1: yes
7:2	/	/	/
1	RW	0	IS_PMU_RST The CPU reset source is digital PMU output. Write 1 clear this bit. 0: no 1: yes
0	RW	0	IS_PWRON_RST The CPU reset source is POWER ON or P_RESETh. Write 1 clear this bit. 0: no 1: yes

3.6.3.53 WLAN HIF override control register

Address : 0x21C			Name: WLAN_HIF_OV_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	Reserved
7	R/W	0	WLAN_IRQ_OVER_HIF: use wlan_irq generated in WLAN_MISC logic to replace the corresponding output of WLAN_HIF 0: not override WLAN_HIF output 1: override WLAN_HIF output
6	R/W	0	WLAN_WUP_OVR_HIF: use WLAN_WUP(bit2) to override the corresponding output of WLAN_HIF 0: not override WLAN_HIF output 1: override WLAN_HIF output
5	R/W	0	RESET_CPU_OVR_HIF: use RESET_CPU(bit1) to override the corresponding output of WLAN_HIF 0: not override WLAN_HIF output 1: override WLAN_HIF output
4	R/W	0	DISABLE_CPU_CLK_OVR_HIF: use DISABLE_CPU_CLK(bit0) to override the corresponding output of WLAN_HIF 0: not override WLAN_HIF output

			1: override WLAN_HIF output
3	/	/	Reserved
2	R/W	0	WLAN_WUP : wake-up wlan from sleep mode, host firmware need to clear it after WLAN waked up
1	R/W	1	RESET_CPU : reset WLAN CPU 0: release WLAN CPU reset 1: WLAN CPU is reset
0	R/W	1	DISABLE_CPU_CLK : disable WLAN CPU clock 0: wlan cpu clock enable 1: wlan cpu clock disable

3.6.3.54 SRAM Bist Configuration Register

Address : 0x0220			Name : SRAM_BIST_CFG Default : 0x22222222
Field	Type	Default	Description
31:24	/	/	/
23:16	RW	0x22	ARM_SINGLE_PORT_SRAM_CFG ram_cfg[7:0] bit [2:0]: connect to EMA bit [4:3]: connect to EMAW others : reserved
15:8	RW	0x22	ARM_DUAL_PORT_SRAM_CFG ram_cfg[7:0] bit [2:0]: connect to EMAA bit [6:4]: connect to EMAB others : reserved
7:0	RW	0x22	SYNOPSISYS_SRAM_CFG ram_cfg[7:0] bit [0]: connect to REME bit [3]: connect to TEST1 bit [7:4]: connect to REM others : reserved

3.7 Clock Control Module (CCM)

3.7.1 Overview

The CCMU module manages the reset and the *HCLK* for each module and provides the configurations for each module to adjust its module clock.

3.7.2 Register List

Module Name	Base Address	
Clock Control Module	0x40040400	
Register Name	Offset Address	Description
CPU_BUS_CLKCFG	0x0000	CPU BUS Clock Configure Register
BUS_CLK_GATING_CTRL	0x0004	Bus Clock Gating Control Register
BUS_DEV_RST_CTRL	0x0008	Bus Device Reset Control Register
SDC1_CLK_CTRL	0x001C	SDC1 Clock Control Register
SPI0_CLK_CTRL	0x0020	SPI0 Clock Control Register
SPI1_CLK_CTRL	0x0024	SPI1 Clock Control Register
SDC_CLK_CTRL	0x0028	SDC Clock Control Register
SS_CLK_CTRL	0x002C	SS Clock Control Register
DAUDIO_CLK_CTRL	0x0034	Digital Audio Clock Control Register
IRRX_CLK_CTRL	0x0038	IRRX Clock Control Register
IRTX_CLK_CTRL	0x003c	IRTX Clock Control Register
SYSTICK_REFCLK_CTRL	0x0040	System Tick Reference Clock Register
SYSTICK_CALIB_CTRL	0x0044	System Tick Clock Calibration Register
DMIC_CLK_CTRL	0x0048	DMIC Clock Control Register
GPADC_CLK_CTRL	0x004c	GPADC Clock Control Register
CSI_DCLK_CTRL	0x0050	CSI Device Clock Control Register
FLASHC_MCLK_CTRL	0x0054	FLASH Controller Module Clock Control Register
AUDIO_CODEC_MCLK_CTRL	0x0058	AUDIO CODEC Module Clock Control Register
PCLK_SPC_MCLK_CTRL	0x005C	Special APB Clock Control Register (UART/GPRCM/RTC APB bus clock)
DEV_RESET_MASK_CTRL	0x0080	
DEV_RESET_MASK	0x0084	

3.7.3 Register Description

3.7.3.1 BUS Clock Configure Register

Address : 0x0000			Name: CPU_BUS_CLKCFG
			Default: 0x0000_0001
Field	Type	Default	Description
31:10	/	/	/
9:8	R/W	0x0	AHB2_CLK_DIV

			00: /1 01: /2 10: /3 11: /4
7:6	/	/	/
5:4	R/W	0x0	APB_CLK_SRC_SEL APB bus clock source select 00: HFCLK 01: LFCLK 1X: AHB2 Clock
3:2	/	/	/
1:0	R/W	1	APB_CLK_DIV 00: /1 01: /2 10: /4 11: /8

3.7.3.2 BUS Clock Gating Control Register

Address : 0x0004			Name: BUS_CLK_GATING_CTRL Default:
Field	Type	Default	Description
31	/	/	/
30	R/W	0	TRNG_CLK_GATING 0: the clock is gated off 1: the clock is running
29	R/W	0	AUDIO_CODEC_CLK_GATING 0: the clock is gated off 1: the clock is running
28	R/W	0	UART2_CLK_GATING 0: the clock is gated off 1: the clock is running
27	R/W	0	GPIO_CLK_GATING 0: the clock is gated off 1: the clock is running

26	/	/	/
25	R/W	0	GPADC_CLK_GATING 0: the clock is gated off 1: the clock is running
24	R/W	0	IRRX_CLK_GATING 0: the clock is gated off 1: the clock is running
23	R/W	0	IRTX_CLK_GATING 0: the clock is gated off 1: the clock is running
22	R/W	0	DAUDIO_CLK_GATING 0: the clock is gated off 1: the clock is running
21	R/W	0	PWM_ECT_CLK_GATING 0: the clock is gated off 1: the clock is running
20	/	/	/
19	R/W	0	TWI1_CLK_GATING 0: the clock is gated off 1: the clock is running
18	R/W	0	TWI0_CLK_GATING 0: the clock is gated off 1: the clock is running
17	R/W	0	UART1_CLK_GATING 0: the clock is gated off 1: the clock is running
16	R/W	0	UART0_CLK_GATING 0: the clock is gated off 1: the clock is running
15:12	/	/	/
11	R/W	0	PSRAM_CTRL_CLK_GATING 0: the clock is gated off 1: the clock is running

10	R/W	0	WLANC_GATING 0: the clock is gated off 1: the clock is running
9	R/W	0	SDC1_CLK_GATING 0: the clock is gated off 1: the clock is running
8:7	/	/	/
6	R/W	0	DMA_CLK_GATING 0: the clock is gated off 1: the clock is running
5	R/W	0	CSI_JPE_CLK_GATING 0: the clock is gated off 1: the clock is running
4	R/W	0	SDC0_CLK_GATING 0: the clock is gated off 1: the clock is running
3	R/W	0	FLASHC_CTRL_CLK_GATING 0: the clock is gated off 1: the clock is running
2	R/W	0	SS_CLK_GATING 0: the clock is gated off 1: the clock is running
1	R/W	0	SPI1_CLK_GATING 0: the clock is gated off 1: the clock is running
0	R/W	0	SPIO_CLK_GATING 0: the clock is gated off 1: the clock is running

3.7.3.3 BUS Module Reset Control Register

Address : 0x0008	Name : BUS_DEV_RST_CTRL Default :
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Field	Type	Default	Description
31	/	/	/
30	R/W	1	TRNG_RST 0: the module is in reset state 1: the module is released to work. The default value is 1, used for TRNG retention registers.
29	R/W	0	AUDIO_CODEC_RST 0: the module is in reset state 1: the module is released to work
28	R/W	0	UART2_RST 0: the module is in reset state 1: the module is released to work
27:26	/	/	/
25	R/W	0	GPADC_CLK_GATING 0: the device is in reset state 1: the clock is running
24	R/W	0	IRRX_RST 0: the device is in reset state 1: the clock is running
23	R/W	0	IRTX_RST 0: the clock is in reset state 1: the clock is running
22	R/W	0	DAUDIO_RST 0: the module is in reset state 1: the module is released to work
21	R/W	0	PWM_ECT_RST 0: the module is in reset state 1: the module is released to work
20	/	/	/
19	R/W	0	TWI1_RST 0: the module is in reset state 1: the module is released to work
18	R/W	0	TWI0_RST

			0: the module is in reset state 1: the module is released to work
17	R/W	0	UART1_RST 0: the module is in reset state 1: the module is released to work
16	R/W	0	UART0_RST 0: the module is in reset state 1: the module is released to work
15:14	/	/	/
13	R/W	1	DCACHE_RST 0: the module is in reset state 1: the module is released to work
12	R/W	1	ICACHE_RST 0: the module is in reset state 1: the module is released to work
11	R/W	0	PSRAM_CTRL_RST 0: the module is in reset state 1: the module is released to work
10	R/W	0	WLANC_RST 0: the module is in reset state 1: the module is released to work
9	R/W	0	SDC1_RST 0: the module is in reset state 1: the module is released to work
8:7	/	/	/
6	R/W	0	DMA_RST 0: the module is in reset state 1: the module is released to work
5	R/W	0	CSI_JPE_RST 0: the module is in reset state 1: the module is released to work
4	R/W	0	SDC0_RST

			0: the module is in reset state 1: the module is released to work
1	R/W	0	FLASHC_CTRL_RST 0: the module is in reset state 1: the module is released to work
2	R/W	0	SS_RST 0: the module is in reset state 1: the module is released to work
1	R/W	0	SPI1_RST 0: the module is in reset state 1: the module is released to work
0	R/W	0	SPIO_RST 0: the module is in reset state 1: the module is released to work

3.7.3.4SDC1 Clock Control Register

Address : 0x001c			Name: SDC1_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: DEVCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2

			10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.5 SPI0 Clock Control Register

Address : 0x0020			Name: SPI0_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work CLKm = CLKsrc / N / M
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: DEVCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.6 SPI1 Clock Control Register

Address : 0x0024	Name: SPI1_CLK_CTRL Default:
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Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: DEVCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.7SDC Clock Control Register

Address : 0x0028			Name: SDC0_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK

			01: DEVCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

Table 3-8 SDC Clock Control Register

3.7.3.8SS Clock Control Register

Address : 0x002c			Name: SS_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: DEVCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N Clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8

15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.9 CSI_JPE Device CLK Control Register

Address : 0x0030			Name: CSI_JPE_DEV_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	CSI_JPE_DEV_CLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: DEVCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.10 DAUDIO Clock Control Register

Address : 0x0034			Name: DAUDIO_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE

			0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:2	/	/	/
1:0	R/W	0	MCLK_SRC_SEL Clock source selection 00: AUDIO_CLK (8X) 01: PLL_AUDIO (8X) / 2 10: PLL_AUDIO (8X) / 4 11: PLL_AUDIO

3.7.3.11 IRRX Clock Control Register

Address : 0x0038			Name: IRRX_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: LFCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M.

			M = value + 1 (1~16)
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3.7.3.12 IRTX Clock Control Register

Address : 0x003c			Name: IRRX_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: LFCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.13 System Tick Reference Clock Control Register

Address : 0x0040			Name: SYSTICK_REFCLK_CTRL Default: 0x00000000
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$

30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: LFCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.14 System Tick Clock Calibration Register

Address : 0x0044			Name: SYSTICK_CALIB_CTRL Default:
Field	Type	Default	Description
31:26	/	/	/
25	R/W	0	ST_NOREF 1: there is no reference clock 0: use the reference clock
24	R/W	0	ST_SKEW <i>Note: set this bit to 0 if the system timer clock, the external reference clock, or FCLK as indicated by ST_NOREF, can guarantee an exact multiple of 10ms. Otherwise, set this bit to 1.</i>
23:0	R/W	0	ST_10MS_COUNTER <i>Note: This field provides an integer value to compute a 10ms (100Hz) delay from either the reference clock, or FCLK if the reference clock is not implemented.</i> <i>For example:</i> <i>No reference clock, FCLK=50MHz, CNT = 0x7A11F(49999)</i>

			REFCLK = 1MHz, CNT = 0x270F(9999)
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3.7.3.15 DMIC Clock Control Register

Address : 0x0048			Name: IRRX_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:0	/	/	/

3.7.3.16 GPADC Clock Control Register

Address : 0x004c			Name: GPADC_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: LFCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M.

			M = value + 1 (1~16)
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3.7.3.17 CSI Output MCLK Control Register

Address : 0x0050			Name: CSI_MCLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	CSI_MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: DEVCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.18 Flash Controller SPI Clock Control Register

Address : 0x0054			Name: FLASH_SPI_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$

30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: DEVCLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.19 AUDIO CODEC Clock Control Register

Address : 0x0058			Name: AUDIO_DODEC_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:0	/	/	/

3.7.3.20 Special APB Clock Control Register

Address : 0x005C			Name: PCLK_SPC_CLK_CTRL Default:
Field	Type	Default	Description
31:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK

			01: DEVCLK 10: FLCLK 11: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8
15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.21 PSRAM Controller SPI/OPI Clock Control Register

Address : 0x0060			Name: PSRAM_OPI_CLK_CTRL Default:
Field	Type	Default	Description
31	R/W	0	MCLK_ENABLE 0: the module clock is gated off 1: the module clock is released to work $CLK_m = CLK_{src} / N / M$
30:26	/	/	/
25:24	R/W	0	MCLK_SRC_SEL Clock source selection 00: HFCLK 01: DEV2CLK 1x: reserved
23:18	/	/	/
17:16	R/W	0	CLK_DIV_RATIO_N clock pre-divide ratio N. 00: N = 1 01: N = 2 10: N = 4 11: N = 8

15:4	/	/	/
3:0	R/W	0	CLK_DIV_RATIO_M Clock divide ratio M. M = value + 1 (1~16)

3.7.3.22 Device Reset Mask Control Register

Address : 0x0080			Name: DEV_RST_MASK_CTRL Default: 0
Field	Type	Default	Description
31:0	R/W	0	DEV_RST_MASK_LOCK 0x44524D4C: DEV_RST_MASK_REG is writeable other: DEV_RST_MASK_REG is locked and cannot be changed. Note: The field of DEV_RST_MASK_REG only can be changed when these bits are set to 0x44524D4C . In one write operation, the GRPCM will check this field whether equals to 0x44524D4C . If not, the value of DEV_RST_MASK_REG is locked and cannot be changed. This field is always read with value 0x0000. Examples: 1. After write 0x44524D4C , then read value = 0x44524D4C and the DEV_RST_MASK_REG is unlocked 2. After write other value, then read value = 0x00000000 and the DEV_RST_MASK_REG is locked

3.7.3.23 Device Reset Mask Register

Address : 0x0084			Name: DEV_RST_MASK Default: When IO_MAPPING = 0, 0x73a707d7 When IO_MAPPING = 1, 0x77ef07f7
Field	Type	Default	Description
31	/	/	/
30	R/W	0	TRNG_RST 0: the module is in reset state 1: the module is released to work
29	R/W	0	AUDIO_CODEC_RST 0: the module is in reset state 1: the module is released to work

28	R/W	0	UART2_RST 0: the module is in reset state 1: the module is released to work
27	/	/	/
26	R/W	0	DMIC_CLK_GATING 0: the device is in reset state 1: the clock is running
25	R/W	0	GPADC_CLK_GATING 0: the device is in reset state 1: the clock is running
24	R/W	0	IRRX_RST 0: the device is in reset state 1: the clock is running
23	R/W	0	IRTX_RST 0: the clock is in reset state 1: the clock is running
22	R/W	0	DAUDIO_RST 0: the module is in reset state 1: the module is released to work
21	R/W	0	PWM_ECT_RST 0: the module is in reset state 1: the module is released to work
20	/	/	/
19	R/W	0	TWI1_RST 0: the module is in reset state 1: the module is released to work
18	R/W	0	TWIO_RST 0: the module is in reset state 1: the module is released to work
17	R/W	0	UART1_RST 0: the module is in reset state 1: the module is released to work

16	R/W	0	UART0_RST 0: the module is in reset state 1: the module is released to work
15:14	/	/	/
13	R/W	0	DCACHE_RST 0: the module is in reset state 1: the module is released to work
12	R/W	0	ICACHE_RST 0: the module is in reset state 1: the module is released to work
11	R/W	0	PSRAMCTRL_RST 0: the module is in reset state 1: the module is released to work
10	R/W	0	WLANC_RST 0: the module is in reset state 1: the module is released to work
9	R/W	0	SDC1_RST 0: the module is in reset state 1: the module is released to work
8	R/W	0	MSGBOX_RST 0: the module is in reset state 1: the module is released to work
7	R/W	0	SPINLOCK_RST 0: the module is in reset state 1: the module is released to work
6	R/W	0	DMA_RST 0: the module is in reset state 1: the module is released to work
5	R/W	0	CSI_JPE_RST 0: the module is in reset state 1: the module is released to work
4	R/W	0	SDC0_RST 0: the module is in reset state 1: the module is released to work
3	R/W	0	FLASHCTRL_RST

			0: the module is in reset state 1: the module is released to work
2	R/W	0	SS_RST 0: the module is in reset state 1: the module is released to work
1	R/W	0	SPI1_RST 0: the module is in reset state 1: the module is released to work
0	R/W	0	SPIO_RST 0: the module is in reset state 1: the module is released to work

3.8 Direct Memory Access Controller (DMA)

3.8.1 Overview

There are 8 AHB DMA channels for this DMA controller. Only one channel can be active and the sequence is according to the priority level.

The DMA controller can support 8-bit/16-bit/32-bit data width. The data width of Source and Destination can be different, but the address should be aligned. Although the increase mode of NDMA should be address aligned, but its byte counter should not be multiple. The DMA Source Address, Destination Address, Byte Counter Registers can be modified even if the DMA is started

3.8.2 Register List

Module Name	Base Address	
DMA Controller	0x40001000	
Register Name	Offset Address	Description
DMA_INT_CTRL_REG	0x00	DMA Interrupt Control Register
DMA_INT_STA_REG	0x04	DMA Interrupt Status Register
DMA_PTY_CFG_REG	0x08	DMA Priority Configure Register
NDMA_CTRL_REG	0x100+N*0x20	Normal DMA Configuration (N=0,1,2,3,4,5,6,7)
NDMA_SRC_ADDR_REG	0x100+N*0x20+0x4	Normal DMA Source Address (N=0,1,2,3,4,5,6,7)
NDMA_DEST_ADDR_REG	0x100+N*0x20+0x8	Normal DMA Destination Address

		(N=0,1,2,3,4,5,6,7)
NDMA_BC_REG	0x100+N*0x20+0xC	Normal DMA Byte Counter (N=0,1,2,3,4,5,6,7)

3.8.3 Register Description

3.8.3.1 DMA IRQ Enable Register

Address : 0x0000			Name: DMA_IRQ_EN_REG Default: 0x0000_0000
Field	Type	Default	Description
31:16	/	/	/
15	R/W	0x0	NDMA7_END_IRQ_EN. Normal DMA 7 End Transfer Interrupt Enable. 0: Disable, 1: Enable.
14	R/W	0x0	NDMA7_HF_IRQ_EN Normal DMA 7 Half Transfer Interrupt Enable. 0: Disable, 1: Enable.
13	R/W	0x0	NDMA6_END_IRQ_EN Normal DMA 6 End Transfer Interrupt Enable. 0: Disable, 1: Enable.
12	R/W	0x0	NDMA6_HF_IRQ_EN Normal DMA 6 Half Transfer Interrupt Enable. 0: Disable, 1: Enable.
11	R/W	0x0	NDMA5_END_IRQ_EN Normal DMA 5 End Transfer Interrupt Enable. 0: Disable, 1: Enable.
10	R/W	0x0	NDMA5_HF_IRQ_EN Normal DMA 5 Half Transfer Interrupt Enable. 0: Disable, 1: Enable.
9	R/W	0x0	NDMA4_END_IRQ_EN Normal DMA 4 End Transfer Interrupt Enable. 0: Disable, 1: Enable.
8	R/W	0x0	NDMA4_HF_IRQ_EN

			Normal DMA 4 Half Transfer Interrupt Enable. 0: Disable, 1: Enable.
7	R/W	0x0	NDMA3_END_IRQ_EN Normal DMA 3 End Transfer Interrupt Enable. 0: Disable, 1: Enable.
6	R/W	0x0	NDMA3_HF_IRQ_EN Normal DMA 3 Half Transfer Interrupt Enable. 0: Disable, 1: Enable.
5	R/W	0x0	NDMA2_END_IRQ_EN Normal DMA 2 End Transfer Interrupt Enable. 0: Disable, 1: Enable.
4	R/W	0x0	NDMA2_HF_IRQ_EN Normal DMA 2 Half Transfer Interrupt Enable. 0: Disable, 1: Enable.
3	R/W	0x0	NDMA1_END_IRQ_EN Normal DMA 1 End Transfer Interrupt Enable. 0: Disable, 1: Enable.
2	R/W	0x0	NDMA1_HF_IRQ_EN Normal DMA 1 Half Transfer Interrupt Enable. 0: Disable, 1: Enable.
1	R/W	0x0	NDMA0_END_IRQ_EN Normal DMA 0 End Transfer Interrupt Enable. 0: Disable, 1: Enable.
0	R/W	0x0	NDMA0_HF_IRQ_EN Normal DMA 0 Half Transfer Interrupt Enable. 0: Disable, 1: Enable.

3.8.3.2DMA IRQ Pending Status Register

Address : 0x0004			Name: DMA_IRQ_PEND_STA_REG Default: 0x0000_0000
Field	Type	Default	Description
31:16	/	/	/

15	R/W	0x0	NDMA7_END_IRQ_PEND. Normal DMA 7 End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
14	R/W	0x0	NDMA7_HF_IRQ_PEND. Normal DMA 7 Half Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
13	R/W	0x0	NDMA6_END_IRQ_PEND. Normal DMA 6 End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
12	R/W	0x0	NDMA6_HF_IRQ_PEND. Normal DMA 6 Half Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
11	R/W	0x0	NDMA5_END_IRQ_PEND. Normal DMA 5 End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
10	R/W	0x0	NDMA5_HF_IRQ_PEND. Normal DMA 5 Half Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
9	R/W	0x0	NDMA4_END_IRQ_PEND. Normal DMA 4 End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
8	R/W	0x0	NDMA4_HF_IRQ_PEND. Normal DMA 4 Half Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
7	R/W	0x0	NDMA3_END_IRQ_PEND.

			Normal DMA 3 End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
6	R/W	0x0	NDMA3_HF_IRQ_PEND. Normal DMA 3 Half Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
5	R/W	0x0	NDMA2_END_IRQ_PEND. Normal DMA 2 End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
4	R/W	0x0	NDMA2_HF_IRQ_PEND. Normal DMA 2 Half Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
3	R/W	0x0	NDMA1_END_IRQ_PEND. Normal DMA 1 End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
2	R/W	0x0	NDMA1_HF_IRQ_PEND. Normal DMA 1 Half Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
1	R/W	0x0	NDMA0_END_IRQ_PEND. Normal DMA 0 End Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.
0	R/W	0x0	NDMA0_HF_IRQ_PEND. Normal DMA 0 Half Transfer Interrupt Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending.

3.8.3.3 Normal DMA Configuration Register

Address : 0x100+N*0x20 (N=0,1,2,3,4,5,6,7)	Name : NDMA_CTRL_REG Default : 0x0000_0000
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Field	Type	Default	Description
31	R/W	0x0	DMA_LOADING. DMA Loading. If set to 1, DMA will start and load the DMA registers to the shadow registers. The bit will hold on until the DMA finished. It will be cleared automatically. Set 0 to the bit will reset the corresponding DMA channel.
30	RO	0x0	DMA Busy Status. 0: DMA idle, 1: DMA busy.
29	R/W	0x0	DMA_CONTI_MODE_EN. DMA Continuous Mode Enable. 0: Disable, 1: Enable.
28:26	R/W	0x0	DMA Wait State. (n DMA clock cycles) 0: 1 1: 2 2: 4 3: 8 4: 16 5: 32 6: 64 7: 128
25:24	R/W	0x0	NDMA_DEST_DATA_WIDTH. Normal DMA Destination Data Width. 00: 8-bit 01: 16-bit 10: 32-bit 11: /
23	R/W	0x0	DMA_DEST_BST_LEN. DMA Destination Burst Length. 00: 1, 01: 4
22	/	/	/
21	R/W	0x0	NDMA_DEST_ADDR_TYPE. Normal DMA Destination Address Type. 0: Increment 1: No Change.
20:16	R/W	0x0	NDMA_DEST_DRQ_TYPE.

			Normal DMA Destination DRQ Type. 00000 : SRAM 00001 : SPI0 TX 00010 : SPI1 TX 00011 : UART0 TX 00100 : UART1 TX 00101 : SS TX 00110 : DAUDIO TX 00111 : FLASHC TX 01000 : Reserved 01001 : UART2 TX 01010 : AUDIO CODEC DAC TX 01011 : Reserved 01100 : PSRAM TX others : /
15	R/W	0x0	BC_MODE_SEL. BC mode select. 0: normal mode (the value read back is equal to the value that is written) 1: remain mode (the value read back is equal to the left bytes to be transferred).
14:10	/	/	/.
9:8	R/W	0x0	NDMA_SRC_DATA_WIDTH. Normal DMA Source Data Width. 00: 8-bit 01: 16-bit 10: 32-bit 11: /
7	R/W	0x0	DMA_SRC_BST_LEN. DMA Source Burst Length. 00: 1 01: 4
6	/	/	/
5	R/W	0x0	NDMA_SRC_ADDR_TYPE. Normal DMA Source Address Type. 0: Increment 1: No Change
4:0	R/W	0x0	NDMA_SRC_DRQ_TYPE.

			Normal DMA Source DRQ Type. 00000 : SRAM 00001 : SPI0 RX 00010 : SPI1 RX 00011 : UART0 RX 00100 : UART1 RX 00101 : SS RX 00110 : DAUDIO RX 00111 : FLASHC RX 01000 : Reserved 01001 : UART2 RX 01010 : AUDIO CODEC ADC RX 01011 : Reserved 01100 : PSRAM RX others : /
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3.8.3.4 Normal DMA Source Address Register

Address : 0x100+N*0x20+0x4 (N=0,1,2,3,4,5,6,7)			Name : NDMA_SRC_ADDR_REG Default : 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	NDMA_SRC_ADDR. Normal DMA Source Address.

3.8.3.5 Normal DMA Destination Address Register

Address : 0x100+N*0x20+0x8 (N=0,1,2,3,4,5,6,7)			Name : NDMA_DEST_ADDR_REG Default : 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	NDMA_DEST_ADDR. Normal DMA Destination Address.

3.8.3.6 Normal DMA Byte Counter Register

Address : 0x100+N*0x20+0xC (N=0,1,2,3,4,5,6,7)			Name : NDMA_BC_REG Default : 0x0000_0000
Field	Type	Default	Description
31:18	/	/	/.
17:0	R/W	0x0	NDMA_BC. Normal DMA Byte Counter.

			Note: If Byte Counter=0, DMA will transfer no byte. The maximum value is 128k.
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3.9 Real Time Clock (RTC)

3.9.1 Overview

The real time clock (RTC) is for calendar usage. It is built around a 30-bit counter and used to count elapsed time in YY-MM-DD and HH-MM-SS. The unit can be operated by the backup battery while the system power is off. It has a built-in leap year generator.

The alarm generates an alarm signal at a specified time in the power-off mode or normal operation mode. In normal operation mode, both the alarm interrupt and the power management wakeup are activated. In power-off mode, the power management wakeup signal is activated. In this section, there are two kinds of alarm. Alarm 0 is a general alarm; its counter is based on second. Alarm 1 is a weekly alarm; its counter is based on the real time.

3.9.2 Register List

Module Name	Base Address	
Real Time Clock	0x40041800	
Register Name	Offset Address	Description
RTC_CTRL	0x00	RTC Control Register
RTC_YYMMDD	0x10	RTC Year Month Day Register
RTC_HHMMSS	0x14	RTC Hour Minute Second Register
ALM0_CNT	0x20	Alarm 0 Counter Register
ALM0_CURVAL	0x24	Alarm 0 Current Value Register
ALM0_EN	0x28	Alarm 0 Enable Register
ALM0_IRQEN	0x2c	Alarm 0 IRQ Enable Register
ALM0_IRQST	0x30	Alarm 0 IRQ Status Register
ALM1_WK_HHMMSS	0x40	Alarm 1 Week Hour-Minute-Second Register
ALM1_EN	0x44	Alarm 1 Enable Register
ALM1_IRQEN	0x48	Alarm 1 IRQ Enable Register
ALM1_IRQST	0x4c	Alarm 1 IRQ Status Register
ALM1_WAKEUP_EN	0x50	Alarm 1 Wakeup Enable Register
FRUN_CNT_L	0x60	Free Running Counter bit[31:0]
FRUN_CNT_H	0x64	Free Running Counter bit[47:32]

3.9.2.1 RTC Control Register

Address : 0x00			Name: RTC_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31	R/W	0x0	RTC_TEST_MODE_CTRL. RTC TEST Mode Control bit.
30	R/W	0x0	RTC_DEBUG. RTC Simulation Control bit 0: No effect. 1: simulation mode
29:3	/	/	/
2	R	0	ALM_DDHHMMSS_ACCE. ALARM DD-HH-MM-SS access. After writing the ALARM DD-HH-MM-SS register, this bit is set and it will be cleared until the real writing operation is finished
1	R	0	RTC_HHMMSS_ACCE. RTC HH-MM-SS access. After writing the RTC HH-MM-SS register, this bit is set and it will be cleared until the real writing operation is finished. After writing the RTC YY-MM-DD register, the YY-MM-DD register will be refreshed for at most one second.
0	R	0	RTC_YYMMDD_ACCE. RTC YY-MM-DD access. After writing the RTC YY-MM-DD register, this bit is set and it will be cleared until the real writing operation is finished. After writing the RTC YY-MM-DD register, the YY-MM-DD register will be refreshed for at most one second.

3.9.2.2 RTC YYMMDD Register

Address : 0x10			Name: RTC_YYMMDD Default: 0x0000_0000
Field	Type	Default	Description
31:25	/	/	/.
24	R/W	0x0	LEAP. Leap Year. 0: not 1: Leap year. This bit cannot be set by hardware. It should be set or cleared by software.

23:16	R/W	x	YEAR. Year. Range from 0~256.
15:12	/	/	/
11:8	R/W	x	MONTH. Month. Range from 1~12.
7:5	/	/	/.
4:0	R/W	x	DAY. Day. Range from 1~31. <i>Note: If the written value is not from 1 to 31, it turns into 31 automatically. The month field and the year field are similar.</i>

3.9.2.3RTC HHMMSS Register

Address : 0x14			Name: RTC_HHMMSS Default: 0x0000_0000
Field	Type	Default	Description
31:29	R/W	0x0	WEEK_NO. Week number. 000: Monday 001: Tuesday 010: Wednesday 011: Thursday 100: Friday 101: Saturday 110: Sunday 111: /
28:21	/	/	/.
20:16	R/W	x	HOUR. Range from 0~23
15:14	/	/	/.
13:8	R/W	x	MINUTE. Range from 0~59

7:6	/	/	/.
5:0	R/W	x	SECOND. Range from 0~59

3.9.2.4 Alarm 0 Counter Register

Address : 0x20			Name: ALMO_CNT Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	ALMO_CNT Alarm 0 Counter is based on second.

3.9.2.5 Alarm 0 Current Value Register

Address : 0x24			Name: ALMO_CURVAL Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	x	ALMO_CURVAL

3.9.2.6 Alarm 0 Enable Register

Address : 0x28			Name: ALMO_CURVAL Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0	ALMO_EN If this bit is set to 1, the alarm 0 counter register's valid bits will down count to 0 and the alarm pending bit will be set to 1 0: disable 1: enable

3.9.2.7 Alarm 0 IRQ Enable Register

Address : 0x2c			Name: ALMO_IRQEN Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0	ALMO_IRQEN 0: disable 1: enable

3.9.2.8 Alarm 0 IRQ Status Register

Address : 0x30			Name : ALM0_IRQST Default : 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0	ALM0_IRQST 0: no effect 1: pending

3.9.2.9 Alarm 1 Week HH-MM-SS Register

Address : 0x40			Name : Alarm 1_WK_HHMMSS Default : 0x0000_0000
Field	Type	Default	Description
31:21	/	/	/
20:16	R/W	X	HOUR Range from 0-23
15:14	/	/	/
13:8	R/W	X	MINUTE Range from 0-59
7:6	/	/	/
5:0	R/W	X	SECOND Range from 0-59

3.9.2.10 Alarm 1 Enable Register

Address : 0x44			Name : ALM1_EN Default : 0x0000_0000
Field	Type	Default	Description
31:7	/	/	/
6	R/W	0	SUN_ALM1_EN Sunday Alarm 1 Enable 0: disable 1: enable If this bit is set to "1", only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC HH-MM-SS bit [31:29] is 6, the alarm IRQ pending bit will be set to "1".
5	R/W	0	SAT_ALM1_EN

			Saturday Alarm 1 Enable 0: disable 1: enable If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC HH-MM-SS bit [31:29] is 5, the alarm IRQ pending bit will be set to “1”.
4	R/W	0	FRI_ALM1_EN Friday Alarm 1 Enable 0: disable 1: enable If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC HH-MM-SS bit [31:29] is 4, the alarm IRQ pending bit will be set to “1”.
3	R/W	0	THU_ALM1_EN Thursday Alarm 1 Enable 0: disable 1: enable If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC HH-MM-SS bit [31:29] is 3, the alarm IRQ pending bit will be set to “1”.
2	R/W	0	WED_ALM1_EN Wednesday Alarm 1 Enable 0: disable 1: enable If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC HH-MM-SS bit [31:29] is 2, the alarm IRQ pending bit will be set to “1”.
1	R/W	0	TUE_ALM1_EN Tuesday Alarm 1 Enable 0: disable 1: enable If this bit is set to “1”, only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC HH-MM-SS bit [31:29] is 1, the alarm IRQ pending bit will be set to “1”.
0	R/W	0	MON_ALM1_EN

			Monday Alarm 1 Enable 0: disable 1: enable If this bit is set to "1", only when the Alarm 1 Week HH-MM-SS register valid bits is equal to RTC HH-MM-SS register and the register RTC HH-MM-SS bit [31:29] is 0, the alarm IRQ pending bit will be set to "1".
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3.9.2.11 Alarm 1 IRQ Enable Register

Address : 0x48			Name: ALM1_IRQEN Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0	ALM1_IRQEN 0: disable 1: enable

3.9.2.12 Alarm 1 IRQ Status Register

Address : 0x4c			Name: ALM1_IRQST Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0	ALM1_WEEK_IRQST Alarm 1 week(0-7) IRQ Pending 0: No effect 1: pending

1.1.3.7 Alarm 1 Wakeup Enable Register

Address : 0x50			Name: ALM1_WAKEUP_EN Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0x0	ALM1_WAKEUP 0: disable alarm1 wakeup output 1: enable alarm1 wakeup output

3.9.2.13 Free Running Counter Low Register

Address : 0x60			Name: FREERUN_CNT_L Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	Free Running Counter bit[31:0] <i>Note: Free running counter is a 48-bit counter which is driven by LFCLK and starts to count as soon as the system reset is released and the LFCLK is ready. Write will reset the counter value.</i>

3.9.2.14 Free Running Counter High Register

Address : 0x64			Name: FREERUN_CNT_H Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	Free Running Counter bit[47:32] <i>Note: Free running counter is a 48-bit counter which is driven by LFCLK and starts to count as soon as the system reset is released and the LFCLK is ready. Write will reset the counter value.</i>

3.10 Timer Controller (Timer)

3.10.1 Overview

Timer 0 and 1 can take their inputs from internal RC oscillator, external 32768Hz crystal or OSC. They provide the operating system's scheduler interrupt. It is designed to offer maximum accuracy and efficient management, even for systems with long or short response time. They provide 24-bit programmable overflow counter and work in auto-reload mode or no-reload mode.

The watch-dog is used to resume the controller operation when it had been disturbed by malfunctions such as noise and system errors. It features a down counter that allows a watchdog period of up to 16 seconds. It can generate a general reset or an interrupt request.

3.10.2 Register List

Module Name	Base Address	
Timer	0x40040800	
Register Name	Offset Address	Description
TMR_IRQ_EN_REG	0x00	Timer IRQ Enable Register
TMR_IRQ_STA_REG	0x04	Timer Status Register

TMR0_CTRL_REG	0x10	Timer 0 Control Register
TMR0_INTV_VALUE_REG	0x14	Timer 0 Interval Value Register
TMR0_CUR_VALUE_REG	0x18	Timer 0 Current Value Register
TMR1_CTRL_REG	0x20	Timer 1 Control Register
TMR1_INTV_VALUE_REG	0x24	Timer 1 Interval Value Register
TMR1_CUR_VALUE_REG	0x28	Timer 1 Current Value Register
WDOG_IRQ_EN_REG	0xA0	Watchdog Irq Enable Register
WDOG_IRQ_STA_REG	0xA4	Watchdog Irq Status Register
WDOG_CTRL_REG	0xB0	Watchdog Control Register
WDOG_CFG_REG	0xB4	Watchdog Configuration Register
WDOG_MODE_REG	0xB8	Watchdog Mode Register
WDOG_OUTPUT_CTRL_REG	0xBC	Watchdog Output Control Register

3.10.3 Register Description

3.10.3.1 Timer IRQ Enable Register

Address : 0x0000			Name : TMR_IRQ_EN_REG Default : 0x0000_0000
Field	Type	Default	Description
31:2	/	/	/.
1	R/W	0x0	TMR1_IRQ_EN. Timer 1 Interrupt Enable. 0: No effect 1: Timer 1 Interval Value reached interrupt enable.
0	R/W	0x0	TMR0_IRQ_EN. Timer 0 Interrupt Enable. 0: No effect 1: Timer 0 Interval Value reached interrupt enable.

3.10.3.2 Timer IRQ Status Register

Address : 0x0004			Name : TMR_IRQ_STA_REG Default : 0x0000_0000
Field	Type	Default	Description
31:2	/	/	/
1	R/W	0x0	TMR1_IRQ_PEND.

			Timer 1 IRQ Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending, timer 1 interval value is reached.
0	R/W	0x0	TMRO_IRQ_PEND. Timer 0 IRQ Pending. Set 1 to the bit will clear it. 0: No effect, 1: Pending, timer 0 interval value is reached.

3.10.3.3 Timer 0 Control Register

Address : 0x0010			Name: TMRO_CTRL_REG Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/.
7	R/W	0x0	TMRO_MODE. Timer 0 mode. 0: Continuous mode. When interval value reached, the timer will not disable automatically. 1: Single mode. When interval value reached, the timer will disable automatically.
6:4	R/W	0x0	TMRO_CLK_PRES. Select the pre-scale of timer 0 clock source. 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128
3:2	R/W	0x1	TMRO_CLK_SRC. Timer 0 Clock Source. 00: LOSC, 01: HOSC. 10: / 11: /.
1	R/W	0x0	TMRO_RELOAD. Timer 0 Reload.

			0: No effect, 1: Reload timer 0 Interval value.
0	R/W	0x0	TMRO_EN. Timer 0 Enable. 0: Stop/Pause 1: Start. If the timer is started, it will reload the interval value to internal register, and the current counter will count from interval value to 0. If the current counter does not reach the zero, the timer enable bit is set to "0"; the current value counter will pause. At least wait for $2 \cdot T_{cycle}$, the start bit can be set to 1. In timer pause state, the interval value register can be modified. If the timer is started again, and the Software hope the current value register to down-count from the new interval value, the reload bit and the enable bit should be set to 1 at the same time.

3.10.3.4 Timer 0 Interval Value Register

Address : 0x0014			Name: TMR0_INTV_VALUE_REG Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	x	TMRO_INTV_VALUE. Timer 0 Interval Value. Note: the value setting should consider the system clock and the timer clock source.

3.10.3.5 Timer 0 Current Value Register

Address : 0x0018			Name: TMR0_CUR_VALUE_REG Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	TMRO_CUR_VALUE. Timer 0 Current Value.

3.10.3.6 Timer 1 Control Register

Address : 0x0020			Name: TMR1_CTRL_REG Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/.

7	R/W	0x0	TMR1_MODE. Timer 1 mode. 0: Continuous mode. When interval value reached, the timer will not disable automatically. 1: Single mode. When interval value reached, the timer will disable automatically.
6:4	R/W	0x0	TMR1_CLK_PRES. Select the pre-scale of timer 1 clock source. 000: /1 001: /2 010: /4 011: /8 100: /16 101: /32 110: /64 111: /128
3:2	R/W	0x1	TMR1_CLK_SRC. Timer 1 Clock Source. 00: LOSC, 01: HOSC. 10: / 11: /.
1	R/W	0x0	TMR1_RELOAD. Timer 1 Reload. 0: No effect 1: Reload timer 1 Interval value.
0	R/W	0x0	TMR1_EN. Timer 1 Enable. 0: Stop/Pause 1: Start. If the timer is started, it will reload the interval value to internal register, and the current counter will count from interval value to 0. If the current counter does not reach the zero, the timer enable bit is set to "0"; the current value counter will pause. At least wait for $2 \cdot T_{cycle}$, the start bit can be set to 1. In timer pause state, the interval value register can be modified. If the timer is started again, and the Software hope the current value register to down-count from the new interval value, the reload bit and the enable bit should be set to 1 at the same time.

3.10.3.7 Timer 1 Interval Value Register

Address : 0x0024			Name: TMR1_INTV_VALUE_REG Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	x	TMR1_INTV_VALUE. Timer 1 Interval Value. Note: the value setting should consider the system clock and the timer clock source.

3.10.3.8 Timer 1 Current Value Register

Address : 0x0028			Name: TMR1_CUR_VALUE_REG Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	TMR1_CUR_VALUE. Timer 1 Current Value.

3.10.3.9 Watchdog IRQ Enable Register

Address : 0x00A0			Name: WDOG_IRQ_EN_REG Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0	WDOG_IRQ_EN 0: No effect 1: Watchdog interrupt enable

3.10.3.10 Watchdog IRQ Status Register

Address : 0x00A4			Name: WDOG_IRQ_EN_REG Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0	WDOG_IRQ_STA Watchdog irq pending. Write 1 to clear it 0: No effect 1: Pending, the interval value is reached.

3.10.3.11 Watchdog Control Register

Address : 0x00B0			Name: WDOG_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:13	/	/	/
12:1	R/W	0	WDOG0_KEY_FIELD. Watchdog 0 Key Field. Should be written at value 0xA57. Writing any other value in this field aborts the write operation.
0	R/W	0	WDOG_RELOAD Watchdog Reload 0: no effect 1: reload the watchdog

3.10.3.12 Watchdog Configuration Register

Address : 0x00B4			Name: WDOG_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:2	/	/	/
8	R/W	0	CLK_SEL 0: 32K 1: 32.768K
7:4	/	/	/
3:2	R/W	0	WDOG0_RST_CPU_MODE 00: when only reset CPU, watchdog resetn to CPU PORESETn. 01: when only reset CPU, watchdog resetn only to CPU SYSRESETn. 1x: when only reset CPU, watchdog resetn to CPU PORESETn/SYSRESETn.
1:0	R/W	1	WDOG0_CONFIG. 00: only reset CPU 01: reset whole system 10: only interrupt 11: /

3.10.3.13 Watchdog Mode Register

Address : 0x00B8			Name: WDOG_MODE Default: 0x0000_0000
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Field	Type	Default	Description
31:8	/	/	/
7:4	R/W	0	WDOG_INTV_VALUE Watchdog Interval Value The watchdog clock source is HOSC/WDOG_PRESCALER. If the HOSC is turned off, the watchdog will not work. 0000: 16000 cycles (0.5s) 0001: 32000 cycles (1s) 0010: 64000 cycles (2s) 0011: 96000 cycles (3s) 0100: 128000 cycles (4s) 0101: 160000 cycles (5s) 0110: 192000 cycles (6s) 0111: 256000 cycles (8s) 1000: 320000 cycles (10s) 1001: 384000 cycles (12s) 1010: 448000 cycles (14s) 1011: 512000 cycles (16s) others: /
3:1	W	0	WDOG_RST_110 When watchdog enable (WDOG_EN=1), written at value 3'b110. The watchdog reset should assert promptly. Other values are no valid.
0	R/W	0	WDOG_EN Watchdog Enable 0: no effect 1: Enable watchdog

3.10.3.14 Watchdog Output Control Register

Address : 0x00BC			Name: WDOG_OUTPUT_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:5	/	/	/
4:0	R/W	0xA	WDOG_OUTPUT_CONTROL 0~31 the number of cycles with which the watch dog holds the reset signal

3.11 Security ID (SID)

3.11.1 Overview

There is a 1024-bit on chip EFUSE, with size up to 128 bits for security chip ID. The other bits are reserved for system parameters. The work clock of Security ID is 26MHz, which clock is sourced from APB clock.

The Security ID includes the following features:

- 1024-bit on chip eFuse
- 128-bit electrical fuses for chip ID
- Other bits are used for system parameter.

3.11.2 Register List

Module Name	Base Address	
Security ID	0x40043C00	
Register Name	Offset Address	Description
SID_PRCTL	0x40	SID Program/Read Control Register
SID_PKEY	0x50	SID Program Key Value Register
SID_RKEY	0x60	SID Read Key Value Register
BURNED_TIMING	0x90	SID Burned Timing Control
SDI_DBG	0x94	SID Debug Register
SID_VALUE	0x100 – 0x1FF	SID Chip ID Register 0

3.11.3 Register Description

3.11.3.1 SID Program/Read Control Register

Address : 0x0040			Name: SID_PR_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:29	/	/	/
			EFUSE_CLOCK_GATE_SW
28	R/W	0	OSC40M (eFuse_clock) clock gate enable. When read or program the effuse, this bit should be set to 1

27:24	/	/	/
23:16	R/W	0	PG_INDEX Program Index The index value of 8-bits electrical fuse hardware macrocell address offset, and the lowest two bits must be zero.
15:8	R/W	0	OP_LOCK Operation Lock The Read Start Bit (bit1) and Program Start bit (bit0) only can be written when these bits (bit[15:8]) set to 0xAC
7:3	/	/	/
2	R	x	HW_READ_STATUS Hardware read status 0: No hardware operation 1: Hardware Reading
1	R/W	0	EFUSE_READ_START Software Read Start Write '1' to start software read and automatically clear to '0' after read.
0	R/W	0	EFUSE_PRG_START Software Program Start Write '1' to start software program and automatically clear to '0' after program.

3.11.3.2 SID Program Key Value Register

Address : 0x0050			Name: SID_PKEY Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	SID_PGKEY_VALUE Program key value

3.11.3.3 SID Read Key Value Register

Address : 0x0060			Name: SID_PKEY Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0	SID_RDKEY_VALUE

			Program key value
--	--	--	-------------------

3.11.3.4 SID Burn Timing Register

Address : 0x0090			Name: SID_BURN_TIMING Default: 0x630B_905C
Field	Type	Default	Description
31:28	R/W	0x6	Control T_{aen} of the eFuse Reading cycle value * T_{osc} $T_{aen} > T_{rd} + 35ns$
28:24	R/W	0x3	Control T_{rd} of the eFuse Reading cycle value * T_{osc} $T_{rd} > 40ns$
23:12	R/W	0X0B9	Control T_{aen} of the eFuse program cycle value * T_{osc} $T_{aen} > T_{pgm} + 2300ns$
11:0	R/W	0X05C	Control T_{pgm} of the eFuse Program cycle value * T_{osc} $T_{pgm} \geq 2300ns$ ($2600ns > T_{pgm} > 2000ns$)

3.11.3.5 SID Debug Register

Address : 0x0094			Name: SID_DBG Default: 0x0000_0000
Field	Type	Default	Description
31:5	/	/	/
4	R/W	0	LDO-EFUSE software switch 0: soft switch off 1: soft switch on <i>Note: Switch on accumulative total time should less than 1 second!!!</i>
3:0	/	/	/

3.11.3.6 SID Register

Address : 0x100 – 0x1FF			Name: SID_VALUE Default: 0xFFFFFFFF
Field	Type	Default	Description
31:0	R	x	SID [32n+31: 32n]

3.12 Secure System (SS)/Crypto Engine (CE)

3.12.1 Overview

The Crypto Engine (CE) is one encrypt/decrypt algorithms accelerator. It is suitable for a variety of applications. Several modes are supported by the CE module. Both CPU mode and DMA method are supported for different applications.

It includes the following features:

- Supports AES, DES, 3DES, SHA-1, MD5, PRNG, CRC32/16, SHA256, TRNG
- Supports ECB, CBC, CTR modes for AES/DES/3DES
- Supports CTS mode for AES
- Supports 128-bits, 192-bits and 256-bit key size for AES
- Supports 160-bits hardware PRNG with 192-bits seed
- 32-word RX FIFO and 32-word TX FIFO for high speed applications
- CPU mode and DMA mode are supported

The Crypto Engine block diagram is shown as below:

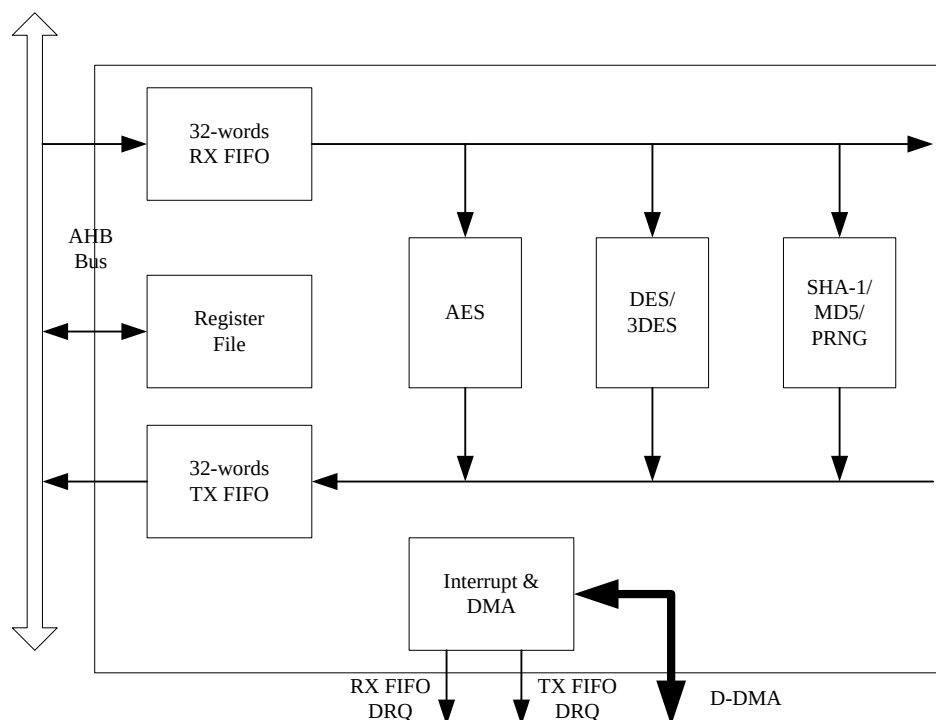


Figure 3-8 Crypto Engine Block Diagram

3.12.2 Register List

Module Name	Base Address	
Secure System Controller	0x40004000	
Register Name	Offset Address	Description
CE_CTL	0x00	Control Register
CE_KEY0	0x04	Input Key 0/ PRNG Seed 0
CE_KEY1	0x08	Input Key 1/ PRNG Seed 1
...	...	...
CE_KEY7	0x20	Input Key 7
CE_IV0	0x24	Initialization Vector 0
CE_IV1	0x28	Initialization Vector 1
CE_IV2	0x2C	Initialization Vector 2
CE_IV3	0x30	Initialization Vector 3
CE_CNT0	0x34	Counter 0/Initialization Vector 4
CE_CNT1	0x38	Counter 1/Initialization Vector 5
CE_CNT2	0x3C	Counter 2/Initialization Vector 6
CE_CNT3	0x40	Counter 3/Initialization Vector 7
CE_FCSR	0x44	FIFO Control/ Status Register
CE_ICSR	0x48	Interrupt Control/ Status Register
CE_MD0	0x4C	SHA1/MD5 Message Digest 0/PRNG Data0
CE_MD1	0x50	SHA1/MD5 Message Digest 1/PRNG Data1
CE_MD2	0x54	SHA1/MD5 Message Digest 2/PRNG Data2
CE_MD3	0x58	SHA1/MD5 Message Digest 3/PRNG Data3
CE_MD4	0x5C	SHA1/MD5 Message Digest 4/PRNG Data4
CE_CTS_LEN	0x60	AES-CTS/CRC text length
CE_CRC_POLY	0x70	CRC Poly Register
CRC_RESULT	0x74	CRC Result Register
CE_MD5	0xa0	SHA256 Message Digest Data 5
CE_MD6	0xa4	SHA256 Message Digest Data 6
CE_MD7	0xa8	SHA256 Message Digest Data 7
CE_RXFIFO	0x200	RX FIFO input port
CE_TXFIFO	0x204	TX FIFO output port

3.12.3 Register Description

3.12.3.1 CE Control Register

Address : 0x00	Name: CE_CTL
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			Default: 0x0000_0000
Field	Type	Default	Description
31	R/W	0	CRC xor out value 0: CRC result as CRC out 1: CRC result xor 0xffffffff(CRC32) or 0xffff(CRC16) as CRC out
30	R/W	0	CRC ref out value 0: output word bit sequence not change 1: output word bit sequence reverse
29	R/W	0	CRC ref in value 0: input data bit sequence not change 1: input byte bit sequence reverse
28	R/W	0	CRC init value 0: init value is 0 1: init value is 0xffffffff for CRC32 or 0xffff for CRC16
27:24	R/W	0	KEY_SELECT AES/DES/3DES key select 0: Select input CE_KEYx (Normal Mode) 1: Select SID_RKEYx from SID 2: / 3-10: Select internal Key n (n from 0 to 7) Others: Reserved
18:16	R	x	DIE_ID Die Bonding ID
15	R/W	0	PRNG_MODE/CRC_CONT PRNG generator mode or CRC Package Continue 0: One-shot mode for PRNG or last package for CRC 1: Continue mode for PRNG or non-last package for CRC
14	R/W	0	IV_MODE IV mode for SHA-1/MD5 constants 0: use initial constants defined in FIPS-180 1: use input IV
13:12	R/W	0	CE_OP_MODE CE Operation Mode 00: Electronic Code Book (ECB) mode 01: Cipher Block Chaining (CBC) mode 10: Counter (CTR) mode 11: Cipher Text Stealing (CTS) mode
11:10	R/W	0	CTR_WIDTH Counter Width for CTR Mode 00: 16-bits Counter 01: 32-bits Counter 10: 64-bits Counter

			11: 128-bits Counter
9:8	R/W	0	AES_KEY_SIZE Key Size for AES 00: 128-bits 01: 192-bits 10: 256-bits 11: Reserved
7	R/W	0	CE_OP_DIR CE Operation Direction 0: Encryption 1: Decryption
6:4	R/W	0	CE_METHOD CE Method 000: AES 001: DES 010: Triple DES (3DES) 011: SHA-1 100: MD5 101: PRNG 110: SHA256 111: CRC
3	R/W	0	CRC_WIDTH 0: width 16, use 16 bits generate nominal 1: width 32, use 32 bits generate nominal
2	R/W	0	SHA1_SHA256_MD5_CRC_END_BIT SHA-1/MD5/SHA256/CRC Data End bit Write '1' to tell SHA-1/MD5/SHA256/CRC engine that the text data ends. If RX FIFO is not empty, the engine would process the data in RX FIFO. After finishing message digest, this bit is clear to '0' by hardware and message digest can be read out from digest registers. Notes: It is only used for SHA-1/MD5/SHA256/CRC engine.
1	R/W	0	PRNG_START PRNG start bit In PRNG one-shot mode, write '1' to start PRNG. After generating one group random data (5 words), this bit is clear to '0' by hardware.
0	R/W	0	CE_ENABLE CE Enable bit 0: Disable: write '0' to tell engine ends 1: Enable: after configuration, set this bit to '1' to start engine

3.12.3.2 CE Key Register

Address : 0x04 +4*n	Name: CE_KEY[n]
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			Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	CE_KEY Key[n] Input Value (n= 0~7)/ PRNG Seed[n] (n= 0~5)

3.12.3.3 CE IV Register

Address : 0x24 +4*n			Name: CE_IV[n] Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	CE_IV_VALUE Initialization Vector (IV[n]) Input Value (n= 0~3)

3.12.3.4 CE Counter Register

Address : 0x34 +4*n			Name: CE_CTR[n] Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	CE_CTR_VALUE/CE_IV_VALUE Counter mode preload Counter Input Value (n= 0~3) Initialization Vector (IV[4+n]) Input Value (n= 0~3)

3.12.3.5 CE FIFO Control/ Status Register

Address : 0x44			Name: CE_FCSR Default: 0x6000_0F0F
Field	Type	Default	Description
31	/	/	/
30	R	0x1	RXFIFO_STATUS RX FIFO Empty 0: No room for new word in RX FIFO 1: More than one room for new word in RX FIFO (>= 1 word)
29:24	R	0x20	RXFIFO_EMP_CNT RX FIFO Empty Space Word Counter
23	/	/	/
22	R	0	TXFIFO_STATUS TX FIFO Data Available Flag 0: No available data in TX FIFO 1: More than one data in TX FIFO (>= 1 word)
21:16	R	0	TXFIFO_AVA_CNT TX FIFO Available Word Counter
15:13	/	/	/

12:8	R/W	0xF	RXFIFO_INT_TRIG_LEVEL RX FIFO Empty Trigger Level Interrupt and DMA request trigger level for RXFIFO normal condition Trigger Level = RXTL + 1 Notes: RX FIFO is used for input the data.
7:5	/	/	/
4:0	R/W	0xF	TXFIFO_INT_TRIG_LEVEL TX FIFO Trigger Level Interrupt and DMA request trigger level for TXFIFO normal condition Trigger Level = TXTL + 1 Notes: TX FIFO is used for output the result data.

3.12.3.6 CE Interrupt Control/ Status Register

Address : 0x48			Name: CE_ICSR Default: 0x0000_0000
Field	Type	Default	Description
31:12	/	/	/
11	R/W	0	HASH_CRC_END_INT_PEND 0: No pending 1: HASH/CRC end pending
10	R/W	0	RXFIFO_EMP_PENDING_BIT RX FIFO Empty Pending bit 0: No pending 1: RX FIFO Empty pending Notes: Write '1' to clear or automatic clear if interrupt condition fails.
9	/	/	/
8	R/W	0	TXFIFO_AVA_PENDING_BIT TX FIFO Data Available Pending bit 0: No TX FIFO pending 1: TX FIFO pending Notes: Write '1' to clear or automatic clear if interrupt condition fails.
7:5	/	/	/
4	R/W	0	DRQ_ENABLE DRQ Enable 0: Disable DRQ (CPU polling mode) 1: Enable DRQ (DMA mode)
3	R/W	0	HASH_CRC_END_INT_ENABLE 0: Disable 1: Enable
2	R/W	0	RXFIFO_EMP_INT_ENABLE RX FIFO Empty Interrupt Enable

			0: Disable 1: Enable Notes: If it is set to '1', when the number of empty room is great or equal ($\geq$) the preset threshold, the interrupt is trigger and the correspond flag is set.
1	/	/	/
0	R/W	0	TXFIFO_AVA_INT_ENABLE TX FIFO Data Available Interrupt Enable 0: Disable 1: Enable Notes: If it is set to '1', when available data number is great or equal ($\geq$) the preset threshold, the interrupt is trigger and the correspond flag is set.

3.12.3.7 CE Message Digest Register

Address : 0x4C +4*n			Name: CE_MD[n] Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0	CE_MD SHA1/ MD5 Message digest for SHA1/MD5 (n= 0~4)

3.12.3.8 CE CTS Length Register

Address : 0x60			Name: CE_CTS_LEN Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	AES-CTS / CRC text length in byte unit The value of '0' means no data.

3.12.3.9 CE CRC Poly Register

Address : 0x70			Name: CE_CRC_POLY Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	CRC_POLY CRC poly nominal configuration

3.12.3.10 CE Result Register

Address : 0x74			Name: CE_RESULT Default: 0x0000_0000
Field	Type	Default	Description

31:0	R/W	0	CRC computation result
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3.12.3.11 CE SHA256 Message Digest Register

Address : 0xA0 + 4*(n-5) n = 5~7			Name: CE_MD[n] Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0	CE_SHA256_MDn SHA256 Message Digest n (n = 5~7)

3.12.3.12 CE RX FIFO Register

Address : 0x200			Name: CE_RX Default: 0x0000_0000
Field	Type	Default	Description
31:0	W	0	CE_RX_FIFO 32-bits RX FIFO for Input

3.12.3.13 CE TX FIFO Register

Address : 0x204			Name: CE_TX Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0	CE_TX_FIFO 32-bits TX FIFO for Output

Chapter 4 Peripherals

4.1 Universal Asynchronous Receivers/Transmitters (UART)

4.1.1 Overview

The UART is used for serial communication with a peripheral, modem (data carrier equipment, DCE) or data set. Data is written from a master (CPU) over the APB bus to the UART and it is converted to serial form and transmitted to the destination device. Serial data is also received by the UART and stored for the master (CPU) to read back.

The UART contains registers to control the character length, baud rate, parity generation/checking, and interrupt generation. Although there is only one interrupt output signal from the UART, there are several prioritized interrupt types that can be responsible for its assertion. Each of the interrupt types can be separately enabled /disabled with the control registers.

The UART has 16450 and 16550 modes of operation, which are compatible with a range of standard software drivers. In 16550 mode, transmit and receive operations are both buffered by FIFOs. In 16450 mode, these FIFOs are disabled.

The UART supports word lengths from five to eight bits, an optional parity bit and 1, 1 ½ or 2 stop bits, and is fully programmable by an AMBA APB CPU interface. A 16-bit programmable baud rate generator and an 8-bit scratch register are included, together with separate transmit and receive FIFOs. Eight modem control lines and a diagnostic loop-back mode are provided.

Interrupts can be generated for a range of TX Buffer/FIFO, RX Buffer/FIFO, Modem Status and Line Status conditions.

For integration in systems where Infrared SIR serial data format is required, the UART can be configured to have a software-programmable IrDA SIR Mode. If this mode is not selected, only the UART (RS232 standard) serial data format is available.

The UART controller includes the following features:

- Compatible with industry-standard 16550 UARTs
- 64-Bytes Transmit and receive data FIFOs
- Support DMA controller interface
- Support Software/ Hardware Flow Control
- Support IrDA 1.0 SIR
- Support RS-485/9-bit mode

The following figures show the UART timing in UART and IrDA modes.

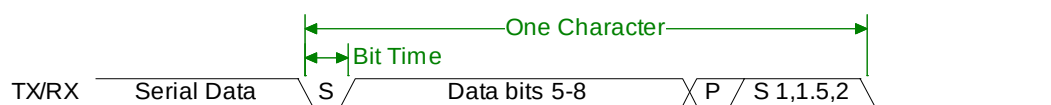


Figure 4-1 UART Serial Data Timing

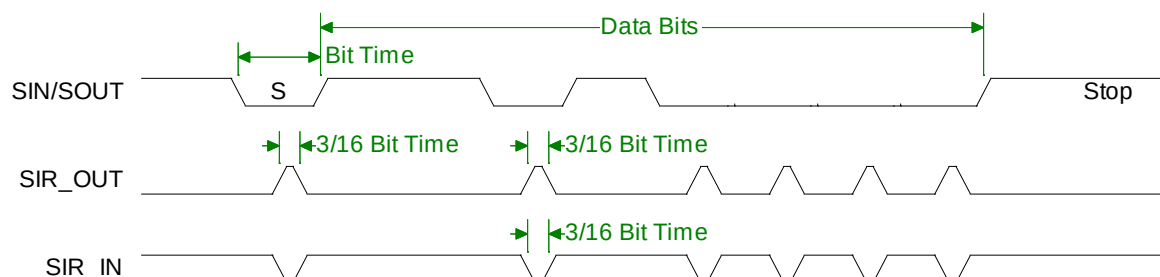


Figure 4-2 UART IrDA Timing

4.1.2 Register List

Module Name	Base Address	
UART Controller 0	0x40040C00	
UART Controller 1	0x40041000	
UART Controller 2	0x40041400	
Register Name	Offset Address	Description
UART_RBR	0x00	UART Receive Buffer Register
UART_THR	0x00	UART Transmit Holding Register
UART_DLL	0x00	UART Divisor Latch Low Register
UART_DLH	0x04	UART Divisor Latch High Register
UART_IER	0x04	UART Interrupt Enable Register
UART_IIR	0x08	UART Interrupt Identity Register
UART_FCR	0x08	UART FIFO Control Register
UART_LCR	0x0C	UART Line Control Register
UART_MCR	0x10	UART Modem Control Register
UART_LSR	0x14	UART Line Status Register
UART_MSR	0x18	UART Modem Status Register
UART_SCH	0x1C	UART Scratch Register
UART_USR	0x7C	UART Status Register
UART_TFL	0x80	UART Transmit FIFO Level
UART_RFL	0x84	UART_RFL
UART_HALT	0xA4	UART Halt TX Register
UART_DBG_DLL	0xB0	UART Debug DLL Register
UART_DBG_DLH	0xB4	UART Debug DLH Register
UART_485_CTL	0xC0	UART RS485 Control and Status Register
RS485_ADDR_MATCH	0xC4	UART RS485 Address Match Register

BUS_IDLE_CHK	0xC8	UART RS485 Bus Idle Check Register
TX_DLY	0xCC	UART TX Delay Register
UART_BDCR	0xD4	UART Baudrate Detection Control Register
UART_BDCLR	0xD8	UART Baudrate Detection Counter Low Register
UART_BDCHR	0xDC	UART Baudrate Detection Counter High Register

4.1.3 Register Description

4.1.3.1 UART Receiver Buffer Register

Address : 0x0000			Name: UART_RBR Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R	0x0	RBR Receiver Buffer Register Data byte received on the serial input port (sin) in UART mode, or the serial infrared input (sir_in) in infrared mode. The data in this register is valid only if the Data Ready (DR) bit in the Line Status Register (LCR) is set. If in FIFO mode and FIFOs are enabled (FCR [0] set to one), this register accesses the head of the receive FIFO. If the receive FIFO is full and this register is not read before the next data character arrives, then the data already in the FIFO is preserved, but any incoming data are lost and an overrun error occurs.

4.1.3.2 UART Transmit Holding Register

Address : 0x0000			Name: UART_THR Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	W	0x0	THR Transmit Holding Register Data to be transmitted on the serial output port (sout) in UART mode or the serial infrared output (sir_out_n) in infrared mode. Data should only be written to the THR when the THR Empty (THRE) bit (LSR[5]) is set. If in FIFO mode and FIFOs are enabled (FCR[0] = 1) and THRE is set, 16 number of characters of data may be written to the THR before the FIFO is

			full. Any attempt to write data when the FIFO is full results in the write data being lost.
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4.1.3.3 UART Divisor Latch Low Register

Address : 0x0000			Name: UART_DLL Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	DLL Divisor Latch Low Lower 8 bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. This register may only be accessed when the DLAB bit (LCR[7]) is set and the UART is not busy (USR[0] is zero). The output baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock freq) / (16 * divisor). Note that with the Divisor Latch Registers (DLL and DLH) set to zero, the baud clock is disabled and no serial communications occur. Also, once the DLL is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

4.1.3.4 UART Divisor Latch High Register

Address : 0x0004			Name: UART_DLH Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	DLH Divisor Latch High Upper 8 bits of a 16-bit, read/write, Divisor Latch register that contains the baud rate divisor for the UART. This register may only be accessed when the DLAB bit (LCR[7]) is set and the UART is not busy (USR[0] is zero). The output baud rate is equal to the serial clock (sclk) frequency divided by sixteen times the value of the baud rate divisor, as follows: baud rate = (serial clock freq) / (16 * divisor). Note that with the Divisor Latch Registers (DLL and DLH) set to zero, the baud clock is disabled and no serial communications occur. Also, once the DLH is set, at least 8 clock cycles of the slowest UART clock should be allowed to pass before transmitting or receiving data.

4.1.3.5 UART Interrupt Enable Register

Address : 0x0004			Name: UART_IER Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7	R/W	0x0	PTIME Programmable THRE Interrupt Mode Enable This is used to enable/disable the generation of THRE Interrupt. 0: Disable 1: Enable
6:5	/	/	/
4	R/W	0x0	RS485_INT_EN RS485 Interrupt Enable 0:Disable 1:Enable
3	R/W	0x0	EDSSI Enable Modem Status Interrupt This is used to enable/disable the generation of Modem Status Interrupt. This is the fourth highest priority interrupt. 0: Disable 1: Enable
2	R/W	0x0	ELSI Enable Receiver Line Status Interrupt This is used to enable/disable the generation of Receiver Line Status Interrupt. This is the highest priority interrupt. 0: Disable 1: Enable
1	R/W	0x0	ETBEI Enable Transmit Holding Register Empty Interrupt This is used to enable/disable the generation of Transmitter Holding Register Empty Interrupt. This is the third highest priority interrupt. 0: Disable 1: Enable
0	R/W	0x0	ERBFI Enable Received Data Available Interrupt

			This is used to enable/disable the generation of Received Data Available Interrupt and the Character Timeout Interrupt (if in FIFO mode and FIFOs enabled). These are the second highest priority interrupts. 0: Disable 1: Enable
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4.1.3.6 UART Interrupt Identity Register

Address : 0x0008			Name: UART_IIR Default: 0x0000_0001
Field	Type	Default	Description
31:8	/	/	/
7:6	R	0x0	FEFLAG FIFOs Enable Flag This is used to indicate whether the FIFOs are enabled or disabled. 00: Disable 11: Enable
5:4	/	/	/
3:0	R	0x1	IID Interrupt ID This indicates the highest priority pending interrupt which can be one of the following types: 0000: modem status 0001: no interrupt pending 0010: THR empty 0011: RS485 Interrupt 0100: received data available 0110: receiver line status 0111: busy detect 1100: character timeout Bit 3 indicates an interrupt can only occur when the FIFOs are enabled and used to distinguish a Character Timeout condition interrupt.

Interrupt ID	Priority Level	Interrupt Type	Interrupt Source	Interrupt Reset
0001	-	None	None	-
0110	Highest	Receiver line status	Overrun/parity/ framing errors or break interrupt	Reading the line status register
0011	Second	RS485 Interrupt	In RS485 mode, receives address data and match setting address	Writes 1 to addr flag to reset

0100	Third	Received data available	Receiver data available (non-FIFO mode or FIFOs disabled) or RCVR FIFO trigger level reached (FIFO mode and FIFOs enabled)	Reading the receiver buffer register (non-FIFO mode or FIFOs disabled) or the FIFO drops below the trigger level (FIFO mode and FIFOs enabled)
1100	Fourth	Character timeout indication	No characters in or out of the RCVR FIFO during the last 4 character times and there is at least 1 character in it during This time.	Reading the receiver buffer register
0010	Fifth	Transmit holding register empty	Transmitter holding register empty (Program THRE Mode disabled) or XMIT FIFO at or below threshold (Program THRE Mode enabled)	Reading the IIR register (if source of interrupt); or, writing into THR (FIFOs or THRE Mode not selected or disabled) or XMIT FIFO above threshold (FIFOs and THRE Mode selected and enabled).
0000	Sixth	Modem status	Clear to send or data set ready or ring indicator or data carrier detect. Note that if auto flow control mode is enabled, a change in CTS (that is, DCTS set) does not cause an interrupt.	Reading the Modem status Register
0111	Seventh	Busy detect indication	UART_16550_COMPATIBLE = NO and master has tried to write to the Line Control Register while the UART is busy (USR[0] is set to one).	Reading the UART status register

Table 4-1 UART Interrupt Identity Priority

4.1.3.7 UART FIFO Control Register

Address : 0x0008			Name: UART_FCR Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:6	W	0x0	RT RCVR Trigger This is used to select the trigger level in the receiver FIFO at which the Received Data Available Interrupt is generated. In auto flow control mode it is used to determine when the rts_n signal is de-asserted. It also determines when the dma_rx_req_n signal is asserted in certain modes of operation. 00: 1 character in the FIFO

			01: FIFO ¼ full 10: FIFO ½ full 11: FIFO-2 less than full
5:4	W	0x0	TFT TX Empty Trigger Writes have no effect when THRE_MODE_USER = Disabled. This is used to select the empty threshold level at which the THRE Interrupts are generated when the mode is active. It also determines when the dma_tx_req_n signal is asserted when in certain modes of operation. 00: FIFO empty 01: 2 characters in the FIFO 10: FIFO ¼ full 11: FIFO ½ full
3	W	0x0	DMAM DMA Mode 0: Mode 0 In this mode, if PTE is high and TX FIFO is enable, the TX DMA request will send when TFL is less than or equals to FIFO Trigger Level. If PTE is high and TX FIFO is disabled, the TX DMA request will send when THRE is empty. If PTE is low, the TX DMA request will send when the TX FIFO is empty. If dma_pte_rx is high and RX FIFO is enabled, the RX DRQ will send when RFL is equals to or more than FIFO Trigger Level. 1: Mode 1 In this mode, if TX FIFO is enabled and the PTE is high, the TX DMA request will send when TFL is less than or equals to FIFO Trigger Level. If PTE is low, the TX DMA request will send when TX FIFO is empty and the request stops only when TX FIFO is empty. If RFL is equals to or more than FIFO Trigger Level, the RX DRQ will be set 1, in otherwise, it will be set 0.
2	W	0x0	XFIFOR XMIT FIFO Reset This resets the control portion of the transmit FIFO and treats the FIFO as empty. This also de-asserts the DMA TX request. It is 'self-clearing'. It is not necessary to clear this bit.
1	W	0x0	RFIFOR RCVR FIFO Reset This resets the control portion of the receive FIFO and treats the FIFO as empty. This also de-asserts the DMA RX request. It is 'self-clearing'. It is not necessary to clear this bit.
0	W	0x0	FIFOE

			Enable FIFOs This enables/disables the transmit (XMIT) and receive (RCVR) FIFOs. Whenever the value of this bit is changed both the XMIT and RCVR controller portion of FIFOs is reset.
--	--	--	---

4.1.3.8 UART Line Control Register

Address : 0x000C			Name: UART_LCR Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7	R/W	0x0	DLAB Divisor Latch Access Bit It is writeable only when UART is not busy (USR[0] is zero) and always readable. This bit is used to enable reading and writing of the Divisor Latch register (DLL and DLH) to set the baud rate of the UART. This bit must be cleared after initial baud rate setup in order to access other registers. 0: Select RX Buffer Register (RBR) / TX Holding Register (THR) and Interrupt Enable Register (IER) 1: Select Divisor Latch LS Register (DLL) and Divisor Latch MS Register (DLM)
6	R/W	0x0	BC Break Control Bit This is used to cause a break condition to be transmitted to the receiving device. If set to one the serial output is forced to the spacing (logic 0) state. When not in Loopback Mode, as determined by MCR[4], the sout line is forced low until the Break bit is cleared. If SIR_MODE = Enabled and active (MCR[6] set to one) the sir_out_n line is continuously pulsed. When in Loopback Mode, the break condition is internally looped back to the receiver and the sir_out_n line is forced low.
5:4	R/W	0x0	EPS Even Parity Select It is writeable only when UART is not busy (USR[0] is zero) and always writable readable. This is used to select between even and odd parity, when parity is enabled (PEN set to one). Setting the LCR[5] is used to reverse the LCR[4]. 00: Odd Parity 01: Even Parity 1X: Reverse LCR[4] In RS485 mode, it is the 9th bit--address bit.

			11:9th bit = 1, indicates that this is a address byte 10:9th bit = 0, indicates that this is a data byte Note: When use this function, PEN(LCR[3]) must set to 1.
3	R/W	0x0	PEN Parity Enable It is writeable only when UART is not busy (USR[0] is zero) and always readable. This bit is used to enable and disable parity generation and detection in transmitted and received serial character respectively. 0: parity disabled 1: parity enabled
2	R/W	0x0	STOP Number of stop bits It is writeable only when UART is not busy (USR[0] is zero) and always readable. This is used to select the number of stop bits per character that the peripheral transmits and receives. If set to zero, one stop bit is transmitted in the serial data. If set to one and the data bits are set to 5 (LCR[1:0] set to zero) one and a half stop bits is transmitted. Other/Wise, two stop bits are transmitted. Note that regardless of the number of stop bits selected the receiver checks only the first stop bit. 0: 1 stop bit 1: 1.5 stop bits when DLS (LCR[1:0]) is zero, else 2 stop bit
1:0	R/W	0x0	DLS Data Length Select It is writeable only when UART is not busy (USR[0] is zero) and always readable. This is used to select the number of data bits per character that the peripheral transmits and receives. The number of bit that may be selected areas follows: 00: 5 bits 01: 6 bits 10: 7 bits 11: 8 bits

4.1.3.9 UART Modem Control Register

Address : 0x0010			Name : UART_MCR Default : 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:6	R/W	0x0	UART_FUNCTION

			Select IrDA or RS485 0:Uart Mode 1:IrDA SIR Mode 2:RS485 Mode 3:Reverse
5	R/W	0x0	AFCE Auto Flow Control Enable When FIFOs are enabled and the Auto Flow Control Enable (AFCE) bit is set, Auto Flow Control features are enabled. 0: Auto Flow Control Mode disabled 1: Auto Flow Control Mode enabled
4	R/W	0x0	LOOP Loop Back Mode 0: Normal Mode 1: Loop Back Mode This is used to put the UART into a diagnostic mode for test purposes. If operating in UART mode (SIR_MODE != Enabled or not active, MCR[6] set to zero), data on the sout line is held high, while serial data output is looped back to the sin line, internally. In this mode all the interrupts are fully functional. Also, in loopback mode, the modem control inputs (dsr_n, cts_n, ri_n, dcd_n) are disconnected and the modem control outputs (dtr_n, rts_n, out1_n, out2_n) are looped back to the inputs, internally. If operating in infrared mode (SIR_MODE == Enabled AND active, MCR[6] set to one), data on the sir_out_n line is held low, while serial data output is inverted and looped back to the sir_in line.
3:2	/	/	/
1	R/W	0x0	RTS Request to Send This is used to directly control the Request to Send (rts_n) output. The Request To Send (rts_n) output is used to inform the modem or data set that the UART is ready to exchange data. When Auto RTS Flow Control is not enabled (MCR[5] set to zero), the rts_n signal is set low by programming MCR[1] (RTS) to a high. In Auto Flow Control, AFCE_MODE == Enabled and active (MCR[5] set to one) and FIFOs enable (FCR[0] set to one), the rts_n output is controlled in the same way, but is also gated with the receiver FIFO threshold trigger (rts_n is inactive high when above the threshold). The rts_n signal is de-asserted when MCR[1] is set low. 0: rts_n de-asserted (logic 1) 1: rts_n asserted (logic 0) Note that in Loopback mode (MCR[4] set to one), the rts_n output is held

			inactive high while the value of this location is internally looped back to an input.
0	R/W	0x0	DTR Data Terminal Ready This is used to directly control the Data Terminal Ready (dtr_n) output. The value written to this location is inverted and driven out on dtr_n. 0: dtr_n de-asserted (logic 1) 1: dtr_n asserted (logic 0) The Data Terminal Ready output is used to inform the modem or data set that the UART is ready to establish communications. Note that in Loopback mode (MCR[4] set to one), the dtr_n output is held inactive high while the value of this location is internally looped back to an input.

4.1.3.10 UART Line Status Register

Address : 0x0014			Name: UART_LSR Default: 0x0000_0060
Field	Type	Default	Description
31:8	/	/	/
7	R	0x0	FIFOERR RX Data Error in FIFO When FIFOs are disabled, this bit is always 0. When FIFOs are enabled, this bit is set to 1 when there is at least one PE, FE, or BI in the RX FIFO. It is cleared by a read from the LSR register provided there are no subsequent errors in the FIFO.
6	R	0x1	TEMT Transmitter Empty If the FIFOs are disabled, this bit is set to "1" whenever the TX Holding Register and the TX Shift Register are empty. If the FIFOs are enabled, this bit is set whenever the TX FIFO and the TX Shift Register are empty. In both cases, this bit is cleared when a byte is written to the TX data channel.
5	R	0x1	THRE TX Holding Register Empty If the FIFOs are disabled, this bit is set to "1" whenever the TX Holding Register is empty and ready to accept new data and it is cleared when the CPU writes to the TX Holding Register. If the FIFOs are enabled, this bit is set to "1" whenever the TX FIFO is empty and it is cleared when at least one byte is written to the TX FIFO.

4	R	0x0	BI Break Interrupt This is used to indicate the detection of a break sequence on the serial input data. If in UART mode (SIR_MODE == Disabled), it is set whenever the serial input, sin, is held in a logic '0' state for longer than the sum of start time + data bits + parity + stop bits. If in infrared mode (SIR_MODE == Enabled), it is set whenever the serial input, sir_in, is continuously pulsed to logic '0' for longer than the sum of start time + data bits + parity + stop bits. A break condition on serial input causes one and only one character, consisting of all zeros, to be received by the UART. In the FIFO mode, the character associated with the break condition is carried through the FIFO and is revealed when the character is at the top of the FIFO. Reading the LSR clears the BI bit. In the non-FIFO mode, the BI indication occurs immediately and persists until the LSR is read.
3	R	0x0	FE Framing Error This is used to indicate the occurrence of a framing error in the receiver. A framing error occurs when the receiver does not detect a valid STOP bit in the received data. In the FIFO mode, since the framing error is associated with a character received, it is revealed when the character with the framing error is at the top of the FIFO. When a framing error occurs, the UART tries to resynchronize. It does this by assuming that the error was due to the start bit of the next character and then continues receiving the other bit i.e. data, and/or parity and stop. It should be noted that the Framing Error (FE) bit (LSR[3]) is set if a break interrupt has occurred, as indicated by Break Interrupt (BI) bit (LSR[4]). 0: no framing error 1: framing error Reading the LSR clears the FE bit.
2	R	0x0	PE Parity Error This is used to indicate the occurrence of a parity error in the receiver if the Parity Enable (PEN) bit (LCR[3]) is set. In the FIFO mode, since the parity error is associated with a character received, it is revealed when the character with the parity error arrives at the top of the FIFO. It should be noted that the Parity Error (PE) bit (LSR[2]) is set if a break interrupt has occurred, as indicated by Break Interrupt (BI) bit (LSR[4]).

			0: no parity error 1: parity error Reading the LSR clears the PE bit.
1	R	0x0	OE Overrun Error This occurs if a new data character was received before the previous data was read. In the non-FIFO mode, the OE bit is set when a new character arrives in the receiver before the previous character was read from the RBR. When this happens, the data in the RBR is overWritten. In the FIFO mode, an overrun error occurs when the FIFO is full and a new character arrives at the receiver. The data in the FIFO is retained and the data in the receive shift register is lost. 0: no overrun error 1: overrun error Reading the LSR clears the OE bit.
0	R	0x0	DR Data Ready This is used to indicate that the receiver contains at least one character in the RBR or the receiver FIFO. 0: no data ready 1: data ready This bit is cleared when the RBR is read in non-FIFO mode, or when the receiver FIFO is empty, in FIFO mode.

4.1.3.11 UART Modem Status Register

Address : 0x0018			Name: UART_MSR Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7	R	0x0	DCD Line State of Data Carrier Detect This is used to indicate the current state of the modem control line dcd_n. This bit is the complement of dcd_n. When the Data Carrier Detect input (dcd_n) is asserted it is an indication that the carrier has been detected by the modem or data set. 0: dcd_n input is de-asserted (logic 1) 1: dcd_n input is asserted (logic 0)
6	R	0x0	RI

			Line State of Ring Indicator This is used to indicate the current state of the modem control line ri_n. This bit is the complement of ri_n. When the Ring Indicator input (ri_n) is asserted it is an indication that a telephone ringing signal has been received by the modem or data set. 0: ri_n input is de-asserted (logic 1) 1: ri_n input is asserted (logic 0)
5	R	0x0	DSR Line State of Data Set Ready This is used to indicate the current state of the modem control line dsr_n. This bit is the complement of dsr_n. When the Data Set Ready input (dsr_n) is asserted it is an indication that the modem or data set is ready to establish communications with UART. 0: dsr_n input is de-asserted (logic 1) 1: dsr_n input is asserted (logic 0) In Loopback Mode (MCR[4] set to one), DSR is the same as MCR[0] (DTR).
4	R	0x0	CTS Line State of Clear To Send This is used to indicate the current state of the modem control line cts_n. This bit is the complement of cts_n. When the Clear to Send input (cts_n) is asserted it is an indication that the modem or data set is ready to exchange data with UART. 0: cts_n input is de-asserted (logic 1) 1: cts_n input is asserted (logic 0) In Loopback Mode (MCR[4] = 1), CTS is the same as MCR[1] (RTS).
3	R	0x0	DDCD Delta Data Carrier Detect This is used to indicate that the modem control line dcd_n has changed since the last time the MSR was read. 0: no change on dcd_n since last read of MSR 1: change on dcd_n since last read of MSR Reading the MSR clears the DDCCD bit. Note: If the DDCCD bit is not set and the dcd_n signal is asserted (low) and a reset occurs (software or other/Wise), then the DDCCD bit is set when the reset is removed if the dcd_n signal remains asserted.
2	R	0x0	TERI Trailing Edge Ring Indicator This is used to indicate that a change on the input ri_n (from an active-low to an inactive-high state) has occurred since the last time the MSR was read.

			0: no change on ri_n since last read of MSR 1: change on ri_n since last read of MSR Reading the MSR clears the TERI bit.
1	R	0x0	DDSR Delta Data Set Ready This is used to indicate that the modem control line dsr_n has changed since the last time the MSR was read. 0: no change on dsr_n since last read of MSR 1: change on dsr_n since last read of MSR Reading the MSR clears the DDSR bit. In Loopback Mode (MCR[4] = 1), DDSR reflects changes on MCR[0] (DTR). Note: If the DDSR bit is not set and the dsr_n signal is asserted (low) and a reset occurs (software or other/Wise), then the DDSR bit is set when the reset is removed if the dsr_n signal remains asserted.
0	R	0x0	DCTS Delta Clear to Send This is used to indicate that the modem control line cts_n has changed since the last time the MSR was read. 0: no change on ctsdsr_n since last read of MSR 1: change on ctsdsr_n since last read of MSR Reading the MSR clears the DCTS bit. In Loopback Mode (MCR[4] = 1), DCTS reflects changes on MCR[1] (RTS). Note: If the DCTS bit is not set and the cts_n signal is asserted (low) and a reset occurs (software or other/Wise), then the DCTS bit is set when the reset is removed if the cts_n signal remains asserted.

4.1.3.12 UART Scratch Register

Address : 0x001C			Name: UART_SCH Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	SCRATCH_REG Scratch Register This register is for programmers to use as a temporary storage space. It has no defined purpose in the UART.

4.1.3.13 UART Status Register

Address : 0x007C	Name: UART_USR
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			Default: 0x0000_0006
Field	Type	Default	Description
31:5	/	/	/
4	R	0x0	RFF Receive FIFO Full This is used to indicate that the receive FIFO is completely full. 0: Receive FIFO not full 1: Receive FIFO Full This bit is cleared when the RX FIFO is no longer full.
3	R	0x0	RFNE Receive FIFO Not Empty This is used to indicate that the receive FIFO contains one or more entries. 0: Receive FIFO is empty 1: Receive FIFO is not empty This bit is cleared when the RX FIFO is empty.
2	R	0x1	TFE Transmit FIFO Empty This is used to indicate that the transmit FIFO is completely empty. 0: Transmit FIFO is not empty 1: Transmit FIFO is empty This bit is cleared when the TX FIFO is no longer empty.
1	R	0x1	TFNF Transmit FIFO Not Full This is used to indicate that the transmit FIFO is not full. 0: Transmit FIFO is full 1: Transmit FIFO is not full This bit is cleared when the TX FIFO is full.
0	R	0x0	BUSY UART Busy Bit 0: Idle or inactive 1: Busy

4.1.3.14 UART Transmit FIFO Level Register

Address : 0x0080			Name: UART_TFL Default: 0x0000_0000
Field	Type	Default	Description
31:7	/	/	/

6:0	R	0x0	TFL Transmit FIFO Level This indicates the number of data entries in the transmit FIFO.
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4.1.3.15 UART Receive FIFO Level Register

Address : 0x0084			Name: UART_RFL Default: 0x0000_0000
Field	Type	Default	Description
31:7	/	/	/
6:0	R	0x0	RFL Receive FIFO Level This indicates the number of data entries in the receive FIFO.

4.1.3.16 UART Halt TX Register

Address : 0x00A4			Name: UART_HALT Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7	R/W	0x0	DMA_PTE_RX In DMA mode 0, if FIFO is enabled and DMA_PTE_RX is set to 1, the RX DMA request will send when RFL is bigger than or equals to FIFO trigger level. Otherwise, the RX DMA request will not send.
6	R/W	0x0	PTE In DMA mode 0, if PTE is high and TX FIFO is enable, the TX DMA request will send when TFL is less than or equals to FIFO Trigger Level. If PTE is high and TX FIFO is disabled, the TX DMA request will send when THRE is empty. If PTE is low, the TX DMA request will send when the TX FIFO is empty. If dma_pte_rx is high and RX FIFO is enabled, the RX DRQ will send when RFL is equals to or more than FIFO Trigger Level. In DMA mode 1, if TX FIFO is enabled and the PTE is high, the TX DMA request will send when TFL is less than or equals to FIFO Trigger Level. If PTE is low, the TX DMA request will send when TX FIFO is empty and the request stops only when TX FIFO is empty. If RFL is equals to or more than FIFO Trigger Level, the RX DRQ will be set 1, in otherwise, it will be set 0.
5	R/W	0x0	SIR_RX_INVERT

			SIR Receiver Pulse Polarity Invert 0: Not invert receiver signal 1: Invert receiver signal
4	R/W	0x0	SIR_TX_INVERT SIR Transmit Pulse Polarity Invert 0: Not invert transmit pulse 1: Invert transmit pulse
3	/	/	/
2	R/WAC	0x0	CHANGE_UPDATE After the user using HALT[1] to change the baud rate or LCR configuration, write 1 to update the configuration and waiting this bit self-clear to 0 to finish update process. Write 0 to this bit has no effect. 1: Update trigger, Self clear to 0 when finish update.
1	R/W	0x0	CHCFG_AT_BUSY This is an enable bit for the user to change LCR register configuration (except for the DLAB bit) and baud rate register (DLH and DLL) when the UART is busy (USB[0] is 1). 1: Enable change when busy
0	R/W	0x0	HALT_TX Halt TX This register is use to halt transmissions for testing, so that the transmit FIFO can be filled by the master when FIFOs are implemented and enabled. 0 : Halt TX disabled 1 : Halt TX enabled Note: If FIFOs are not enabled, the setting of the halt TX register has no effect on operation.

4.1.3.17 UART TX Delay

Address : 0x00CC			Name: UART_TXDLY Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	DLY The delay time between the last stop bit and the next start bit. The unit is Tclk. It is use to control the space between two bytes in TX.

4.1.3.18 UART Baudrate Detection Control Register

Address : 0x0D4			Name: UART_BDC Default: 0x0000_0000
Field	Type	Default	Description
31:2	/	/	/
1	R/W	0x0	Auto Baudrate Detect Enable 1: enable 0: disable <i>Note: Set this bit to 1 to start the UART Baudrate Detection. The detection circuit will detect the start bit and count the cycles of PCLK in one bit duration. The number of the cycles will be written in the PCLK_NUM field. Set this bit to 0 to stop the UART Baudrate Detection.</i>
0	R/W	0x0	Mode 1: Normal mode 0: Compare mode <i>Note:</i> <i>In normal mode, when the detection circuit is enabled, the number of the PCLK cycles will be latched in each one bit duration and be written into the PCLK_NUM field.</i> <i>In compare mode, the number of the PCLK cycles will be latched in each one bit duration, but only the value which equals to the value latched in last time will be written into the PCLK_NUM field.</i>

4.1.3.19 UART Baudrate Detection Counter Low Register

Address : 0x0D8			Name: UART_BDCL Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	PCLK_NUM_LOW The low 8 bits of the number of the cycles counted in one bit duration.

4.1.3.20 UART Baudrate Detection Counter High Register

Address : 0x0DC			Name: UART_ABCR Default: 0x0000_0000
Field	Type	Default	Description

31:8	/	/	/
7:0	R/W	0x0	PCLK_NUM_HIGH The high 8 bits of the number of the cycles counted in one bit duration.

4.2 Serial Peripheral Interface (SPI)

4.2.1 Overview

The SPI supports 4 different formats for data transfer. Software can select one of the four modes in which the SPI works by setting the bit1 (Polarity) and bit0 (Phase) of SPI Transfer Control Register. The SPI controller master uses the SPI_SCLK signal to transfer data in and out of the shift register. Data is clocked using any one of four programmable clock phase and polarity combinations.

During Phase 0, Polarity 0 and Phase 1, Polarity 1 operations, output data changes on the falling clock edge and input data is shifted in on the rising edge.

During Phase 1, Polarity 0 and Phase 0, Polarity 1 operations, output data changes on the rising edges of the clock and is shifted in on falling edges.

The POL defines the signal polarity when SPI_SCLK is in idle state. The SPI_SCLK is high level when POL is '1' and it is low level when POL is '0'. The PHA decides whether the leading edge of SPI_SCLK is used for setup or sample data. The leading edge is used for setup data when PHA is '1' and for sample data when PHA is '0'. The four kinds of modes are listed Table 1-3:

SPI Mode	POL	PHA	Leading Edge	Trailing Edge
0	0	0	Rising, Sample	Falling, Setup
1	0	1	Rising, Setup	Falling, Sample
2	1	0	Falling, Sample	Rising, Setup
3	1	1	Falling, Setup	Rising, Sample

Table 4-2 SPI Modes with Clock Polarity and Phase

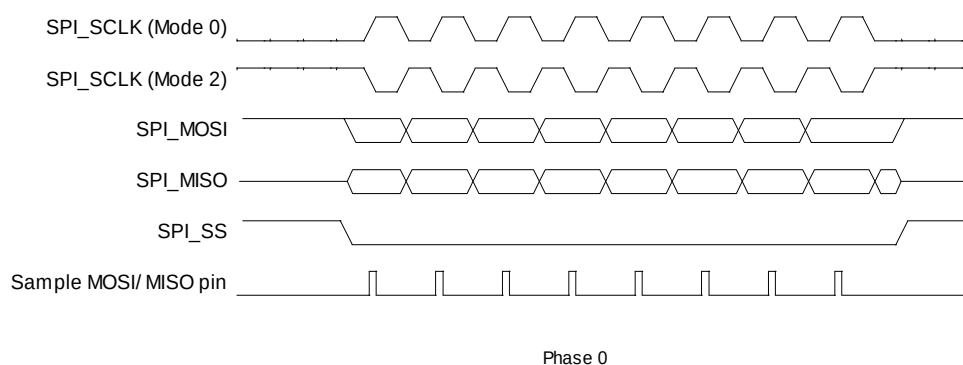


Figure 4-3 SPI Phase 0 Transfer Format

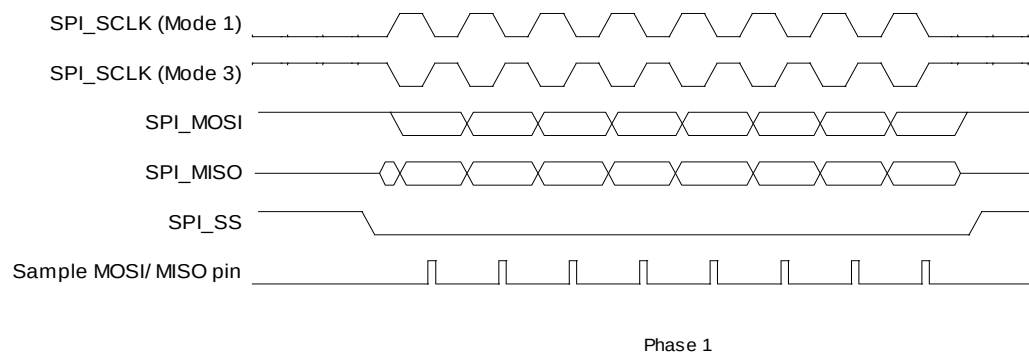


Figure 4-4 SPI Phase 1 Transfer Format

4.2.2 Register List

Module Name	Base Address	
SPI Controller 0	0x40005000	
SPI Controller 1	0x40006000	
Register Name	Offset Address	Description
SPI_VER	0x0000	SPI Version Number Register
SPI_CTRL	0x0004	SPI Global Control Register
SPI_TCTRL	0x0008	SPI Transfer Control register
/	0x000c	reserved
SPI_IER	0x0010	SPI Interrupt Control register
SPI_STA	0x0014	SPI Interrupt Status register
SPI_FCTL	0x0018	SPI FIFO Control register
SPI_FST	0x001c	SPI FIFO Status register
SPI_WAIT	0x0020	SPI Wait Clock Counter register
SPI_CCTR	0x0024	SPI Clock Rate Control register
/	0x0028	reserved
/	0x002c	reserved
SPI_BC	0x0030	SPI Burst Counter register
SPI_TC	0x0034	SPI Transmit Counter Register
SPI_BCC	0x0038	SPI Burst Control register
SPI_NDMA_MODE_CTRL	0x0088	SPI Normal DMA Mode Control Register
SPI_TXD	0x0200	SPI TX Data register
SPI_RXD	0x0300	SPI RX Data register

4.2.3 Register Description

4.2.3.1 SPI Version Number Register

Address : 0x0000			Name: SPI_VER Default: 0x0009_0000
Field	Type	Default	Description
31:0	R	0x90000	SPI_VER

4.2.3.2 SPI Global Control Register

Address : 0x0004			Name: SPI_CTRL Default: 0x0000_0080
Field	Type	Default	Description
31	R/WAC	0x0	SPI_RST SPI Software Reset Write 1 to this bit will clear the SPI controller, and auto clear to 0 when reset operation completes. Write 0 has no effect.
30:8	/	/	/
7	R/W	0x1	TP_EN Transmit Pause Enable In master mode, it is used to control transmit state machine to stop smart burst sending when RX FIFO is full. 1: stop transmit data when RXFIFO full 0: normal operation Note: cannot be written when XCH=1.
6:2	/	/	/
1	R/W	0x0	SPI_MODE Master Slave selection 0: slave mode 1: master mode Note: cannot be written when XCH=1.
0	R/W	0x0	SPI_EN SPI Module Enable Control 0: SPI Controller Disable 1: SPI Controller Enable

4.2.3.3 SPI Transfer Control Register

Address : 0x0008			Name: SPI_TCTRL Default: 0x0000_0087
Field	Type	Default	Description
31	R/W	0x0	XCH Exchange Burst In master mode it is used to start SPI burst. 0: Idle 1: Initiates exchange Write 1 to this bit will start the SPI burst and this bit will auto clear after finishing the bursts transfer specified by BC(Burst Counter). Write 1 to SPI_RST will also clear this bit. Write 0 to this bit has no effect.
30:15	/	/	/
14	R/W	0x0	SDDM Sending Data Delay Mode 0: normal sending 1: delay sending Set this bit to 1 to make the data that should be send with a delay of half cycle of SPI_CLK in dual io mode for SPI mode 0.
13	R/W	0x0	SDM Master Sample Data Mode 1: normal sample mode 0: delay sample mode In normal sample mode, SPI master samples the data at the correct edge for each SPI mode. In delay sample mode, SPI master samples data at the edge that is half cycle delayed by the correct edge defined in respective SPI mode.
12	R/W	0x0	FBS First Transmit Bit Select 0: MSB first 1: LSB first
11	R/W	0x0	SDC Master Sample Data Control Set this bit to 1 to make the internal read sample point with a delay of half

			cycle of SPI_CLK. It is used in high speed(>48MHz) read operation to reduce the error caused by the time delay of SPI_CLK propagating between master and slave. 0: normal operation, do not delay internal read sample point 1: delay internal read sample point
10	R/W	0x0	RPSM Rapids mode select Select Rapids mode for high speed write. 0: normal write mode 1: rapids write mode Note: Can't be written when XCH=1.
9	R/W	0x0	DDB Dummy Burst Type 0: The bit value of dummy SPI burst is zero 1: The bit value of dummy SPI burst is one Note: Can't be written when XCH=1.
8	R/W	0x0	DHB Discard Hash Burst In master mode it controls whether discarding unused SPI bursts 0: Receiving all SPI bursts in BC period 1: Discard unused SPI bursts, only fetching the SPI bursts during dummy burst period. The bursts number is specified by TC(Transmit Counter). Note: Can't be written when XCH=1.
7	R/W	0x1	SS_LEVEL When control SS signal manually (SPI_CTRL_REG.SS_OWNER==1), set this bit to '1' or '0' to control the level of SS signal. 0: set SS to low 1: set SS to high Note: Can't be written when XCH=1.
6	R/W	0x0	SS_OWNER SS Output Owner Select Usually, controller sends SS signal automatically with data together. When this bit is set to 1, software must manually write SPI_CTL_REG.SS_LEVEL to 1 or 0 to control the level of SS signal. 0: SPI controller 1: Software Note: Can't be written when XCH=1.

5:4	R/W	0x0	SS_SEL SPI Chip Select Select one of four external SPI Master/Slave Devices 00: SPI_SS0 will be asserted 01: SPI_SS1 will be asserted 10: SPI_SS2 will be asserted 11: SPI_SS3 will be asserted Note: Can't be written when XCH=1.
3	R/W	0x0	SSCTL In master mode, this bit selects the output wave form for the SPI_SSx signal. Only valid when SS_OWNER = 0. 0: SPI_SSx remains asserted between SPI bursts 1: Negate SPI_SSx between SPI bursts Note: Can't be written when XCH=1.
2	R/W	0x1	SPOL SPI Chip Select Signal Polarity Control 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle) Note: Can't be written when XCH=1.
1	R/W	0x1	CPOL SPI Clock Polarity Control 0: Active high polarity (0 = Idle) 1: Active low polarity (1 = Idle) Note: Can't be written when XCH=1.
0	R/W	0x1	CPHA SPI Clock/Data Phase Control 0: Phase 0 (Leading edge for sample data) 1: Phase 1 (Leading edge for setup data) Note: Can't be written when XCH=1.

4.2.3.4 SPI Interrupt Control Register

Address : 0x0010			Name: SPI_IER Default: 0x0000_0000
Field	Type	Default	Description
31:14	R	0x0	Reserved.
13	R/W	0x0	SS_INT_EN

			SSI Interrupt Enable Chip Select Signal (SSx) from valid state to invalid state 0: Disable 1: Enable
12	R/W	0x0	TC_INT_EN Transfer Completed Interrupt Enable 0: Disable 1: Enable
11	R/W	0x0	TF_UDR_INT_EN TXFIFO under run Interrupt Enable 0: Disable 1: Enable
10	R/W	0x0	TF_OVF_INT_EN TX FIFO Overflow Interrupt Enable 0: Disable 1: Enable
9	R/W	0x0	RF_UDR_INT_EN RXFIFO under run Interrupt Enable 0: Disable 1: Enable
8	R/W	0x0	RF_OVF_INT_EN RX FIFO Overflow Interrupt Enable 0: Disable 1: Enable
7	R	0x0	Reserved
6	R/W	0x0	TF_FUL_INT_EN TX FIFO Full Interrupt Enable 0: Disable 1: Enable
5	R/W	0x0	TX_EMP_INT_EN TX FIFO Empty Interrupt Enable 0: Disable 1: Enable
4	R/W	0x0	TX_ERQ_INT_EN

			TX FIFO Empty Request Interrupt Enable 0: Disable 1: Enable
3	R	0x0	Reserved
2	R/W	0x0	RF_FUL_INT_EN RX FIFO Full Interrupt Enable 0: Disable 1: Enable
1	R/W	0x0	RX_EMP_INT_EN RX FIFO Empty Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	RF_RDY_INT_EN RX FIFO Ready Request Interrupt Enable 0: Disable 1: Enable

4.2.3.5 SPI Interrupt Status Register

Address : 0x0014			Name: SPI_STA Default: 0x0000_0032
Field	Type	Default	Description
31:14	/	0x0	/
13	R/W1C	0x0	SSI SS Invalid Interrupt When SSI is 1, it indicates that SS has changed from valid state to invalid state. Writing 1 to this bit clears it.
12	R/W1C	0x0	TC Transfer Completed In master mode, it indicates that all bursts specified by BC has been exchanged. In other condition, When set, this bit indicates that all the data in TXFIFO has been loaded in the Shift register, and the Shift register has shifted out all the bits. Writing 1 to this bit clears it. 0: Busy 1: Transfer Completed
11	R/W1C	0x0	TF_UDF TXFIFO under run

			This bit is set when if the TXFIFO is under run. Writing 1 to this bit clears it. 0: TXFIFO is not under run 1: TXFIFO is under run
10	R/W1C	0x0	TF_OVF TXFIFO Overflow This bit is set when if the TXFIFO is overflow. Writing 1 to this bit clears it. 0: TXFIFO is not overflow 1: TXFIFO is overflowed
9	R/W1C	0x0	RX_UDF RXFIFO Under run When set, this bit indicates that RXFIFO has under run. Writing 1 to this bit clears it.
8	R/W1C	0x0	RX_OVF RXFIFO Overflow When set, this bit indicates that RXFIFO has overflowed. Writing 1 to this bit clears it. 0: RXFIFO is available 1: RXFIFO has overflowed
7	/	/	/
6	R/W1C	0x0	TX_FULL TXFIFO Full This bit is set when if the TXFIFO is full. Writing 1 to this bit clears it. 0: TXFIFO is not Full 1: TXFIFO is Full
5	R/W1C	0x1	TX_EMP TXFIFO Empty This bit is set if the TXFIFO is empty. Writing 1 to this bit clears it. 0: TXFIFO contains one or more words 1: TXFIFO is empty
4	R/W1C	0x1	TX_READY TXFIFO Ready 0: TX_WL > TX_TRIG_LEVEL 1: TX_WL <= TX_TRIG_LEVEL This bit is set any time if TX_WL <= TX_TRIG_LEVEL. Writing "1" to this bit clears it. Where TX_WL is the water level of RXFIFO.
3	/	/	/

2	R/W1C	0x0	RX_FULL RXFIFO Full This bit is set when the RXFIFO is full. Writing 1 to this bit clears it. 0: Not Full 1: Full
1	R/W1C	0x1	RX_EMP RXFIFO Empty This bit is set when the RXFIFO is empty. Writing 1 to this bit clears it. 0: Not empty 1: empty
0	R/W1C	0x0	RX_RDY RXFIFO Ready 0: RX_WL < RX_TRIG_LEVEL 1: RX_WL >= RX_TRIG_LEVEL This bit is set any time if RX_WL >= RX_TRIG_LEVEL. Writing “1” to this bit clears it. Where RX_WL is the water level of RXFIFO.

4.2.3.6 SPI FIFO Control Register

Address : 0x0018			Name : SPI_FCTL Default : 0x0040_0001
Field	Type	Default	Description
31	R/WAC	0x0	TX_FIFO_RST TX FIFO Reset Write ‘1’ to this bit will reset the control portion of the TX FIFO and auto clear to ‘0’ when completing reset operation, write to ‘0’ has no effect.
30	R/W	0x0	TF_TEST_ENB TX Test Mode Enable 0: disable 1: enable Note: In normal mode, TX FIFO can only be read by SPI controller, write ‘1’ to this bit will switch TX FIFO read and write function to AHB bus. This bit is used to test the TX FIFO, don’t set in normal operation and don’t set RF_TEST and TF_TEST at the same time.
29:25	/	/	/
24	R/W	0x0	TF_DRQ_EN TX FIFO DMA Request Enable

			0: Disable 1: Enable
23:16	R/W	0x40	TX_TRIG_LEVEL TX FIFO Empty Request Trigger Level
15	R/WAC	0x0	RF_RST RXFIFO Reset Write '1' to this bit will reset the control portion of the receiver FIFO, and auto clear to '0' when completing reset operation, write '0' to this bit has no effect.
14	R/W	0x0	RF_TEST RX Test Mode Enable 0: Disable 1: Enable Note: In normal mode, RX FIFO can only be written by SPI controller, write '1' to this bit will switch RX FIFO read and write function to AHB bus. This bit is used to test the RX FIFO, don't set in normal operation and don't set RF_TEST and TF_TEST at the same time.
13:9	/	/	/
8	R/W	0x0	RF_DRQ_EN RX FIFO DMA Request Enable 0: Disable 1: Enable
7:0	R/W	0x1	RX_TRIG_LEVEL RX FIFO Ready Request Trigger Level

4.2.3.7 SPI FIFO Status Register

Address : 0x001C			Name: SPI_FST Default: 0x0000_0000
Field	Type	Default	Description
31	R	0x0	TB_WR TX FIFO Write Buffer Write Enable
30:28	R	0x0	TB_CNT TX FIFO Write Buffer Counter These bits indicate the number of words in TX FIFO Write Buffer.

27:24	R	0x0	Reserved
23:16	R	0x0	TF_CNT TX FIFO Counter These bits indicate the number of words in TX FIFO 0: 0 byte in TX FIFO 1: 1 byte in TX FIFO ... 64 64 bytes in TX FIFO other: Reserved
15	R	0x0	RB_WR RX FIFO Read Buffer Write Enable
14:12	R	0x0	RB_CNT RX FIFO Read Buffer Counter These bits indicate the number of words in RX FIFO Read Buffer.
11:8	R	0x0	Reserved
7:0	R	0x0	RF_CNT RX FIFO Counter These bits indicate the number of words in RX FIFO. 0: 0 byte in RX FIFO 1: 1 byte in RX FIFO ... 64: 64 bytes in RX FIFO other: Reserved

4.2.3.8 SPI Wait Clock Register

Address : 0x0020			Name: SPI_WAIT Default: 0x0000_0000
Field	Type	Default	Description
31:20	/	/	/
19:16	R/W	0x0	SWC Dual mode direction switch wait clock counter (for master mode only). 0: No wait states inserted n: n SPI_SCLK wait states inserted Note: These bits control the number of wait states to be inserted before start dual data transfer in dual SPI mode. The SPI module counts SPI_SCLK by SWC for delaying next word data transfer.

			Note: Can't be written when XCH=1.
15:0	R/W	0x0	WCC Wait Clock Counter (In Master mode) These bits control the number of wait states to be inserted in data transfers. The SPI module counts SPI_SCLK by WCC for delaying next word data transfer. 0: No wait states inserted N: N SPI_SCLK wait states inserted

4.2.3.9 SPI Clock Control Register

Address : 0x0024			Name: SPI_CCTR Default: 0x0000_0002
Field	Type	Default	Description
31:13	/	/	/
12	R/W	0x0	DRS Divide Rate Select (Master Mode Only) 0: Select Clock Divide Rate 1 1: Select Clock Divide Rate 2
11:8	R/W	0x0	CDR1 Clock Divide Rate 1 (Master Mode Only) The SPI_SCLK is determined according to the following equation: $SPI_CLK = SRC_CLK / 2^n$.
7:0	R/W	0x2	CDR2 Clock Divide Rate 2 (Master Mode Only) The SPI_SCLK is determined according to the following equation: $SPI_CLK = SRC_CLK / (2^{(n+1)})$.

4.2.3.10 SPI Master Burst Counter Register

Address : 0x0030			Name: SPI_BC Default: 0x0000_0000
Field	Type	Default	Description
31:24	/	/	/
23:0	R/W	0x0	MBC Master Burst Counter In master mode, this field specifies the total burst number. 0: 0 burst

			1: 1 burst ... N: N bursts
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4.2.3.11 SPI Transmit Counter Register

Address : 0x0034			Name: SPI_TC Default: 0x0000_0000
Field	Type	Default	Description
31:24	/	/	/
23:0	R/W	0x0	MWTC Master Write Transmit Counter In master mode, this field specifies the burst number that should be sent to TXFIFO before automatically sending dummy burst. For saving bus bandwidth, the dummy burst (all zero bits or all one bits) is sent by SPI Controller automatically. 0: 0 burst 1: 1 burst ... N: N bursts

4.2.3.12 SPI Master Burst Control Register

Address : 0x0038			Name: SPI_BCC Default: 0x0000_0000
Field	Type	Default	Description
31:29	R	0x0	Reserved
28	R/W	0x0	DRM Master Dual Mode RX Enable 0: RX use single-bit mode 1: RX use dual mode Note: Can't be written when XCH=1.
27:24	R/W	0x0	DBC Master Dummy Burst Counter In master mode, this field specifies the burst number that should be sent before receive in dual SPI mode. The data is don't care by the device. 0: 0 burst 1: 1 burst ... N: N bursts

			Note: Can't be written when XCH=1.
23:0	R/W	0x0	STC Master Single Mode Transmit Counter In master mode, this field specifies the burst number that should be sent in single mode before automatically sending dummy burst. This is the first transmit counter in all bursts. 0: 0 burst 1: 1 burst ... N: N bursts Note: Can't be written when XCH=1.

4.2.3.13 SPI Normal DMA Mode Control

Address : 0x0088			Name: SPI_NDMA_MODE_CTRL Default: 0x0000_00a5
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0xA5	NDMA_MODE_CTL 0xEA: NDMA handshake mode Note: NDMA wait mode doesn't care this value. 0xA5 can be used in handshake mode, but 0xEA is better.

4.2.3.14 SPI TX Data Register

Address : 0x0200			Name: SPI_TXD Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	TDATA Transmit Data This register can be accessed in byte, half-word or word unit by AHB. In byte accessing method, if there are rooms in TXFIFO, one burst data is written to TXFIFO and the depth is increased by 1. In half-word accessing method, two SPI burst data are written and the TXFIFO depth is increase by 2. In word accessing method, four SPI burst data are written and the TXFIFO depth is increased by 4. Note: This address is writing-only if TF_TEST is '0', and if TF_TEST is set to '1', this address is readable and writable to test the TX FIFO through the AHB bus.

4.2.3.15 SPI RX Data Register

Address : 0x0300			Name: SPI_RXD Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	RDATA Receive Data This register can be accessed in byte, half-word or word unit by AHB. In byte accessing method, if there are data in RXFIFO, the top word is returned and the RXFIFO depth is decreased by 1. In half-word accessing method, two SPI bursts are returned and the RXFIFO depth is decrease by 2. In word accessing method, the four SPI bursts are returned and the RXFIFO depth is decreased by 4. Note: This address is read-only if RF_TEST is '0', and if RF_TEST is set to '1', this address is readable and writable to test the RX FIFO through the AHB bus.

4.3 Two Wire Interface Controller (TWI)

4.3.1 Overview

This TWI Controller is designed to be used as an interface between CPU host and the serial TWI bus. It can support all the standard TWI transfer, including Slave and Master. The communication to the TWI bus is carried out on a byte-wise basis using interrupt or polled handshaking. This TWI Controller can be operated in standard mode (100K bps) or fast-mode, supporting data rate up to 400K bps. Multiple Masters and 10-bit addressing Mode are supported for this specified application. General Call Addressing is also supported in Slave mode.

The TWI controller includes the following features:

- Software-programmable for Slave or Master
- Supports Repeated START signal
- Multi-master systems supported
- Allows 10-bit addressing with TWI bus
- Performs arbitration and clock synchronization
- Own address and General Call address detection
- Interrupt on address detection
- Supports speeds up to 400Kbits/s ('fast mode')
- Allows operation from a wide range of input clock frequencies

Data transferred are always in a unit of 8-bit (byte), followed by an acknowledge bit. The number of bytes that can be transmitted per transfer is unrestricted. Data is transferred in serial with the MSB first. Between each byte of data transfer, a receiver device will hold the clock line SCL low to force the transmitter into a wait state while

waiting the response from microprocessor.

Data transfer with acknowledge is obligatory. The clock line is driven by the master all the time, including the acknowledge-related clock cycle, except for the SCL holding between each bytes. After sending each byte, the transmitter releases the SDA line to allow the receiver to pull down the SDA line and send an acknowledge signal (or leave it high to send a "not acknowledge") to the transmitter.

When a slave receiver doesn't acknowledge the slave address (unable to receive because of no resource available), the data line must be left high by the slave so that the master can then generate a STOP condition to abort the transfer. Slave receiver can also indicate not to want to send more data during a transfer by leave the acknowledge signal high. And the master should generate the STOP condition to abort the transfer.

The figure provides an illustration the relation of SDA signal line and SCL signal line on the TWI serial bus.

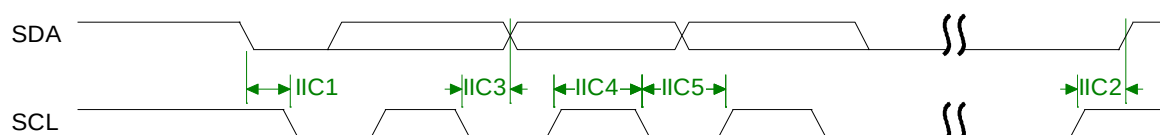


Figure 4-5 TWI Timing Diagram

4.3.2 Register List

Module Name	Base Address	
TWI Controller 0	0x40041c00	
TWI Controller 1	0x40042000	
Register Name	Offset Address	Description
TWI_ADDR	0x0000	TWI Slave address
TWI_XADDR	0x0004	TWI Extended slave address
TWI_DATA	0x0008	TWI Data byte
TWI_CNTR	0x000C	TWI Control register
TWI_STAT	0x0010	TWI Status register
TWI_CCR	0x0014	TWI Clock control register
TWI_SRST	0x0018	TWI Software reset
TWI_EFR	0x001C	TWI Enhance Feature register
TWI_LCR	0x0020	TWI Line Control register

4.3.3 Register Description

4.3.3.1 TWI Slave Address Register

Address : 0x0000	Name: TWI_ADDR Default: 0x0000_0000
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Field	Type	Default	Description
31:8	/	/	/
7:1	R/W	0x0	SLA Slave address - 7-bit addressing SLA6, SLA5, SLA4, SLA3, SLA2, SLA1, SLA0 - 10-bit addressing 1, 1, 1, 1, 0, SLAX[9:8] For 7-bit addressing: SLA6 – SLA0 is the 7-bit address of the TWI when in slave mode. When the TWI receives this address after a START condition, it will generate an interrupt and enter slave mode. (SLA6 corresponds to the first bit received from the TWI bus.) If GCE is set to '1', the TWI will also recognize the general call address (00h). For 10-bit addressing: When the address received starts with 11110b, the TWI recognizes this as the first part of a 10-bit address and if the next two bits match ADDR[2:1] (i.e. SLAX9 and SLAX8 of the device's extended address), it sends an ACK. (The device does not generate an interrupt at this point.) If the next byte of the address matches the XADDR register (SLAX7 – SLAX0), the TWI generates an interrupt and goes into slave mode.
0	R/W	0x0	GCE General call address enable 0: Disable 1: Enable

4.3.3.2 TWI Extend Address Register

Address : 0x0004			Name: TWI_XADDR Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	SLAX Extend Slave Address SLAX[7:0]

4.3.3.3 TWI Data Register

Address : 0x0008			Name: TWI_DATA Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0	TWI_DATA Data byte for transmitting or received

4.3.3.4 TWI Control Register

Address : 0x000C			Name: TWI_CTR Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7	R/W	0x0	INT_EN Interrupt Enable 1'b0: The interrupt line always low 1'b1: The interrupt line will go high when INT_FLAG is set
6	R/W	0x0	BUS_EN TWI Bus Enable 1'b0: The TWI bus inputs ISDA/ISCL are ignored and the TWI Controller will not respond to any address on the bus 1'b1: The TWI will respond to calls to its slave address – and to the general call address if the GCE bit in the ADDR register is set Notes: In master operation mode, this bit should be set to '1'.
5	R/WAC	0x0	M_STA Master Mode Start When M_STA is set to '1', TWI Controller enters master mode and will transmit a START condition on the bus when the bus is free. If the M_STA bit is set to '1' when the TWI Controller is already in master mode and one or more bytes have been transmitted, then a repeated START condition will be sent. If the M_STA bit is set to '1' when the TWI is being accessed in slave mode, the TWI will complete the data transfer in slave mode then enter master mode when the bus has been released. The M_STA bit is cleared automatically after a START condition has been sent: writing a '0' to this bit has no effect.

4	R/W1 C	0x0	M_STP Master Mode Stop If M_STP is set to '1' in master mode, a STOP condition is transmitted on the TWI bus. If the M_STP bit is set to '1' in slave mode, the TWI will behave as if a STOP condition has been received, but no STOP condition will be transmitted on the TWI bus. If both M_STA and M_STP bits are set, the TWI will first transmit the STOP condition (if in master mode) then transmit the START condition. The M_STP bit is cleared automatically: writing a '0' to this bit has no effect.
3	R/W1 C	0x0	INT_FLAG Interrupt Flag INT_FLAG is automatically set to '1' when any of 28 (out of the possible 29) states is entered (see 'STAT Register' below). The only state that does not set INT_FLAG is state F8h. If the INT_EN bit is set, the interrupt line goes high when IFLG is set to '1'. If the TWI is operating in slave mode, data transfer is suspended when INT_FLAG is set and the low period of the TWI bus clock line (SCL) is stretched until '1' is written to INT_FLAG. The TWI clock line is then released and the interrupt line goes low.
2	R/W	0x0	A_ACK Assert Acknowledge When A_ACK is set to '1', an Acknowledge (low level on SDA) will be sent during the acknowledge clock pulse on the TWI bus if: 1. Either the whole of a matching 7-bit slave address or the first or the second byte of a matching 10-bit slave address has been received. 2. The general call address has been received and the GCE bit in the ADDR register is set to '1'. 3. A data byte has been received in master or slave mode. When A_ACK is '0', a Not Acknowledge (high level on SDA) will be sent when a data byte is received in master or slave mode. If A_ACK is cleared to '0' in slave transmitter mode, the byte in the DATA register is assumed to be the 'last byte'. After this byte has been transmitted, the TWI will enter state C8h then return to the idle state (status code F8h) when INT_FLAG is cleared. The TWI will not respond as a slave unless A_ACK is set.
1:0	/	/	/

4.3.3.5 TWI Status Register

Address : 0x0010			Name: TWI_STAT Default: 0x0000_00F8
Field	Type	Default	Description
31:8	/	/	/
7:0	R	0xF8	STA Status Information Byte Code Status 0x00: Bus error 0x08: START condition transmitted 0x10: Repeated START condition transmitted 0x18: Address + Write bit transmitted, ACK received 0x20: Address + Write bit transmitted, ACK not received 0x28: Data byte transmitted in master mode, ACK received 0x30: Data byte transmitted in master mode, ACK not received 0x38: Arbitration lost in address or data byte 0x40: Address + Read bit transmitted, ACK received 0x48: Address + Read bit transmitted, ACK not received 0x50: Data byte received in master mode, ACK transmitted 0x58: Data byte received in master mode, not ACK transmitted 0x60: Slave address + Write bit received, ACK transmitted 0x68: Arbitration lost in address as master, slave address + Write bit received, ACK transmitted 0x70: General Call address received, ACK transmitted 0x78: Arbitration lost in address as master, General Call address received, ACK transmitted 0x80: Data byte received after slave address received, ACK transmitted 0x88: Data byte received after slave address received, not ACK transmitted 0x90: Data byte received after General Call received, ACK transmitted 0x98: Data byte received after General Call received, not ACK transmitted 0xA0: STOP or repeated START condition received in slave mode 0xA8: Slave address + Read bit received, ACK transmitted 0xB0: Arbitration lost in address as master, slave address + Read bit received, ACK transmitted 0xB8: Data byte transmitted in slave mode, ACK received 0xC0: Data byte transmitted in slave mode, ACK not received 0xC8: Last byte transmitted in slave mode, ACK received 0xD0: Second Address byte + Write bit transmitted, ACK received 0xD8: Second Address byte + Write bit transmitted, ACK not received 0xF8: No relevant status information, INT_FLAG=0

			Others: Reserved
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4.3.3.6 TWI Clock Register

Address : 0x0014			Name: TWI_CCR Default: 0x0000_0000
Field	Type	Default	Description
31:7	/	/	/
6:3	R/W	0	CLK_M
2:0	R/W	0	CLK_N The TWI bus is sampled by the TWI at the frequency defined by F0: $F_{\text{samp}} = F_0 = F_{\text{in}} / 2^{\text{CLK_N}}$ The TWI OSCL output frequency, in master mode, is F1 / 10: $F_1 = F_0 / (\text{CLK_M} + 1)$ $F_{\text{oscl}} = F_1 / 10 = F_{\text{in}} / (2^{\text{CLK_N}} * (\text{CLK_M} + 1) * 10)$ For Example: Fin = 48Mhz (APB clock input) For 400kHz full speed 2Wire, CLK_N = 2, CLK_M=2 $F_0 = 48\text{M} / 2^2 = 12\text{Mhz}$, $F_1 = F_0 / (10 * (2+1)) = 0.4\text{Mhz}$ For 100Khz standard speed 2Wire, CLK_N=2, CLK_M=11 $F_0 = 48\text{M} / 2^2 = 12\text{Mhz}$, $F_1 = F_0 / (10 * (11+1)) = 0.1\text{Mhz}$

4.3.3.7 TWI Soft Reset Register

Address : 0x0018			Name: TWI_SRST Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/WAC	0x0	SOFT_RST Soft Reset Write '1' to this bit to reset the TWI and clear to '0' when completing Soft Reset operation.

4.3.3.8 TWI Enhance Feature Register

Address : 0x001C			Name: TWI_EFR Default: 0x0000_0000
Field	Type	Default	Description
31:2	/	/	/
0:1	R/W	0x0	DBN

			Data Byte number follow Read Command Control No Data Byte to be wrote after read command Only 1 byte data to be wrote after read command 2 bytes data can be wrote after read command 3 bytes data can be wrote after read command
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4.3.3.9 TWI Line Control Register

Address : 0x0020			Name: TWI_LCR Default: 0x0000_003A
Field	Type	Default	Description
31:6	/	/	/
5	R	0x1	SCL_STATE Current state of TWI_SCL 0 – low 1 - high
4	R	0x1	SDA_STATE Current state of TWI_SDA 0 – low 1 - high
3	R/W	0x1	SCL_CTL TWI_SCL line state control bit When line control mode is enabled (bit[2] set), value of this bit decide the output level of TWI_SCL. 0 – output low level 1 – output high level
2	R/W	0x0	SCL_CTL_EN TWI_SCL line state control enable When this bit is set, the state of TWI_SCL is control by the value of bit[3]. 0-disable TWI_SCL line control mode 1-enable TWI_SCL line control mode
1	R/W	0x1	SDA_CTL TWI_SDA line state control bit When line control mode is enabled (bit[0] set), value of this bit decide the output level of TWI_SDA. 0 – output low level 1 – output high level

0	R/W	0x0	SDA_CTL_EN TWI_SDA line state control enable When this bit is set, the state of TWI_SDA is control by the value of bit[1]. 0-disable TWI_SDA line control mode 1-enable TWI_SDA line control mode
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4.4 SDMMC/SDIO Controller (SDC)

4.4.1 Overview

The SD3.0 controller can be configured either as a Secure Digital Multimedia Card controller, which simultaneously supports Secure Digital memory (SD Memo), UHS-1 Card, Secure Digital I/O (SDIO), Multimedia Cards (MMC), eMMC Card and Consumer Electronics Advanced Transport Architecture (CE-ATA).

The SD3.0 controller includes the following features:

- Supports Secure Digital memory protocol commands (up to SD3.0)
- Supports Secure Digital I/O protocol commands
- Supports Multimedia Card protocol commands (up to MMC4.41)
- Supports CE-ATA digital protocol commands
- Supports eMMC boot operation and alternative boot operation
- Supports Command Completion signal and interrupt to host processor and Command Completion Signal disable feature
- Supports one SD (Version 1.0 to 3.0) or MMC (Version 3.3 to 4.41) or CE-ATA device
- Supports hardware CRC generation and error detection
- Supports programmable baud rate
- Supports host pull-up control
- Supports SDIO interrupts in 1-bit and 4-bit modes
- Supports SDIO suspend and resume operation
- Supports SDIO read wait
- Supports block size of 1 to 65535 bytes
- Supports descriptor-based internal DMA controller
- Internal 128 bytes FIFO for data transfer

4.4.2 Register List

Module Name	Base Address	
SDMMC/SDIO Controller 0	0x40002000	
Register Name	Offset Address	Description

SD_GCTL	0x00	Control register
SD_CKCR	0x04	Clock Control register
SD_TMOR	0x08	Time out register
SD_BWDR	0x0C	Bus Width register
SD_BKSR	0x10	Block size register
SD_BYCR	0x14	Byte count register
SD_CMDR	0x18	Command register
SD_CAGR	0x1c	Command argument register
SD_RESP0	0x20	Response 0 register
SD_RESP1	0x24	Response 1 register
SD_RESP2	0x28	Response 2 register
SD_RESP3	0x2C	Response 3 register
SD_IMKR	0x30	Interrupt mask register
SD_MISR	0x34	Masked interrupt status register
SD_RISR	0x38	Raw interrupt status register
SD_STAR	0x3C	Status register
SD_FWLR	0x40	FIFO Water Level register
SD_FUNS	0x44	FIFO Function Select register
SD_CBCR	0x48	Transferred byte count between controller and card
SD_BBCR	0x4C	Transferred byte count between host memory and internal FIFO
SD_DBGC	0x50	Current Debug Control address
SD_A12A	0x58	Auto command 12 argument
SD_NTSR	0x5C	SD New Timing Set Register
SD_SDBG	0x60	SD New Timing Set Debug Register
SD_HWRST	0x78	Hardware Reset Register
SD_DMAC	0x80	BUS Mode Control
SD_DLBA	0x84	Descriptor List Base Address
SD_IDST	0x88	DMAC Status
SD_IDIE	0x8C	DMAC Interrupt Enable
SD_CHDA	0x90	Current host descriptor address
SD_CBDA	0x94	Current buffer descriptor address
SD_THLDC	0x100	Card Threshold Control register
SD_DSBD	0x10C	eMMC4.5 DDR Start Bit Detection Control
SD_RESP_CRC	0x110	Response CRC
SD_D7_CRC	0x114	DATA7 CRC
SD_D6_CRC	0x118	DATA7 CRC
SD_D5_CRC	0x11C	DATA7 CRC
SD_D4_CRC	0x120	DATA7 CRC
SD_D3_CRC	0x124	DATA7 CRC
SD_D2_CRC	0x128	DATA7 CRC
SD_D1_CRC	0x12C	DATA7 CRC

SD_DO_CRC	0x130	DATA7 CRC
SD_WCRC_STA	0x134	Write DATA CRC Status
SD_DRV_DL	0x140	Drive delay control register
SD_SMAP_DL	0x144	sample delay control register
SD_DS_DL	0x148	Data strobe signal delay control register
SD_FIFO	0x200	Read/ Write FIFO

4.4.3 Register Description

4.4.3.1SD Global Control Register

Address : 0x0000			Name: SD_GCTRL Default: 0x0000_0300
Field	Type	Default	Description
31	R/W	0x0	FIFO_ACCESS_MODE 0: DMA bus 1: AHB bus
30:11	/	/	/
10	R/W	0x0	DDR_MODE DDR Mode Select 0: SDR mode 1: DDR mode
9	/	/	/
8	R/W	0x1	CARD_DBC_ENB Card Detect (D3 irq) De-bounce Enable 0: disable de-bounce 1: enable de-bounce
7:6	/	/	/
5	R/W	0x0	IDMA_ENB Internal DMA Enable 0: disable IDMA to transfer data 1: enable IDMA to transfer data
4	R/W	0x0	INT_ENB Global Interrupt Enable

			0: disable interrupts 1: enable interrupts
3	/	/	/
2	R/WAC	0x0	IDMA_RST 0: no effect 1: Internal DMA Reset This bit is auto-cleared after reset operation.
1	R/WAC	0x0	FIFO_RST 0: no effect 1: FIFO Reset This bit is auto-cleared after reset operation.
0	R/WAC	0x0	SOFT_RST 0: no effect 1: Software Reset This bit is auto-cleared after reset operation.

4.4.3.2SD Clock Control Register

Address : 0x0004			Name: SD_CLK_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31	R/W	0x0	MASK_DATA0 0: ignore the data0 status when update card clock 1: consider the data0 status when update card clock
20:18	/	/	/
17	R/W	0x0	CCLK_CTRL 0: card clock always on 1: card clock is turned off when FSM is in IDLE state
16	R/W	0x0	CCLK_ENB 0: card clock is off 1: card clock is on
15:8	/	/	/
7:0	R/W	0x0	CCLK_DIV Card Clock Divider n: the card source clock is divided by 2 * n (n = 0 ~ 255)

4.4.3.3SD Timeout Register

Address : 0x0008			Name: SD_TMOUT Default: 0xFFFF_FF40
Field	Type	Default	Description
31:8	R/W	0FFFFFFF	DATA_TMO_LIMIT Data Timeout Limit
7:0	R/W	0x40	RSP_TMO_LIMIT Response Timeout Limit

4.4.3.4SD Card Width Register

Address : 0x000C			Name: SD_WIDTH Default: 0x0000_0000
Field	Type	Default	Description
31:2	/	/	/
1:0	R/W	0x0	CARD_WIDTH Card Width 0: 1-bit width 1: 4-bit width 2'b1x: 8-bit width

4.4.3.5SD Block Size Register

Address : 0x0010			Name: SD_BLK_SZ Default: 0x0000_0200
Field	Type	Default	Description
31:16	/	/	/
15:0	R/W	0x200	BLK_SIZE Transfer Block Size

4.4.3.6SD Byte Counter Register

Address : 0x0014			Name: SD_BYTE_CNT Default: 0x0000_0200
Field	Type	Default	Description
31:0	R/W	0x200	BYTE_CNT

			Transfer Byte Counter The value is the number of bytes to be transferred and must be integer multiple of BLK_SIZE for block transfers.
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4.4.3.7 SD Command Register

Address : 0x0018			Name: SD_CMD Default: 0x0000_0000
Field	Type	Default	Description
31	R/W	0x0	CMD_LOAD Start Command This bit is auto-cleared when current command is sent. If there is no any response error happened, a command complete interrupt bit, CMD_OVER, will be set in SD_INTR register. You should not write any other command before this bit is cleared. Or a command busy interrupt bit, CMD_BUSY, will be set in SD_INTR register.
30:29	/	/	/
28	R/W	0x0	VOL_SWITCH Voltage Switch 0: this command is a normal command 1: this command is a voltage switch command, only set for CMD11
27	R/W	0x0	BOOT_ABT Boot Abort Setting this bit will terminate the boot operation.
26	R/W	0x0	EXP_BOOT_ACK Expect the Boot Acknowledge When software sets this bit along in mandatory boot operation, controller expects a boot acknowledge start pattern of 0-1-0 from the selected card.
25:24	R/W	0x0	BOOT_MODE 2'b00: normal command 2'b01: Mandatory Boot Operation 2'b10: Alternate Boot Operation 2'b11: reserved
23	R/W	0x0	CCS_EXP 0 – Interrupts are not enabled in CE-ATA device (nIEN = 1 in ATA control register), or command does not expect CCS from device 1 – Interrupts are enabled in CE-ATA device (nIEN = 0), and RW_BLK

			command expects command completion signal from CE-ATA device If the command expects Command Completion Signal, CCS, from CE-ATA device, the software should set this bit. SD/MMC sets DATA_OVER bit in SD_INTR register and generates interrupt to host if Data Transfer Over is not masked.
22	R/W	0x0	RD_CEATA_DEV 0 – Host is not performing read access (RW_REG or RW_BLK) towards CE-ATA device 1 – Host is performing read access (RW_REG or RW_BLK) towards CE-ATA device
21	R/W	0x0	UPD_CLK Update Clock 0: normal command 1: Update Card Clock When this bit is set, controller will update clock domain and clock output state. No command will be sent in this situation.
20:16	/	/	/
15	R/W	0x0	SEND_INIT_SEQ 0: normal command 1: Send initialization sequence before sending this command
14	R/W	0x0	STOP_ABT_CMD 0: normal command 1: Send stop or abort command to stop current data transfer in progress. CMD12, CMD52 for writing I/O Abort in SDIO CCCR
13	R/W	0x0	WAIR_PRE_OVER Wait Data Transfer Over 0: send command at once, do not care of data transferring 1: wait for data transfer completion before sending current command
12	R/W	0x0	STOP_CMD_FLAG Send Stop Command Automatically, CMD12 0: do not send CMD12 at the end of data transfer 1: send CMD12 automatically at the end of data transfer
11	R/W	0x0	TRANS_MODE 0: Block mode

			1: Stream mode
10	R/W	0x0	TRANS_DIR 0: Read operation 1: Write operation
9	R/W	0x0	DATA_TRANS 0: no data transfer 1: with data transfer
8	R/W	0x0	CHK_RESP_CRC 0: do not check the response CRC 1: check the response CRC
7	R/W	0x0	LONG_RESP 0: short response 1: long response
6	R/W	0x0	RESP_RCV 0: command without a response 1: command with a response
5:0	R/W	0x0	CMD_INDEX Command Index

4.4.3.8SD Command Argument Register

Address : 0x001C			Name: SD_CMD_ARG Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	CMD_ARG Command Argument

4.4.3.9SD Response 0 Register

Address : 0x0020			Name: SD_RESP0 Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	CMD_RESP0 Command Response 0

4.4.3.10 SD Response 1 Register

Address : 0x0024			Name: SD_RESP1 Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	CMD_RESP1 Command Response 1

4.4.3.11 SD Response 2 Register

Address : 0x0028			Name: SD_RESP2 Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	CMD_RESP2 Command Response 2

4.4.3.12 SD Response 3 Register

Address : 0x002C			Name: SD_RESP3 Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	CMD_RESP3 Command Response 3

4.4.3.13 SD Interrupt Mask Register

Address : 0x0030			Name: SD_INT_MASK Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	INT_MASK 0: interrupt is disabled 1: interrupt is enabled Bit field defined as follows: bit 31: card removed bit 30: card inserted bit 17~19: reserved bit 16: SDIO interrupt

			bit 15: Data End-bit Error bit 14: Auto Stop Command Done bit 13: Data Start Bit Error bit 12: Command Busy and illegal write bit 11: FIFO under run or overflow bit 10: Data Starvation Timeout/V1.8 Switch Done bit 9: Data Timeout/Boot Data Start bit 8: Response Timeout/Boot ACK Received bit 7: Data CRC Error bit 6: Response CRC Error bit 5: Data Receive Request bit 4: Data Transmit Request bit 3: Data Transfer complete bit 2: Command complete bit 1: Response Error, no response or CRC error bit 0: reserved
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4.4.3.14 SD Masked Interrupt Status Register

Address : 0x0034			Name : SD_MINTR Default : 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	MSKD_ISTA Interrupt status. Enabled only when corresponding bit in INT_MASK register is set. Bit field defined as follows: bit 31: card removed bit 30: card inserted bit 17~19: reserved bit 16: SDIO interrupt bit 15: Data End-bit Error bit 14: Auto Stop Command Done bit 13: Data Start Bit Error bit 12: Command Busy and illegal write bit 11: FIFO under run or overflow bit 10: Data Starvation Timeout/V1.8 Switch Done bit 9: Data Timeout/Boot Data Start bit 8: Response Timeout/Boot ACK Received bit 7: Data CRC Error bit 6: Response CRC Error bit 5: Data Receive Request

			bit 4: Data Transmit Request bit 3: Data Transfer complete bit 2: Command complete bit 1: Response Error, no response or CRC error bit 0: reserved
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4.4.3.15 SD Raw Interrupt Status Register

Address : 0x0038			Name: SD_RINTR Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W1C	0x0	RAW_ISTA Raw Interrupt status. These bits are write-1-to-clear bits. Bit field defined as follows: bit 31: card removed bit 30: card inserted bit 17~19: reserved bit 16: SDIO interrupt bit 15: Data End-bit Error bit 14: Auto Stop Command Done bit 13: Data Start Bit Error bit 12: Command Busy and illegal write bit 11: FIFO under run or overflow bit 10: Data Starvation Timeout/V1.8 Switch Done bit 9: Data Timeout/Boot Data Start bit 8: Response Timeout/Boot ACK Received bit 7: Data CRC Error bit 6: Response CRC Error bit 5: Data Receive Request bit 4: Data Transmit Request bit 3: Data Transfer complete bit 2: Command complete bit 1: Response Error, no response or CRC error bit 0: reserved

4.4.3.16 SD Status Register

Address : 0x003C			Name: SD_STATUS Default: 0x0000_0006
Field	Type	Default	Description
31	R	0x0	DMA_REQ

			DMA Request Signal State
30:23	/	/	/
22:17	R	0x0	FIFO_LEVEL Number of filled locations in FIFO
16:11	R	0x0	RESP_IDX Index of previous response, including the auto-stop command sent by controller.
10	R	0x0	FSM_BUSY Data transmit or receive state machine is busy.
9	R	0x0	CARD_BUSY Inverted version of DATA0 0: card is not busy 1: card is busy
8	R	0x0	CARD_PRESENT Status of DATA3 0: card not present 1: card present
7:4	R	0x0	FSM_STA Command FSM States: 0: Idle 1: Send init sequence 2: TX CMD start bit 3: TX CMD tx bit 4: TX CMD index/arg 5: TX CMD CRC7 6: TX CMD End bit 7:RX RESP start bit 8: RX RESP IRQ response 9: RX RESP tx bit 10: RX RESP CMD Index 11: RX RESP data 12: RX RESP CRC7 13: RX RESP end bit 14: CMD path wait NCC 15: Wait, CMD-to-RESP turnaround

3	R	0x0	FIFO_FULL 0: FIFO not full 1: FIFO full
2	R	0x1	FIFO_EMPTY 0: FIFO not empty 1: FIFO empty
1	R	0x1	FIFO_TX_LEVEL FIFO TX Water Level Flag 0: FIFO does not reach the transmit trigger level 1: FIFO reaches transmit trigger level
0	R	0x0	FIFO_RX_LEVEL FIFO RX Water Level Flag 0: FIFO does not reach the receive trigger level 1: FIFO reaches receive trigger level

4.4.3.17 SD Water Level Register

Address : 0x0040			Name: SD_FIFO_WL Default: 0x000F_0000
Field	Type	Default	Description
31	/	/	/
30:28	R/W	0x0	BSIZE_OF_TRANS Burst size of multiple transaction 000: 1 transfers 001: 4 010: 8 011: 16 100: 32 101: 64 110: 128 111: 256 Should be set same as DMA controller multiple transaction size. The units for transfers are the DWORD. A single transfer would be signaled based on this value. Value should be sub-multiple of RX_TL +1 and FIFO_DEPTH – TX_TL. Recommended: MSize = 8, TX_TL = 16, RX_TL = 15
27:21	/	/	/
20:16	R/W	0xF	RX_TL RX Trigger Level 0x0 ~ 0x1e: 0 ~ 30

			0x1f: reserved FIFO threshold when FIFO request host to receive data from FIFO. When FIFO data level is greater than this value, DMA request is raised if DMA enabled, or RX interrupt bit is set if interrupt enabled. At the end of packet, if the last transfer is less than this level, the value will be ignored and relative request will be raised as usual. Recommended: 15
15:5	/	/	/
4:0	R/W	0xF	TX_TL TX Trigger Level 0x1 ~ 0x1f: 1 ~ 31 0x0: reserved FIFO threshold when FIFO request host to transmit data from FIFO. When FIFO data level is less than or equal to this value, DMA RX request is raised if DMA enabled, or TX interrupt bit is set if interrupt enabled. At the end of packet, if the last transfer is less than this level, the value will be ignored and relative request will be raised as usual. Recommended: 16

4.4.3.18 SD Function Select Register

Address : 0x0044			Name : SD_FUNC_SEL Default : 0x0000_0000
Field	Type	Default	Description
31:16	R/W	0x0	CEATA_EN CEATA Support ON/OFF 0xceaa – CEATA support on. All hidden CEATA relative bits are accessible normally and these 16 bits return value of 0x1 when be read. Other value – CEATA support off. All hidden CEATA relative bits cannot be access and these 16 bits return value of 0 when be read.
15:11	/	/	/
10	R/W	0x0	CEATA_ISTA CEATA device interrupt status 0 – Interrupts not enabled in CE-ATA device (nIEN = 1 in ATA control register) 1 – Interrupts are enabled in CE-ATA device (nIEN = 0 in ATA control register) Software should appropriately write to this bit after power-on reset or any other reset to CE-ATA device. After reset, usually CE-ATA device interrupt is

			disabled (nIEN = 1). If the host enables CE-ATA device interrupt, then software should set this bit.
9	R/W	0x0	SEND_AUTO_STOP_CCSD send auto stop ccscd 0 – Clear bit if SD/MMC does not reset the bit 1 – Send internally generated STOP after sending CCSD to CEATA device When set, SD/MMC automatically sends internally generated STOP command (CMD12) to CE-ATA device. After sending internally-generated STOP command, Auto Command Done (ACD) bit in RINTSTS is set and generates interrupt to host if Auto Command Done interrupt is not masked. After sending the CCSD, SD/MMC automatically clears send_auto_stop_ccscd bit.
8	R/W	0x0	SEND_CCSD send ccscd 0 – Clear bit if SD/MMC does not reset the bit 1 – Send Command Completion Signal Disable (CCSD) to CE-ATA device When set, SD/MMC sends CCSD to CE-ATA device. Software sets this bit only if current command is expecting CCS (that is, RW_BLK) and interrupts are enabled in CE-ATA device. Once the CCSD pattern is sent to device, SD/MMC automatically clears send_ccscd bit. It also sets Command Done (CD) bit in RINTSTS register and generates interrupt to host if Command Done interrupt is not masked.
7:3	/	/	/
2	R/W	0x0	ABT_RDATA Abort Read Data 0 – Ignored 1 –After suspend command is issued during read-transfer, software polls card to find when suspend happened. Once suspend occurs, software sets bit to reset data state-machine, which is waiting for next block of data. Used in SDIO card suspends sequence. This bit is auto-cleared once controller reset to idle state.
1	R/W	0x0	READ_WAIT Read Wait 0 – Clear SDIO read wait 1 – Assert SDIO read wait
0	R/W	0x0	HOST_SEND_MMC_IRQRESQ Host Send MMC IRQ Response

			0 – Ignored 1 – Send auto IRQ response When host is waiting MMC card interrupt response, setting this bit will make controller cancel wait state and return to idle state, at which time, controller will receive IRQ response sent by itself. This bit is auto-cleared after response is sent.
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4.4.3.19 SD Transferred CIU Card Byte Counter Register

Address : 0x0048			Name: SD_TCBCNT Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	TRANS_BYTE_CNT_TO_CARD Number of bytes transferred by CIU unit to card. The register should be accessed in full to avoid read-coherency problems and read only after data transfer completes.

4.4.3.20 SD Transferred Host to BIU-FIFO Byte Counter Register

Address : 0x004c			Name: SD_TBBCNT Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	TRANS_BYTE_CNT_ON_BUS Number of bytes transferred by Host/DMA memory and BIU FIFO. The register should be accessed in full to avoid read-coherency problems and read only after data transfer completes.

4.4.3.21 SD Debug Control Register

Address : 0x0050			Name: SD_DBG Default: 0x0000_0000
Field	Type	Default	Description
31:12	/	/	/
11:0	R/W	0x0	DBG_ON 0xdeb – Debug on. ALL hidden register bits will be accessible and these 12 bits return 0x1 when be read. Other values – Debug off. All hidden register bits will not be accessed and these 12 bits return 0 when be read.

4.4.3.22 SD Auto Command CMD12 Argument Register

Address : 0x0058			Name: SD_A12A Default: 0x0000_FFFF
Field	Type	Default	Description
31:16	/	/	/
0:15	R/W	0xFFFF	SD_A12A SD_A12A set the argument of command 12 automatically send by controller.

4.4.3.23 SD New Timing Set Register

Address : 0x005C			Name: SD_NTSTR Default: 0x8171_0000
Field	Type	Default	Description
31	R/W	0x1	MODE_SELEC 0: Old mode of sample/output timing 1: New mode of sample/output timing
30:28	/	/	/
27	R/W	0x0	DATA0_BYPASS Select data0 input async or bypass sample logic, used to check card busy or not 0: disable 1: enable
26:25	/	/	/
24	R/W	0x1	CMD_DAT_RX_PHASE_CLR During update clock, command and data rx phase clear 0: disable 1: enable
23	/	/	/
22	R/W	0x1	DAT_CRC_STATUS_RX_PHASE_CLR Before receiving CRC status, data rx phase clear 0: disable 1: enable
21	R/W	0x1	DAT_TRANS_RX_PHASE_CLR Before transferring data, data rx phase clear 0: disable 1: enable
20	R/W	0x1	DAT_RECV_RX_PHASE_CLR Before receiving data, data rx phase clear 0: disable 1: enable

19:17	/	/	/
16	R/W	0x1	CMD_SEND_RX_PHASE_CLR Before send command, command rx phase clear 0: disable 1: enable
15:6	/	/	/
5:4	R/W	0x0	SAMPLE_TIMING_PHASE 00: sample timing phase offset 90° 01: sample timing phase offset 180° 10: sample timing phase offset 270° 11: ignore
3:0	/	/	/

4.4.3.24 SD Hardware Reset Register

Address : 0x0078			Name: SD_HRST Default: 0x0000_0001
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0x1	HW_RST 1 – Active mode 0 – Reset These bits cause the cards to enter pre-idle state, which requires them to be re-initialized.

4.4.3.25 SD DMAC Control Register

Address : 0x0080			Name: SD_DMA_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31	W	0x0	DES_LOAD_CTRL When DMAC fetches a descriptor, if the valid bit of a descriptor is not set, DMAC FSM will go to the suspend state. Setting this bit will make DMAC re-fetch descriptor again and do the transfer normally.
30:11	/	/	/
10:8	R	0x0	PRG_BURST_LEN Programmable Burst Length These bits indicate the maximum number of beats to be performed in one IDMAC transaction. The IDMAC will always attempt to burst as specified in PBL each time it starts a Burst transfer on the host bus. The permissible

			values are 1, 4, 8, 16, 32, 64, 128 and 256. This value is the mirror of MSIZE of FIFOTH register. In order to change this value, write the required value to FIFOTH register. This is an encode value as follows. 000 – 1 transfers 001 – 4 transfers 010 – 8 transfers 011 – 16 transfers 100 – 32 transfers 101 – 64 transfers 110 – 128 transfers 111 – 256 transfers Transfer unit is 16, 32, or 64 bits, based on HDATA_WIDTH. PBL is a read-only value.
7	R/W	0x0	IDMAC_ENB IDMAC Enable When set, the IDMAC is enabled. DE is read/write.
6:2	R/W	0x0	DES_SKIP_LEN Descriptor Skip Length Specifies the number of Word to skip between two unchained descriptors. This is applicable only for dual buffer structure. Default value is set to 4 DWORD.
1	R/W	0x0	FIX_BUST_CTRL Fixed Burst Controls whether the AHB Master interface performs fixed burst transfers or not. When set, the AHB will use only SINGLE, INCR4, INCR8 during start of normal burst transfers. When reset, the AHB will use SINGLE and INCR burst transfer operations.
0	R/W	0x0	IDMAC_RST DMA Reset When set, the DMA Controller resets all its internal registers. SWR is read/write. It is automatically cleared after 1 clock cycle.

4.4.3.26 SD Descriptor List Base Address Register

Address : 0x0084			Name: SD_DLBA_REG Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	DES_BASE_ADDR

			Start of Descriptor List Contains the base address of the First Descriptor. The LSB bits [1:0] are ignored and taken as all-zero by the IDMAC internally. Hence these LSB bits are read-only.
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4.4.3.27 SD DMAC Status Register

Address : 0x0088			Name: SD_DMATA Default: 0x0000_0000
Field	Type	Default	Description
31:17	/	/	/
16:13	R	0x0	DMAC_FSM_STA DMAC FSM present state 0 – DMA_IDLE 1 – DMA_SUSPEND 2 – DESC_RD 3 – DESC_CHK 4 – DMA_RD_REQ_WAIT 5 – DMA_WR_REQ_WAIT 6 – DMA_RD 7 – DMA_WR 8 – DESC_CLOSE This bit is read-only.
12:10	R	0x0	DMAC_ERR_STA Error Bits Indicates the type of error that caused a Bus Error. Valid only with Fatal Bus Error bit (IDSTS[2]) set. This field does not generate an interrupt. 3'b001 – Host Abort received during transmission 3'b010 – Host Abort received during reception Others: Reserved EB is read-only.
9	R/W	0x0	ABN_INT_SUM Abnormal Interrupt Summary Logical OR of the following: IDSTS[2] – Fatal Bus Interrupt IDSTS[4] – DU bit Interrupt IDSTS[5] – Card Error Summary Interrupt Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes AIS to be set is cleared. Writing a 1 clears this bit.

8	R/W	0x0	NOR_INT_SUM Normal Interrupt Summary Logical OR of the following: IDSTS[0] – Transmit Interrupt IDSTS[1] – Receive Interrupt Only unmasked bits affect this bit. This is a sticky bit and must be cleared each time a corresponding bit that causes NIS to be set is cleared. Writing a 1 clears this bit.
7:6	/	/	/
5	R/W	0x0	ERR_FLAG_SUM Card Error Summary Indicates the status of the transaction to/from the card; also present in RINTSTS. Indicates the logical OR of the following bits: EBE – End Bit Error RTO – Response Timeout/Boot Ack Timeout RCRC – Response CRC SBE – Start Bit Error DRTO – Data Read Timeout/BDS timeout DCRC – Data CRC for Receive RE – Response Error Writing a 1 clears this bit.
4	R/W	0x0	DES_UNAVL_INT Descriptor Unavailable Interrupt This bit is set when the descriptor is unavailable due to OWN bit = 0 (DES0[31] = 0). Writing a 1 clears this bit.
3	/	/	/
2	R/W	0x0	FATAL_BERR_INT Fatal Bus Error Interrupt Indicates that a Bus Error occurred (IDSTS[12:10]). When this bit is set, the DMA disables all its bus accesses. Writing a 1 clears this bit.
1	R/W	0x0	RX_INT Receive Interrupt Indicates the completion of data reception for a descriptor. Writing a 1 clears this bit.
0	R/W	0x0	TX_INT Transmit Interrupt

			Indicates that data transmission is finished for a descriptor. Writing a '1' clears this bit.
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4.4.3.28 SD DMAC Interrupt Enable Register

Address : 0x008C			Name: SD_IDIE Default: 0x0000_0000
Field	Type	Default	Description
31:10	/	/	/
9	R/W	0x0	ABN_INT_ENB Abnormal Interrupt Summary Enable When set, an abnormal interrupt is enabled. This bit enables the following bits: IDINTEN[2] – Fatal Bus Error Interrupt IDINTEN[4] – DU Interrupt IDINTEN[5] – Card Error Summary Interrupt
8	R/W	0x0	NOR_INT_ENB Normal Interrupt Summary Enable When set, a normal interrupt is enabled. When reset, a normal interrupt is disabled. This bit enables the following bits: IDINTEN[0] – Transmit Interrupt IDINTEN[1] – Receive Interrupt
7:6	/	/	/
5	R/W	0x0	ERR_SUM_INT_ENB Card Error summary Interrupt Enable When set, it enables the Card Interrupt summary.
4	R/W	0x0	DES_UNAVL_INT_ENB Descriptor Unavailable Interrupt When set along with Abnormal Interrupt Summary Enable, the DU interrupt is enabled.
3	/	/	/
2	R/W	0x0	FERR_INT_ENB Fatal Bus Error Enable When set with Abnormal Interrupt Summary Enable, the Fatal Bus Error Interrupt is enabled. When reset, Fatal Bus Error Enable Interrupt is disabled.
1	R/W	0x0	RX_INT_ENB

			Receive Interrupt Enable When set with Normal Interrupt Summary Enable, Receive Interrupt is enabled. When reset, Receive Interrupt is disabled.
0	R/W	0x0	TX_INT_ENB Transmit Interrupt Enable When set with Normal Interrupt Summary Enable, Transmit Interrupt is enabled. When reset, Transmit Interrupt is disabled.

4.4.3.29 SD Current Host Descriptor Address Register

Address : 0x0090			Name: SD_CHDAR Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	CUR_DES_ADDR Host Descriptor Address Pointer Cleared on reset. Pointer updated by IDMAC during operation. This register points to the start address of the current descriptor read by the IDMAC.

4.4.3.30 SD Current Buffer Descriptor Address Register

Address : 0x009C			Name: SD_CLK_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	CUR_BUFF_ADDR Host Buffer Address Pointer Cleared on Reset. Pointer updated by IDMAC during operation. This register points to the current Data Buffer Address being accessed by the IDMAC.

4.4.3.31 SD Card Threshold Control Register

Address : 0x0100			Name: SD_THLD_CTRL Default: 0x0000_0000
Field	Type	Default	Description
31:28	/	/	/
27:16	R/W	0x0	CARD_RD_THLD Card Read Threshold Size
15:1	/	/	/
0	R/W	0x0	CARD_RD_THLD_ENB

			Card Read Threshold Enable 0 – Card Read Threshold Disable 1 - Card Read Threshold Enable Host controller initiates Read Transfer only if CARD_RD_THLD amount of space is available in receive FIFO
--	--	--	---

4.4.3.32 eMMC4.5 DDR Start Bit Detection Control Register

Address : 0x010C			Name: EMMC_DDR_SBIT_DET Default: 0x0000_0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0x0	HALF_START_BIT Control for start bit detection mechanism inside mstorage based on duration of start bit. For eMMC 4.5, start bit can be: 0 - Full cycle 1 - Less than one full cycle Set HALF_START_BIT=1 for eMMC 4.5 and above; set to 0 for SD applications.

4.4.3.33 SD Response CRC Register

Address : 0x0110			Name: SD_RESP_CRC Default: 0x0000_0000
Field	Type	Default	Description
31:7	/	/	/
6:0	R	0x0	RESP_CRC

4.4.3.34 SD DATA7 CRC Register

Address : 0x0114			Name: SD_DATA7_CRC Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	DATA7_CRC

4.4.3.35 SD DATA6 CRC Register

Address : 0x0118			Name: SD_DATA6_CRC Default: 0x0000_0000
Field	Type	Default	Description

31:0	R	0x0	DATA6_CRC
------	---	-----	-----------

4.4.3.36 SD DATA5 CRC Register

Address : 0x011C			Name: SD_DATA5_CRC Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	DATA5_CRC

4.4.3.37 SD DATA4 CRC Register

Address : 0x0120			Name: SD_DATA4_CRC Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	DATA4_CRC

4.4.3.38 SD DATA3 CRC Register

Address : 0x0124			Name: SD_DATA3_CRC Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	DATA3_CRC

4.4.3.39 SD DATA2 CRC Register

Address : 0x0128			Name: SD_DATA2_CRC Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	DATA2_CRC

4.4.3.40 SD DATA1 CRC Register

Address : 0x012C			Name: SD_DATA1_CRC Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	DATA1_CRC

4.4.3.41 SD DATA0 CRC Register

Address : 0x0130			Name: SD_DATA0_CRC Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	DATA0_CRC

4.4.3.42 SD CRC Status Register

Address : 0x0134			Name: SD_CRC_STA Default: 0x0000_0000
Field	Type	Default	Description
31:3	/	/	/
2:0	R	0x0	CRC_STATUS

4.4.3.43 SD FIFO Register

Address : 0x0200			Name: SD_FIFO Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	TX/RX_FIFO Data FIFO

4.4.4 SD/MMC DMA Controller Description

SD3.0 controller has an internal DMA controller (IDMAC) to transfer data between host memory and SDMMC port. With a descriptor, IDMAC can efficiently move data from source to destination by automatically loading next DMA transfer arguments, which need less CPU intervention. Before transfer data in IDMAC, host driver should construct a descriptor list, configure arguments of every DMA transfer, then launch the descriptor and start the DMA. IDMAC has an interrupt controller, when enabled, it can interrupt the HOST CPU in situations such as data transmission completed or some errors happened.

4.4.4.1 IDMAC Descriptor Structure

The IDMAC uses a descriptor with a chain structure, and each descriptor points to a unique buffer and the next descriptor.

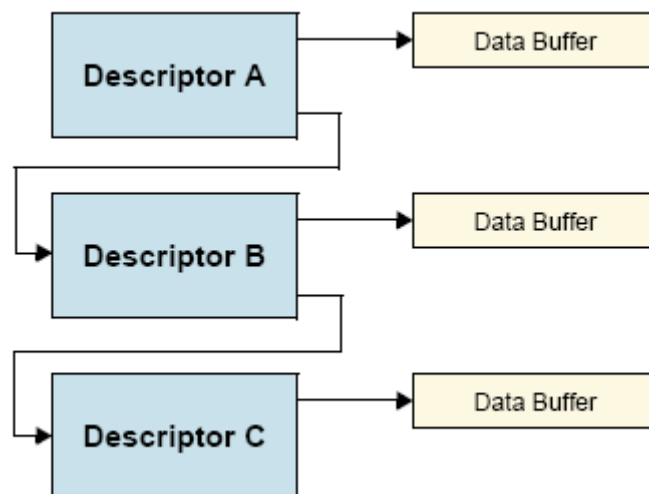


Figure 4-6 SDC IDMAC Chain Descriptor

This figure illustrates the internal formats of a descriptor. The descriptor addresses must be aligned to the bus width used for 32-bit buses. Each descriptor contains 16 bytes of control and status information.

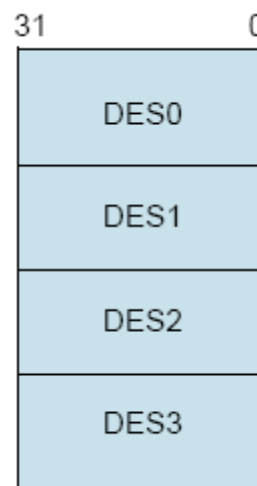


Figure 4-7 SDC IDMAC Fix Step Descriptor

DES0 is a notation used to denote the [31:0] bits, DES1 to denote [63:32] bits, DES2 to denote [95:64] bits, and DES3 to denote [127:96] bits in a descriptor.

4.4.4.2 DES0 definition

Bits	Name	Descriptor
31	HOLD	DES_OWN_FLAG When set, this bit indicates that the descriptor is owned by the IDMAC. When this bit is reset, it indicates that the descriptor is owned by the host. This bit is

		cleared when transfer is over.
30	ERROR	ERR_FLAG When some error happened in transfer, this bit will be set.
29:5	/	/
4	Chain Flag	CHAIM_MOD When set, this bit indicates that the second address in descriptor is the next descriptor address. Must be set 1.
3	First DES Flag	FIRST_FLAG When set, this bit indicates that this descriptor contains the first buffer of data. Must be set to 1 in first DES.
2	Last DES Flag	LAST_FLAG When set, this bit indicates that the buffers pointed to by this descriptor are the last data buffer.
1	Disable Interrupt on completion	CUR_TXRX_OVER_INT_DIS When set, this bit will prevent the setting of the TX/RX interrupt bit of the IDMAC status register for data that ends in the buffer pointed to by this descriptor.
0	/	/

4.4.4.3DES1 definition

Bits	Name	Descriptor
31:13	/	/
12:0	Buffer size	BUFF_SIZE These bits indicate the data buffer byte size, which must be a multiple of 4 bytes. If this filed is 0, the DMA ignores this buffer and proceeds to the next descriptor.

4.4.4.4DES2 definition

Bits	Name	Descriptor
31:0	Buffer address pointer	BUFF_ADDR These bits indicate the physical address of data buffer. The IDMAC ignores DES2 [1:0], corresponding to the bus width of 32.

4.4.4.5 DES3 definition

Bits	Name	Descriptor
31:0	Next descriptor address	NEXT_DESP_ADDR These bits indicate the pointer to the physical memory where the next descriptor is present.

4.5 Pulse Width Modulation/Event Capture Controller (PWM/ECT)

4.5.1 Overview

PWM controller has eight channels (PWM0, PWM1, PWM2, PWM3, PWM4, PWM5, PWM6, PWM7, PWM8), and divides to 4 PWM pairs: PWM01 pair, PWM23 pair, PWM45 pair, PWM67 pair.

The PWM controller includes the following features:

- 8 PWM channels
- Supports pulse, cycle and complementary pair output
- Supports capture input
- Programming dead zone output
- Build-in the programmable dead-time generator, controllable dead-time
- Three kinds of output waveform: continuous waveform, pulse waveform and complementary pair
- Output frequency range: 0~ 24MHz/100MHz
- Various duty-cycle: 0% ~100%
- Minimum resolution: 1/65536
- Interrupt generation of PWM output and capture input

4.5.2 Block Diagram

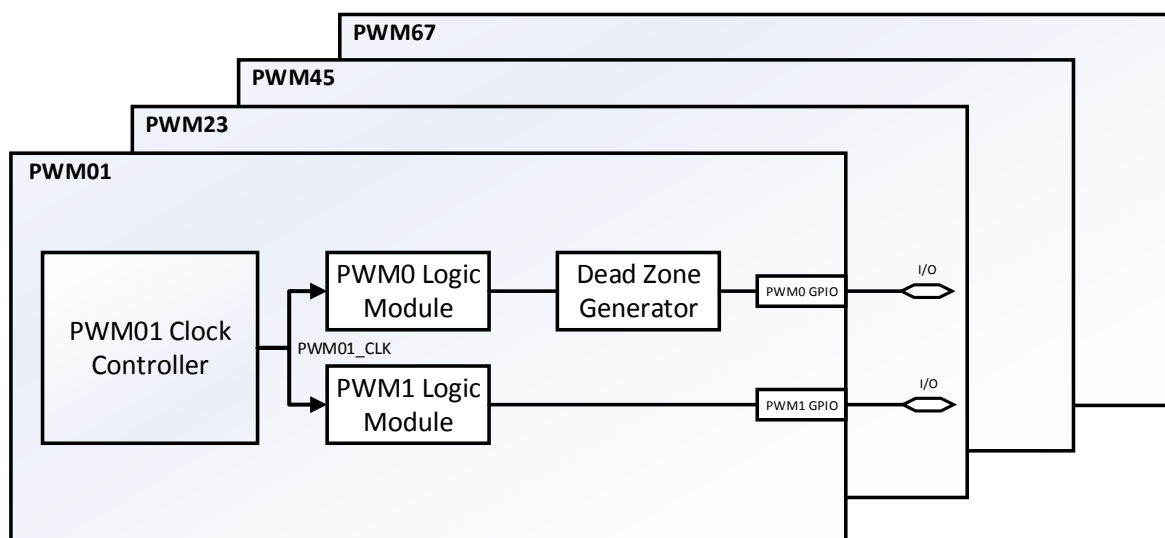


Figure 4-8 PWM01 Block Diagram

Each PWM pair consists of 1 clock module, 2 timer logic module and 1 programmable dead-time generator.

4.5.3 Operations and Functional Descriptions

4.5.3.1 Clock Controller

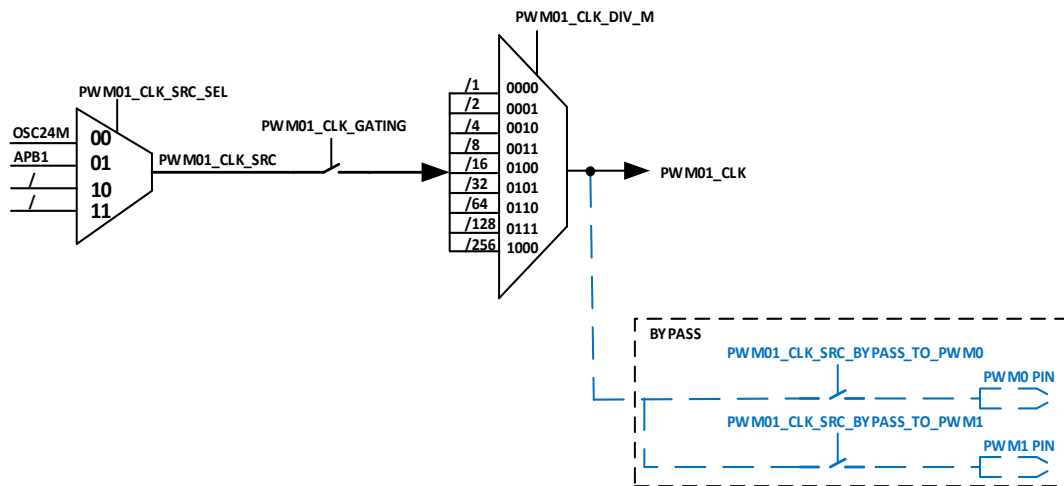


Figure 4-9 PWM01 Clock Controller Diagram

The clock controller of each PWM pair includes clock source select (PWM01_CLK_SRC_SEL), 1~256 scaler (PWM01_CLK_DIV_M), clock source bypass (CLK_SRC_BYPASS) and clock switch (PWM01_CLK_GATING).

The clock sources of PWM have OSC24M and APB1 Bus. OSC24M comes from external high frequency oscillator, APB1 is APB1 bus clock, usually is 100MHz.

4.5.3.2 PWM Output

Using PWM01 as an example, Figure 4-10 indicates PWM01 output logic module diagram. Other PWM pairs (PWM23, PWM45, PWM67) logic module diagrams are the same as PWM01, PWM8 logic module diagram is the same as PWM1.

PWM Timer Logic consists of one 16-bit up-counter and one 16-bit comparator. The up-counter is used to control period, and the comparator is used to control duty-cycle. The up-counter and the comparator support cache-loading, PWM output is enabled, the register value of the counter and the comparator can be changed at any time, the changed value is cached to the cache register, when the value of up-counter is equal to **PWM ENTIRE CYCLE**, the value of the cache register is loaded to the counter and the comparator. Cache-loading is good to avoid unstable PWM output waveform with burred feature when updating the counter value and the comparator value.

PWM supports cycle and pulse waveform output.

Cycle mode: When the value of up-counter reaches **PWM ENTIRE CYCLE**, the value of up-counter is loaded automatically to 0 and the up-counter continues to count, then the output waveform is a continuous waveform.

Pulse mode: When the value of up-counter reaches **PWM_ENTIRE_CYCLE**, the value of up-counter is loaded automatically to 0 and the up-counter stops counting, then the output waveform is a pulse waveform.

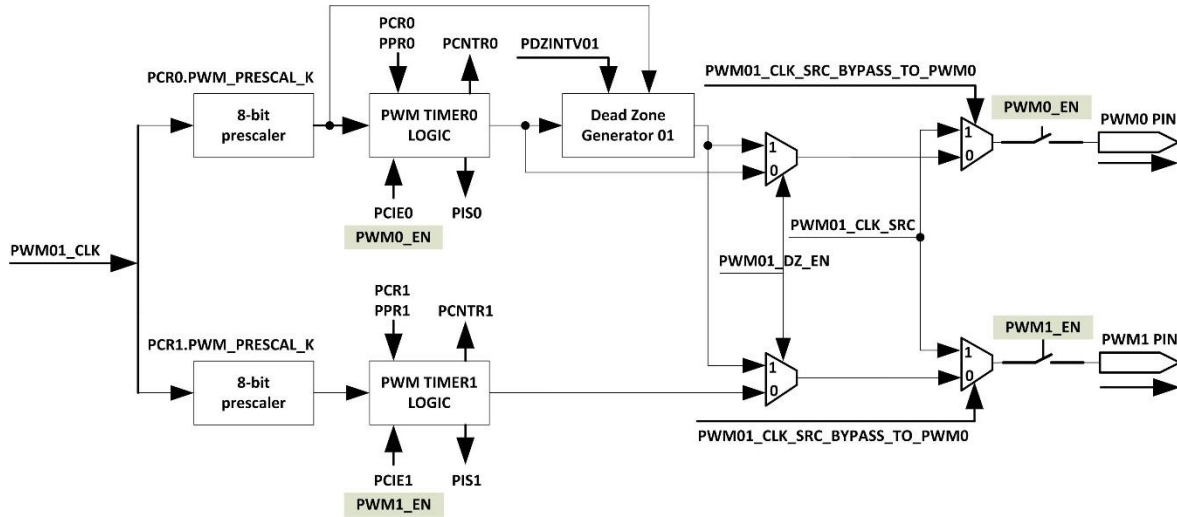


Figure 4-10 PWM01 Output Logic Module Diagram

4.5.3.3 Up-Counter and Comparator

The period, duty-cycle and active state of PWM output waveform are decided by the up-counter and comparator. The rule of the comparator is as follows.

$PCNTR \geq (PWM_ENTIRE_CYCLE - PWM_ACT_CYCLE)$, output "active state"

$PCNTR < (PWM_ENTIRE_CYCLE - PWM_ACT_CYCLE)$, output "~ (active state)"

- **Active state of PWM0 channel is high level (PCR0. PWM_ACT_STA = 1)**

When $PCNTR0 \geq (PPR0. PWM_ENTIRE_CYCLE - PPR0. PWM_ACT_CYCLE)$, then PWM0 outputs 1.

When $PCNTR0 < (PPR0. PWM_ENTIRE_CYCLE - PPR0. PWM_ACT_CYCLE)$, then PWM0 outputs 0.

The formula of PWM output period and duty-cycle is as follows.

$$T_{\text{period}} = (PWM01_CLK / PWM0_PRESCALE_K)^{-1} * PPR0. PWM_ENTIRE_CYCLE$$

$$T_{\text{high-level}} = (PWM01_CLK / PWM0_PRESCALE_K)^{-1} * PPR0. PWM_ACT_CYCLE$$

$$T_{\text{low-level}} = (PWM01_CLK / PWM0_PRESCALE_K)^{-1} * (PPR0. PWM_ENTIRE_CYCLE - PPR0. PWM_ACT_CYCLE)$$

$$\text{Duty-cycle} = (\text{high level time}) / (1 \text{ period time})$$

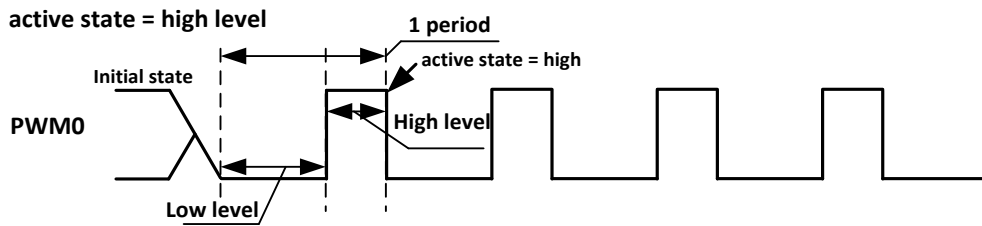


Figure 4-11 PWM0 High Level Active State

- **Active state of PWM0 channel is low level (PCR0. PWM_ACT_STA = 0)**

When PCNTRO >= (PPR0. PWM_ENTIRE_CYCLE - PPR0. PWM_ACT_CYCLE), then PWM0 outputs 0.

When PCNTRO < (PPR0. PWM_ENTIRE_CYCLE - PPR0. PWM_ACT_CYCLE), then PWM0 outputs 1.

The formula of PWM output period and duty-cycle is as follows.

$$T_{\text{period}} = (\text{PWM01_CLK} / \text{PWM0_PRESCALE_K})^{-1} * \text{PPR0. PWM_ENTIRE_CYCLE}$$

$$T_{\text{high-level}} = (\text{PWM01_CLK} / \text{PWM0_PRESCALE_K})^{-1} * (\text{PPR0. PWM_ENTIRE_CYCLE} - \text{PPR0. PWM_ACT_CYCLE})$$

$$T_{\text{low-level}} = (\text{PWM01_CLK} / \text{PWM0_PRESCALE_K})^{-1} * \text{PPR0. PWM_ACT_CYCLE}$$

$$\text{Duty-cycle} = (\text{high level time}) / (\text{1 period time})$$

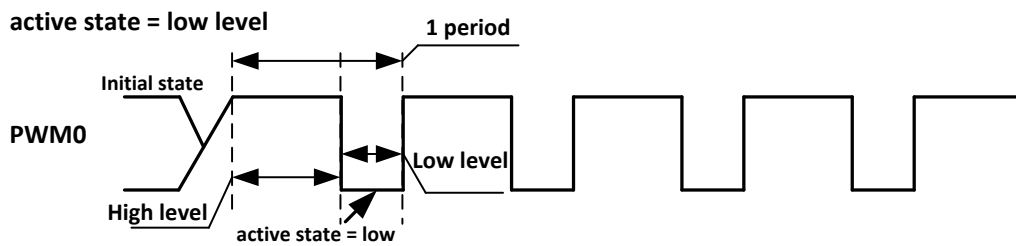


Figure 4-12 PWM0 Low Level Active State

4.5.3.4 Pulse Mode and Cycle Mode

PWM output supports pulse mode and cycle mode. PWM in pulse mode outputs one pulse waveform, but PWM in cycle mode outputs continuous waveform. Figure 4-13. shows the PWM output waveform in pulse mode and cycle mode.

and comparator, namely the value of up-counter and comparator will be overloaded in the next cycle.

Take Figure 5.(active state =low level) as an example.

In cycle mode, the initial PPR0.PWM_ENTIRE_CYCLE value is 10, the initial PPR0. PWM _ACT_CYCLE value is 3. At some time, the value of PPR0. PWM _ACT_CYCLE value is modified to 1, during the current cycle, the modified PPR0 values is cached to PPR0 buffer register, at the beginning of the next cycle, the value of PPR0 buffer register is loaded into up-counter and comparator, then up-counter starts to work.

In pulse mode, the initial value of PPR0. PWM _ACT_CYCLE is 3, in the generation process of a single pulse, the value of PPR0. PWM _ACT_CYCLE is modified to 1, during the current cycle, the modified PPR0 values is cached to PPR0 buffer register, when the value of up-counter reaches PPR0. PWM_ENTIRE_CYCLE, then the pulse waveform output ends, the value of PPR0 buffer register is loaded into up-counter and comparator, at the next time, after PCR0. PWM_PUL_START is set to 1, PPR0 modified value has taken effect.

4.5.3.5 Interrupt

PWM supports interrupt generation when PWM channel is configured to PWM output or capture input .

For PWM output function, whether pulse mode or cycle mode, if the value of the up-counter reaches **PWM_ENTIRE_CYCLE**, the timer logic module will automatically set the PIS(PWM Interrupt Status) bit to 1 by hardware. But the PIS bit is cleared by software.

For capture input function, when the timer logic module of the capture channel0 captures rising edge, and **CRIE0** is 1, then **CRIS0** is set to 1; when the timer logic module of the capture channel0 captures falling edge, and **CFIE0** is 1, then **CFIS0** is set to 1.

4.5.4 Register List

Module Name	Base Address	
PWM	0x40042800	
Register Name	Offset Address	Description
PIER	0x00	PWM IRQ Enable Register
PISR	0x04	PWM IRQ Status Register
CIER	0x10	Capture IRQ Enable Register
CISR	0x14	Capture IRQ Status Register
PCCR01	0x20	PWM01 Clock Configuration Register
PCCR23	0x24	PWM23 Clock Configuration Register
PCCR45	0x28	PWM45 Clock Configuration Register
PCCR67	0x2C	PWM67 Clock Configuration Register
PDZCR01	0x30	PWM01 Dead Zone Control Register
PDZCR23	0x34	PWM23 Dead Zone Control Register
PDZCR45	0x38	PWM45 Dead Zone Control Register

PDZCR67	0x3C	PWM67 Dead Zone Control Register
PER	0x40	PWM Enable Register
CER	0x44	Capture Enable Register
PCR	0x60+0x0+N*0x20(N= 0~7)	PWM Control Register
PPR	0x60+0x4+N*0x20(N= 0~7)	PWM Period Register
PCNTR	0x60+0x8+N*0x20(N= 0~7)	PWM Count Register
CCR	0x60+0xC+N*0x20(N= 0~7)	Capture Control Register
CRLR	0x60+0x10+N*0x20(N= 0~7)	Capture Rise Lock Register
CFLR	0x60+0x14+N*0x20(N= 0~7)	Capture Fall Lock Register

4.5.5 Register Description

4.5.5.1 PWM IRQ Enable Register

Address : 0x00			Name: PIER Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7	R/W	0x0	PCIE7 PWM channel 7 Interrupt Enable 0: PWM channel 7 Interrupt Disable 1: PWM channel 7 Interrupt Enable
6	R/W	0x0	PCIE6 PWM channel 6 Interrupt Enable 0: PWM channel 6 Interrupt Disable 1: PWM channel 6 Interrupt Enable
5	R/W	0x0	PCIE5 PWM channel 5 Interrupt Enable 0: PWM channel 5 Interrupt Disable 1: PWM channel 5 Interrupt Enable
4	R/W	0x0	PCIE4 PWM channel 4 Interrupt Enable 0: PWM channel 4 Interrupt Disable 1: PWM channel 4 Interrupt Enable
3	R/W	0x0	PCIE3 PWM channel 3 Interrupt Enable 0: PWM channel 3 Interrupt Disable 1: PWM channel 3 Interrupt Enable
2	R/W	0x0	PCIE2 PWM channel 2 Interrupt Enable 0: PWM channel 2 Interrupt Disable

			1: PWM channel 2 Interrupt Enable
1	R/W	0x0	PCIE1 PWM channel 1 Interrupt Enable 0: PWM channel 1 Interrupt Disable 1: PWM channel 1 Interrupt Enable
0	R/W	0x0	PCIE0 PWM channel 0 Interrupt Enable 0: PWM channel 0 Interrupt Disable 1: PWM channel 0 Interrupt Enable

4.5.5.2 PWM IRQ Status Register

Address : 0x04			Name: PISR Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7	R/W1 C	0x0	PIS7 PWM channel 7 Interrupt Status. When PWM channel 7 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 7 interrupt is not pending 1: PWM channel 7 interrupt is pending Writes 0: no effect 1: Clear PWM channel 7 interrupt status
6	R/W1 C	0x0	PIS6 PWM channel 6 Interrupt Status. When PWM channel 6 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 6 interrupt is not pending 1: PWM channel 6 interrupt is pending Writes 0: no effect 1: Clear PWM channel 6 interrupt status
5	R/W1 C	0x0	PIS5 PWM channel 5 Interrupt Status. When PWM channel 5 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 5 interrupt is not pending 1: PWM channel 5 interrupt is pending Writes 0: no effect 1: Clear PWM channel 5 interrupt status
4	R/W1 C	0x0	PIS4. PWM channel 4 Interrupt Status. When PWM channel 4 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 4 interrupt is not pending. 1: PWM channel 4 interrupt is pending. Writes 0: no effect.

			1: Clear PWM channel 4 interrupt status.
3	R/W1 C	0x0	PIS3 PWM channel 3 Interrupt Status. When PWM channel 3 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 3 interrupt is not pending 1: PWM channel 3 interrupt is pending Writes 0: no effect 1: Clear PWM channel 3 interrupt status
2	R/W1 C	0x0	PIS2 PWM channel 2 Interrupt Status. When PWM channel 2 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 2 interrupt is not pending 1: PWM channel 2 interrupt is pending Writes 0: no effect 1: Clear PWM channel 2 interrupt status
1	R/W1 C	0x0	PIS1 PWM channel 1 Interrupt Status. When PWM channel 1 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 1 interrupt is not pending 1: PWM channel 1 interrupt is pending Writes 0: no effect 1: Clear PWM channel 1 interrupt status
0	R/W1 C	0x0	PIS0 PWM channel 0 Interrupt Status. When PWM channel 0 counter reaches Entire Cycle Value, this bit is set 1 by hardware. Write 1 to clear this bit. Reads 0: PWM channel 0 interrupt is not pending 1: PWM channel 0 interrupt is pending Writes 0: no effect 1: Clear PWM channel 0 interrupt status

4.5.5.3 Capture IRQ Enable Register

Address :0x10			Name: CIER Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:16	/	/	/
15	R/W	0x0	CFIE7 If this enable bit is set 1, when capture channel 7 captures falling edge, it generates a capture channel 7 pending. 0: Capture channel 7 fall lock Interrupt disable 1: Capture channel 7 fall lock Interrupt enable
14	R/W	0x0	CRIE7 If this enable bit is set 1, when capture channel 7 captures rising edge, it

			generates a capture channel 7 pending. 0: Capture channel 7 rise lock Interrupt disable 1: Capture channel 7 rise lock Interrupt enable
13	R/W	0x0	CFIE6 If this enable bit is set 1, when capture channel 6 captures falling edge, it generates a capture channel 6 pending. 0: Capture channel 6 fall lock Interrupt disable 1: Capture channel 6 fall lock Interrupt enable
12	R/W	0x0	CRIE6 If this enable bit is set 1, when capture channel 6 captures rising edge, it generates a capture channel 6 pending. 0: Capture channel 6 rise lock Interrupt disable 1: Capture channel 6 rise lock Interrupt enable
11	R/W	0x0	CFIE5 If this enable bit is set 1, when capture channel 5 captures falling edge, it generates a capture channel 5 pending. 0: Capture channel 5 fall lock Interrupt disable 1: Capture channel 5 fall lock Interrupt enable
10	R/W	0x0	CRIE5 If this enable bit is set 1, when capture channel 5 captures rising edge, it generates a capture channel 5 pending. 0: Capture channel 5 rise lock Interrupt disable 1: Capture channel 5 rise lock Interrupt enable
9	R/W	0x0	CFIE4 If this enable bit is set 1, when capture channel 4 captures falling edge, it generates a capture channel 4 pending. 0: Capture channel 4 fall lock Interrupt disable 1: Capture channel 4 fall lock Interrupt enable
8	R/W	0x0	CRIE4 If this enable bit is set 1, when capture channel 4 captures rising edge, it generates a capture channel 4 pending. 0: Capture channel 4 rise lock Interrupt disable 1: Capture channel 4 rise lock Interrupt enable
7	R/W	0x0	CFIE3 If this enable bit is set 1, when capture channel 3 captures falling edge, it generates a capture channel 3 pending. 0: Capture channel 3 fall lock Interrupt disable 1: Capture channel 3 fall lock Interrupt enable
6	R/W	0x0	CRIE3 If this enable bit is set 1, when capture channel 3 captures rising edge, it generates a capture channel 3 pending. 0: Capture channel 3 rise lock Interrupt disable

			1: Capture channel 3 rise lock Interrupt enable
5	R/W	0x0	CFIE2 If this enable bit is set 1, when capture channel 2 captures falling edge, it generates a capture channel 2 pending. 0: Capture channel 2 fall lock Interrupt disable 1: Capture channel 2 fall lock Interrupt enable
4	R/W	0x0	CRIE2 If this enable bit is set 1, when capture channel 2 captures rising edge, it generates a capture channel 2 pending. 0: Capture channel 2 rise lock Interrupt disable 1: Capture channel 2 rise lock Interrupt enable
3	R/W	0x0	CFIE1 If this enable bit is set 1, when capture channel 1 captures falling edge, it generates a capture channel 1 pending. 0: Capture channel 1 fall lock Interrupt disable 1: Capture channel 1 fall lock Interrupt enable
2	R/W	0x0	CRIE1 If this enable bit is set 1, when capture channel 1 captures rising edge, it generates a capture channel 1 pending. 0: Capture channel 1 rise lock Interrupt disable 1: Capture channel 1 rise lock Interrupt enable
1	R/W	0x0	CFIE0 If this enable bit is set 1, when capture channel 0 captures falling edge, it generates a capture channel 0 pending. 0: Capture channel 0 fall lock Interrupt disable 1: Capture channel 0 fall lock Interrupt enable
0	R/W	0x0	CRIE0 If this enable bit is set 1, when capture channel 0 captures rising edge, it generates a capture channel 0 pending. 0: Capture channel 0 rise lock Interrupt disable 1: Capture channel 0 rise lock Interrupt enable

4.5.5.4 Capture IRQ Status Register

Address: 0x14			Name: CISR Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:8	/	/	/
15	R/W1C	0x0	CFIS7 Capture channel 7 falling lock interrupt status When capture channel 7 captures falling edge, if capture channel 7 fall lock interrupt (CFIE7) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit

			Reads 0: Capture channel 7 interrupt is not pending 1: Capture channel 7 interrupt is pending Writes 0: no effect 1: Clear capture channel 7 interrupt status
14	R/W1C	0x0	CRIS7 Capture channel 7 rising lock interrupt status When capture channel 7 captures rising edge, if capture channel 7 rise lock interrupt (CRIE7) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 7 interrupt is not pending 1: Capture channel 7 interrupt is pending Writes 0: no effect 1: Clear capture channel 7 interrupt status
13	R/W1C	0x0	CFIS6 Capture channel 6 falling lock interrupt status When capture channel 6 captures falling edge, if capture channel 6 fall lock interrupt (CFIE6) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 6 interrupt is not pending 1: Capture channel 6 interrupt is pending Writes 0: no effect 1: Clear capture channel 6 interrupt status
12	R/W1C	0x0	CRIS6 Capture channel 6 rising lock interrupt status When capture channel 6 captures rising edge, if capture channel 6 rise lock interrupt (CRIE6) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 6 interrupt is not pending 1: Capture channel 6 interrupt is pending Writes 0: no effect 1: Clear capture channel 6 interrupt status
11	R/W1C	0x0	CFIS5 Capture channel 5 falling lock interrupt status When capture channel 5 captures falling edge, if capture channel 5 fall lock interrupt (CFIE5) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 5 interrupt is not pending 1: Capture channel 5 interrupt is pending Writes 0: no effect 1: Clear capture channel 5 interrupt status
10	R/W1C	0x0	CRIS5 Capture channel 5 rising lock interrupt status When capture channel 5 captures rising edge, if capture channel 5 rise

			lock interrupt (CRIE5) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 5 interrupt is not pending 1: Capture channel 5 interrupt is pending Writes 0: no effect 1: Clear capture channel 5 interrupt status
9	R/W1C	0x0	CFIS4 Capture channel 4 falling lock interrupt status When capture channel 4 captures falling edge, if capture channel 4 fall lock interrupt (CFIE4) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 4 interrupt is not pending 1: Capture channel 4 interrupt is pending Writes 0: no effect 1: Clear capture channel 4 interrupt status
8	R/W1C	0x0	CRIS4 Capture channel 4 rising lock interrupt status When capture channel 4 captures rising edge, if capture channel 4 rise lock interrupt (CRIE4) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 4 interrupt is not pending 1: Capture channel 4 interrupt is pending Writes 0: no effect 1: Clear capture channel 4 interrupt status
7	R/W1C	0x0	CFIS3 Capture channel 3 falling lock interrupt status When capture channel 3 captures falling edge, if capture channel 3 fall lock interrupt (CFIE3) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 3 interrupt is not pending 1: Capture channel 3 interrupt is pending Writes 0: no effect 1: Clear capture channel 3 interrupt status
6	R/W1C	0x0	CRIS3 Capture channel 3 rising lock interrupt status When capture channel 3 captures rising edge, if capture channel 3 rise lock interrupt (CRIE3) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 3 interrupt is not pending 1: Capture channel 3 interrupt is pending Writes 0: no effect 1: Clear capture channel 3 interrupt status
5	R/W1C	0x0	CFIS2

			Capture channel 2 falling lock interrupt status When capture channel 2 captures falling edge, if capture channel 2 fall lock interrupt (CFIE2) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 2 interrupt is not pending 1: Capture channel 2 interrupt is pending Writes 0: no effect 1: Clear capture channel 2 interrupt status
4	R/W1C	0x0	CRIS2 Capture channel 2 rising lock interrupt status When capture channel 2 captures rising edge, if capture channel 2 rise lock interrupt (CRIE2) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 2 interrupt is not pending 1: Capture channel 2 interrupt is pending Writes 0: no effect 1: Clear capture channel 2 interrupt status
3	R/W1C	0x0	CFIS1 Capture channel 1 falling lock interrupt status When capture channel 1 captures falling edge, if capture channel 1 fall lock interrupt (CFIE1) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 1 interrupt is not pending 1: Capture channel 1 interrupt is pending Writes 0: no effect 1: Clear capture channel 1 interrupt status
2	R/W1C	0x0	CRIS1 Capture channel 1 rising lock interrupt status When capture channel 1 captures rising edge, if capture channel 1 rise lock interrupt (CRIE1) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 1 interrupt is not pending 1: Capture channel 1 interrupt is pending Writes 0: no effect 1: Clear capture channel 1 interrupt status
1	R/W1C	0x0	CFIS0 Capture channel 0 falling lock interrupt status When capture channel 0 captures falling edge, if capture channel 0 fall lock interrupt (CFIE0) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 0 interrupt is not pending 1: Capture channel 0 interrupt is pending Writes 0: no effect

			1: Clear capture channel 0 interrupt status
0	R/W1C	0x0	CRIS0 Capture channel 0 rising lock interrupt status When capture channel 0 captures rising edge, if capture channel 0 rise lock interrupt (CRIE0) is enabled, this bit is set 1 by hardware. Write 1 to clear this bit Reads 0: Capture channel 0 interrupt is not pending 1: Capture channel 0 interrupt is pending Writes 0: no effect 1: Clear capture channel 0 interrupt status

4.5.5.5 PWM01 Clock Configuration Register

Address: 0x20			Name: PCCR01 Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:9	/	/	/
8:7	R/W	0x0	PWM01_CLK_SRC Select PWM01 clock source 00: 24M_PLL 01: APB1 Others: /
6	R/W	0x0	PWM01_CLK_SRC_BYPASS_TO_PWM1 Bypass PWM01 clock source to PWM1 output 0: not bypass 1: bypass
5	R/W	0x0	PWM01_CLK_SRC_BYPASS_TO_PWM0 Bypass PWM01 clock source to PWM0 output 0: not bypass 1: bypass
4	R/W	0x0	PWM01_CLK_GATING Gating clock for PWM01 0: Mask 1: Pass
0:3	R/W	0x0	PWM01_CLK_DIV_M PWM01 clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16

			0101: /32 0110: /64 0111: /128 1000: /256 others: /
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4.5.5.6 PWM23 Clock Configuration Register

Address: 0x24			Name: PCCR23 Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:9	/	/	/
8:7	R/W	0x0	PWM23_CLK_SRC_SEL Select PWM23 clock source 00: 24M_PLL 01: APB1 Others: /
6	R/W	0x0	PWM23_CLK_SRC_BYPASS_TO_PWM3 Bypass PWM23 clock source to PWM3 output 0: not bypass 1: bypass
5	R/W	0x0	PWM23_CLK_SRC_BYPASS_TO_PWM2 Bypass PWM23 clock source to PWM2 output 0: not bypass 1: bypass
4	R/W	0x0	PWM23_CLK_GATING Gating clock for PWM23 0: Mask 1: Pass
0:3	R/W	0x0	PWM23_CLK_DIV_M PWM23 clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

4.5.5.7 PWM45 Clock Configuration Register

Address: 0x28			Name: PCCR45 Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:9	/	/	/
8:7	R/W	0x0	PWM45_CLK_SRC_SEL Select PWM45 clock source 00: 24M_PLL 01: APB1 Others: /
6	R/W	0x0	PWM45_CLK_SRC_BYPASS_TO_PWM5 Bypass PWM45 clock source to PWM5 output 0: not bypass 1: bypass
5	R/W	0x0	PWM45_CLK_SRC_BYPASS_TO_PWM4 Bypass PWM45 clock source to PWM4 output 0: not bypass 1: bypass
4	R/W	0x0	PWM45_CLK_GATING Gating clock for PWM45 0: Mask 1: Pass
0:3	R/W	0x0	PWM45_CLK_DIV_M PWM45 clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

4.5.5.8 PWM67 Clock Configuration Register

Address: 0x2C			Name: PCCR67 Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:9	/	/	/

8:7	R/W	0x0	PWM67_CLK_SRC_SEL Select PWM67 clock source 00: 24M_PLL 01: APB1 Others: /
6	R/W	0x0	PWM67_CLK_SRC_BYPASS_TO_PWM7 Bypass PWM67 clock source to PWM7 output 0: not bypass 1: bypass
5	R/W	0x0	PWM67_CLK_SRC_BYPASS_TO_PWM6 Bypass PWM67 clock source to PWM6 output 0: not bypass 1: bypass
4	R/W	0x0	PWM67_CLK_GATING Gating clock for PWM67 0: Mask 1: Pass
0:3	R/W	0x0	PWM67_CLK_DIV_M PWM67 clock divide M 0000: /1 0001: /2 0010: /4 0011: /8 0100: /16 0101: /32 0110: /64 0111: /128 1000: /256 others: /

4.5.5.9 PWM01 Dead Zone Control Register

Address: 0x30			Name: PDZCR01 Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PDZINTV01 PWM01 Dead Zone interval value
7:1	/	/	/
0	R/W	0x0	PWM01_DZ_EN PWM01 Dead Zone enable 0: Dead Zone disable 1: Dead Zone enable

4.5.5.10 PWM23 Dead Zone Control Register

Address: 0x34			Name: PDZCR23 Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PWM23_DZ_INTV PWM23 Dead Zone interval value
7:1	/	/	/
0	R/W	0x0	PWM23_DZ_EN PWM23 Dead Zone enable 0: Dead Zone disable 1: Dead Zone enable

4.5.5.11 PWM45 Dead Zone Control Register

Address: 0x38			Name: PDZCR45 Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PWM45_DZ_INTV PWM45 Dead Zone interval value
7:1	/	/	/
0	R/W	0x0	PWM45_DZ_EN PWM45 Dead Zone enable 0: Dead Zone disable 1: Dead Zone enable

4.5.5.12 PWM67 Dead Zone Control Register

Address: 0x3C			Name: PDZCR67 Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:16	/	/	/
15:8	R/W	0x0	PWM67_DZ_INTV PWM67 Dead Zone interval value
7:1	/	/	/
0	R/W	0x0	PWM67_DZ_EN PWM67 Dead Zone enable 0: Dead Zone disable 1: Dead Zone enable

4.5.5.13 PWM Enable Register

Address: 0x40			Name: PER Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:1	/	/	/
7	R/W	0x0	PWM7_EN When enable PWM, the 16-bit up-counter starts working and PWM channel7 is permitted to output PWM waveform. 0: PWM disable 1: PWM enable
6	R/W	0x0	PWM6_EN When enable PWM, the 16-bit up-counter starts working and PWM channel6 is permitted to output PWM waveform. 0: PWM disable 1: PWM enable
5	R/W	0x0	PWM5_EN When enable PWM, the 16-bit up-counter starts working and PWM channel5 is permitted to output PWM waveform. 0: PWM disable 1: PWM enable
4	R/W	0x0	PWM4_EN When enable PWM, the 16-bit up-counter starts working and PWM channel4 is permitted to output PWM waveform. 0: PWM disable 1: PWM enable
3	R/W	0x0	PWM3_EN When enable PWM, the 16-bit up-counter starts working and PWM channel3 is permitted to output PWM waveform. 0: PWM disable 1: PWM enable
2	R/W	0x0	PWM2_EN When enable PWM, the 16-bit up-counter starts working and PWM channel2 is permitted to output PWM waveform. 0: PWM disable 1: PWM enable
1	R/W	0x0	PWM1_EN. When enable PWM, the 16-bit up-counter starts working and PWM channel1 is permitted to output PWM waveform. 0: PWM disable 1: PWM enable.
0	R/W	0x0	PWM0_EN

			When enable PWM, the 16-bit up-counter starts working and PWM channel0 is permitted to output PWM waveform. 0: PWM disable 1: PWM enable
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4.5.5.14 Capture Enable Register

Address: 0x44			Name: CER Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:8	/	/	/
7	R/W	0x0	CAP7_EN When enable capture function, the 16-bit up-counter starts working and capture channel7 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable
6	R/W	0x0	CAP6_EN When enable capture function, the 16-bit up-counter starts working and capture channel6 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable
5	R/W	0x0	CAP5_EN When enable capture function, the 16-bit up-counter starts working and capture channel5 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable
4	R/W	0x0	CAP4_EN When enable capture function, the 16-bit up-counter starts working and capture channel4 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable
3	R/W	0x0	CAP3_EN When enable capture function, the 16-bit up-counter starts working and capture channel3 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable
2	R/W	0x0	CAP2_EN When enable capture function, the 16-bit up-counter starts working and

			capture channel2 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable
1	R/W	0x0	CAP1_EN When enable capture function, the 16-bit up-counter starts working and capture channel1 is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable
0	R/W	0x0	CAPO_EN When enable capture function, the 16-bit up-counter starts working and capture channel is permitted to capture external falling edge or rising edge. 0: Capture disable 1: Capture enable

4.5.5.15 PWM Control Register

Address: 0x60+0x0+N*0x20(N= 0~7)			Name: PCR Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:13	/	/	/
11	R	0x0	PWM_PERIOD_RDY PWM period register ready 0: PWM period register is ready to write 1: PWM period register is busy
10	R/W1S	0x0	PWM_PUL_START PWM pulse output start 0: no effect 1: output 1 pulse After finishing configuring for outputting pulse, set this bit once and then PWM would output one pulse. After the pulse is finished, the bit will be cleared automatically.
9	R/W	0x0	PWM_MODE PWM output mode select 0: cycle mode 1: pulse mode
8	R/W	0x0	PWM_ACT_STA PWM active state 0: Low Level 1: High Level
7:0	R/W	0x0	PWM_PRESCAL_K

			PWM pre-scale K, actual pre-scale is (K+1) K = 0, actual pre-scale: 1 K = 1, actual pre-scale: 2 K = 2, actual pre-scale: 3 K = 3, actual pre-scale: 4 K = 255, actual pre-scale: 256
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4.5.5.16 PWM Period Register

Address: 0x60+0x4+N*0x20(N= 0~7)			Name: PPR Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:16	R/W	0x0	PWM_ENTIRE_CYCLE Number of the entire cycles in the PWM clock 0 = 1 cycle 1 = 2 cycles N = N+1 cycles If the register need to be modified dynamically, the PCLK should be faster than the PWM CLK .
15:0	R/W	0x0	PWM_ACT_CYCLE Number of the active cycles in the PWM clock 0 = 0 cycle 1 = 1 cycles N = N cycles

4.5.5.17 PWM Counter Register

Address: 0x60+0x8+N*0x20(N= 0~7)			Name: PCNTR Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	On PWM output or capture input, reading this register could get the current value of the PWM 16bit up-counter.

4.5.5.18 Capture Control Register

Address: 0x60+0xC+N*0x20(N= 0~7)			Name: CCR Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:3	/	/	/

2	R/W	0x0	CRLF When capture channel captures rising edge, the 16-bit up-counter's current value is latched to CRLR and then this bit is set 1 by hardware. Write 1 to clear this bit.
1	R/W	0x0	CFLF When capture channel captures falling edge, the 16-bit up-counter's current value is latched to CFLR and then this bit is set 1 by hardware. Write 1 to clear this bit.
0	R/W	0x0	CAPINV Inversing the signal inputted from capture channel before capture channel's 16bit counter. 0: not inverse 1: inverse

4.5.5.19 Capture Rise Lock Register

Address: 0x60+0x10+N*0x20(N= 0~7)			Name: CRLR Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	When capture channel captures rising edge, the 16-bit up-counter's current value is latched to this register.

4.5.5.20 Capture Fall Lock Register

Address: 0x60+0x14+N*0x20(N= 0~7)			Name: CFLR Default: 0x0000_0000
Bit	R/W	Default/Hex	Description
31:16	/	/	/
15:0	R	0x0	When capture channel captures falling edge, the 16-bit up-counter's current value is latched to this register.

4.6 CIR Receiver Controller (IRRX)

4.6.1 Overview

The CIR includes the following features:

- Full physical layer implementation
- Support CIR for remote control
- 64x8bits FIFO for data buffer
- Programmable FIFO thresholds

For saving CPU resource, CIR receiver is implemented in hardware. The CIR receiver samples the input signal on

the programmable frequency and records these samples into RX FIFO when one CIR signal is found on the air. The CIR receiver uses Run-Length Code (RLC) to encode pulse width. The encoded data is buffered in a 64 levels and 8-bit width RX FIFO; the MSB bit is used to record the polarity of the receiving CIR signal. The high level is represented as '1' and the low level is represented as '0'. The rest 7 bits are used for the length of RLC. The maximum length is 128. If the duration of one level (high or low level) is more than 128, another byte is used.

In the air, there is always some noise. One threshold can be set to filter the noise to reduce system loading and improve the system stability.

4.6.2 Register List

Module Name	Base Address	
CIR Receiver	0x40043800	
Register Name	Offset Address	Description
CIR_ACTL	0x00	CIR Active Control Register
CIR_CTL	0x04	CIR Control Register
CIR_RXCTL	0x10	CIR Receiver Configure Register
CIR_RXFIFO	0x20	CIR Receiver FIFO Register
CIR_RXINT	0x2C	CIR Receiver Interrupt Control Register
CIR_RXSTA	0x30	CIR Receiver Status Register
CIR_CONFIG	0x34	CIR Configure Register

4.6.3 Register Description

4.6.3.1 CIR Receiver Control Register

Address : 0x0000			Name: CIR_CTL Default: 0x0000_0000
Field	Type	Default	Description
31:9	/	/	/
8	R/W	0x0	CGPO General Program Output (GPO) Control in CIR mode for TX Pin 0: Low level 1: High level
7:6	R/W	0x0	Active Pulse Accept Mode 00: Start to accept data after detect de-bounce pulse edge 01: Start to accept data after detect de-bounce pulse edge 10: Start to accept data after detect low voltage de-bounce pulse 11: Start to accept data after detect high voltage de-bounce pulse
5:4	R/W	0x0	CIR ENABLE 00~10: Reserved

			11: CIR mode enable
3:2	/	/	/
1	R/W	0x0	RXEN Receiver Block Enable 0: Disable 1: Enable
0	R/W	0x0	GEN Global Enable A disable on this bit overrides any other block or channel enables and flushes all FIFOs. 0: Disable 1: Enable

4.6.3.2 CIR Receiver Configure Register

Address : 0x0010			Name: IR_RXCTL Default: 0x0000_0004
Field	Type	Default	Description
31:3	/	/	/
2	R/W	0x1	RPPI Receiver Pulse Polarity Invert 0: Not invert receiver signal 1: Invert receiver signal
1:0	/	/	/

4.6.3.3 CIR Receiver FIFO Register

Address : 0x0020			Name: IR_RXFIFO Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R	0x0	Receiver Byte FIFO

4.6.3.4 CIR Receiver Interrupt Control Register

Address : 0x002C			Name: IR_RXINT Default: 0x0000_0000
Field	Type	Default	Description
31:16	/	/	/
15:8	R/W	0x0	RAL RX FIFO Available Received Byte Level for interrupt and DMA request. TRIGGER_LEVEL = RAL + 1
5	R/W	0x0	DRQ_EN

			RX FIFO DMA Enable 0: Disable 1: Enable When set to '1', the Receiver FIFO DRQ is asserted if reaching RAL. The DRQ is de-asserted when condition fails.
4	R/W	0x0	RAI_EN RX FIFO Available Interrupt Enable 0: Disable 1: Enable When set to '1', the Receiver FIFO IRQ is asserted if reaching RAL. The IRQ is de-asserted when condition fails.
3:2	/	/	/
1	R/W	0x0	RPEI_EN Receiver Packet End Interrupt Enable 0: Disable 1: Enable
0	R/W	0x0	ROI_EN Receiver FIFO Overrun Interrupt Enable 0: Disable 1: Enable

4.6.3.5 CIR Receiver Status Register

Address : 0x0030			Name: IR_RXSTA Default: 0x0000_0000
Field	Type	Default	Description
31:15	/	/	/
14:8	R	0x0	RAC RX FIFO Available Counter 0: No available data in RX FIFO 1: 1 byte available data in RX FIFO 2: 2 byte available data in RX FIFO ... 64: 64 byte available data in RX FIFO
7	R	0x0	STAT Status of CIR 0x0 – Idle 0x1 – busy
6:5	/	/	/
4	R/W	0x0	RA RX FIFO Available 0: RX FIFO not available according its level

			1: RX FIFO available according its level This bit is cleared by writing a '1'.
3:2	/	/	/
1	R/W	0x0	RPE Receiver Packet End Flag 0: STO was not detected. In CIR mode, one CIR symbol is receiving or not detected. 1: STO field or packet abort symbol (7'b0000,000 and 8'b0000,0000 for MIR and FIR) is detected. In CIR mode, one CIR symbol is received. This bit is cleared by writing a '1'.
0	R/W	0x0	ROI Receiver FIFO Overrun 0: Receiver FIFO not overrun 1: Receiver FIFO overrun This bit is cleared by writing a '1'.

4.6.3.6 CIR Receiver Controller Configure Register

Address : 0x0034			Name: IR_CIR Default: 0x0000_1828
Field	Type	Default	Description
31:25	/	/	/
24	R/W	0x0	SCS2 Bit2 of Sample Clock Select for CIR This bit is defined by SCS bits below.
23	R/W	0x0	ATHC Active Threshold Control for CIR 0x0 –ATHR in Unit of (Sample Clock) 0x1 –ATHR in Unit of (128*Sample Clocks)
22:16	R/W	0x0	ATHR Active Threshold for CIR These bits control the duration of CIR from Idle to Active State. The duration can be calculated by ((ATHR + 1)*(ATHC? Sample Clock: 128*Sample Clock)).
15:8	R/W	0x18	ITHR Idle Threshold for CIR The Receiver uses it to decide whether the CIR command has been received. If there is no CIR signal on the air, the receiver is staying in IDLE status. One active pulse will bring the receiver from IDLE status to Receiving status. After the CIR is end, the inputting signal will keep the specified level (high or low level) for a long time. The receiver can use this idle signal duration to decide that it has received the CIR command. The corresponding flag is asserted. If the corresponding interrupt is enable, the interrupt line is asserted to CPU. When the duration of signal keeps one status (high or low level) for the

			specified duration ((ITHR + 1)*128 sample_clk), this means that the previous CIR command has been finished.																																				
7:2	R/W	0xA	NTHR Noise Threshold for CIR When the duration of signal pulse (high or low level) is less than NTHR, the pulse is taken as noise and should be discarded by hardware. 0: all samples are recorded into RX FIFO. 1: If the signal is only one sample duration, it is taken as noise and discarded. 2: If the signal is less than ($\leq$) two sample duration, it is taken as noise and discarded. ... 61: if the signal is less than ($\leq$) sixty-one sample duration, it is taken as noise and discarded.																																				
1:0	R/W	0x0	SCS Sample Clock Select for CIR <table border="1"> <thead> <tr> <th>SCS2</th><th>SCS[1]</th><th>SCS[0]</th><th>Sample Clock</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>0</td><td>ir_clk/64</td></tr> <tr> <td>0</td><td>0</td><td>1</td><td>ir_clk/128</td></tr> <tr> <td>0</td><td>1</td><td>0</td><td>ir_clk/256</td></tr> <tr> <td>0</td><td>1</td><td>1</td><td>ir_clk/512</td></tr> <tr> <td>1</td><td>0</td><td>0</td><td>ir_clk</td></tr> <tr> <td>1</td><td>0</td><td>1</td><td>Reserved</td></tr> <tr> <td>1</td><td>1</td><td>0</td><td>Reserved</td></tr> <tr> <td>1</td><td>1</td><td>1</td><td>Reserved</td></tr> </tbody> </table>	SCS2	SCS[1]	SCS[0]	Sample Clock	0	0	0	ir_clk/64	0	0	1	ir_clk/128	0	1	0	ir_clk/256	0	1	1	ir_clk/512	1	0	0	ir_clk	1	0	1	Reserved	1	1	0	Reserved	1	1	1	Reserved
SCS2	SCS[1]	SCS[0]	Sample Clock																																				
0	0	0	ir_clk/64																																				
0	0	1	ir_clk/128																																				
0	1	0	ir_clk/256																																				
0	1	1	ir_clk/512																																				
1	0	0	ir_clk																																				
1	0	1	Reserved																																				
1	1	0	Reserved																																				
1	1	1	Reserved																																				

Register Name	Offset Address	Description
CIR_TGLR	0x00	CIR Transmit Global Register
CIR_TMCR	0x04	CIR Transmit Modulation Control Register
CIR_TCR	0x08	CIR Transmit Control Register
CIR_IDC_H	0x0C	CIR Transmit Idle Duration threshold Register
CIR_IDC_L	0x10	CIR Transmit Idle Duration threshold Register
CIR_TICR_H	0x14	CIR Transmit Idle Counter Register
CIR_TICR_L	0x18	CIR Transmit Idle Counter Register
CIR_TEL	0x20	CIR TX FIFO empty Level Register
CIR_TXINT	0x24	CIR Transmit Interrupt Control Register
CIR_TAC	0x28	CIR Transmit FIFO Available Counter Register
CIR_TXSTA	0x2C	CIR Transmit Status Register
CIR_TXT	0x30	CIR Transmit Threshold Register
CIR_DMA	0x34	CIR DMA Control Register
CIR_TXFIFO	0x80	CIR Transmit FIFO Data Register

4.7.3 Register Description

4.7.3.1 CIR Transmit Global Register

Address : 0x0000			Name: CIR_TGLR Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7	R/W	0x0	IMS Internal Modulation Select 0: the transmitting signal is not modulated 1: the transmitting signal is modulated internally
6:5	R/W	0x0	DRMC duty ratio of modulated carrier is high level /low level. 0: low level is the one time of high level 1: low level is the two times of high level 2: low level is the three times of high level 3: reserved
4:3	/	/	/
2	R/W	0x0	TPPI Transmit Pulse Polarity Invert 0: Not invert transmit pulse 1: Invert transmit pulse
1	R/W	0x0	TR Transmit Reset

			When this bit is set, the transmitting is reset. The FIFO will be flush, the TIC filed and CSS field will be clean during Transmit Reset. This field will automatically clean when the Transmit Reset is finished, and the CIR transmitter will state Idle.
0	R/W	0x0	TXEN Transmit Block Enable 0: Disable the CIR Transmitter 1: Enable the CIR Transmitter

4.7.3.2 CIR Transmit Modulation Control Register

Address : 0x0004			Name: CIR_TMCR Default: 0x0000_009E
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x9E	RFMC Reference Frequency of modulated carrier Reference Frequency of modulated carrier based on a division of a fixed functional clock (FCLK). The range of the modulated carrier is usually 30KHz to 60KHz. The most consumer electronics is 38KHz. The default modulated carrier is 38KHz when FCLK is 6MHz. RFMC= FCLK/N.

4.7.3.3 CIR Transmit Control Register

Address : 0x0008			Name: CIR_TCR Default: 0x0000_0002
Field	Type	Default	Description
31:8	/	/	/
7	R/W	0x0	CSS Cyclical Pulse Start/Stop Control Start to transmit when set to '1'. 0: Stop when cleared to '0'. From start to stop, all data in FIFO must be transmitted. 1: Start
6:4	/	/	/
3:1	R/W	0x1	RCS Reference Clock Select for CIR Transmit(>=1) 001: CIR Transmit reference clock is ir_clk/2 010: CIR Transmit reference clock is ir_clk/4 011: CIR Transmit reference clock is ir_clk/8 100: CIR Transmit reference clock is ir_clk/64 101: CIR Transmit reference clock is ir_clk/128 110: CIR Transmit reference clock is ir_clk/256

			111: CIR Transmit reference clock is ir_clk/512
0	R/W	0x0	TTS Type of the transmission signal 0: The transmitting wave is single non-cyclical pulse 1: The transmitting wave is cyclical short-pulse

4.7.3.4 CIR Transmit Idle Duration Counter Register

Address : 0x000C			Name: CIR_IDC_H Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
3:0	R/W	0x0	IDC_H Idle Duration Counter threshold(High 4 bit) Idle Duration = 128*IDC*Ts (IDC = 0~4095)

4.7.3.5 CIR Transmit Idle Duration Counter Register

Address : 0x0010			Name: CIR_IDC_L Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	IDC_L Idle Duration Counter threshold(Low 8 bit) Idle Duration = 128*IDC*Ts (IDC = 0~4095)

4.7.3.6 CIR Transmit Idle Counter Register

Address : 0x0014			Name: CIR_TICR_H Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	TIC_H Transmit Idle Counter_H(High 8 bit) Count in 128*Ts (Sample Duration, 1/Fs) when transmitter is idle, and it should be reset when the transmitter active. When this counter reaches the maximum value (0xFFFF), it will stop automatically, and should not be cleared to zero.

4.7.3.7 CIR Transmit Idle Counter Register

Address : 0x0018			Name: CIR_TICR_L Default: 0x0000_0000
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Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	TIC_L Transmit Idle Counter_L(Low 8 bit) Count in 128*Ts (Sample Duration, 1/Fs) when transmitter is idle, and it should be reset when the transmitter active. When this counter reach the maximum value (0xFFFF), it will stop automatically, and should not be cleared to zero.

4.7.3.8 CIR Transmit FIFO Empty Register

Address : 0x0020			Name: CIR_TEL Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	TEL TX FIFO empty Level for DRQ and IRQ TRIGGER_LEVEL = TEL + 1

4.7.3.9 CIR Transmit Interrupt Control Register

Address : 0x0024			Name: CIR_TXINT Default: 0x0000_0000
Field	Type	Default	Description
31:3	/	/	/
2	R/W	0x0	DRQ_EN TX FIFO DMA Enable 0: Disable 1: Enable When set to '1', the Tx FIFO DRQ is asserted if the number of the transmitting data in the FIFO is less then the RAL. The DRQ is de-asserted when condition fails.
1	R/W	0x0	TAI_EN TX FIFO Available Interrupt Enable 0:Disable 1:Enable
0	R/W	0x0	TPEI_EN Transmit Packet End Interrupt Enable for Cyclical Pulse 0:Disable 1:Enable TUI_EN Transmitter FIFO under run Interrupt Enable for Non-cyclical Pulse

			0: Disable 1: Enable
--	--	--	-------------------------

4.7.3.10 CIR Transmit FIFO Available Counter Register

Address : 0x0028			Name: CIR_TAC Default: 0x0000_0000
Field	Type	Default	Description
31:6	/	/	/
7:0	R	0x80	TAC TX FIFO Available Space Counter 0x0: No available space in TX FIFO 0x1: 1 byte available space in TX FIFO 0x2: 2 byte available space in TX FIFO ... 0x80: 128 byte available space in TX FIFO

4.7.3.11 CIR Transmit Status Register

Address : 0x002C			Name: CIR_TXSTA Default: 0x0000_0000
Field	Type	Default	Description
31:4	/	/	/
3	R	0x0	STCT Status of CIR Transmitter 0x0:Idle 0x1:Active This bit will automatically set when the controller begins transmit the data in the FIFO. The "1" will last when the data in the FIFO. It will automatically be cleaned to "0" when all data in the FIFO is transmitted. The bit is for debug.
2	R	0x0	DRQ DMA Request Flag When set to '1', the Tx FIFO DRQ is asserted if the number of the transmitting data in the FIFO is less then the RAL. The DRQ is de-asserted when condition fails. This bit is for debug.
1	R/W	0x0	TAI TX FIFO Available Interrupt Flag 0: TX FIFO not available by its level 1: TX FIFO available by its level This bit is cleared by writing a '1'.

0	R/W	0x0	TPE Transmitter Packet End Flag for Cyclical Pulse 0: Transmissions of address, control and data fields not completed 1: Transmissions of address, control and data fields completed TUR Transmitter FIFO Under Run Flag for Non-cyclical Pulse 0: No transmitter FIFO under run 1: Transmitter FIFO under run This bit is cleared by writing a '1'.
---	-----	-----	---

4.7.3.12 CIR Transmit Threshold Register

Address : 0x0030			Name: CIR_TXT Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7:0	R/W	0x0	NCTT Non-cyclical Pulse Transmit Threshold The controller will trigger transmitting the data in the FIFO when the data byte number has reach the Transmit Threshold set in this field.

4.7.3.13 CIR DMA Control Register

Address : 0x0034			Name: CIR_DMA_CTL Default: 0x0000_00A5
Field	Type	Default	Description
31:8	/	/	/
7:6	R/W	0x2	00:dma_active is low 01:dma_active is high 10:dma_req 11:active is control by controller
5	R/W	0x1	0: active fall do not care ack 1: active fall must after detect ack is high
4:0	R/W	0x5	dma_active high after last signal is high

4.7.3.14 CIR Transmit FIFO Data Register

Address : 0x0080			Name: CIR_TXFIFO Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/

7:0	W	0x0	Transmit Byte FIFO When the transmitting is trigger, the data in the FIFO will be transmitted until the data number has been transmitted finished.
-----	---	-----	---

4.8 General Purpose ADC (GPADC)

4.8.1 Overview

The GPADC is a 12-bit sampling analog to digital converter with 9 channels multiplexer. This ADC is a type of successive approximation register (SAR) converter.

- 12-bit Resolution and 10-bit effective SAR type A/D converter
- 9-channel multiplexer and the 9th channel is used for VBAT voltage detection
- 64 FIFO depth of data register
- Power Supply Voltage: 1.62~3.6V (from VDDIO), Analog Input Range: 0 to 2.5V
- Maximum Sampling frequency: 1 MHz
- Support self-calibration
- Support data compare and interrupt
- Support four operation mode
 - Single conversion mode
 - Single-cycle conversion mode
 - Continuous conversion mode
 - Outbreak conversion mode

4.8.2 GPADC Block Diagram

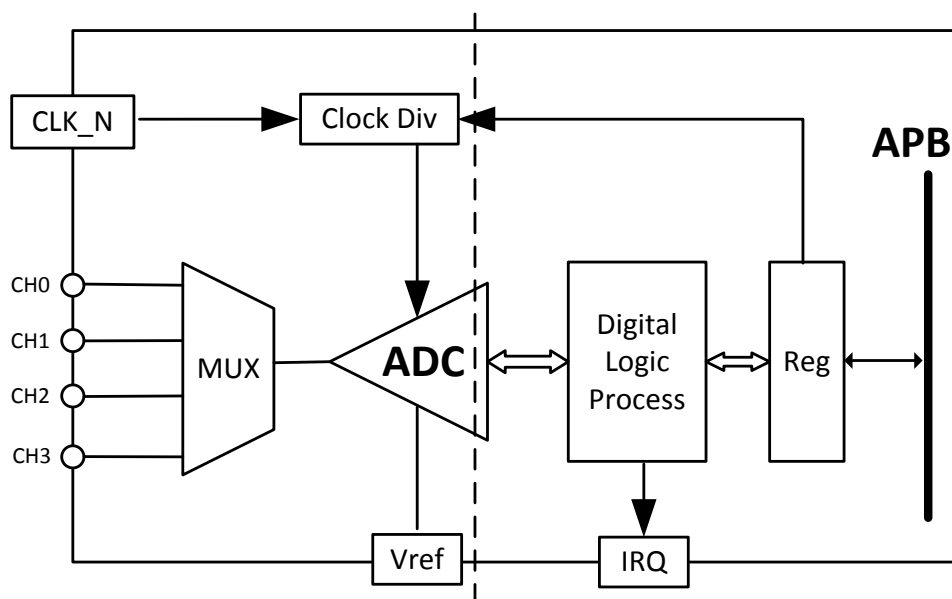


Figure 4-16 GPADC Block Diagram

4.8.3 Operations and Functional Descriptions

4.8.3.1 External Signals

Signal	Description	Pin Number	Type
GPADC0	Analog Input	IO	Analog
GPADC1	Analog Input	IO	Analog
GPADC2	Analog Input	IO	Analog
GPADC3	Analog Input	IO	Analog
GPADC4	Analog Input	IO	Analog
GPADC5	Analog Input	IO	Analog
GPADC6	Analog Input	IO	Analog
GPADC7	Analog Input	IO	Analog
GPADC8	VBAT Input	VBAT	Analog

Table 4-3 ADC Signals

4.8.4 Register List

Module Name	Base Address	
GPADC	0x40043000	
Register Name	Offset Address	Description
GP_SR_CON	0x00	GPADC Sample Rate Configure Register
GP_CTRL	0x04	GPADC Control Register
GP_CS_EN	0x08	GPADC Compare and Select Enable Register
GP_FIFO_INTC	0x0C	GPADC FIFO Interrupt Control Register
GP_FIFO_INTS	0x10	GPADC FIFO Interrupt Status Register
GP_FIFO_DATA	0x14	GPADC FIFO Data Register
GP_CDATA	0x18	GPADC Calibration Data Register
GP_DATAH_INTC	0x20	GPADC Data Low Interrupt Configure Register
GP_DATAH_INTS	0x24	GPADC Data High Interrupt Configure Register
GP_DATA_INTC	0x28	GPADC Data Interrupt Configure Register
GP_DATAH_INTS	0x30	GPADC Data Low Interrupt Status Register
GP_DATAH_INTS	0x34	GPADC Data High Interrupt Status Register
GP_DATA_INTS	0x38	GPADC Data Interrupt Status Register
GP_CHn_CMP_DATA	$0x40 + 4 * n$	GPADC CH n Compare Data Register
GP_CHn_DATA	$0x80 + 4 * n$	GPADC CH n Data Register

4.8.5 Register Description

4.8.5.1 GPADC Sample Rate Configure Register

Address : 0x0000			Name: GP_SR_CON Default: 0x01DF_002F
Field	Type	Default	Description
31:16	R/W	0x207(50K)	FS_DIV. ADC Sample Frequency Divider CLK_IN/(n+1) <i>Note: HOSC = 26MHz, n = 0x207; HOSC = 40MHz, n = 0x320</i>
15:0	R/W	0x34(2uS)	TACQ. ADC acquire time CLK_IN/(n+1) <i>Note: HOSC = 26MHz, n = 0x34; HOSC = 40MHz, n = 0x50</i>

4.8.5.2 GPADC Control Register

Address : 0x0004			Name: GP_CTRL Default: 0x00800000
Field	Type	Default	Description
31:24	R/ W	0x0	ADC_FIRST_DLY ADC First Convert Delay setting, ADC conversion of each channel is delayed by N samples.
23:22	/	/	/
21:20	R/W	0x0	ADC_OP_BIAS.(Adjust the bandwidth of the ADC amplifier) ADC OP Bias
19:18	R/W	0x0	GPADC Work Mode 00: Single conversion mode 01: Single-cycle conversion mode 10: Continuous conversion mode 11: Burst conversion mode
17	R/W	0x0	ADC_CALI_EN ADC Calibration 1: start Calibration, it is clear to 0 after calibration
16	R/W	0x0	ADC_EN ADC Function Enable 0: Disable 1: Enable <i>Note: the work mode and the channel number must be set before the ADC_EN bit being set.</i>

15:5	/	/	/
4	R/W	0x0	VBAT_DET_EN 0: disable 1: enable <i>Note: the 8th channel is designed for VBAT Voltage Detection and this bit should be set to 1 before enable the CH8</i>
3:1	R/W	0x0	VREF_MODE_SEL 000: Verf is 1.4V from BG, choose when vdd is 1.6V~3V 001: Verf is 2.5V from BG, choose when vdd is 3V~3.6V 010: Verf is 2.5V divider of EXT_LDO, choose when vdd = 3.3V 011: Verf is 2.5V divider of EXT_LDO, choose when vdd = 3.1V Others: Reserved
0	R/W	0x0	ADC_LDO_EN 0: disable 1: enable <i>Note: before enabling the ADC function, you must enable the LDO-ADC firstly.</i>

4.8.5.3 GPADC Compare and Select Enable Register

Address : 0x0008			Name: GP_CS_EN Default: 0x00000000
Field	Type	Default	Description
31:24	/	/	/
[n+16] (n=0~8)	R/W	0x0	ADC_CHn_CMP_EN Channel n Compare Enable 0: Disable 1: Enable
15:8	/	/	/
[n] (n = 0~8)	R/W	0x0	ADC_CHn_SELECT Analog input channel n Select 0: Disable 1: Enable

4.8.5.4 GPADC FIFO Interrupt Control Register

Address : 0x000C			Name: GP_FIFO_INTC Default: 0x0000_1F00
Field	Type	Default	Description
31:19	/	/	/
18	R/W	0x0	FIFO_DATA_DRQ_EN ADC FIFO Data DRQ Enable 0: Disable

			1:Enable
17	R/W	0x0	FIFO_OVERRUN_IRQ_EN ADC FIFO Over Run IRQ Enable 0: Disable 1: Enable
16	R/W	0x0	FIFO_DATA_IRQ_EN ADC FIFO Data Available IRQ Enable 0: Disable 1: Enable
15:14	/	/	/
13:8	R/W	0x1F	FIFO_TRIG_LEVEL Interrupt and DMA request trigger level for ADC Trigger Level = TXTL + 1
7:5	/	/	/
4	R/W1 C	0x0	FIFO_FLUSH ADC FIFO Flush Write '1' to flush TX FIFO, self clear to '0'.
3:0	/	/	/

4.8.5.5 GPADC FIFO Interrupt Status Register

Address : 0x0010			Name: GP_FIFO_INTS Default: 0x00000000
Field	Type	Default	Description
31:18	/	/	/
17	R/W1 C	0x0	FIFO_OVERRUN_PENDING ADC FIFO Over Run IRQ pending 0: No Pending IRQ 1: FIFO Overrun Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails.
16	R/W1 C	0x0	FIFO_DATA_PENDING ADC FIFO Data Available pending Bit 0: NO Pending IRQ 1: FIFO Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails.
15:14	/	/	/
13:8	R	0x0	RXA_CNT ADC FIFO available Sample Word Counter
7:0	/	/	/

4.8.5.6 GPADC FIFO Data Register

Address : 0x0014			Name: GP_FIFO_DATA Default: 0x00000000
Field	Type	Default	Description
31:12	/	/	/
11:0	R/W	0xX	GP_FIFO_DATA GPADC Data in FIFO

4.8.5.7 GPADC Calibration Data Register

Address : 0x0018			Name: GP_CDATA Default: 0x00000000
Field	Type	Default	Description
11:0	R/W	0x0	GP_CDATA GPADC Calibration Data

4.8.5.8 GPADC Low Interrupt Configure Register

Address : 0x0020			Name: GP_DATA_L_INT Default: 0x00000000
Field	Type	Default	Description
31:8	/	/	/
[n] (n = 0~8)	RW	0x0	CHn_LOW_IRQ_EN 0: Disable 1: Enable

4.8.5.9 GPADC HIGH Interrupt Configure Register

Address : 0x0024			Name: GP_DATA_H_INT Default: 0x00000000
Field	Type	Default	Description
31:8	/	/	/
[n] (n = 0~8)	RW	0	CHn_HIGH_IRQ_EN 0: Disable 1: Enable

4.8.5.10 GPADC DATA Interrupt Configure Register

Address : 0x0028			Name: GP_DATA_INT Default: 0x00000000
Field	Type	Default	Description

31:8	/	/	/
[n] (n = 0~8)	RW	0	CHn_DATA_IRQ_EN 0: Disable 1: Enable

4.8.5.11 GPADC Low Interrupt Status Register

Address : 0x0030			Name: GP_DATA_INTS Default: 0x00000000
Field	Type	Default	Description
31:8	/	/	/
[n] (n = 0~8)	RW	0	CHn_LOW_PENDGING 1: Channel n Voltage Low Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails.

4.8.5.12 GPADC High Interrupt Status Register

Address : 0x0034			Name: GP_DATAH_INTS Default: 0x00000000
Field	Type	Default	Description
31:8	/	/	/
[n] (n = 0~8)	RW	0	CHn_HIGH_PENDGING 1: Channel n Voltage High Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails.

4.8.5.13 GPADC Data Interrupt Status Register

Address : 0x0038			Name: Default: 0x00000000
Field	Type	Default	Description
31:8	/	/	/
[n] (n = 0~8)	RW	0	CHn_DATA_PENDGING 0: NO Pending IRQ 1: Channel n Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails.

4.8.5.14 GPADC CHn Compare Data Register

Address : 0x0040 + 4 * n (n = 0~8)			Name: GP_CHn_CMP_DATA Default: 0x0BFF_0400
Field	Type	Default	Description
31:28	/	/	/
27:16	R/W	0xBFF	CHn_CMP_HIG_DATA

			Channel n Voltage High Value
15:12	/	/	/
11:0	R/W	0x400	CHn_CMP_LOW_DATA Channel n Voltage Low Value

4.8.5.15 GPADC CHn Data Register

Address : 0x0080 + 4 * n (n = 0~8)			Name : GP_CHn_DATA Default : 0x00000000
Field	Type	Default	Description
31:12	/	/	/
11:0	R	0x0	GP_CHn_DATA Channel n Data

Chapter 5 Audio Interface

5.1 Digital Audio Controller (DAUDIO)

5.1.1 Overview

The Digital Audio Interface Controller has been designed to transfer streaming audio-data between the system memory and the codec chip. The controller supports standard I2S format, Left-justified Mode format, Right-justified Mode format, PCM Mode format and TDM Mode format.

Features:

- Compliant with standard Philips Inter-IC sound (I2S) bus specification
- Compliant with Left-justified, Right-justified, PCM mode, and TDM (Time Division Multiplexing) format
- Support different sample period width in each interface when using LRCK and LRCKR at the same time
- Support full-duplex synchronous work mode
- Support Master / Slave mode
- Support adjustable interface voltage
- Support clock up to 100MHz
- Support adjustable audio sample resolution from 8-bit to 32-bit.
- Support up to 8 slots which has adjustable width from 8-bit to 32-bit.
- Support sample rate from 8KHz to 192KHz
- Support up to 4 data output pin
- Support 8-bits u-law and 8-bits A-law companded sample
- One 128 depth x 32-bit width FIFO for data transmit, one 64 depth x 32-bit width FIFO for data receive
- Support programmable PCM frame width: 1 BCLK width (short frame) and 2 BCLKs width (long frame)

5.1.2 Interface Block Diagram

5.1.2.1 Typical Applications

The Digital Audio Interface is the extended of I2S and pcm which provides a serial bus interface for stereo and multichannel audio data. This interface is most commonly used by consumer audio market, including compact disc,

digital audio tape, digital sound processors, and digital TV-sound.

5.1.2.2 Functional Block Diagram

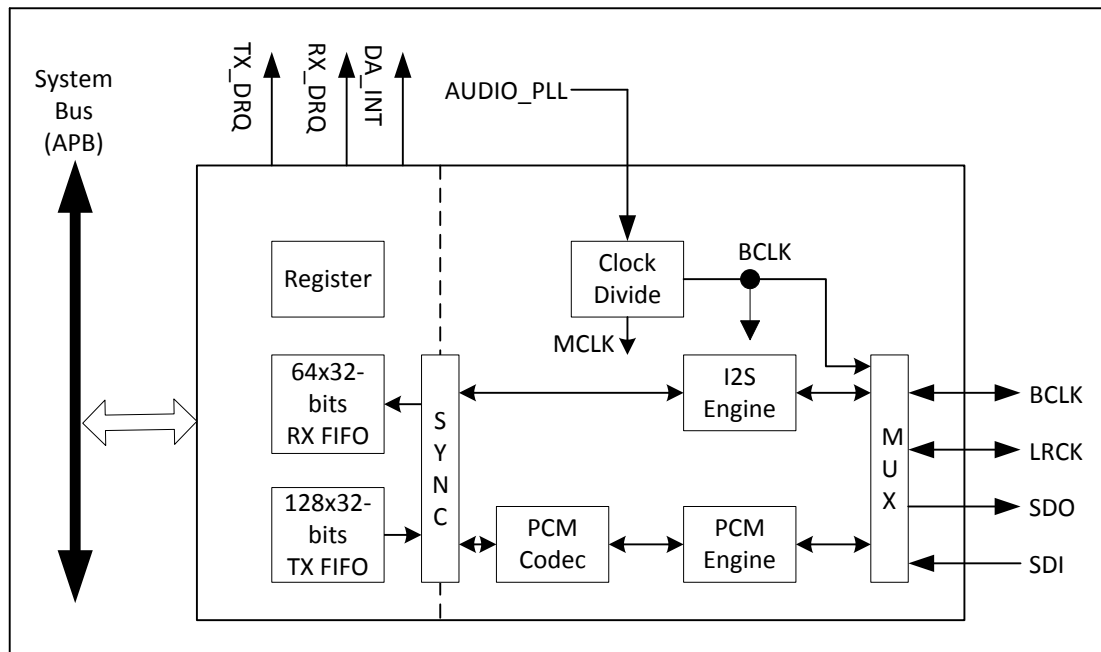


Figure 5-1 DAUDIO Function Block Diagram

5.1.2.3 Digital Audio Interface Electrical Characteristics

The timing of DAI in Master mode is showed in the following Figure.

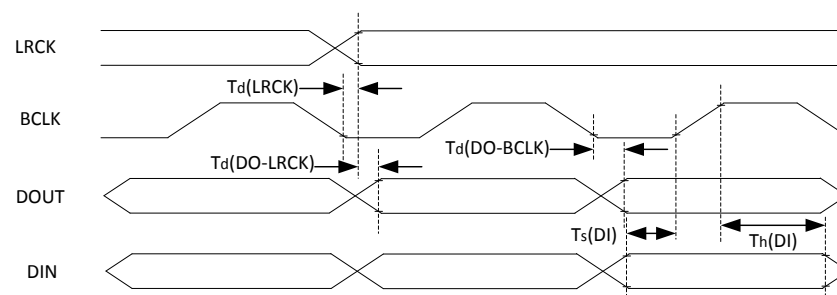


Figure 5-2 DAI Timing in Master Mode

The timing parameters of DAI in Master mode are showed in the following Table.

PARAMETER		MIN	MAX	UNITS
$T_d(\text{LRCK})$	LRCK delay		10	ns
$T_d(\text{DO-LRCK})$	LRCK to DOUT delay(For LJF)		10	ns
$T_d(\text{DO-BCLK})$	BCLK to DOUT delay		10	ns

$T_s(DI)$	DIN setup	4		ns
$T_h(DI)$	DIN hold	4		ns
T_r	BCLK Rise time		8*	ns
T_f	BCLK Fall time		8	ns

Note *容性负载的大小有关

Table 5-1 DAI Timing Constants at 3.3V in Master Mode

The timing of DAI in Slave mode is showed in the following Figure

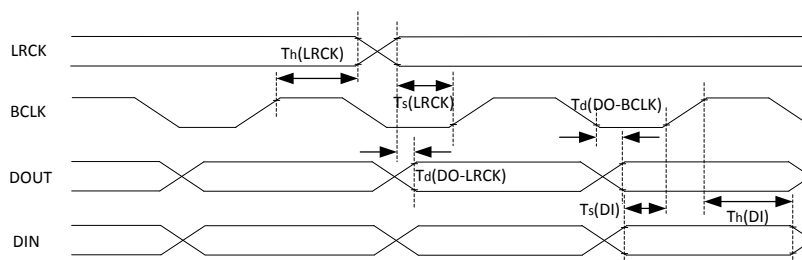


Figure 5-3 DAI Timing in Slave Mode

The timing parameters of DAI in Slave mode are showed in the following Table

PARAMETER		MIN	MAX	UNITS
$T_s(LRCK)$	LRCK setup	4		ns
$T_h(LRCK)$	LRCK hold	4		ns
$T_d(DO-LRCK)$	LRCK to DOUT delay(For LJF)		10	ns
$T_d(DO-BCLK)$	BCLK to DOUT delay		10	ns
$T_s(DI)$	DIN setup	4		ns
$T_h(DI)$	DIN hold	4		ns
T_r	BCLK Rise time		4	ns
T_f	BCLK Fall time		4	ns

Table 5-2 DAI Timing Constants in Slave Mode

5.1.3 Operations and Functional Descriptions

5.1.3.1 External Signals

The following table describes the external signals of DAI. BCLK and LRCK are bidirectional I/O, when DAI is configured as Master device, BCLK and LRCK is output pin; when DAI is configure as slave device, BCLK and LRCK is input pin. MCLK is a output pin for external device. SDO is always the serial data output pin, and SDI is the serial data input. For information about General Purpose I/O port, see Port Controller.

Signal	Description	Type
MCLK	Digital Audio Interface MCLK Output	O
LRCK	Digital Audio Interface Serial Clock	I/O
BCLK	Digital Audio Interface Sample Rate Clock/Sync	I/O
SDO	Digital Audio Interface Serial Data Output	O
SDI	Digital Audio Interface Serial Data input	I

Table 5-3 Digital Audio Interface External Signals

5.1.3.2 Clock Sources

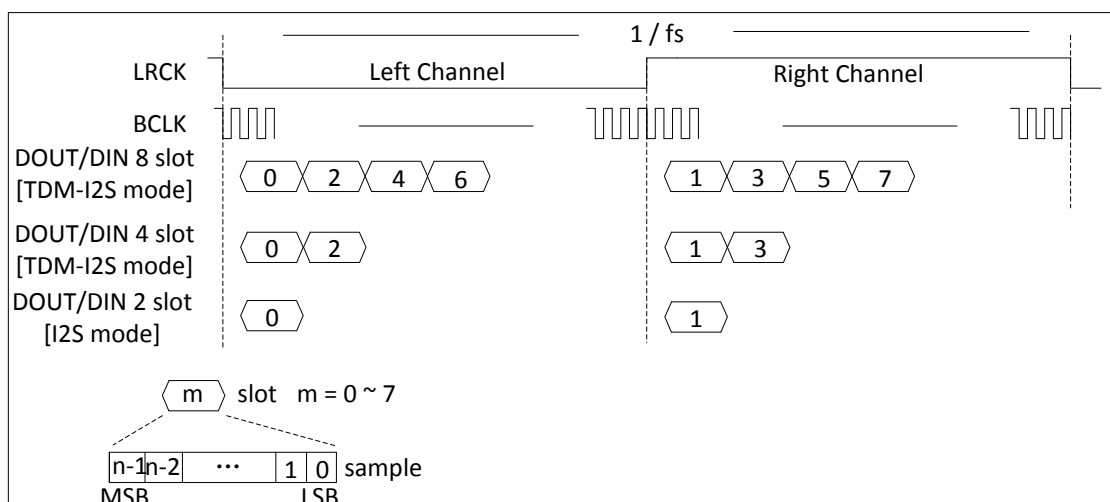
Each DAI controller get three different clocks. Users can select one of them to make DAI Clock Source. The table describes the clock sources for DAI. Users can see Clock Controller Unit for clock setting, configuration and gating information.

Clock Sources	Description
Audio_PLL	24.576Mhz or 22.5792Mhz generated by AUDIO-PLL to produce 48KHz or 44.1KHz serial frequency.

Table 5-4 Digital Audio Interface Clock Sources

5.1.4 Digital Audio Interface Transmit Format

The DAI support standard I2S mode, Left-justified I2S mode, Right-justified I2S mode, PCM mode and TDM mode. Software can select one of them in which the DAI works by setting the DAI Control Register.


Figure 5-4 Timing diagram for standard I2S/TDM-I2S mode

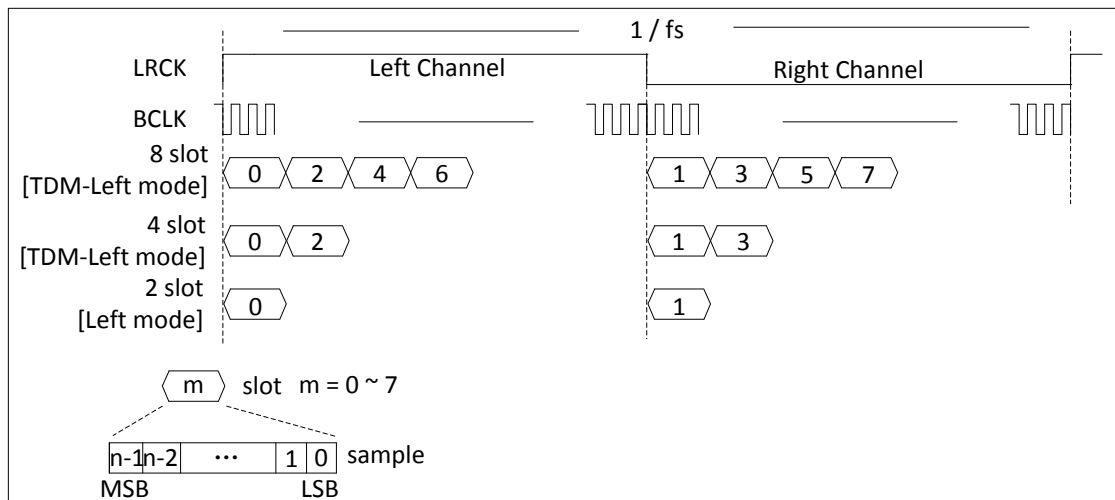


Figure 5-5 Timing diagram for Left-justified/TDM-Left mode

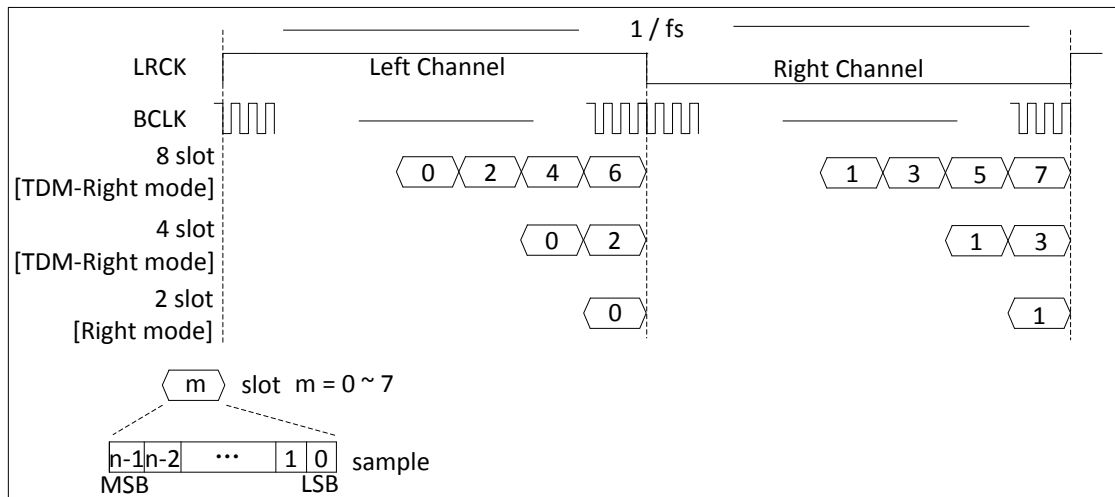


Figure 5-6 Timing diagram for Right-justified/TDM-Right mode

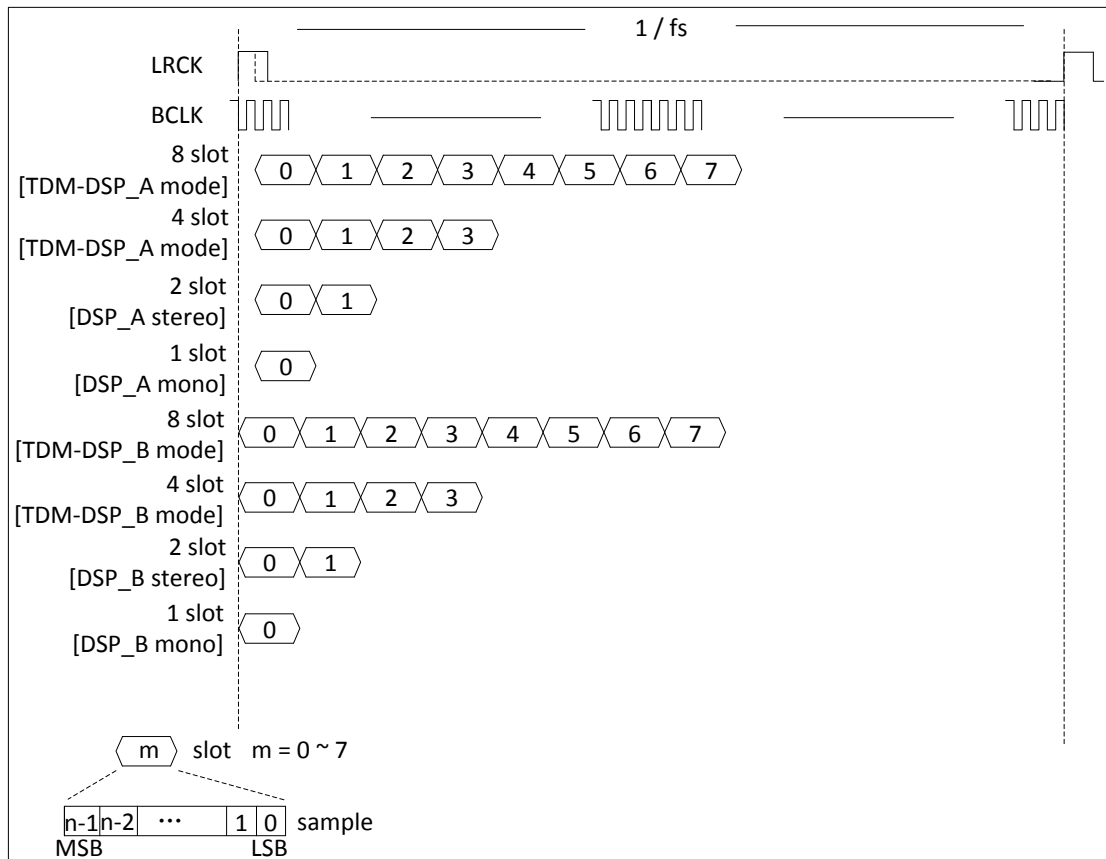


Figure 5-7 Timing diagram for PCM/TDM-PCM mode

5.1.5 Operation Modes

The software operation of the DAI is divided into five steps: system setup, DAI initialization, the channel setup, DMA setup and Enable/Disable module. These five setup are described in detail in the following sections.

5.1.5.1 System setup and DAI initialization

The first step In the system setup is properly programming the GPIO. Because the DAI port is a multiplex pin. You can find the function in the pin multiplex specification. The clock source for the DAI should be followed. At first you must reset the audio PLL though the PLL_ENABLE bit of PLL_AUDIO_CTRL_REG in the CCMU. The second step, you must setup the frequency of the audio pll in the PLL_AUDIO_CTRL_REG. The configuration of audio pll can be found in the chapter 1.3.2. After that, you must open the DAI gating though the DAI0_CLK_REG/DAI1_CLK_REG when you checkout that the LOCK bit of PLL_AUDIO_CTRL_REG become 1. At last, you must reset the DAI the BUS_RST_REG3's bit[13:12] and open the DAI bus gating in the BUS_GATING_REG2's bit[13:12].

After the system setup, the register of DAI can be setup. At first, you should initialization the DAI. You should closed the globe enable bit(DAI_CTL[0]), TX enable bit(DAI_CTL[2]) and RX enable bit(DAI_CTL[1]) by write 0 to it. After that, you must clear the TX/RX FIFO by write 0 to register DAI_FCTL[25:24]. At last, you can clear the TX/RX FIFO counter by write 0 to DAI_TXCNT/DAI_RXCNT.

5.1.5.2 The channel setup and DMA setup

Before the usage and control of DAI, you must configure the TWI. The configuration of TWI will not be described in this chapter. But you can only configure DAI master and slave through the TWI. In the following, you can setup the DAI of master and slave. The configuration can be referred to the protocol of DAI. Then, you can set the translation mode, the sample precision, the width of slot, the frame mode and the trigger level. The register set can be found in the spec.

The DAI supports three methods to transfer the data. The most common way is DMA, the set of DMA can be found in the DMA spec. In this module, you just need to enable the DRQ.

5.1.5.3 Enable and disable the DAI

To enable the function, you can enable TX/RX by writing the DAI_CTL[2:1]. After that, you must enable DAI by writing the Global Enable bit to 1 in the DAI_CTL. The disable process is writing the Global Enable to 0.

5.1.6 Register List

Module Name	Base Address	
Digital Audio	0x40042C00	
Register Name	Offset Address	Description
DA_CTL	0x00	Digital Audio Control Register
DA_FMT0	0x04	Digital Audio Format Register 0
DA_FMT1	0x08	Digital Audio Format Register 1
DA_ISTA	0x0C	Digital Audio Interrupt Status Register
DA_RXFIFO	0x10	Digital Audio RX FIFO Register
DA_FCTL	0x14	Digital Audio FIFO Control Register
DA_FSTA	0x18	Digital Audio FIFO Status Register
DA_INT	0x1C	Digital Audio DMA & Interrupt Control Register
DA_TXFIFO	0x20	Digital Audio TX FIFO Register
DA_CLKD	0x24	Digital Audio Clock Divide Register
DA_TXCNT	0x28	Digital Audio TX Sample Counter Register
DA_RXCNT	0x2C	Digital Audio RX Sample Counter Register
DA_CHCFG	0x30	Digital Audio Channel Configuration register
DA_TX0CHSEL	0x34	Digital Audio TX0 Channel Select Register
DA_TX1CHSEL	0x38	Digital Audio TX1 Channel Select Register
DA_TX2CHSEL	0x3C	Digital Audio TX2 Channel Select Register
DA_TX3CHSEL	0x40	Digital Audio TX3 Channel Select Register
DA_TX0CHMAP	0x44	Digital Audio TX0 Channel Mapping Register
DA_TX1CHMAP	0x48	Digital Audio TX1 Channel Mapping Register
DA_TX2CHMAP	0x4C	Digital Audio TX2 Channel Mapping Register

DA_TX3CHMAP	0x50	Digital Audio TX3 Channel Mapping Register
DA_RXCHSEL	0x54	Digital Audio RX Channel Select register
DA_RXCHMAP	0x58	Digital Audio RX Channel Mapping Register

5.1.7 Register Description

5.1.7.1 Digital Audio Control Register

Address : 0x0000			Name: DA_CTL Default: 0x0006_0000
Field	Type	Default	Description
31:19	/	/	/
18	R/W	1	BCLK_OUT 0: input 1: output
17	R/W	1	LRCK_OUT 0: input 1: output
16:9	/	/	/
8	R/W	0	SDO0_EN 0: Disable, Hi-Z state 1: Enable
7	/	/	/
6	R/W	0	OUT Mute 0: normal transfer 1: force DOUT to output 0
5:4	R/W	0	MODE_SEL Mode Selection 0: PCM mode (offset 0: DSP_B; offset 1: DSP_A) 1: Left mode (offset 0: LJ mode; offset 1: I2S mode) 2: Right-Justified mode 3: Reserved
3	R/W	0	LOOP Loop back test 0: Normal mode 1: Loop back test When set '1', connecting the SDO0 with the SDI
2	R/W	0	TXEN Transmitter Block Enable 0: Disable 1: Enable

1	R/W	0	RXEN Receiver Block Enable 0: Disable 1: Enable
0	R/W	0	GEN Globe Enable A disable on this bit overrides any other block or channel enables. 0: Disable 1: Enable

5.1.7.2 Digital Audio Format Register 0

Address : 0x0004			Name: DA_FMT0 Default: 0x0000_0033
Field	Type	Default	Description
31	/	/	/
30	R/W	0	LRCK_WIDTH (only apply in PCM mode) LRCK width 0: LRCK = 1 BCLK width (short frame) 1: LRCK = 2 BCLK width (long frame)
29:20	/	/	/
19	R/W	0	LRCK_POLARITY When apply in I2S / Left-Justified / Right-Justified mode: 0: Left channel when LRCK is low 1: Left channel when LRCK is high When apply in PCM mode: 0: PCM LRCK asserted at the negative edge 1: PCM LRCK asserted at the positive edge
18	/	/	/
17:8	R/W	0	LRCK_PERIOD It is used to program the number of BCLKs per channel of sample frame. This value is interpreted as follow: PCM mode: Number of BCLKs within (Left + Right) channel width I2S / Left-Justified / Right-Justified mode: Number of BCLKs within each individual channel width (Left or Right) N+1 For example: n = 7: 8 BCLK width ... n = 1023: 1024 BCLKs width
7	R/W	0	BCLK_POLARITY 0: normal mode, negative edge drive and positive edge sample

			1: invert mode, positive edge drive and negative edge sample
6:4	R/W	3	SR Sample Resolution 0: Reserved 1: 8-bit 2: 12-bit 3: 16-bit 4: 20-bit 5: 24-bit 6: 28-bit 7: 32-bit
3	R/W	0	EDGE_TRANSFER 0: SDO drive data and SDI sample data at the different BCLK edge 1: SDO drive data and SDI sample data at the same BCLK edge BCLK_POLARITY = 0, use negative edge BCLK_POLARITY = 1, use positive edge
2:0	R/W	0x3	SW Slot Width Select 0: Reserved 1: 8-bit 2: 12-bit 3: 16-bit 4: 20-bit 5: 24-bit 6: 28-bit 7: 32-bit

5.1.7.3 Digital Audio Format Register 1

Address : 0x0008			Name : DA_FMT1 Default : 0x0000_0030
Field	Type	Default	Description
31:8	/	/	
7	R/W	0	RX MLS MSB / LSB First Select 0: MSB First 1: LSB First
6	R/W	0	TX MLS MSB / LSB First Select 0: MSB First 1: LSB First
5:4	R/W	3	SEXT

			Sign Extend in slot [sample resolution < slot width] 0: Zeros or audio gain padding at LSB position 1: Sign extension at MSB position 2: Reserved 3: Transfer 0 after each sample in each slot
3:2	R/W	0	RX_PDM PCM Data Mode 0: Linear PCM 1: reserved 2: 8-bits u-law 3: 8-bits A-law
1:0	R/W	0	TX_PDM PCM Data Mode 0: Linear PCM 1: reserved 2: 8-bits u-law 3: 8-bits A-law

5.1.7.4 Digital Audio Interrupt Status Register

Address : 0x000C			Name: DA_ISTA Default: 0x0000_0010
Field	Type	Default	Description
31:7	/	/	/
6	R/W	0	TXU_INT TX FIFO Under run Pending Interrupt 0: No Pending Interrupt 1: FIFO Under run Pending Interrupt Write 1 to clear this interrupt
5	R/W	0	TXO_INT TX FIFO Overrun Pending Interrupt 0: No Pending Interrupt 1: FIFO Overrun Pending Interrupt Write '1' to clear this interrupt
4	R/W	1	TXE_INT TX FIFO Empty Pending Interrupt 0: No Pending IRQ 1: FIFO Empty Pending Interrupt when data in TX FIFO are less than TX trigger level Write '1' to clear this interrupt or automatic clear if interrupt condition fails.
3	/	/	/

2	R/W	0	RXU_INT RX FIFO Under run Pending Interrupt 0: No Pending Interrupt 1:FIFO Under run Pending Interrupt Write 1 to clear this interrupt
1	R/W	0	RXO_INT RX FIFO Overrun Pending Interrupt 0: No Pending IRQ 1: FIFO Overrun Pending IRQ Write '1' to clear this interrupt
0	R/W	0	RXA_INT RX FIFO Data Available Pending Interrupt 0: No Pending IRQ 1: Data Available Pending IRQ when data in RX FIFO are more than RX trigger level Write '1' to clear this interrupt or automatic clear if interrupt condition fails.

5.1.7.5 Digital Audio RX FIFO register

Address : 0x0010			Name: DA_RXFIFO Default: 0x0000_0000
Field	Type	Default	Description
31:0	R	0	RX_DATA RX Sample Host can get one sample by reading this register. The left channel sample data is first and then the right channel sample.

5.1.7.6 Digital Audio FIFO Control Register

Address : 0x0014			Name: DA_FCTL Default: 0x0004_00F0
Field	Type	Default	Description
31	R/W	0	HUB_EN Audio Hub Enable 0 : Disable 1 : Enable
30:26	/	/	/
25	R/W	0	FTX Write '1' to flush TX FIFO, self clear to '0' .
24	R/W	0	FRX

			Write '1' to flush RX FIFO, self clear to '0'.
23:19	/	/	/
18:12	R/W	0x40	TXTL TX FIFO Empty Trigger Level Interrupt and DMA request trigger level for TXFIFO normal condition Trigger Level = TXTL
11:10	/	/	/
9:4	R/W	0xF	RXTL RX FIFO Trigger Level Interrupt and DMA request trigger level for RXFIFO normal condition Trigger Level = RXTL + 1
3	/	/	/
2	R/W	0	TXIM TX FIFO Input Mode (Mode 0, 1) 0: Valid data at the MSB of TXFIFO register 1: Valid data at the LSB of TXFIFO register Example for 20-bits transmitted audio sample: Mode 0: FIFO_I[31:0] = {APB_WDATA[31:12], 12' h0} Mode 1: FIFO_I[31:0] = {APB_WDATA[19:0], 12' h0}
1:0	R/W	0	RXOM RX FIFO Output Mode (Mode 0, 1, 2, 3) 00: Expanding '0' at LSB of DA_RXFIFO register. 01: Expanding received sample sign bit at MSB of DA_RXFIFO register. 10: Truncating received samples at high half-word of DA_RXFIFO register and low half-word of DA_RXFIFO register is filled by '0'. 11: Truncating received samples at low half-word of DA_RXFIFO register and high half-word of DA_RXFIFO register is expanded by its sign bit. Example for 20-bits received audio sample: Mode 0: APB_RDATA[31:0] = {FIFO_O[31:12], 12' h0} Mode 1: APB_RDATA [31:0] = {12{FIFO_O[31]}, FIFO_O[31:12]} Mode 2: APB_RDATA [31:0] = {FIFO_O[31:16], 16' h0} Mode 3: APB_RDATA [31:0] = {16{FIFO_O[31]}, FIFO_O[31:16]}

5.1.7.7 Digital Audio FIFO Status Register

Address : 0x0018			Name : DA_FSTA Default : 0x1080_0000
Field	Type	Default	Description
31:29	/	/	/
28	R	1	TXE TX FIFO Empty 0: No room for new sample in TX FIFO

			1: More than one room for new sample in TX FIFO (≥ 1 word)
27:24	/	/	/
23:16	R	0x80	TXE_CNT TX FIFO Empty Space Word Counter
15:9	/	/	/
8	R	0	RXA RX FIFO Available 0: No available data in RX FIFO 1: More than one sample in RX FIFO (≥ 1 word)
7	/	/	/
6:0	R	0	RXA_CNT RX FIFO Available Sample Word Counter

5.1.7.8 Digital Audio DMA & Interrupt Control Register

Address : 0x001C			Name: DA_INT Default: 0x0000_0000
Field	Type	Default	Description
31:8	/	/	/
7	R/W	0	TX_DRQ TX FIFO Empty DRQ Enable 0: Disable 1: Enable
6	R/W	0	TXUI_EN TX FIFO Under run Interrupt Enable 0: Disable 1: Enable
5	R/W	0	TXOI_EN TX FIFO Overrun Interrupt Enable 0: Disable 1: Enable When set to '1', an interrupt happens when writing new audio data if TX FIFO is full.
4	R/W	0	TXEI_EN TX FIFO Empty Interrupt Enable 0: Disable 1: Enable
3	R/W	0	RX_DRQ RX FIFO Data Available DRQ Enable 0: Disable 1: Enable When set to '1', RXFIFO DMA Request line is asserted if Data is

			available in RX FIFO.
2	R/W	0	RXUI_EN RX FIFO Under run Interrupt Enable 0: Disable 1: Enable
1	R/W	0	RXOI_EN RX FIFO Overrun Interrupt Enable 0: Disable 1: Enable
0	R/W	0	RXAI_EN RX FIFO Data Available Interrupt Enable 0: Disable 1: Enable

5.1.7.9 Digital Audio TX FIFO register

Address : 0x0020			Name: DA_TXFIFO Default: 0x0000_0000
Field	Type	Default	Description
31:0	W	0	TX_DATA TX Sample Transmitting left, right channel sample data should be written this register one by one. The left channel sample data is first and then the right channel sample.

5.1.7.10 Digital Audio Clock Divide Register

Address : 0x0024			Name: DA_CLKD Default: 0x0000_0000
Field	Type	Default	Description
31:9	/	/	/
8	R/W	0	MCLKO_EN 0: Disable MCLK Output 1: Enable MCLK Output Notes: Whether in Slave or Master mode, when this bit is set to 1, MCLK should be output.
7:4	R/W	0	BCLKDIV BCLK Divide Ratio from PLL2 0: reserved 1: Divide by 1 2: Divide by 2

			3: Divide by 4 4: Divide by 6 5: Divide by 8 6: Divide by 12 7: Divide by 16 8: Divide by 24 9: Divide by 32 10: Divide by 48 11: Divide by 64 12: Divide by 96 13: Divide by 128 14: Divide by 176 15: Divide by 192
3:0	R/W	0	MCLKDIV MCLK Divide Ratio from PLL2 Output 0: reserved 1: Divide by 1 2: Divide by 2 3: Divide by 4 4: Divide by 6 5: Divide by 8 6: Divide by 12 7: Divide by 16 8: Divide by 24 9: Divide by 32 10: Divide by 48 11: Divide by 64 12: Divide by 96 13: Divide by 128 14: Divide by 176 15: Divide by 192

5.1.7.11 Digital Audio TX Counter register

Address : 0x0028			Name: DA_TXCNT Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	TX_CNT TX Sample Counter The audio sample number of sending into TXFIFO. When one sample is put into TXFIFO by DMA or by host IO, the TX sample counter register increases by one. The TX sample counter register can be set to any initial

			valve at any time. After been updated by the initial value, the counter register should count on base of this initial value.
--	--	--	--

5.1.7.12 Digital Audio RX Counter register

Address : 0x002C			Name: DA_RXCNT Default: 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0	RX_CNT RX Sample Counter The audio sample number of writing into RXFIFO. When one sample is written by Digital Audio Engine, the RX sample counter register increases by one. The RX sample counter register can be set to any initial valve at any time. After been updated by the initial value, the counter register should count on base of this initial value.

5.1.7.13 Digital Audio Channel Configuration register

Address : 0x0030			Name: DA_CHCFG Default: 0x0000_0000
Field	Type	Default	Description
31:10	/	/	/
9	R/W	0	TX_SLOT_HIZ 0: normal mode for the last half cycle of BCLK in the slot 1: turn to hi-z state for the last half cycle of BCLK in the slot
8	R/W	0	TXn_STATE 0: transfer level 0 when not transferring slot 1: turn to hi-z state (TDM) when not transferring slot
7	/	/	/
6:4	R/W	0	RX_SLOT_NUM RX Channel/Slot Number which between CPU/DMA and FIFO 0: 1 channel or slot ... 7: 8 channels or slots
3	/	/	/
2:0	R/W	0	TX_SLOT_NUM TX Channel/Slot Number which between CPU/DMA and FIFO 0: 1 channel or slot ... 7: 8 channels or slots

5.1.7.14 Digital Audio TXn Channel Select register

Address : 0x34 + N*4 (N = 0~3)			Name : DA_TXnCHSEL Default : 0x0000_0000
Field	Type	Default	Description
31:13	/	/	/
12	R/W	0	TXn_OFFSET TXn offset tune, TXn data offset to LRCK 0: no offset 1: data is offset by 1 BCLKs to LRCK
11:4	R/W	0	TXn_CHEN TXn Channel (slot) enable, bit[11:4] refer to slot [7:0]. When one or more slot(s) is(are) disabled, the affected slot(s) is(are) set to disable state 0: disable 1: enable
3	/	/	/
2:0	R/W	0	TXn_CHSEL TXn Channel (slot) number Select for each output 0: 1 channel / slot ... 7: 8 channels / slots

5.1.7.15 Digital Audio TXn Channel Mapping Register

Address : 0x44 + n*4 (n = 0, 1, 2, 3)			Name : DA_TXnCHMAP Default : 0x0000_0000
Field	Type	Default	Description
31	/	/	/
30:28	R/W	0	TXn_CH7_MAP TXn Channel7 Mapping 0: 1st sample ... 7: 8th sample
27	/	/	/
26:24	R/W	0	TXn_CH6_MAP TXn Channel6 Mapping 0: 1st sample ... 7: 8th sample
23	/	/	/
22:20	R/W	0	TXn_CH5_MAP

			TXn Channel5 Mapping 0: 1st sample ... 7: 8th sample
19	/	/	/
18:16	R/W	0	TXn_CH4_MAP TXn Channel4 Mapping 0: 1st sample ... 7: 8th sample
15	/	/	/
14:12	R/W	0	TXn_CH3_MAP TXn Channel3 Mapping 0: 1st sample ... 7: 8th sample
11	/	/	/
10:8	R/W	0	TXn_CH2_MAP TXn Channel2 Mapping 0: 1st sample ... 7: 8th sample
7	/	/	/
6:4	R/W	0	TXn_CH1_MAP TXn Channel1 Mapping 0: 1st sample ... 7: 8th sample
3	/	/	/
2:0	R/W	0	TXn_CH0_MAP TXn Channel0 Mapping 0: 1st sample ... 7: 8th sample

5.1.7.16 Digital Audio RX Channel Select register

Address : 0x0054			Name : DA_RXCHSEL Default : 0x0000_0000
Field	Type	Default	Description
31:13	/	/	/
12	R/W	0	RX_OFFSET

			RX offset tune, RX data offset to LRCK 0: no offset 1: data is offset by 1 BCLKs to LRCK
11:3	/	/	
2:0	R/W	0	RX_CHSEL RX Channel (slot) number Select for input 0: 1 channel / slot ... 7: 8 channels / slots

5.1.7.17 Digital Audio RX Channel Mapping Register

Address : 0x0058			Name: DA_RXCHMAP Default: 0x0000_0000
Field	Type	Default	Description
31	/	/	/
30:28	R/W	0	RX_CH7_MAP RX Channel7 Mapping 0: 1st sample ... 7: 8th sample
27	/	/	/
26:24	R/W	0	RX_CH6_MAP RX Channel6 Mapping 0: 1st sample ... 7: 8th sample
23	/	/	/
22:20	R/W	0	RX_CH5_MAP RX Channel5 Mapping 0: 1st sample ... 7: 8th sample
19	/	/	/
18:16	R/W	0	RX_CH4_MAP RX Channel4 Mapping 0: 1st sample ... 7: 8th sample
15	/	/	/
14:12	R/W	0	RX_CH3_MAP RX Channel3 Mapping

			0: 1st sample ... 7: 8th sample
11	/	/	/
10:8	R/W	0	RX_CH2_MAP RX Channel2 Mapping 0: 1st sample ... 7: 8th sample
7	/	/	/
6:4	R/W	0	RX_CH1_MAP TX Channel1 Mapping 0: 1st sample ... 7: 8th sample
3	/	/	/
2:0	R/W	0	RX_CH0_MAP RX Channel0 Mapping 0: 1st sample ... 7: 8th sample

5.1.7.18 Digital Audio DBG register

Address : 0x005C			Name : DA_DBG Default : 0x0000_0000
Field	Type	Default	Description
31:0	/	/	/

5.2 Audio CODEC

5.2.1 Overview

The embedded Audio Codec is a high-quality audio codec.

- One audio digital-to-analog(DAC) channel.
 - SNR 99dB $\pm$ 2dB (A-Weighting).
 - Supports DAC Sample Rates from 8KHz to 192KHz.
- Analog low-power loop from microphone input to lineout output.
- One audio analog-to-digital(ADC) channel

- SNR 94dB ± 2 dB (A-Weighting).
- Supports ADC Sample Rates from 8KHz to 48KHz.
- One 128x24-bits FIFO for DAC data transmit, one 128x24-bits FIFO for ADC data receive.
- One low-noise analog microphone bias output.
- Programmable FIFO thresholds.
- DMA and Interrupt Support.

5.2.2 Block Diagram

Figure 5-1 shows a block diagram of the Audio Codec.

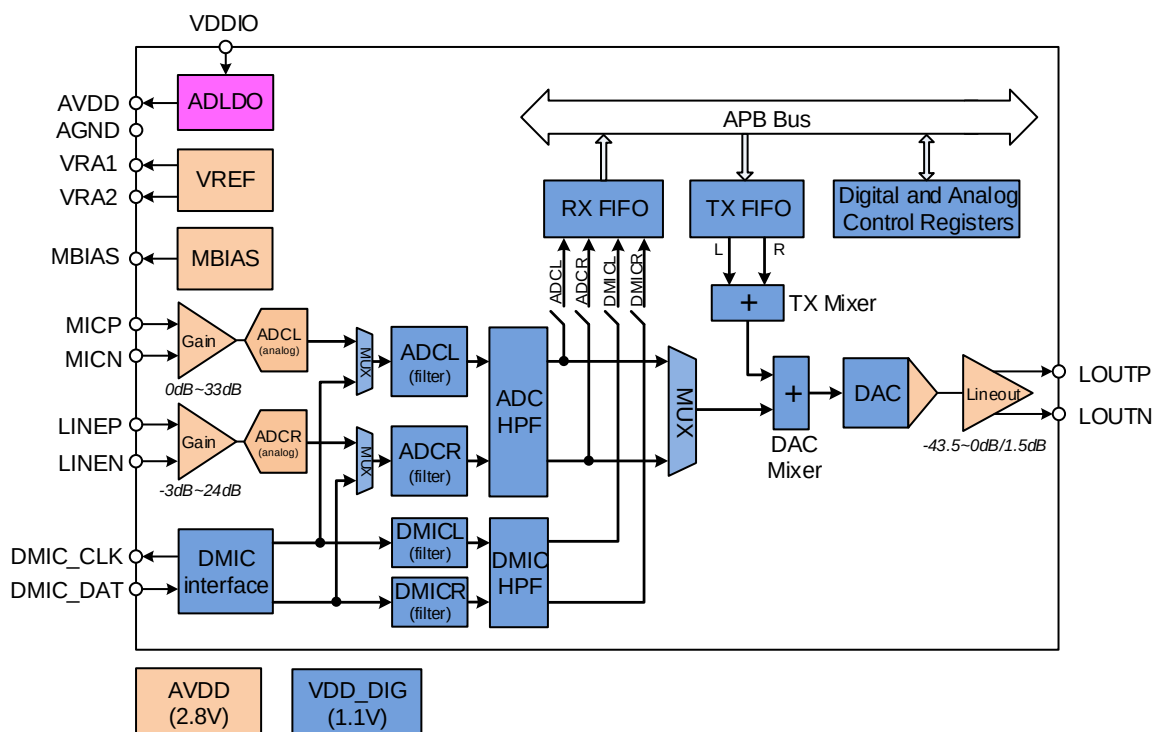


Figure 5-8 Audio Codec Block Diagram

5.2.3 Power and Signal Description

● Analog I/O Pins

Name	Type	Pin	Description
MICP	I	Yes	Positive differential input for MIC
MICN	I	Yes	Negative differential input for MIC

LINEP	I	Yes	Positive differential input for Linein
LINEN	I	Yes	Negative differential input for Linein
LOUTP	O	Yes	Left single-end output for lineout
LOUTN	O	Yes	Right single-end output for lineout

● Reference

Name	Type	Pin	Description
MBIAS	O	Yes	First bias voltage output for main microphone
VRA1	O	Yes	Internal reference voltage
VRA2	O	Yes	Internal reference voltage
VRP	O	To AVCC	Internal reference voltage

● Power/Ground

Name	Type	Pin	Description
AVDD	P	Yes	Analog power 2.8V
AGND	G	Yes	Analog ground

5.2.4 Operations and Functional Descriptions

5.2.4.1 Clock Sources

Figure 1-3 describes the Audio Codec clock source. Users can see **Clock Controller Management Unit(CCMU)** for clock setting, configuration and gating information.

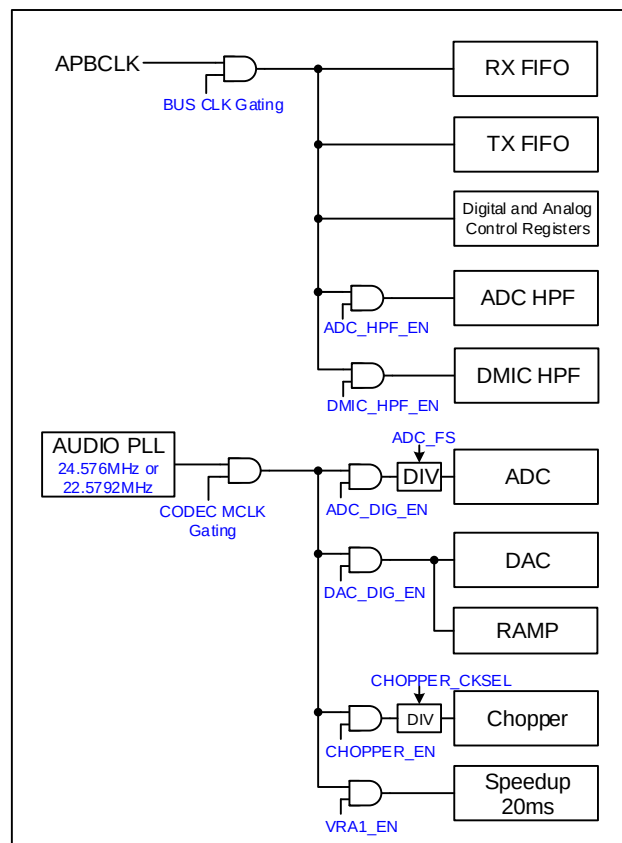


Figure 5-9 Audio Codec clock Diagram

5.2.4.2 Reset System

The SYS_RST will be provided by the VDD_SYS domain, which is come from VDD_SYS domain and produced by RTC domain which controlled by CCU when power on. Each domain has the de-bounce to confirm the reset system is strong. The codec register part, MIX will be reset by the SYS_RST during the power on or the system soft writing the reset control logic. The other parts will be reset by the soft configure through writing register.

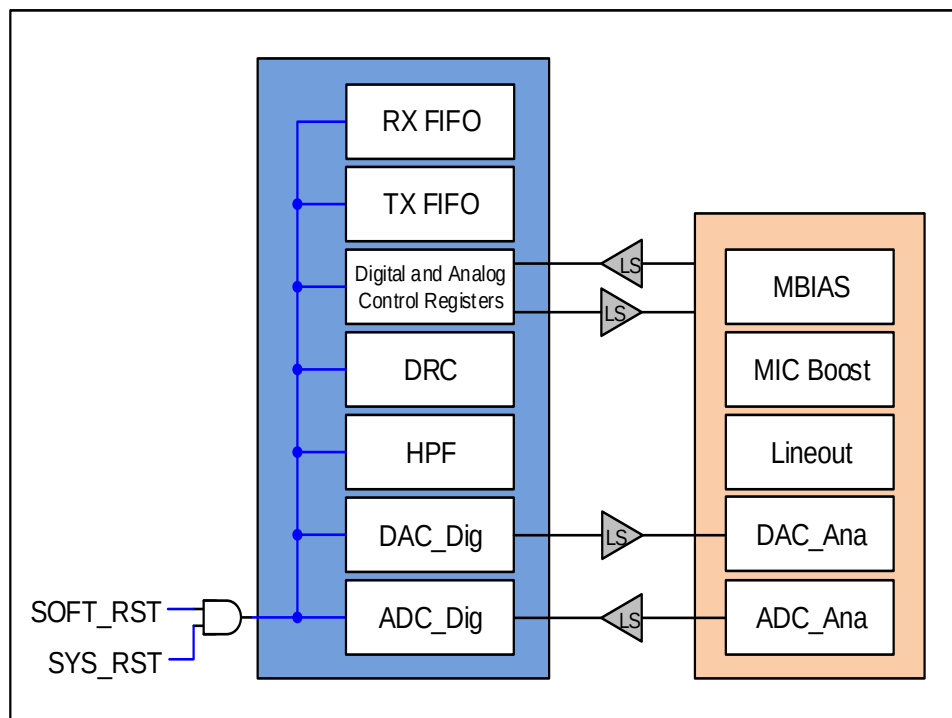


Figure 5-10 Audio Codec Reset System

When power on AVCC, which will send the AVCC_POR signal. It will through the level shift and RC filter part and send to ADDA Logic Core, which will reset the AVCC Analog part

5.2.4.3 Mono ADC

The three ADC is used for recording stereo sound and a reference signal. The sample rate of the three ADC is independent of DAC sample rate. In order to save power, the left and right analog ADC part can be enabled/disabled separately by setting register AC_ADC_CTRL Bit7:5. The digital ADC part can be enabled/disabled by AC_ADC_FIFOC Bit28.

5.2.4.4 Mono DAC

The stereo DAC sample rate can be configured by setting the register. In order to save power, the left and right DAC can be enabled/disabled separately by setting register DAC_PA_SRC Bit7:6. The digital DAC part can be enabled/ disabled by AC_DAC_DPC Bit31.

5.2.4.5 Analog Audio Input Port

The Codec supports one analogue audio Input path and one analogue audio output path :

- MICINP/N
- LINEOUT

● Microphone Input

MICINP/N provide differential input that can be mixed into the ADC record mixer,. MICIN is high impedance, low capacitance input suitable for connection to a wide range of differential microphones of different dynamics and sensitive. The gain for each pre-amplifier can be set independently. MBIAS provide reference voltage for electret condenser type(ECM) microphones. The MICBIAS output provides a low noise reference voltage suitable for biasing electrets type microphones and the associated external resistor biasing network. Refer to the Applications Information section for recommended external components.

● LINEOUT

LINEOUT provides one stereo output to drive line level signals to external audio equipment .The LINEOUTL output source can be selected from left output mixer or (left+right) output mixer. The LINEOUTR output source can be selected from right output mixer or left output mixer differential output. The volume control is logarithmic with a 43.5dB rang in 1.5dB step from -43.5dB to 0dB. The LINEOUT output buffer power up or down by register LINEOUT_ MIC2 _CTRL Bit7:6 .

5.2.4.6Interrupt

The Codec has two interrupt. Figure 1-6 describes the Audio Codec Interrupt System.

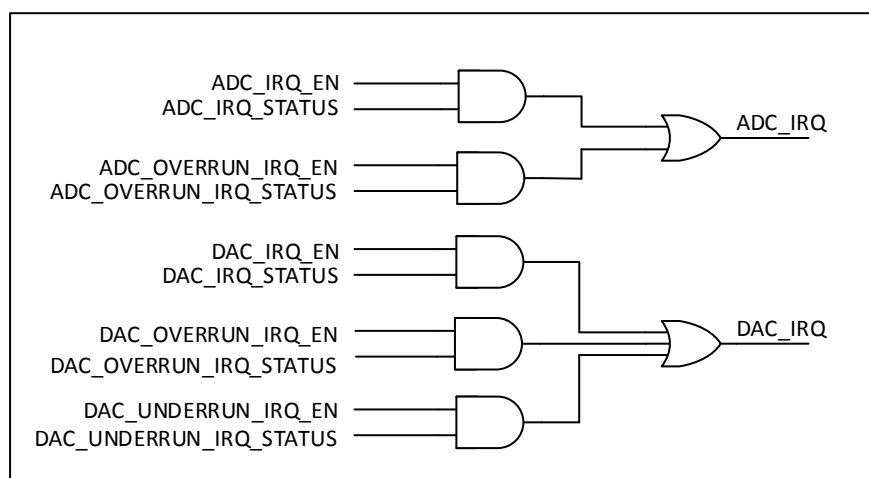


Figure 5-11 Interrupt System

5.2.4.7HPF Function

The ADC has individual channel high pass filter (HPF, -3dB cutoff < 1Hz) that can be enable and disable. The filter cutoff frequency is less than 1 Hz that can be remove DC offset from ADC recording. The HPF can also be bypassed.

5.2.5 Audio Codec Register List

Module Name	Base Address	
Audio Codec	0x40044000	
Register Name	Offset Address	Description
AC_DAC_DIG_CTRL	0x00	DAC Digital Part Control Register
AC_DAC_FIFO_CTRL	0x04	DAC FIFO Control Register
AC_DAC_FIFO_STA	0x08	DAC FIFO Status Register
AC_DAC_TXDATA	0x0C	DAC TX DATA Register
AC_DAC_TXCNT	0x10	DAC TX FIFO Counter Register
AC_DAC_DEBUG	0x14	DAC Debug Register
AC_ADC_FIFO_CTRL	0x18	ADC FIFO Control Register
AC_ADC_FIFO_STA	0x1C	ADC FIFO Status Register
AC_ADC_RXDATA	0x20	ADC RX Data Register
AC_ADC_RXCNT	0x24	ADC RX Counter Register
AC_ADC_HPF_CTRL	0x28	ADC HPF Control Register
AC_ADC_HPF_COEF	0x2C	ADC HPF Coef Register
AC_ADC_HPF_GAIN	0x30	ADC HPF Gain Coef Register
AC_ADDA_DBG_CTRL	0x34	ADDA Debug Control Register
AC_POWER_CTRL	0x38	Audio Codec Power Control Register
AC_MBIAS_CTRL	0x3C	MBIAS Control Register
AC_ADC_ANA_CTRL	0x40	ADC Analog Control Register
AC_DAC_ANA_CTRL	0x44	DAC Analog Control Register

5.2.6 Audio Codec Register Description

5.2.6.1 DAC Digital Part Control Register

Address: 0x00			Register Name: AC_DAC_DIG_CTRL Default Value : 0x0000_0003
Field	Type	Default	Description
31	R/W	0x0	DAC_DIG_EN. DAC Digital Part Enable 0: Disable 1: Enable
30:29	/	/	/
28:25	R/W	0x0	MODQU. Internal DAC Quantization Levels Levels= $[7 * (21 + \text{MODQU}[3:0])]/128$ Default levels= $7 * 21/128 = 1.15$
24	R/W	0x0	DWA. DWA Function Disable 0: Enable 1: Disable
23:19	/	/	/
18	R/W	0x0	HPF_EN. High Pass Filter Enable 0: Disable 1: Enable
17:12	R/W	0x0	DIG_VOL. Digital volume control: DVC, ATT=DVC[5:0]*(-1.16dB) 64 steps, -1.16dB/step
11:4	/	/	/
3:2	R/W	0x0	DAC_MIX_CTRL 00: TX_MIX_OUT 01: ADC_DAT 10: TX_Mixer_OUT+ADC_DAT 11: (TX_Mixer_OUT+ADC_DAT)/2
1:0	R/W	0x3	TX_MIX_CTRL 00: TXL 01: TXR 10: TXL+TXR 11: (TXL+TXR)/2

5.2.6.2 DAC FIFO Control Register

Address: 0x04			Register Name: AC_DAC_FIFO_CTRL Default Value : 0x0000_4000
Field	Type	Default	Description
31:29	R/W	0x0	DAC_FS. Sample Rate of DAC (Audio PLL=24.576MHz) 000: 48KHz 001: 32KHz 010: 24KHz 011: 16KHz 100: 12KHz 101: 8KHz 110: 192KHz 111: 96KHz Sample Rate of DAC (Audio PLL=22.5792MHz) 000: 44.1KHz 001: Reserved 010: 22.05KHz 011: Reserved 100: 11.025KHz 101: Reserved 110: Reserved 111: Reserved
28	R/W	0	FIR_VER. FIR Version 0: 64-Tap FIR 1: 32-Tap FIR
27	/	/	/
26	R/W	0x0	SEND_LASAT. Audio sample select when TX FIFO underrun 0: Sending zero 1: Sending last audio sample
25	/	/	/
24	R/W	0x0	FIFO_MODE. For 24-bits transmitted audio sample: 0: FIFO_I[23:0] = {TXDATA[31:8]} 1: Reserved For 16-bits transmitted audio sample: 0: FIFO_I[23:0] = {TXDATA[15:0], 8'b0} 1: FIFO_I[23:0] = {TXDATA[31:16], 8'b0}
23	/	/	/

22:21	R/W	0x0	DAC_DRQ_CLR_CNT. When TX FIFO available room less than or equal N, DRQ Request will be de-asserted. N is defined here: 00: IRQ/DRQ De-asserted when WLEVEL > TXTL 01: 4 10: 8 11: 16
20:15	/	/	/
14:8	R/W	0x40	TX_TRIG_LEVEL. TX FIFO Empty Trigger Level (TXTL[6:0]) Interrupt and DMA request trigger level for TX FIFO normal condition. IRQ/DRQ Generated when WLEVEL ≤ TXTL Notes: WLEVEL represents the number of valid samples in the TX FIFO Only TXTL[6:0] valid when TXMODE = 0
7	/	/	/
6	R/W	0x0	DAC_MONO_EN. DAC Mono Enable 0: Stereo, 64 levels FIFO 1: mono, 128 levels FIFO When enabled, L & R channel send same data
5	R/W	0x0	TX_SAMPLE_BITS. Transmitting Audio Sample Resolution 0: 16 bits 1: 24 bits
4	R/W	0x0	DAC_DRQ_EN. DAC FIFO Empty DRQ Enable 0: Disable 1: Enable
3	R/W	0x0	DAC_IRQ_EN. DAC FIFO Empty IRQ Enable 0: Disable 1: Enable
2	R/W	0x0	FIFO_UNDERRUN_IRQ_EN. DAC FIFO Under Run IRQ Enable 0: Disable 1: Enable
1	R/W	0x0	FIFO_OVERRUN_IRQ_EN. DAC FIFO Over Run IRQ Enable 0: Disable 1: Enable
0	R/WC	0x0	FIFO_FLUSH.

			DAC FIFO Flush Write '1' to flush TX FIFO, self clear to '0'
--	--	--	---

5.2.6.3 DAC FIFO Status Register

Address: 0x08			Register Name: AC_DAC_FIFO_STA Default Value : 0x0080_8008
Field	Type	Default	Description
31:24	/	/	/
23	R	0x1	TX_EMPTY. TX FIFO Empty 0: More than one sample point in TX FIFO (>= 1 word) 1: No anyone sample point in TX FIFO
22:16	/	/	/
15:8	R	0x80	TXE_CNT. TX FIFO Empty Space Word Counter
7:4	/	/	/
3	R/WC	0x1	TXE_INT. TX FIFO Empty Pending Interrupt 0: No Pending IRQ 1: FIFO Empty Pending Interrupt Write '1' to clear this interrupt or automatic clear if interrupt condition fails.
2	R/WC	0x0	TXU_INT. TX FIFO Underrun Pending Interrupt 0: No Pending Interrupt 1: FIFO Underrun Pending Interrupt Write '1' to clear this interrupt
1	R/WC	0x0	TXO_INT. TX FIFO Overrun Pending Interrupt 0: No Pending Interrupt 1: FIFO Overrun Pending Interrupt Write '1' to clear this interrupt
0	/	/	/

5.2.6.4 DAC TX DATA Register

Address: 0x0C			Register Name: AC_DAC_TXDATA Default Value : 0x0000_0000
Field	Type	Default	Description
31:0	W	0x0	TX_DATA.

			Transmitting left, right channel sample data should be written this register one by one. The left channel sample data is first and then the right channel sample.
--	--	--	---

5.2.6.5 DAC TX Counter Register

Address: 0x10			Register Name: AC_DAC_TXCNT Default Value : 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	TX_CNT. TX Sample Counter The audio sample number of sending into TXFIFO. When one sample is put into TXFIFO by DMA or by host IO, the TX sample counter register increases by one. The TX sample counter register can be set to any initial value at any time. After been updated by the initial value, the counter register should count on base of this initial value. Notes: It is used for Audio/ Video Synchronization

5.2.6.6 DAC Debug Register

Address: 0x14			Register Name: AC_DAC_DEBUG Default Value : 0x0000_0000
Field	Type	Default	Description
31:12	/	/	/
11	R/W	0x0	DAC_MODU_SELECT. DAC Modulator Debug 0: DAC Modulator Normal Mode 1: DAC Modulator Debug Mode
10:9	R/W	0x0	DAC_PATTERN_SELECT. DAC Pattern Select 00: Normal (Audio Sample from TX FIFO) 01: -6 dB Sin wave 10: -60 dB Sin wave 11: silent wave(zero data)
8	R/W	0x0	CODEC_CLK_SELECT. CODEC Clock Source Select: cksel 0: CODEC Clock from Audio PLL 1: CODEC Clock from OSC (for Debug)
7:1	/	/	/
0	R/W	0x0	ADDA_LOOP_MODE ADDA Loop MODE SELECT

			0: Disable 1: ADDA LOOP MODE DAC connect to ADC
--	--	--	--

5.2.6.7 ADC FIFO Control Register

Address: 0x18			Register Name: AC_ADC_FIFO_CTRL Default Value : 0x0000_0400
Field	Type	Default	Description
31:29	R/W	0x0	ADC_FS. Sample Rate of ADC (Audio PLL=24.576MHz) 000: 48KHz 001: 32KHz 010: 24KHz 011: 16KHz 100: 12KHz 101: 8KHz 110: Reserved 111: Reserved Sample Rate of ADC (Audio PLL=22.5792MHz) 000: 44.1KHz 001: Reserved 010: 22.05KHz 011: Reserved 100: 11.025KHz 101: Reserved 110: Reserved 111: Reserved
28	R/W	0x0	ADC_DIG_EN. ADC Digital Part Enable 0: Disable 1: Enable
27	R/W	0x0	DMIC_DIG_EN. DMIC Digital Part Enable 0: Disable 1: Enable
26:25	R/W	0x0	ADCFDT ADC FIFO Delay Time For writing Data after ADC_DIG_EN 00:5ms 01:10ms 10:20ms 11:30ms
24	R/W	0x0	ADC_DFEN

			ADC FIFO Delay Function For writing Data after EN_AD 0: Disable 1: Enable
23	R/W	0x0	RX_FIFO_MODE. RX FIFO Output Mode (Mode 0, 1) 0: Expanding '0' at LSB of TX FIFO register 1: Expanding received sample sign bit at MSB of TX FIFO register For 24-bits received audio sample: Mode 0: RXDATA[31:0] = {FIFO_O[23:0], 8'h0} Mode 1: Reserved For 16-bits received audio sample: Mode 0: RXDATA[31:0] = {16{FIFO_O[23]}, FIFO_O[23:8]} Mode 1: RXDATA[31:0] = {FIFO_O[23:8], 16'h0}
22	R/W	0x0	ADC_DMIC_EN ADC DMIC Enable 0: Disable 1: Enable
21	R/W	0x0	ADC_DMIC_MONO_EN ADC DMIC Mono Mode Enable 0: Disable 1: Enable
20:18	/	/	/
17	R/W	0x0	RX_SAMPLE_BITS. Receiving Audio Sample Resolution 0: 16-bit 1: 24-bit
16	R/W	0x0	A2D_CHSEL 0: ADC Left channel 1: ADC Right channel
15	R/W	0x0	AD_SWP. ADC output channel swap enable (for digital filter) 0: disable 1: enable
14	R/W	0x0	DMICR_EN DMIC Right channel enable 0: Disable 1: Enable
13	R/W	0x0	DMICL_EN DMIC Left channel enable 0: Disable 1: Enable
12	R/W	0x0	ADCR_EN

			ADC Right channel enable 0: Disable 1: Enable
11	R/W	0x0	ADCL_EN ADCL Left channel enable 0: Disable 1: Enable
10:4	R/W	0x40	RX_FIFO_TRG_LEVEL. RX FIFO Trigger Level (RXTL[6:0]) Interrupt and DMA request trigger level for RX FIFO normal condition IRQ/DRQ Generated when WLEVEL > RXTL[6:0] Notes: WLEVEL represents the number of valid samples in the RX FIFO
3	R/W	0x0	ADC_DRQ_EN. ADC FIFO Data Available DRQ Enable. 0: Disable 1: Enable
2	R/W	0x0	ADC_IRQ_EN. ADC FIFO Data Available IRQ Enable. 0: Disable 1: Enable
1	R/W	0x0	ADC_OVERRUN_IRQ_EN. ADC FIFO Over Run IRQ Enable 0: Disable 1: Enable
0	R/WC	0x0	ADC_FIFO_FLUSH. ADC FIFO Flush. Write '1' to flush TX FIFO, self clear to '0'.

5.2.6.8 ADC FIFO Status Register

Address: 0x1C			Register Name: AC_ADC_FIFO_STA Default Value : 0x0080_0000
Field	Type	Default	Description
31:24	/	/	/
23	R	0x1	RX_EMPTY. RX FIFO Empty 0: More than one sample point in RX FIFO (>= 1 word) 1: No anyone sample point in RX FIFO
22:16	/	/	/
15:8	R	0x0	RXA_CNT. RX FIFO Available Sample Word Counter

7:4	/	/	/
3	R/WC	0x0	RXA_INT. RX FIFO Data Available Pending Interrupt 0: No Pending IRQ 1: Data Available Pending IRQ Write '1' to clear this interrupt or automatic clear if interrupt condition fails.
2	/	/	/
1	R/WC	0x0	RXO_INT. RX FIFO Overrun Pending Interrupt 0: No Pending IRQ 1: FIFO Overrun Pending IRQ Write '1' to clear this interrupt
0	/	/	/

5.2.6.9 ADC RX DATA Register

Address: 0x20			Register Name: AC_ADC_RXDATA Default Value : 0x0000_0000
Field	Type	Default	Description
31:0	R	0x0	RX_DATA. RX Sample Host can get one sample by reading this register. The left channel sample data is first and then the right channel sample.

5.2.6.10 ADC RX Counter Register

Address: 0x24			Register Name: AC_ADC_RXCNT Default Value : 0x0000_0000
Field	Type	Default	Description
31:0	R/W	0x0	RX_CNT. RX Sample Counter The audio sample number of writing into RXFIFO. When one sample is written by Digital Audio Engine, the RX sample counter register increases by one. The RX sample counter register can be set to any initial value at any time. After been updated by the initial value, the counter register should count on base of this initial value. Notes: It is used for Audio/ Video Synchronization

5.2.6.11 ADC HPF Control Register

Address: 0x28			Register Name: AC_ADC_HPF_CTRL Default Value : 0x0000_0000
Field	Type	Default	Description
31:2	/	/	/
1	R/W	0x0	DMIC_HPF_EN DMIC HPF enable 0:disable 1:enable
0	R/W	0x0	ADC_HPF_EN ADC HPF enable 0:disable 1:enable

5.2.6.12 ADC HPF Coef Register

Address: 0x2C			Register Name: AC_ADC_HPF_COEF Default Value: 0x00FF_FAC1
Field	Type	Default	Description
31:27	/	/	/
26:0	R/W	0x0FF FAC1	HPF coefficient setting and the data is 3.24 format.

5.2.6.13 ADC HPF Gain Coef Register

Address: 0x30			Register Name: AC_ADC_HPF_GAIN Default Value: 0x0100_0000
Field	Type	Default	Description
31:27	/	/	/
26:0	R/W	0x1000000	The gain of the HPF coefficient setting which format is 3.24.(gain = 1)

5.2.6.14 ADDA Debug Control Register

Address: 0x34			Register Name: AC_ADDA_DBG_CTL Default Value: 0x0000_0000
Field	Type	Default	Description
31:16	W	0x0	ADDA Debug Key Write '0x1825' to enable ADDA Debug enable to be writable
15:5	/	/	/
4	R/W	0x0	ADC_PTN_SEL

			ADC Pattern Select 0: Normal 1: 0x5A5A5A/0x12345(24-bit)
3	R/W	0x0	ADDA Debug enable 0: Disable 1: Enable
2	R/W	0x0	ADC Debug select 0: ADC analog debug(sdm_clk input, sdm_dat output) 1: ADC digital debug(sdm_clk output, sdm_dat input)
1	R/W	0x0	DAC Debug select 0: DAC analog debug(sdm_clk and sdm_dat input) 1: DAC digital debug(sdm_clk and sdm_dat output)
0	/	/	/

5.2.6.15 Audio Codec Power Control Register

Address: 0x38			Register Name: AC_POWER_CTRL Default Value: 0x0030_002A
Field	Type	Default	Description
31:23	/	/	/
22:20	R/W	0x3	Ramp_time 000: 3us 001: 6us 010: 12us 011: 24us 100: 48us 101: 60us 110: 96us 111: 120us
19	R	0x0	BG_SPEEDUP_STA (delay 20ms after VRA1_EN) 0: Speed up not done 1: Speed up done
18	R/W	0x0	ADLDO_EN ADLDO enable 0: Disable 1: Enable
17:15	R/W	0x0	ADLDO_VCTRL ADLDO Output Voltage Control When Vref=1.4V: 000: 2.8V 001: 2.5V 010: 2.575V

			011: 2.65V 100: 2.725V 101: 2.875V 110: 2.95V 111: 3.025V When Vref=0.9V: 000: 1.8V 001: 1.6V 010: 1.65V 011: 1.7V 100: 1.75V 101: 1.85V 110: 1.9V 111: 1.95V
14	R/W	0x0	ADLDO_BYPASS adldo bypass mode enable 0: adldo in normal mode 1: adldo in bypass mode
13	R/W	0x0	VRA1_EN (BG_EN) Enable VRA1 0: Disable 1: Enable
12:11	R/W	0x0	VRA1_VCTRL (BG_VSEL) VRA1 Output Voltage Control 00: vref=1.4V 01: vref=1.25V 10: vref=1.1V 11: vref=0.9V
10:7	R/W	0x0	BG_TRIM Trim of Output Voltage of BG 0000: Min 1111: Max
6:5	R/W	0x1	IOPVRS OPVR bias control 00: Min 11: Max
4:3	R/W	0x1	IOPDACS DAC OP bias control 00: Min 11: Max
2:1	R/W	0x1	ILINEOUTS LINEOUT OP bias control

			00: Min 11: Max
0	R/W	0x0	RSWITCH 0: OUTPUT of RAMP DAC 1: VRA1

5.2.6.16 MBIAS Control Register

Address: 0x3C			Register Name: AC_MBIAS_CTRL Default Value: 0x0000_0009
Field	Type	Default	Description
31:6	/	/	/
5	R/W	0x0	MBIAS_CHOPPER_EN MICBIAS CHOPPER Enable 0: Disable 1: Enable
4:3	R/W	0x1	MBIAS_CHOPPER_CKSEL MICBIAS CHOPPER Clock select 00: 250KHz 01: 500KHz 10: 1MHz 11: 2MHz
2	R/W	0x0	MBIAS_EN MICBIAS Enable 0: Disable 1: Enable
1:0	R/W	0x1	MBIAS_VCTRL MBIAS Output Voltage Control 00: 1.88V 01: 2.09V 10: 2.33V 11: 2.50V

5.2.6.17 ADC Analog Control Register

Address: 0x40			Register Name: AC_ADC_ANA_CTRL Default Value: 0x4455_4055
Field	Type	Default	Description
31:30	R/W	0x1	DITH_AMP 00: min 11: max

29	R/W	0x0	LOUT_LOOPB_EN Lineout(p/n) loop back to Linein(p/n) 0: disconnect 1: connect to linein
28:26	R/W	0x1	LINEIN_GAIN OPAMP gain control 000: -3dB 001: 0dB 010: 3dB 011: 6dB 100: 9dB 101: 12dB 110: 18dB 111: 24dB
25	R/W	0x0	LINEIN_OPEN OPAMP enable 0: Disable 1: Enable
24	R/W	0x0	ADCR_ANA_EN ADC Right Channel Analog Part Enable 0: Disable 1: Enable
23:22	R/W	0x1	IOPADC1S ADC OP bias control1 00: Min 11: Max
21:20	R/W	0x1	IOPADC2S ADC OP bias control2 00: Min 11: Max
19:18	R/W	0x1	IOPMICS MICAMP bias control1 00: Min 11: Max
17:16	R/W	0x1	IOPMIXS MICAMP bias control2 00: Min 11: Max
15	/	/	/
14	R/W	0x1	DITHER_EN Dither enable 0: Disable

			1: Enable
13	R/W	0x0	MIC_MODE MIC Input Mode Select 0: Differential 1: Single-ended
12:10	R/W	0x0	MIC_PGA_GAIN MICAMP gain control 000: 0dB 001: 21dB 010: 24dB 011: 27dB 100: 30dB 101: 33dB 110: 36dB 111: 39dB
9	R/W	0x0	MIC_PGA_EN MICAMP enable 0: Disable 1: Enable
8	R/W	0x0	ADCL_ANA_EN ADC Left Channel Analog Part enable 0: Disable 1: Enable
7:6	R/W	0x1	IOPADC1S ADC OP bias control1 00: Min 11: Max
5:4	R/W	0x1	IOPADC2S ADC OP bias control2 00: Min 11: Max
3:2	R/W	0x1	IOPMICS MICAMP bias control1 00: Min 11: Max
1:0	R/W	0x1	IOPMIXS MICAMP bias control2 00: Min 11: Max

5.2.6.18 DAC Analog Control Register

Address: 0x44			Register Name: AC_DAC_ANA_CTRL Default Value: 0x0000_005F
Field	Type	Default	Description
31:18	/	/	/
17	R/W	0x0	RAMP_MAN_EN 0: Disable 1: Enable
16	R/W	0x0	PLAY_ANA_EN Playback Path Analog Part enable(DAC Analog+Lineout+Antipop) 0: Disable 1: Enable
15	R/W	0x0	DAC_ANA_EN DAC Analog enable 0: Disable 1: Enable
14:10	/	/	/
9	R/W	0x0	ADDA_BIAS_EN 0: Disable 1: Enable
8	R/W	0x0	RAMP_EN RAMP enable 0: Disable 1: Enable
7	R/W	0x0	LOUT_EN Lineout enable 0: Disable 1: Enable
6	R/W	0x1	LOUT_DIFF_EN Lineout Differential Mode Enable 0: Single-ended mode 1: Differential mode
5	R/W	0x0	LOUT_MUTE Lineout Mute control 0:Mute 1:Normal
4:0	R/W	0x1F	LOUT_GAIN Lineout OP Gain control 00000: Mute 00001: Mute 00010: -43.5dB (Min)

			00011: -42.0dB 11110: -1.5dB 11111: 0dB (Max) LINEOUT Volume control, Total 30level from 0x1F to 0x02 with the volume 0dB to -43.5dB, -1.5dB/step. Mute when 00000 &00001
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Chapter 6 Camera

6.1 Camera Serial Interface (CSI)

6.1.1 Overview

The Camera Serial Interface (CSI) is a parallel image input interface which can receive image or video data via digital camera(DC)interface, CCIR656 interface.

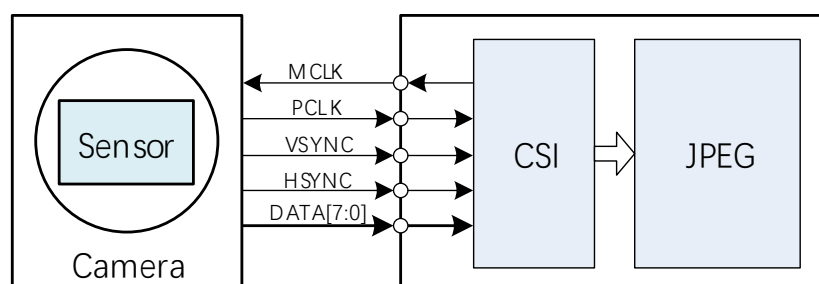


Figure 6-1 CSI Hardware Interface

It includes the following features:

- 8 bits input data
- Support digital camera(DC) interface and CCIR656 interface
- Pass raw data direct to memory
- Pass jpeg data direct to memory
- Pass YUV420 data to JPEG
- Convert YUV422 data to YUV420
- support X/Y scale 1/2 and cropwin

6.1.2 Block Diagram

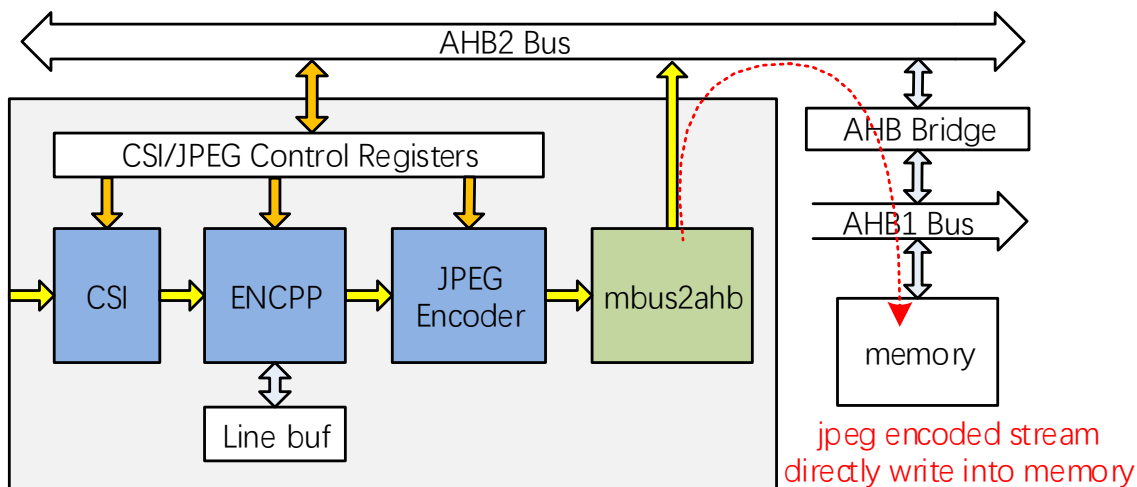
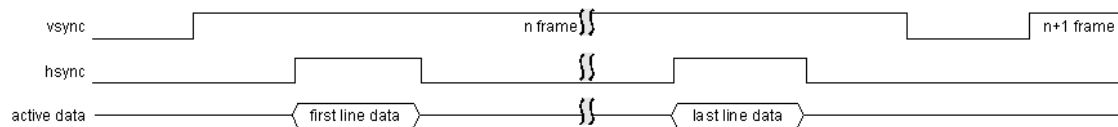
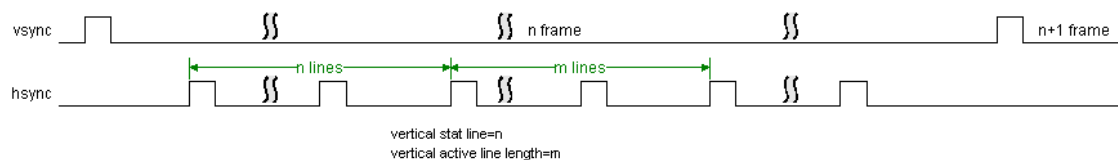


Figure 6-2 CSI Block Diagram

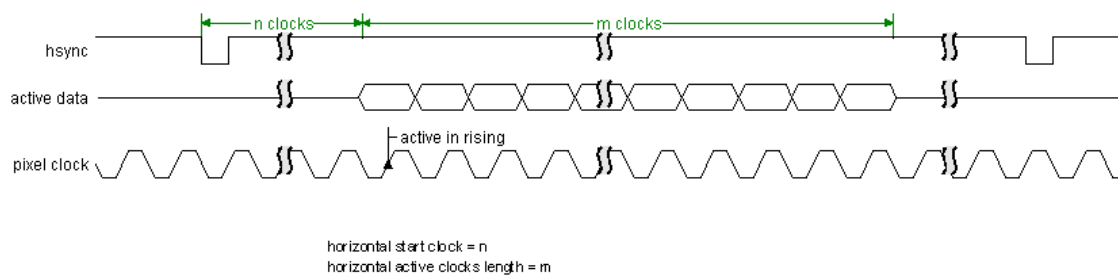
6.1.3 CSI Timing



Vref= positive; Href= positive



Vertical size setting



Horizontal size setting and pixel clock timing (Href= positive)

6.1.4 Register List

Module Name	Base Address	
CSI Controller	0x40007000	
Register Name	Offset Address	Description
CSI_EN_REG	0x0000	CSI Enable Register
CSI_CFG_REG	0x0004	CSI Configuration Register
CSI_CAP_REG	0x0008	CSI Capture Register
CSI_SIGNAL_STA	0x000C	CSI Signal Status Register
CSI_OUT_HOR_SIZE	0x0010	CSI Output Horizontal Size Register
CSI_OUT_VER_SIZE	0x0014	CSI Output Vertical Size Register
CSI_INPUT_SIZE	0x0018	CSI Input Size Register
CSI_INT_EN_REG	0x001C	CSI Interrupt Enable Register
CSI_INT_STA_REG	0x0020	CSI Interrupt Status Register
CSI_EN_REG	0x0000	CSI Enable Register
CSI_CFG_REG	0x0004	CSI Configuration Register
CSI_CAP_REG	0x0008	CSI Capture Register

6.1.5 Register Description

6.1.5.1 CSI Enable Register

Address : 0x40007000 + 0x0000			Name: CSI_EN_REG Default: 0x0000 0000
Field	Type	Default	Description
31:12	R/W	0x0	TIMEOUT_THD CSI Timeout threshold
11:5	/	/	/
4	R/W	0x0	TIMEOUT_EN CSI Timeout enable 0:disable 1:enable
3	/	/	/
2	R/W	0x0	PCLK_EN 0:Gate pclk input 1:Enable pclk input
1	R/W	0x0	CSIC_EN 0: Reset and disable the CSIC module 1: Enable the CSIC module
0	R/W	0x0	PRS_EN 0: Reset and disable the parser module

			1: Enable the parser module
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6.1.5.2 CSI Configuration Register

Address : 0x0004			Name: CSI_CFG_REG Default: 0x0000 0a8a
Field	Type	Default	Description
31:13	/	/	/
12	R/W	0x0	YUV420 Mask Valid when Output_Mode set 'YUV422 to YUV420' 0: mask UV in odd lines(1,3,5.....) 1:mask UV in even lines(2,4,6.....)
11	R/W	0x1	Output_Mode 0: original output 1:YUV422 to YUV420
10	R/W	0x0	YUV420_LINE_ORDER 0: YUV420 input in Y-YC-Y-YC Line Order 1: YUV420 input in YC-Y-YC-Y Line Order
9:8	R/W	0x2	INPUT_SEQ Input data sequence, only valid for YUV422 and YUV420 input format. 00: YUYV 01: YVYU 10: UYVY 11: VYUY
7:6	R/W	0x2	INPUT_FMT Input data format 0x: RAW stream 10: YUV422 11: YUV420
5:4	/	/	/
3	R/W	0x1	VREF_POL Vref polarity 0: negative 1: positive This register is not apply to CCIR656 interface.
2	R/W	0x0	HERF_POL Href polarity 0: negative 1: positive This register is not apply to CCIR656 interface.
1	R/W	0x1	CLK_POL Data clock type

			0: active in rising edge 1: active in falling edge
0	R/W	0x0	CSI_IF 0: separate syncs(DC) 1: embedded syncs(CCIR656)

6.1.5.3 CSI Capture Register

Address : 0x0008			Name: CSI_CAP_REG Default: 0x0000 0000
Field	Type	Default	Description
31:7	/	/	/
6	R/W	0x0	CSI_HMASK Frame rate half down.
5:2	R/W	0x0	CSI_CAP_MASK Vsync number masked before capture.
1	R/W	0x0	CSI_VCAP_ON Video capture control: Capture the video image data stream. 0: Disable video capture If video capture is in progress, the CSI stops capturing image data at the end of the current frame, and all of the current frame data is wrote to output FIFO. 1: Enable video capture The CSI starts capturing image data at the start of the next frame.
0	RC/W	0x0	CSI_SCAP_ON Still capture control: Capture a single still image frame. 0: Disable still capture. 1: Enable still capture The CSI module starts capturing image data at the start of the next frame. The CSI module captures only one frame of image data. This bit is self clearing and always reads as a 0.

6.1.5.4 CSI Signal Status Register

Address : 0x000C			Name: CSI_SIGNAL_STA Default: 0x0000 0000
Field	Type	Default	Description
31:23	/	/	/
22:20	R	0x0	PCLK_CNT_STA Indicates the pclk status
19	R	0x0	VSYNC_STA Indicates the Vsync status 0:low

			1:high
18	R	0x0	HSYNC_STA Indicates the Hsync status 0:low 1:high
17:8	/	/	/
7:0	R	0x0	DATA_STA Indicates the Dn status(n=0~7),MSB for D7,LSB for D0 0:low 1:high

6.1.5.5 CSI Output Horizontal Size Register

Address : 0x0010			Name: CSI_OUT_HOR_SIZE Default: 0x0500 0000
Field	Type	Default	Description
31:30	/	/	/
29:16	R/W	0x500	HOR_LEN Horizontal pixel unit length. Valid pixel of a line.
15:14	/	/	/
13:0	R/W	0x0	HOR_START Horizontal pixel unit start. Pixel is valid from this pixel.

6.1.5.6 CSI Output Vertical Size Register

Address : 0x0014			Name: CSI_OUT_VER_SIZE Default: 0x02d0 0000
Field	Type	Default	Description
31:30	/	/	/
29:16	R/W	0x2d0	VER_LEN Valid line number of a frame.
15:14	/	/	/
13:0	R/W	0x0	VER_START Vertical line start. data is valid from this line.

6.1.5.7 CSI Input Size Register

Address : 0x0018			Name: CSI_INPUT_SIZE Default: 0x0000 0000
Field	Type	Default	Description
31	/	/	/
30:16	R	0x0	INPUT_Y_SIZE
15	/	/	/

14:0	R	0x0	INPUT_X_SIZE
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6.1.5.8 CSI Interrupt Enable Register

Address : 0x001C			Name: CSI_INT_EN_REG Default: 0x0000 0000
Field	Type	Default	Description
31:3	/	/	/
2	R/W	0x0	INPUT_TIMEOUT_INT_EN Input timeout interrupt enable 0:disable 1:enable
1	R/W	0x0	INPUT_FRM_END_INT_EN Input frame end interrupt enable 0:disable 1:enable
0	R/W	0x0	INPUT_SIZE_CHG_INT_EN Input frame size(X or Y) change interrupt enable 0:disable 1:enable

6.1.5.9 CSI Interrupt Status Register

Address : 0x0020			Name: CSI_INT_STA_REG Default: 0x0000 0000
Field	Type	Default	Description
31:2	/	/	/
4	R/W1C	0x0	INPUT_Y_SIZE_CHG_PD CSI input Y size change pending When the csi input Y size changed,this flag set to "1". Write "1" to clear.
3	R/W1C	0x0	INPUT_X_SIZE_CHG_PD CSI input X size change pending When the csi input X size changed,this flag set to "1". Write "1" to clear.
2	R/W1C	0x0	INPUT_TIMEOUT_PD CSI timeout pending When no valid data input timeout,this flag set to "1". Write "1" to clear.
1	R/W1C	0x0	INPUT_FRM_END_PD CSI input frame end pending When the captured frame end,this flag set to "1". Write "1" to clear.
0	R/W1C	0x0	INPUT_SIZE_CHG_PD CSI input X or Y size change pending When the parser input size changed,this flag set to "1". Write "1" to clear.

6.2 Video Encoder (JPEG)

6.2.1 Overview

The Video Encoder only supports encoding the CSI data into a JPEG image.

It includes the following features:

- Supports JPEG base line
- Supports online encoding mode with CSI
- Supports nv12 input format in offline encoding mode
- Supports configurable picture resolutions
- Minimum picture resolution: 32x32
- Offline maximum picture resolution: 1920x1088
- Online maximum picture resolution: 1920x1088
- Supports online 640x480@60fps and online 1280x720@40fps
- Supports offline 640x480@60fps and offline 1280x720@20fps
- Supports bitrate control through quantization table

6.2.2 Register List

Module Name	Base Address	
VE_TOP	0x40007400	
Register Name	Offset Address	Description
VE_MODE_REG	0x0000	VE Mode Control Register
VE_RESET_REG	0x0004	VE Reset Control Register
VE_CNT_REG	0x0008	VE Counter Control Register
VE_OVERTIME_REG	0x000C	VE Overtime Register
VE_INT_STA_REG	0x001C	VE Interrupt Status Register
CSI_OUTPUT_ADDR_Y	0x0020	CSI Output Y Address Register
CSI_OUTPUT_ADDR_UV	0x0024	CSI Output UV Address Register
CSI_OUTPUT_STRIDE	0x0028	CSI Output Stride Control Register
Module Name	Base Address	
ENCPP	0x40007800	
Register Name	Offset Address	Description
JPEG_INPUT_SIZE	0x0000	JPEG Input Size Register
JPEG_INPUT_STRIDE0	0x0004	JPEG Input Stride0 Control Register

JPEG_INPUT_STRIDE1	0x0014	JPEG Input Stride1 Control Register
JPEG_INPUT_ADDR_Y	0x0078	JPEG Input Y Address Register
JPEG_INPUT_ADDR_UV	0x007C	JPEG Input UV Address Register
Module Name	Base Address	
JPEG	0x40007C00	
JPEG_PARA_REG	0x0004	JPEG Para Config Register
JPEG_BITRATE_CTRL	0x0008	JPEG Bitrate Control Register
VE_INT_EN_REG	0x0014	VE Interrupt Enable Register
VE_START_TRIG	0x0018	VE Start Trigger Register
VE_STA_REG	0x001C	VE Status Register
VE_PUTBITS_DATA	0x0020	VE Putbits Data Register
MBLEVEL_OVERTIME	0x0024	Macroblock Level Overtime Register
OUTSTM_START_ADDR	0x0080	Output Stream Start address Register
OUTSTM_END_ADDR	0x0084	Output Stream End address Register
OUTSTM_OFFSET	0x0088	Output Stream Offset Register
OUTSTM_VSIZE	0x008C	Output Stream Valid Size Register
OUTSTM_LEN	0x0090	Output Stream Length Register
VE_QMINPUT_INDEX	0x00E0	VE Quantiser Matrix Input Index Register
VE_QMINPUT_DATA	0x00E4	VE Quantiser Matrix Input Data Register

6.2.3 Register Description

6.2.3.1 VE Mode Control Register

Address : 0x40007400 + 0x0000			Name: VE_MODE_REG Default: 0x0000 0000
Field	Type	Default	Description
31:18	/	/	/
17	W	0x0	MEM_PARTITION_TAKE Output stream memory partition take away Software reading value always zero. Software write "1" to this bit after taking away one memory partition.
16	R/W	0x0	MEM_PARTITION_MODE Output stream memory partition mode enable 0: disable memory partition mode 1: enable memory partition mode
15:14	R/W	0x0	MEM_PARTITION_NUM Output stream memory partition number 00: 2 memory blocks 01: 4 memory blocks 10: 8 memory blocks

			11: Reserved
13	R/W	0x0	AHB_WRITE_BL_8W 0: encoder use burst 4, access 4 words each time; 1: encoder use burst 8, access 8 words each time; This bit influence bandwidth only. It does not have an effect on the hardware result.
12	R/W	0x0	AHB_READ_BL_8W 0: encoder use burst 4, access 4 words each time; 1: encoder use burst 8, access 8 words each time; This bit influence bandwidth only. It does not have an effect on the hardware result.
11	R/W	0x0	JPE_HEIGHT_HALF Used in online mode only. 0: JPEG encoder input picture height is the same with CSI input picture height; 1: JPEG encoder input picture height equals to half of CSI input picture height, CSI input picture height should align to 32.
10	R/W	0x0	JPE_WIDTH_HALF Used in online mode only. 0: JPEG encoder input picture width is the same with CSI input picture height; 1: JPEG encoder input picture width equals to half of CSI input picture height, CSI input picture width should align to 32.
9	R/W	0x0	ONLINE_MODE_EN 0: Off line mode; JPEG encoder need a whole frame as input; 1: On line mode; No need to set JPEG encoder input data buffer.
8	R/W	0x0	SENSOR_OUT_JPEG 0: CSI output data format is yuv420 data (NV12 format). 1: CSI output data is JPEG data. Used in offline mode. Some sensors have the ability to output JPEG picture data. In this case, driver should set this flag as 1 to inform CSI to process JPEG data.
7	R/W	0x0	ENC_CLK_EN Enable encoder top level clock. 0: disable 1: enable
6	R/W	0x0	JPE_CLK_EN Enable jpeg encoder level clock. 0: disable 1: enable
5	R/W	0x0	CLOCK_GATING_DISABLE This is a black door bit.

			0: Use JPE_ENABLE and ENC_ENABLE to control clock. 1: Enable top and jpeg encoder clock at the same time. The same function as ENC_CLK_EN equals one and JPE_CLK_EN equals one. ENC_CLK_EN and JPE_CLK_EN are invalid in this case.
4:0	/	/	/

6.2.3.2VE Reset Control Register

Address : 0x40007400 + 0x0004			Name: VE_RESET_REG Default: 0x0000 0000
Field	Type	Default	Description
31:1	/	/	/
0	R/W	0x0	VE_RESET 0: Assert 1: De-assert Write "1" then write "0" to finish reset operation.

6.2.3.3VE Counter Control Register

Address : 0x40007400 + 0x0008			Name: VE_CNT_REG Default: 0x0000 0000
Field	Type	Default	Description
31	R/W	0x0	VE_CNT_EN 0: counter stop 1: counter enable VE_CNT will be clear to 0 when write "1" to this bit.
30:0	R/W	0x0	VE_CNT JPEG encoder counter.

6.2.3.4VE Overtime Register

Address : 0x40007400 + 0x000C			Name: VE_OVERTIME_REG Default: 0x0000 0000
Field	Type	Default	Description
31	/	/	/
30:8	R/W	0x0	VE_OVERTIME When VE_CNT > VE_OVERTIME, VE_OVER_FLAG will be set, and VE_CNT will stop until VE_CNT be initial. VE_OVER_FLAG will be send to all modules for interrupt output.
7:0	/	/	Hardware wired to 8'h00

6.2.3.5 VE Interrupt Status Register

Address : 0x40007400 + 0x001C			Name: VE_INT_STA_REG Default: 0x0000 0000
Field	Type	Default	Description
31:26	/	/	/
25	R	0x0	AHB_SYNC_IDLE Read only. 0: some data still in ahb bus; 1: all data has been written to memory;
24	R/W1C	0x0	CSI_WB_FINISH_INT_PD CSI write back finish interrupt pending. Software write “1” to clear this pending. Only used in csi offline mode.
23	R/W1C	0x0	FIFO_OVERFLOW_INT_PD CSI&JPEG fifo overflow interrupt pending. Software write “1” to clear this pending.
22	R	0x0	CSI_ERROR CSI path error flag. Read only.
21	R	0x0	CSI_TIMEOUT CSI timeout flag. Read only.
20:11	/	/	/
10	R/W1C	0x0	MEM_PARTITION_OVERFLOW All memory blocks are full. To avoid overflow, software should take away a memory block data when MEM_PARTITION_INTERRUPT equals “1”, and write “1” to OUTMEM_PARTITION_TAKE (VE_MODE_REG[17]).
9	R/W1C	0x0	MEM_PARTITION_INTERRUPT The bit is set “1” by hardware when one memory block data is available for software.
8	R/W	0x0	VE_TIMEOUT_INT_EN VE timeout interrupt enable. 0: disable 1: enable
7	R/W1C	0x0	MB_OVERTIME_INT_PD Marcoblock overtime interrupt pending. Software write “1” to clear this pending.
6	R/W1C	0x0	BS_STALL_INT_PD Bitstream stall interrupt pending. Software write “1” to clear this pending.
5:4	/	/	/
3	R/W1C	0x0	ENC_FINISH_INT_PD JPEG Encode finish interrupt pending. Software write “1” to clear this pending.

2	R	0x0	CSI_FRAME_END CSI frame end flag. Read only.
1	R	0x0	CSI_SIZE_CHG CSI size change flag. Read only.
0	R/W1C	0x0	VE_TIMEOUT_INT_PD VE timeout interrupt pending. Software write "1" to clear this pending.

6.2.3.6 CSI Output Y Address Register

Address : 0x40007400 + 0x0020			Name : CSI_OUTPUT_ADDR_Y Default : 0x0000 0000
Field	Type	Default	Description
31:0	R/W	0x0	CSI output Luma(Y) data address. NV12 format. For offline mode only. When SENSOR_OUTPUT_JPEG equals one, JPEG data is stored in CSI_OUTPUT_ADDR_Y, and CSI_OUTPUT_ADDR_UV would be ignored.

6.2.3.7 CSI Output UV Address Register

Address : 0x40007400 + 0x0024			Name : CSI_OUTPUT_ADDR_UV Default : 0x0000 0000
Field	Type	Default	Description
31:0	R/W	0x0	CSI output Chroma(UV) data address. NV12 format. For offline mode only. When SENSOR_OUTPUT_JPEG equals one, JPEG data is stored in CSI_OUTPUT_ADDR_Y, and CSI_OUTPUT_ADDR_UV would be ignored.

6.2.3.8 CSI Output Stride Control Register

Address : 0x40007400 + 0x0028			Name : CSI_OUTPUT_STRIDE Default : 0x0000 0000
Field	Type	Default	Description
31:27	/	/	/
26:16	R/W	0x0	CSI_OUTPUT_Y_STRIDE_DIV16 CSI output Luma stride in 16x16 block. For offline mode only.
15:12	/	/	/
11:0	R/W	0x0	CSI_OUTPUT_UV_STRIDE_DIV8 CSI output Chroma stride in 8x8 block. For offline mode only.

6.2.3.9 JPEG Input Size Register

Address : 0x40007800 + 0x0000	Name : JPEG_INPUT_SIZE
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			Default: 0x0000 0000
Field	Type	Default	Description
31:27	/	/	/
26:16	R/W	0x0	PIC_WIDTH_IN_8x8 Block The width of each picture in units of 8X8 macroblock.
15:11	/	/	/
10:0	R/W	0x0	PIC_HEIGHT_IN_8x8 Block The height of each picture in units of 8X8 macroblock.

6.2.3.10 JPEG Input Stride Control Register

Address : 0x40007800 + 0x0004			Name: JPEG_INPUT_STRIDE Default: 0x0000 0000
Field	Type	Default	Description
31:27	/	/	/
26:16	R/W	0x0	JPEG_INPUT_STRIDE_DIV16 JPEG input picture stride in macroblocks. This parameter should be set to even to get better memory efficiency. If input mode has two or three planar, this parameter only used for Y planar.
15:0	/	/	/

6.2.3.11 JPEG Input Stride1 Control Register

Address : 0x40007800 + 0x0014			Name: JPEG_INPUT_STRIDE1 Default: 0x0000 0000
Field	Type	Default	Description
31:12	/	/	/
11:0	R/W	0x0	JPEG_INPUT_C_STRIDE_DIV8 JPEG input picture stride of Chroma planar. If input mode only has one planar, this parameter is not used.

6.2.3.12 JPEG Input Y Address Register

Address : 0x40007800 + 0x0078			Name: JPEG_INPUT_ADDR_Y Default: 0x0000 0000
Field	Type	Default	Description
31:4	R/W	0x0	JPEG_INPUT_ADDR_Y[31:4] JPEG input Luma(Y) data address. NV12 format. For offline mode only.
3:0	R	0x0	JPEG_INPUT_ADDR_Y[3:0] wired to 4h'0.

6.2.3.13 JPEG Input UV Address Register

Address : 0x40007800 + 0x007C			Name: JPEG_INPUT_ADDR_UV Default: 0x0000 0000
Field	Type	Default	Description
31:3	R/W	0x0	JPEG_INPUT_ADDR_UV[31:3] JPEG input Chroma(UV) data address.NV12 format.For offline mode only.
2:0	R	0x0	JPEG_INPUT_ADDR_UV[2:0] wired to 3h'0.

6.2.3.14 JPEG Para Control Register

Address : 0x40007C00 + 0x0004			Name: JPEG_PARA_REG Default: 0x0000 0000
Field	Type	Default	Description
31	/	/	/
30	R/W	0x0	STUFF_ZERO_EN Stuffing Zero Insert Disable 0: Disable insert stuffing zero after 0xff 1: Insert stuffing zero after 0xff
29:27	/	/	/
26:16	R/W	0x0	DC_CHRO JPEG Encoder Chroma Default DC
15:11	/	/	/
10:0	R/W	0x0	DC_LUMA JPEG Encoder Luma Default DC

6.2.3.15 JPEG Bitrate Control Register

Address : 0x40007C00 + 0x0008			Name: JPEG_BITRATE_CTRL Default: 0x0000 0000
Field	Type	Default	Description
31	R/W	0x0	run_length_opt_disable 0: run_length_th enable 1: run_length_th disable Suggested val is 1.
30	R/W	0x0	coef_down_sample_disable 0: classify_th enable 1: classify_th disable Suggested val is 1.
29:12	/	/	/
11:8	R/W	0x0	run_length_th

			Value: 0~15. The larger run_length_th val, the larger bit stream size. Suggested val is 8.
7:0	R/W	0x0	classify_th The larger classify_th val, the smaller bit stream size. Suggested val is 48.

6.2.3.16 VE Interrupt Enable Register

Address : 0x40007C00 + 0x0014			Name : VE_INT_EN_REG Default : 0x0000 0000
Field	Type	Default	Description
31:24	R	0x0	MB_X_CURR Readonly (for debug)
23:16	R	0x0	MB_Y_CURR Readonly (for debug)
15:3	/	/	/
2	R/W	0x0	MB_OVERTIME_INT_EN Marcolever overtime interrupt enable 0: disable 1: enable
1	R/W	0x0	BS_STALL_INT_EN Bitstream stall interrupt enable 0: disable 1: enable
0	R/W	0x0	VE_FINISH_INT_EN VE encode finish interrupt enable 0: disable 1: enable

6.2.3.17 VE Start Trigger Register

Address : 0x40007C00 + 0x0018			Name : VE_START_TRIG Default : 0x0000 0000
Field	Type	Default	Description
31:14	/	/	/
13:8	R/W	0x0	PUTBITS_LEN The bits number to put.
7:4	/	/	/
3:0	R/W	0x0	1000: VE encode start Others: reserved

6.2.3.18 VE Status Register

Address : 0x40007C00 + 0x001C			Name : VE_STA_REG
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			Default: 0x0000 0000
Field	Type	Default	Description
31	R	0x0	PIC_ENC_BUSY Readonly (for debug)
30:24	/	/	/
23	/	/	READ_MB_BUSY Readonly (for debug)
22:14	/	/	/
13	R	0x0	IPTIT_BUSY Readonly (for debug)
12	R	0x0	VLC_BUSY Readonly (for debug)
11:10	/	/	/
9	R	0x0	PUTBITS_STATUS 0: Put bits busy 1: Put bits free
8	/	/	/
7	R	0x0	ENC_MEM_BUSY Readonly (for debug)
6	R	0x0	ENC_ISP Readonly (for debug)
5:0	/	/	/

6.2.3.19 VE Putbits Data Register

Address : 0x40007C00 + 0x0020			Name: VE_PUTBITS_DATA Default: 0x0000 0000
Field	Type	Default	Description
31:0	R/W	0x0	PUTBITS_DATA The bits data to put.

6.2.3.20 Macroblock Level Overtime Register

Address : 0x40007C00 + 0x0024			Name: MBLEVEL_OVERTIME Default: 0x0000 0000
Field	Type	Default	Description
31:16	/	/	/
15	R/W	0x0	MBOVERTIME_EN 0: disable mblevel overtime counter 1: enable mblevel overtime counter
14:12	R/W	0x0	MBOVERTIME_THRESHOLD[14:12] Macroblock level encode overflow threshold, if mb level encode time

			overflow, it will set MBOVERTIME_INTERRUPT to "1", but it continue encode next macroblock if it can go on.
11:0	R	0x0	MBOVERTIME_THRESHOLD[11:0] Hardware wired to 12'h0.

6.2.3.21 Output Stream Start address Register

Address : 0x40007C00 + 0x0080			Name: OUTSTM_START_ADDR Default: 0x0000 0000
Field	Type	Default	Description
31:10	R/W	0x0	STREAM_START_ADDR[31:10] The start address of JPEG encoded output stream.It must configure OUTSTRM_OFFSET register before set this register.
9:0	R	0x0	STREAM_START_ADDR[9:0] Hardware wired to 10'h0.

6.2.3.22 Output Stream End address Register

Address : 0x40007C00 + 0x0084			Name: OUTSTM_END_ADDR Default: 0xFFFF FFFF
Field	Type	Default	Description
31:10	R/W	0x0	STREAM_END_ADDR[31:10] The end address of JPEG encoded output stream.
9:0	R	0x3FF	STREAM_END_ADDR[9:0] Hardware wired to 10'h3FF.

6.2.3.23 Output Stream Offset Register

Address : 0x40007C00 + 0x0088			Name: OUTSTM_OFFSET Default: 0x0000 0000
Field	Type	Default	Description
31:28	/	/	/
27:0	R	0x0	STREAM_OFFSET JPEG Encoded bitstream bit offset Software configuration register, but it's read value hardware_offset register. It must be 256 bit align.

6.2.3.24 Output Stream Valid Size Register

Address : 0x40007C00 + 0x008C			Name: OUTSTM_VSIZE Default: 0x0FFF 0000
Field	Type	Default	Description
31:28	/	/	/
27:16	R/W	0x0	STREAM_VALID_SIZE

			JPEG Encoded bitstream valid output memory space size.
15:0	/	/	/

6.2.3.25 Output Stream Length Register

Address : 0x40007C00 + 0x0090			Name : OUTSTM_LEN Default : 0x0000 0000
Field	Type	Default	Description
31:28	/	/	/
27:0	R	0x0	OUTSTM_LEN JPEG Encoded stream bit offset. Attention of that it is always great than STREAMOFFSET even it is overflow. Output bitstream bit size is OUTSTM_LEN minus STREAM_OFFSET.

6.2.3.26 VE Quantiser Matrix Input Index Register

Address : 0x40007C00 + 0x00E0			Name : VE_QMINPUT_INDEX Default : 0x0000 0000
Field	Type	Default	Description
31:7	/	/	/
6:0	R/W	0x0	QM_INDEX[6:0] The Index of Quantiser Matrix.

6.2.3.27 VE Quantiser Matrix Input Data Register

Address : 0x40007C00 + 0x00E4			Name : VE_QMINPUT_DATA Default : 0x0000 0000
Field	Type	Default	Description
31:0	R/W	0x0	QM_DATA[31:0] The Quantiser Matrix Data.

Chapter 7 GPIO

The XR872 GPIO unit provides as many as 33 GPIO (General Purpose IO) pins. All ports are brought out of the device using alternate function multiplexing. The GPIO function can be multiplexed on a multi-function I/O pin by selecting the GPIO alternate function in the GPIO Controller registers.

There are two types of GPIO designs in XR872: GPIO and AGPIO. Each GPIO can be configured with the following options:

- Input / Output / Floating(Hi-Z) mode
- Input mode: Pull-up or Pull-down
- Output mode: Active driving
- Pull-up/down control: the pull-up and pull-down resistance is 90KΩ with $\pm 30\%$ variation over PVT condition
- External Interrupt IO with 5 trigger modes: high-level, low-level, rising edge, falling edge, double edge
- 9 WAKEUP IOs can be set to wake system by external interrupt at HIBERNATION mode (RTC on only)
- All IOs can be set to wake system by external interrupt in STANDBY mode (RTC and OA domain on)

The digital IO AGPIO function is equivalent to GPIO as shown above. A dedicated internal control signal is used to select between the digital and analog functions. These IOs are multiplexed with 8 channels ADC (1 is internal connected to measure VBAT voltage).

GPIO PA23 has a special function which is used to enter test mode when it is high on first power-up. So we need to keep it from pulling high (floating or tie low) to have whole chip power up correctly.

7.1.1 Register List

Module Name	Base Address	
GPIO Controller	0x40050000	
Register Name	Offset Address	Description
Pn_CFG0	$n * 0x24 + 0x00$	Port n Configure Register 0 ($n = 0,1$)
Pn_CFG1	$n * 0x24 + 0x04$	Port n Configure Register 1 ($n = 0,1$)
Pn_CFG2	$n * 0x24 + 0x08$	Port n Configure Register 2 ($n = 0,1$)
Pn_CFG3	$n * 0x24 + 0x0C$	Port n Configure Register 3 ($n = 0,1$)
Pn_DATA	$n * 0x24 + 0x10$	Port n Data Register ($n = 0,1$)
Pn_DRV0	$n * 0x24 + 0x14$	Port n Multi-Driving Register 0 ($n = 0,1$)
Pn_DRV1	$n * 0x24 + 0x18$	Port n Multi-Driving Register 1 ($n = 0,1$)
Pn_PULL0	$n * 0x24 + 0x1C$	Port n Pull Register 0 ($n = 0,1$)
Pn_PULL1	$n * 0x24 + 0x20$	Port n Pull Register 1 ($n = 0,1$)

Pn_INT_CFG0	0x200 + n*0x20 + 0x00	PIO Interrupt Configure Register 0
Pn_INT_CFG1	0x200 + n*0x20 + 0x04	PIO Interrupt Configure Register 1
Pn_INT_CFG2	0x200 + n*0x20 + 0x08	PIO Interrupt Configure Register 2
Pn_INT_CFG3	0x200 + n*0x20 + 0x0C	PIO Interrupt Configure Register 3
Pn_INT_CTL	0x200 + n*0x20 + 0x10	PIO Interrupt Control Register
Pn_INT_STA	0x200 + n*0x20 + 0x14	PIO Interrupt Status Register
Pn_INT_DBC	0x200 + n*0x20 + 0x18	PIO Interrupt Debounce Register

7.1.2 Register Description

7.1.2.1 Pn Configure Register 0

Address : n*0x24 + 0x00			Name : Pn_CFG0 Default :
Field	Type	Default	Description
[4*m+3:4*m] (m = 0~7)	R/W	0x7	GPIO Group n IO m Config

7.1.2.2 Pn Configure Register 1

Address : n*0x24 + 0x04			Name : Pn_CFG1 Default :
Field	Type	Default	Description
[4*(m-8)+3:4*(m-8)] (m = 8~15)	R/W	0x7	GPIO Group n IO m Config <i>Note: The function numbers please refer to the AW1735 Pin Multiplexing. If this IO is not exist, this field is reserved.</i>

7.1.2.3 Pn Configure Register 2

Address : n*0x24 + 0x08			Name : Pn_CFG2 Default :
Field	Type	Default	Description
[4*(m-16)+3:4*(m-16)] (m = 16~23)	R/W	0x7	GPIO Group n IO m Config <i>Note: The function numbers please refer to the AW1735 Pin Multiplexing. If this IO is not exist, this field is reserved.</i>

7.1.2.4 Pn Configure Register 3

Address : n*0x24 + 0x0C			Name : Pn_CFG3
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			Default:
Field	Type	Default	Description
$[4*(m-24)+3:4*(m-24)]$ ($m = 24 \sim 31$)	R/W	0x7	GPIO Group n IO m Config <i>Note: The function numbers please refer to the AW1735 Pin Multiplexing. If this IO is not exist, this field is reserved.</i>

7.1.2.5 Pn Data Register

Address : $n*0x24 + 0x10$			Name: Pn_DATA Default:
Field	Type	Default	Description
31:0	R/W	0	DATA If the port is configured as input, the corresponding bit is the pin state. If the port is configured as output, the pin state is the same as the corresponding bit. The read bit value is the value setup by software. If the port is configured as functional pin, the undefined value will be read.

7.1.2.6 Pn Multi-Driving Register 0

Address : $n*0x24 + 0x14$			Name: Pn_DRV0 Default:
Field	Type	Default	Description
$[2*m+1:2*m]$ ($m = 15 \sim 0$)	R/W	0x1	DRV_STRENGTH 00: level 0 01: level 1 10: level 2 11: level 3 <i>Note: This field is only valid when the related IO is exist.</i>

7.1.2.7 Pn Multi-Driving Register 1

Address : $n*0x24 + 0x18$			Name: Pn_DRV1 Default:
Field	Type	Default	Description
$[2*(m-16)+1:2*(m-16)]$ ($m = 31 \sim 16$)	R/W	0x1	DRV_STRENGTH 00: level 0 01: level 1 10: level 2 11: level 3 <i>Note: This field is only valid when the related IO is exist.</i>

7.1.2.8 Pn Pull Register 0

Address : $n*0x24 + 0x1C$	Name: Pn_PULL0
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			Default:
Field	Type	Default	Description
[2*m+1:2*m] (m = 15~0)	R/W	0x0	PULL 00: NO PULL 01: Pull-Up 10: Pull-Down 11: / <i>Note: This field is only valid when the related IO is exist.</i>

7.1.2.9 Pn Pull Register 1

Address : n*0x24 + 0x20			Name: Pn_PULL0 Default:
Field	Type	Default	Description
[2*(m-16)+1:2*(m-16)] (m = 31~16)	R/W	0x0	PULL 00: NO PULL 01: Pull-Up 10: Pull-Down 11: / <i>Note: This field is only valid when the related IO is exist.</i>

7.1.2.10 Pn External Interrupt Configure 0 Register

Address : 0x200 + 0x20*n + 0x00			Name: Pn_EINT_CFG0 Default:
Field	Type	Default	Description
[4*m+3:4*m] (m = 0~7)	R/W	0x0	EINT_MODE 0: Positive Edge 1: Negative Edge 2: High Level 3: Low Level 4: Double Edge others: / <i>Note: This field is only valid when the related IO is exist.</i>

7.1.2.11 Pn External Interrupt Configure 1 Register

Address : 0x200 + 0x20*n + 0x04			Name: Pn_EINT_CFG1 Default:
Field	Type	Default	Description
[4*(m-8)+3:4*(m-8)] (m = 8~15)	R/W	0x0	EINT_MODE 0: Positive Edge

			1: Negative Edge 2: High Level 3: Low Level 4: Double Edge others: / <i>Note: This field is only valid when the related IO is exist.</i>
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7.1.2.12 Pn External Interrupt Configure 2 Register

Address : 0x200 + 0x20*n + 0x08			Name: Pn_EINT_CFG2 Default:
Field	Type	Default	Description
[4*(m-16)+3:4 *(m-16)] (m = 16~23)	R/W	0x0	EINT_MODE 0: Positive Edge 1: Negative Edge 2: High Level 3: Low Level 4: Double Edge others: / <i>Note: This field is only valid when the related IO is exist.</i>

7.1.2.13 Pn External Interrupt Configure 3 Register

Address : 0x200 + 0x20*n + 0x0C			Name: Pn_EINT_CFG3 Default:
Field	Type	Default	Description
[4*(m-24)+3:4 *(m-24)] (m = 24~31)	R/W	0x0	EINT_MODE 0: Positive Edge 1: Negative Edge 2: High Level 3: Low Level 4: Double Edge others: / <i>Note: This field is only valid when the related IO is exist.</i>

7.1.2.14 Pn External Interrupt Control Register

Address : 0x200 + 0x20*n + 0x10	Name: Pn_EINT_CTRL
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			Default:
Field	Type	Default	Description
m (m = 31~0)	R/W	0x0	EINT_IRQEN 0: disable 1: enable <i>Note: This field is only valid when the related IO is exist.</i>

7.1.2.15 Pn External Interrupt Status Register

Address : 0x200 + 0x20*n + 0x14			Name: Pn_EINT_ST Default:
Field	Type	Default	Description
[m] (m = 31~0)	R/W	0x0	EINT_IRQST 0: nothing 1: irq pending write 1 to clear <i>Note: This field is only valid when the related IO is exist.</i>

7.1.2.16 Pn External Interrupt Debounce Register

Address : 0x200 + 0x20*n + 0x18			Name: Pn_DBC Default:
Field	Type	Default	Description
31:7	/	/	/
6:4	R/W	0	Debounce Clock Pre-Scale n The selected clock is prescaled by 2^n
3:1	/	/	/
0	R/W	0x0	PIO_INT_CLK_SEL 0: LFCLK 1: HFCLK

7.1.3 Pin Multiplexing

GPIO	FUNC2	FUNC3	FUNC4	FUNC5	FUNC6
PA00	SPI1_MOSI	SD_CMD	TWI1_SCL	CSI_D0	EINTA0
PA01	SPI1_MISO	SD_DATA0	TWI1_SDA	CSI_D1	EINTA1
PA02	SPI1_CLK	SD_CLK	UART0_TX	CSI_D2	EINTA2
PA03	SPI1_CS0	SD_DATA1	UART0_RX	CSI_D3	EINTA3
PA04/WUPIO0	UART1_RTS	SD_DATA2	TWI0_SCL	CSI_D4	EINTA4

PA05/WUPIO1	UART1_CTS	SD_DATA3	TWI0_SDA	CSI_D5	EINTA5
PA06/WUPIO2	UART1_RX	SPI1_CS1	TWI0_SCL	CSI_D6	EINTA6
PA07/WUPIO3	UART1_TX	SPI1_CS2	TWI0_SDA	CSI_D7	EINTA7
PA08	FEM_CTRL1	PWM0/ECT0	TWI1_SCL	CSI_PCLK	EINTA8
PA09	FEM_CTRL2	PWM1/ECT1	TWI1_SDA	CSI_MCLK	EINTA9
PA10	ADC_CH0	PWM2/ECT2	DMIC_CLK	CSI_HSYNC	EINTA10
PA11	ADC_CH1	PWM3/ECT3	DMIC_DATA	CSI_VSYNC	EINTA11
PA12	ADC_CH2	PWM4/ECT4	I2S_MCLK	IR_TX	EINTA12
PA13	ADC_CH3	PWM5/ECT5	I2S_BCLK	UART1_TX	EINTA13
PA14	ADC_CH4	PWM6/ECT6	I2S_DI	UART1_RX	EINTA14
PA15	ADC_CH5	PWM7/ECT7	I2S_DO	UART1_CTS	EINTA15
PA16	ADC_CH6	IR_RX	I2S_LRCLK	UART1_RTS	EINTA16
PA17	TWI0_SCL	IR_RX	/	FEM_CTRL1	EINTA17
PA18	TWI0_SDA	IR_TX	FEM_CTRL2	IR_RX	EINTA18
PA19/WUPIO5	UART2_RTS	TWI0_SCL	PWM0/ECT0	SPI1_MOSI	EINTA19
PA20/WUPIO6	UART2_CTS	TWI0_SDA	PWM1/ECT1	SPI1_MISO	EINTA20
PA21/WUPIO7	UART2_RX	DMIC_CLK	PWM2/ECT2	SPI1_CLK	EINTA21
PA22/WUPIO8	UART2_TX	DMIC_DATA	PWM3/ECT3	SPI1_CS0	EINTA22
PA23/WUPIO9/TEST	EXT_DCDC_PUP	/	FEM_CTRL1	FEM_CTRL2	EINTA23
PB00	UART0_TX	JTAG_TMS	PWM4/ECT4	SWD_TMS	EINTB0
PB01	UART0_RX	JTAG_TCK	PWM5/ECT5	SWD_TCK	EINTB1
PB02	SWD_TMS	JTAG_TDO	PWM6/ECT6	FLASH_WP/IO2	EINTB2
PB03	SWD_TCK	JTAG_TDI	PWM7/ECT7	FLASH_HOLD/IO	EINTB3
PB04	SPI0_MOSI	SD_CMD	UART1_TX	FLASH_MOSI/IO0	EINTB4
PB05	SPI0_MISO	SD_DATA0	UART1_RX	FLASH_MISO/IO1	EINTB5
PB06	SPI0_CS0	FEM_CTRL2	UART1_CTS	FLASH_CS	EINTB6
PB07	SPI0_CLK	SD_CLK	UART1_RTS	FLASH_CLK	EINTB7
PB16	/	SD_CMD	/	/	EINTB16
PB17	/	SD_DATA0	/	/	EINTB17
PB18	/	SD_CLK	/	/	EINTB18

Table 7-1 Pin Multiplexing

